

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User (Modify in Config->Save Options)

Job File : ...\\120-C-001 DBEop case 1 REV.01.pvdb

Date of Analysis : Sep 28,2026 12:59pm

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1800.000 mm ID	
Vessel Design Length, Tangent to Tangent	38000.00 mm
Distance of Bottom Tangent above Grade	6300.00 mm
Distance of Base above Grade	300.00 mm
Specified Datum Line Distance	6000.00 mm
Internal Design Temperature	125 °C
Internal Design Pressure	2.157 MPa
External Design Temperature	190 °C
External Design Pressure	0.049 MPa
Maximum Allowable Working Pressure	2 MPa
External Max. Allowable Working Pressure	0 MPa
Shop Test Pressure	3.185 MPa
Field Test Pressure	3.185 MPa
Required Minimum Design Metal Temperature	-25.0 °C
Warmest Computed Minimum Design Metal Temperature	-17.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
 Earthquake Design Code Response Spectrum

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Skirt	SA-36	...	...	K02600	
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	+IT
Nozzle	SA-516 70	...	...	K02700	
Re-Pad	SA-516 70	...	...	K02700	+IT
Nozzle Flg	SA-105	...	...	K03504	+IT
Basering	SA-516 70	...	...	K02700	
Base Bolting	SA-36	...	...	K02600	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Bottom skirt	...	...	...	1.0000	N/A	No
Top skirt	...	...	...	1.0000	N/A	No
Bottom head	2.182	0.05	4.000	3.0000	No	No
Shell 1	2.179	0.05	4.000	3.0000	N/A	No
Shell 2	2.164	0.05	4.000	3.0000	N/A	No
Shell 3	2.157	0.05	4.000	3.0000	N/A	No
Shell 4	2.157	0.05	4.000	3.0000	N/A	No
Shell 5	2.157	0.05	4.000	3.0000	N/A	No
Shell 6	2.157	0.05	4.000	3.0000	N/A	No
Shell 7	2.157	0.05	4.000	3.0000	N/A	No
Shell 8	2.157	0.05	4.000	3.0000	N/A	No
Shell 9	2.157	0.05	4.000	3.0000	N/A	No



Shell 10	2.157	0.05	4.000	3.0000	N/A	No
Shell 11	2.157	0.05	3.000	3.0000	N/A	No
Shell 12	2.157	0.05	3.000	3.0000	N/A	No
Shell 13	2.157	0.05	3.000	3.0000	N/A	No
Top head	2.157	0.05	2.000	3.0000	No	No

Liquid Level: 4750.00 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 0.520

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Skirt	-3000.0	3000.0	20.0	20.0	...	...	0.70	0.70
Skirt	0.0	3000.0	28.0	28.0	...	...	0.70	0.49
Ellipse	100.0	100.0	32.0	27.5	17.2	6.4	1.00	1.00
Cylinder	3100.0	3000.0	30.0	30.0	17.4	15.9	1.00	1.00
Cylinder	6100.0	3000.0	30.0	30.0	17.3	15.9	1.00	1.00
Cylinder	9100.0	3000.0	30.0	30.0	17.3	15.9	1.00	1.00
Cylinder	12100.0	3000.0	30.0	30.0	17.3	15.9	1.00	1.00
Cylinder	15100.0	3000.0	30.0	30.0	17.3	15.9	1.00	1.00
Cylinder	18100.0	3000.0	30.0	30.0	17.3	15.9	1.00	1.00
Cylinder	21100.0	3000.0	28.0	28.0	17.3	15.9	1.00	1.00
Cylinder	24100.0	3000.0	28.0	28.0	17.3	15.9	1.00	1.00
Cylinder	27100.0	3000.0	28.0	28.0	17.3	15.9	1.00	1.00
Cylinder	30100.0	3000.0	28.0	28.0	17.3	15.9	1.00	1.00
Cylinder	33100.0	3000.0	25.0	25.0	17.3	15.8	1.00	1.00
Cylinder	36100.0	3000.0	25.0	25.0	17.3	15.8	1.00	1.00
Cylinder	37900.0	1800.0	22.0	22.0	17.3	15.8	1.00	1.00
Ellipse	38000.0	100.0	22.0	19.0	17.1	6.3	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Trunnion	WRC-107/537	0.437	n/a	Passed
Trunnion	WRC-107/537	0.437	n/a	Passed
Trunnion	WRC-107/537	0.437	n/a	Passed
Trunnion	WRC-107/537	0.437	n/a	Passed
Trunnion	WRC-107/537	0.437	n/a	Passed
Trunnion	WRC-107/537	0.437	n/a	Passed
Trunnion	WRC-107/537	0.436	n/a	Passed
Trunnion	WRC-107/537	0.436	n/a	Passed
Trunnion	WRC-107/537	0.435	n/a	Passed
Trunnion	WRC-107/537	0.434	n/a	Passed
Trunnion	WRC-107/537	0.433	n/a	Passed
Trunnion	WRC-107/537	0.432	n/a	Passed
Trunnion	WRC-107/537	0.431	n/a	Passed
Trunnion	WRC-107/537	0.430	n/a	Passed
Trunnion	WRC-107/537	0.428	n/a	Passed
Trunnion	WRC-107/537	0.427	n/a	Passed
Trunnion	WRC-107/537	0.425	n/a	Passed
Trunnion	WRC-107/537	0.423	n/a	Passed
Trunnion	WRC-107/537	0.421	n/a	Passed
Trunnion	WRC-107/537	0.419	n/a	Passed
Trunnion	WRC-107/537	0.417	n/a	Passed
Trunnion	WRC-107/537	0.415	n/a	Passed
Trunnion	WRC-107/537	0.413	n/a	Passed
Trunnion	WRC-107/537	0.410	n/a	Passed
Trunnion	WRC-107/537	0.408	n/a	Passed
Trunnion	WRC-107/537	0.405	n/a	Passed
Trunnion	WRC-107/537	0.402	n/a	Passed
Trunnion	WRC-107/537	0.399	n/a	Passed
Trunnion	WRC-107/537	0.396	n/a	Passed
Trunnion	WRC-107/537	0.393	n/a	Passed
Trunnion	WRC-107/537	0.390	n/a	Passed
Trunnion	WRC-107/537	0.387	n/a	Passed
Trunnion	WRC-107/537	0.383	n/a	Passed



Trunnion	WRC-107/537	0.380	n/a	Passed
Trunnion	WRC-107/537	0.376	n/a	Passed
Trunnion	WRC-107/537	0.373	n/a	Passed
Trunnion	WRC-107/537	0.369	n/a	Passed
Trunnion	WRC-107/537	0.365	n/a	Passed
Trunnion	WRC-107/537	0.361	n/a	Passed
Trunnion	WRC-107/537	0.357	n/a	Passed
Trunnion	WRC-107/537	0.353	n/a	Passed
Trunnion	WRC-107/537	0.348	n/a	Passed
Trunnion	WRC-107/537	0.344	n/a	Passed
Trunnion	WRC-107/537	0.340	n/a	Passed
Trunnion	WRC-107/537	0.335	n/a	Passed
Trunnion	WRC-107/537	0.331	n/a	Passed
Trunnion	WRC-107/537	0.326	n/a	Passed
Trunnion	WRC-107/537	0.321	n/a	Passed
Trunnion	WRC-107/537	0.316	n/a	Passed
Trunnion	WRC-107/537	0.311	n/a	Passed
Trunnion	WRC-107/537	0.306	n/a	Passed
Trunnion	WRC-107/537	0.301	n/a	Passed
Trunnion	WRC-107/537	0.296	n/a	Passed
Trunnion	WRC-107/537	0.291	n/a	Passed
Trunnion	WRC-107/537	0.286	n/a	Passed
Trunnion	WRC-107/537	0.281	n/a	Passed
Trunnion	WRC-107/537	0.275	n/a	Passed
Trunnion	WRC-107/537	0.270	n/a	Passed
Trunnion	WRC-107/537	0.264	n/a	Passed
Trunnion	WRC-107/537	0.259	n/a	Passed
Trunnion	WRC-107/537	0.253	n/a	Passed
Trunnion	WRC-107/537	0.248	n/a	Passed
Trunnion	WRC-107/537	0.242	n/a	Passed
Trunnion	WRC-107/537	0.236	n/a	Passed
Trunnion	WRC-107/537	0.232	n/a	Passed
Trunnion	WRC-107/537	0.234	n/a	Passed
Trunnion	WRC-107/537	0.236	n/a	Passed
Trunnion	WRC-107/537	0.237	n/a	Passed
Trunnion	WRC-107/537	0.239	n/a	Passed
Trunnion	WRC-107/537	0.241	n/a	Passed
Trunnion	WRC-107/537	0.243	n/a	Passed
Trunnion	WRC-107/537	0.244	n/a	Passed
Trunnion	WRC-107/537	0.246	n/a	Passed
Trunnion	WRC-107/537	0.248	n/a	Passed
Trunnion	WRC-107/537	0.250	n/a	Passed
Trunnion	WRC-107/537	0.252	n/a	Passed
Trunnion	WRC-107/537	0.254	n/a	Passed
Trunnion	WRC-107/537	0.256	n/a	Passed
Trunnion	WRC-107/537	0.259	n/a	Passed
Trunnion	WRC-107/537	0.262	n/a	Passed
Trunnion	WRC-107/537	0.265	n/a	Passed
Trunnion	WRC-107/537	0.268	n/a	Passed
Trunnion	WRC-107/537	0.272	n/a	Passed
Trunnion	WRC-107/537	0.277	n/a	Passed
Trunnion	WRC-107/537	0.282	n/a	Passed
Trunnion	WRC-107/537	0.289	n/a	Passed
Trunnion	WRC-107/537	0.298	n/a	Passed
Trunnion	WRC-107/537	0.311	n/a	Passed
Trunnion	WRC-107/537	0.330	n/a	Passed
Trunnion	WRC-107/537	0.361	n/a	Passed
Trunnion	WRC-107/537	0.423	n/a	Passed
Trunnion	WRC-107/537	0.437	n/a	Passed
B2 10"	WRC-107/537	0.538	n/a	Passed
B1 4 "ASSUME #	WRC-107/537	0.524	n/a	Passed
F3 #300	WRC-297	0.529	Shell	Passed
F1 4" #300	WRC-297	0.573	Shell	Passed
F2 6" #300	WRC-297	0.747	Shell	Passed
T 8" #300	WRC-107/537	0.794	n/a	Passed
V1 4" #300	WRC-107/537	0.794	n/a	Passed



From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1500	218755	237234	5.469E+09	5.553E+09
20	30	4500	208285	233693	4.828E+09	4.965E+09
30	40	6050	193609	227262	4.225E+09	4.415E+09
40	50	7600	190488	220473	4.206E+09	4.398E+09
50	60	10600	183354	205810	3.645E+09	3.900E+09
60	70	13600	170566	189277	3.114E+09	3.454E+09
70	80	16600	156447	170262	2.623E+09	3.060E+09
80	90	19600	141486	154661	2.176E+09	2.707E+09
90	100	22600	133324	140964	1763848320	2.385E+09
100	110	25600	118436	129873	1386055040	2081589504
110	120	28600	102802	121944	1054063872	1783863936
120	130	31600	86466	118695	770047232	1476182528
130	140	34600	77459	117596	524059968	1152891008
140	150	37600	61164	115212	316041088	817244608
150	160	40600	44442.5	106078	157567056	476962720
160	170	43000	27288.3	86240.3	49927292	159060688
170	180	43950	21641.3	37640.2	5872747	3765551

Abs Max of the all of the Stress Ratio's : 0.7704

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	55.0000	mm
Inside Diameter of Basing	3000.9998	mm
Outside Diameter of Basing	3451.0002	mm
Nominal Diameter of Bolts	64.0000	mm
Diameter of Bolt Circle	3299.0002	mm
Number of Bolts	20	

Thickness of Gusset Plates	20.0000	mm
Average Width of Gusset Plates	165.0000	mm
Height of Gussets	300.0000	mm
Distance between Gussets	110.0000	mm
Thickness of Top Plate or Ring	45.0000	mm
Radial Width of the Top Plate	165.0000	mm

Loads for Foundation/Support Design:

Total Wind Shear on Support	234312.	N
Total Earthquake Shear on Support	237234.	N
Wind Moment on Support	6184817664.	N-mm
Earthquake Moment on Support	5569087488.	N-mm
Wind Load Case Test Moment on Support	2051464064.	N-mm
Earthquake Load Case Test Moment on Support	15633091.	N-mm
Test Case Wind Shear on Support	77323.	N
Maximum Initial Bolt Load	300258.	N

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	85048.4	kgm
Shop Test - Fabricated + water (full)	188231.0	kgm
Shipping - Fab. + rem. intls.+ shipping app.	140448.3	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	145448.3	kgm
Empty - Fab. + intls. + details + wghts.	150348.4	kgm
Operating - Empty + operating liquid (no ca)	162030.4	kgm
Field Test - Empty weight + test medium (full)	243537.6	kgm



Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
B2 10"	4	OK	...	OK	11.10	OK	Passed
B1 4 "ASSUME #3	4	OK	...	OK	8.26	OK	Passed
Noz L1B #300		...	...	OK	7.80	OK	No Calc[*]
Noz L2B #300		...	...	OK	7.80	OK	No Calc[*]
Noz L3B #300		...	...	OK	7.80	OK	No Calc[*]
Noz L4B #300		...	...	OK	7.80	OK	No Calc[*]
Noz U2 #300		...	...	OK	7.80	OK	No Calc[*]
Noz U1 #300		...	...	OK	7.80	OK	No Calc[*]
Noz MW5 24" #30		...	...	...		OK	No Calc[*]
Noz L1A #300		...	...	OK	7.80	OK	No Calc[*]
Noz L2A #300		...	...	OK	7.80	OK	No Calc[*]
Noz L3A #300		...	...	...		OK	No Calc[*]
Noz L4A #300		...	...	...		OK	No Calc[*]
MW4 #300	4	OK	...	...		OK	Passed
P1 #300	4	...	...	OK	7.80	OK	No Calc[*]
P2 #300	4	...	...	OK	7.80	OK	No Calc[*]
F3 #300	4	OK	...	OK	11.10	OK	Passed
T3 #300	4	...	...	OK	7.80	OK	No Calc[*]
MW3 24" #300	4	OK	...	...		OK	Passed
T2 #300	4	...	...	OK	7.80	OK	No Calc[*]
F1 4" #300	3	OK	...	OK	8.26	OK	Passed
MW2 24" #300	3	OK	...	...		OK	Passed
T1 #300	3	...	...	OK	7.80	OK	No Calc[*]
MW1 24" #300	3	OK	...	...		OK	Passed
F2 6" #300	3	OK	...	OK	9.22	OK	Passed
T 8" #300	2	OK	...	OK	10.16	OK	Passed
V1 4" #300	2	OK	...	OK	8.02	OK	Passed
NV2 2" #300	7	...	...	OK	4.50	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 2 Nozzle : V1 4" #300
 Minimum MAWP Shells/Flanges : 2 Element : Top head
 Minimum MAPnc Shells/Flanges : 3 Element : Top head

Computed Vessel M.A.W.P. : 2 MPa

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
30	B2 10"	0.000	180.000	505.293
30	B1 4 "ASSUME #30	0.000	0.000	196.099
40	Noz L1B #300	350.000	180.000	138.400
40	Noz L2B #300	350.000	90.000	138.400
40	Noz L3B #300	350.000	270.000	138.400
40	Noz L4B #300	350.000	0.000	138.400
40	Noz U2 #300	150.000	180.000	138.400
40	Noz U1 #300	150.000	0.000	138.400
40	Noz MW5 24" #300	1600.000	180.000	1183.200
50	Noz L1A #300	4350.000	180.000	138.400
50	Noz L2A #300	4350.000	90.000	138.400
50	Noz L3A #300	4350.000	270.000	138.400
50	Noz L4A #300	4350.000	0.000	138.400



60	MW4	#300	7599.999	90.000	1183.200
60	P1	#300	7849.998	180.000	138.400
60	P2	#300	7649.998	0.000	138.400
60	F3	#300	6999.998	180.000	505.293
70	T3	#300	11760.000	180.000	138.400
80	MW3	24" #300	13899.999	270.000	1183.200
110	T2	#300	23750.002	180.000	134.400
120	F1	4" #300	25000.004	180.000	186.650
150	MW2	24" #300	34900.004	90.000	1183.200
160	T1	#300	36360.000	180.000	122.400
160	MW1	24" #300	37100.000	180.000	1167.200
160	F2	6" #300	37700.000	0.000	276.800
170	T	8" #300	0.000	0.000	370.400
170	V1	4" #300	0.000	180.000	167.200
170	NV2	2" #300	0.000	270.000	110.501

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Thk mm	Cut Length mm	Flg Class
Noz L1B #300	2.000 in	Actual	LWN	3.323	16.800	...	...	280.99	300
Noz L2B #300	2.000 in	Actual	LWN	3.323	16.800	...	...	280.99	300
Noz L3B #300	2.000 in	Actual	LWN	3.323	16.800	...	...	280.99	300
Noz L4B #300	2.000 in	Actual	LWN	3.323	16.800	...	...	280.99	300
Noz U2 #300	2.000 in	Actual	LWN	3.323	16.800	...	...	280.99	300
Noz U1 #300	2.000 in	Actual	LWN	3.323	16.800	...	...	280.99	300
Noz L1A #300	2.000 in	Actual	LWN	3.323	16.800	...	...	280.99	300
Noz L2A #300	2.000 in	Actual	LWN	3.323	16.800	...	...	280.99	300
Noz L3A #300	2.000 in	Actual	LWN	3.323	16.800	...	...	280.99	300
Noz L4A #300	2.000 in	Actual	LWN	3.323	16.800	...	...	280.99	300
P1 #300	2.000 in	Actual	LWN	3.323	16.800	...	...	258.76	300
P2 #300	2.000 in	Actual	LWN	3.323	16.800	...	...	258.76	300
T3 #300	2.000 in	Actual	LWN	3.323	16.800	...	...	280.99	300
T2 #300	2.000 in	Actual	LWN	3.323	16.800	...	...	278.99	300
T1 #300	2.000 in	Actual	LWN	3.323	16.800	...	...	272.99	300
NV2 2" #300	2.000 in	80	WNF	2.375	5.537	...	...	276.39	300
B1 4 "ASSUME #30	4.000 in	120	WNF	4.500	11.125	240.00	30.00	278.49	300
F1 4" #300	4.000 in	160	WNF	4.500	13.487	224.00	18.00	279.82	300
V1 4" #300	4.000 in	Actual	WNF	4.000	12.000	240.00	22.00	287.18	300
F2 6" #300	6.000 in	Actual	WNF	6.000	10.000	345.00	20.00	175.23	300
T 8" #300	8.000 in	Actual	WNF	8.000	12.000	440.00	22.00	322.16	300
B2 10"	10.000 in	80	WNF	10.750	15.088	540.00	28.00	283.19	300
F3 #300	10.000 in	80	WNF	10.750	15.088	540.00	28.00	290.42	300
Noz MW5 24" #300	24.000 in	Actual	WNF	24.000	12.000	1220.00	28.00	383.18	300
MW4 #300	24.000 in	Actual	WNF	24.000	12.000	1220.00	28.00	383.18	300
MW3 24" #300	24.000 in	Actual	WNF	24.000	12.000	1220.00	28.00	383.18	300
MW2 24" #300	24.000 in	Actual	WNF	24.000	12.000	1220.00	26.00	378.18	300
MW1 24" #300	24.000 in	Actual	WNF	24.000	16.000	1220.00	18.00	375.18	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
Noz L1B #3	SA-106 B	18.000	10.000	...	...	...
Noz L2B #3	SA-106 B	18.000	10.000	...	...	...
Noz L3B #3	SA-106 B	18.000	10.000	...	...	...
Noz L4B #3	SA-106 B	18.000	10.000	...	...	...
Noz U2 #300	SA-106 B	18.000	10.000	...	...	...
Noz U1 #30	SA-106 B	18.000	10.000	...	...	...
Noz L1A #30	SA-106 B	18.000	10.000	...	...	...
Noz L2A #30	SA-106 B	18.000	10.000	...	...	...
Noz L3A #30	SA-106 B	18.000	10.000	...	...	...
Noz L4A #30	SA-106 B	18.000	10.000	...	...	...
P1 #300	SA-106 B	28.000	10.000	...	...	...
P2 #300	SA-106 B	28.000	10.000	...	...	...
T3 #300	SA-106 B	30.000	10.000	...	...	...
T2 #300	SA-106 B	28.000	10.000	...	...	...
T1 #300	SA-106 B	22.000	20.000	...	...	...
NV2 2" #300	SA-106 B	18.000	10.000	...	...	...
B1 4 "ASSUM	SA-106 B	24.000	10.000	16.000	30.000	...
F1 4" #300	SA-106 B	18.000	10.000	14.000	18.000	...



V1 4" #300	SA-106 B	19.000	10.000	14.000	22.000	...
F2 6" #300	SA-106 B	18.000	15.000	14.000	18.000	...
T 8" #300	SA-106 B	19.000	10.000	14.000	22.000	...
B2 10"	SA-106 B	24.000	10.000	14.000	28.000	...
F3 #300	SA-106 B	28.000	10.000	14.000	28.000	...
Noz MW5 24"	SA-516 70	28.000	20.000	15.000	28.000	...
MW4 #300	SA-516 70	30.000	20.000	15.000	28.000	...
MW3 24" #3	SA-516 70	30.000	20.000	14.000	28.000	...
MW2 24" #3	SA-516 70	18.000	20.000	14.000	26.000	...
MW1 24" #30	SA-106 B	22.000	20.000	16.000	18.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
Noz L1B #300	350.000	180.0	250.00	0.00	Shell 1
Noz L2B #300	350.000	90.0	250.00	0.00	Shell 1
Noz L3B #300	350.000	270.0	250.00	0.00	Shell 1
Noz L4B #300	350.000	0.0	250.00	0.00	Shell 1
Noz U2 #300	150.000	180.0	250.00	0.00	Shell 1
Noz U1 #300	150.000	0.0	250.00	0.00	Shell 1
Noz L1A #300	4350.000	180.0	250.00	0.00	Shell 2
Noz L2A #300	4350.000	90.0	250.00	0.00	Shell 2
Noz L3A #300	4350.000	270.0	250.00	0.00	Shell 2
Noz L4A #300	4350.000	0.0	250.00	0.00	Shell 2
P1 #300	7849.999	180.0	227.77	0.00	Shell 3
P2 #300	7649.998	0.0	227.77	0.00	Shell 3
T3 #300	11759.999	180.0	250.00	0.00	Shell 4
T2 #300	23750.000	180.0	250.00	0.00	Shell 8
T1 #300	36359.996	180.0	250.00	0.00	Shell 13
NV2 2" #300	...	270.0	250.00	0.00	V1 4" #300
B1 4 "ASSUME #30	...	0.0	250.00	0.00	Bottom head
F1 4" #300	25000.004	180.0	250.00	0.00	Shell 9
V1 4" #300	...	180.0	250.00	0.00	Top head
F2 6" #300	37700.000	0.0	150.00	0.00	Shell 13
T 8" #300	...	0.0	300.00	0.00	Top head
B2 10"	...	180.0	250.00	0.00	Bottom head
F3 #300	6999.998	180.0	250.00	0.00	Shell 3
Noz MW5 24" #300	1600.000	180.0	300.00	0.00	Shell 1
MW4 #300	7599.999	90.0	300.00	0.00	Shell 3
MW3 24" #300	13899.999	270.0	300.00	0.00	Shell 5
MW2 24" #300	34900.004	90.0	300.00	0.00	Shell 12
MW1 24" #300	37100.000	180.0	300.00	0.00	Shell 13

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	ELLIPTICAL HEAD: 2.0 X 1, 32.0mm THK X 1800.0mm ID X 100.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 22.0mm THK X 1800.0mm ID X 100.0mm	SA-516 70
1	INSULATION: 3000mm X 50mm THK	
2	LINING: 3000.000mm X 50.000mm THK	...
11	PLATFORM: 1200mm X 1500mm WIDE	
1	LADDER: 3500.0mm	...
1	INSULATION: 3000mm X 50mm THK	Fire proofing
1	LADDER: 2000.0mm	...
2	INSULATION: 550mm X 60mm THK	HOT INSULATION
12	INSULATION: 3000mm X 60mm THK	HOT INSULATION
9	LADDER: 4000.0mm	...
41	TRAYS # 1 SPACING 23.622047 mm	
4	TRAYS # 1 SPACING 39.370079 mm	
1	INSULATION: 1800mm X 60mm THK	HOT INSULATION
2	TRUNNIONS:	SA-516 70
2	TRUNNION PADS: 700.0mm OD X 406.0mm ID X 20.0mm THK	
1	PLATFORM: 1200mm X 7839mm WIDE	
1	LADDER: 2350.0mm	...
5	CLASS 300 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	CLASS 300 GR 1.1, 2.0" BLIND FLANGE(S)	SA-105
1	BASERING: 3451.0mm OD X 3001.0 ID X 55.0mm	SA-516 70
20	BASERING BOLTS: 64.0mm DIA	SA-36
20	BASERING BOLT NUTS: 64.0mm DIA	...
1	BASERING TOP RING: 2820.8mm OD X 2490.8 ID X 45.0mm	...
40	BASERING GUSSET PLATES: 165.0mm X 165.0mm X 300.0mm X 20.0mm	...
1	CON SKIRT SEGMENT 20.0mm THK X 3999.3mm X 3000.0mm X 5027.4mm	SA-36
1	CON SKIRT SEGMENT 20.0mm THK X 3699.9mm X 3000.0mm X 4651.0mm	SA-36
1	CON SKIRT SEGMENT 28.0mm THK X 3053.5mm X 3000.0mm X 4121.8mm	SA-516 70
1	CON SKIRT SEGMENT 28.0mm THK X 2612.8mm X 3000.0mm X 3526.9mm	SA-516 70
6	CYLINDRICAL SEGMENT 30.0mm THK X 3047.3mm X 3000.0mm	SA-516 70
6	CYLINDRICAL SEGMENT 30.0mm THK X 2607.5mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 28.0mm THK X 2607.5mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 28.0mm THK X 3047.3mm X 3000.0mm	SA-516 70



1	CYLINDRICAL SEGMENT 28.0mm THK X 1979.2mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm THK X 3675.6mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm THK X 1272.3mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm THK X 4382.4mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 25.0mm THK X 722.6mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 25.0mm THK X 4932.2mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 25.0mm THK X 298.4mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 25.0mm THK X 5356.3mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 22.0mm THK X 78.5mm X 1800.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 22.0mm THK X 5576.2mm X 1800.0mm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Bottom head	[10]	D	-32	-48		0.660	27.500	1.00	Yes
Bottom head	[7]	D	-29	-48		0.565	32.000	1.00	Yes
Shell 1	[8]	B	4	-18		0.606	30.000	1.00	Yes
Shell 2	[8]	B	4	-18		0.602	30.000	1.00	Yes
Shell 3	[8]	B	4	-18		0.601	30.000	1.00	Yes
Shell 4	[8]	B	4	-18		0.601	30.000	1.00	Yes
Shell 5	[8]	B	4	-18		0.601	30.000	1.00	Yes
Shell 6	[8]	B	4	-18		0.601	30.000	1.00	Yes
Shell 7	[8]	B	2	-17		0.649	28.000	1.00	Yes
Shell 8	[8]	B	2	-17		0.649	28.000	1.00	Yes
Shell 9	[8]	B	2	-17		0.649	28.000	1.00	Yes
Shell 10	[8]	B	2	-17		0.649	28.000	1.00	Yes
Shell 11	[8]	B	-1	-16	-29	0.737	25.000	1.00	Yes
Shell 12	[8]	B	-1	-16	-29	0.737	25.000	1.00	Yes
Shell 13	[8]	B	-5	-13	-29	0.853	22.000	1.00	Yes
Top head	[10]	D	-41	-41	-29	1.000	19.000	1.00	Yes
Top head	[7]	D	-38	-46	-29	0.853	22.000	1.00	Yes
B2 10"	[1]	D	-32	-48		0.524	27.500	1.00	Yes
Nozzle Flg	[5]	!	-18	-62					
B1 4 "ASSUME #	[1]	D	-32	-48		0.524	27.500	1.00	Yes
Nozzle Flg	[5]	!	-18	-62					
Noz L1B #300	[1]	B	-13	-104	-29	0.038	16.800	1.00	Yes
Nozzle Flg	[5]	!	-25	-25					
Noz L2B #300	[1]	B	-13	-104	-29	0.038	16.800	1.00	Yes
Nozzle Flg	[5]	!	-25	-25					
Noz L3B #300	[1]	B	-13	-104	-29	0.038	16.800	1.00	Yes
Nozzle Flg	[5]	!	-25	-25					
Noz L4B #300	[1]	B	-13	-104	-29	0.038	16.800	1.00	Yes
Nozzle Flg	[5]	!	-25	-25					
Noz U2 #300	[1]	B	-13	-104	-29	0.038	16.800	1.00	Yes
Nozzle Flg	[5]	!	-25	-25					
Noz U1 #300	[1]	B	-13	-104	-29	0.038	16.800	1.00	Yes
Nozzle Flg	[5]	!	-25	-25					
Noz MW5 24" #3	[1]	B	2	-27		0.532	28.000	1.00	Yes
Nozzle Flg	[5]	D	-18	-48					
Noz L1A #300	[1]	B	-13	-104	-29	0.038	16.800	1.00	Yes
Nozzle Flg	[5]	!	-25	-25					
Noz L2A #300	[1]	B	-13	-104	-29	0.038	16.800	1.00	Yes
Nozzle Flg	[5]	!	-25	-25					
Noz L3A #300	[1]	B	-13	-104	-29	0.038	16.800	1.00	Yes
Nozzle Flg	[5]	!	-25	-25					
Noz L4A #300	[1]	B	-13	-104	-29	0.038	16.800	1.00	Yes
Nozzle Flg	[5]	!	-25	-25					
MW4 #300	[1]	B	2	-27		0.528	28.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
P1 #300	[1]	B	-13	-104	-29	0.038	16.800	1.00	Yes
Nozzle Flg	[5]	!	-25	-25					
P2 #300	[1]	B	-13	-104	-29	0.038	16.800	1.00	Yes
Nozzle Flg	[5]	!	-25	-25					
F3 #300	[1]	B	2	-27		0.528	28.000	1.00	Yes
Nozzle Flg	[5]	!	-18	-63					
T3 #300	[1]	B	-13	-104	-29	0.038	16.800	1.00	Yes
Nozzle Flg	[5]	!	-25	-25					
MW3 24" #300	[1]	B	2	-27		0.528	28.000	1.00	Yes
Nozzle Flg	[5]	D	-18	-48					
T2 #300	[1]	B	-13	-104	-29	0.038	16.800	1.00	Yes
Nozzle Flg	[5]	!	-25	-25					
F1 4" #300	[1]	B	-11	-35	-29	0.570	18.000	1.00	Yes
Nozzle Flg	[5]	!	-18	-63					
MW2 24" #300	[1]	B	-1	-29	-29	0.648	25.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
T1 #300	[1]	B	-13	-104	-29	0.038	16.800	1.00	Yes
Nozzle Flg	[5]	!	-25	-25					



MW1 24" #300	[1]	B	-11	-29	-29	0.751	18.000	1.00	Yes
Nozzle Flg	[5]	!	-18	-63					
F2 6" #300	[1]	B	-7	-29	-29	0.751	20.000	1.00	Yes
Nozzle Flg	[5]	!	-18	-63					
T 8" #300	[1]	D	-41	-48	-29	0.793	19.000	1.00	Yes
Nozzle Flg	[5]	!	-18	-63					
V1 4" #300	[1]	D	-41	-48	-29	0.793	19.000	1.00	Yes
Nozzle Flg	[5]	!	-18	-63					
NV2 2" #300	[1]	!	-25	-104		0.297	4.845	1.00	Yes
Nozzle Flg	[5]	!	-18	-63					
Nozl. Bolting[22]			-48			1.000	63.500	1.00	Yes

Warmest MDMT: 4 -13

Required Minimum Design Metal Temperature -25.0 °C
 Warmest Computed Minimum Design Metal Temperature -17.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and
 UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
 The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note	20	40	There is a high jump in the Joint Eff.
Note	20	30	The wind diameter multiplier is very big.
Note	30	40	The wind diameter multiplier is very big.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	2.1575	MPa
Design Internal Temperature	125.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-25.0	°C
Type of Construction	Welded	
Special Service	Lethal	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	Y	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	4	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class B	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.0	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	Spectrum
Range Type	Period
Ordinate Type	Acceleration
Include Missing Mass Components	Yes
Shock Scale Factor X Direction	1.0000
Shock Scale Factor Y Direction	0.6700
Combination Method	Group
Spectrum Name	User Defined

User Defined Spectrum Data Points:

Acceleration	Period
0.1600	0.0010
0.3920	0.0500
0.5600	0.1000
0.5600	0.1300
0.5600	0.2000
0.5600	0.2700
0.5600	0.3000
0.5600	0.4000
0.5600	0.5000
0.5600	0.5500
0.5130	0.6000
0.4600	0.6700
0.3850	0.8000
0.3080	1.0000
0.2570	1.2000
0.2200	1.4000
0.1930	1.6000
0.1710	1.8000
0.1540	2.0000
0.1280	2.4000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	No
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No



Use the MAWP to compute the MDMT Yes
 For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
 For PWHT & P1 Materials the MDMT can be < -55F (-48C) No
 Using Metric Material Databases, ASME II D No
 Calculate B31.3 type stress for Nozzles with Loads Yes
 Reduce the MDMT due to lower membrane stress Yes
 Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/17
Element From Node	10
Element To Node	20
Element Type	Skirt Support
Description	Bottom skirt
Distance "FROM" to "TO"	3000 mm
Skirt Outside Diameter	2491 mm
Diameter of Skirt at Base	3121 mm
Skirt Thickness	20 mm
Internal Corrosion Allowance	0.5 mm
Nominal Thickness	20 mm
External Corrosion Allowance	0.5 mm
Design Temperature Internal Pressure	125 °C
Design Temperature External Pressure	190 °C
Effective Diameter Multiplier	1.3
Material Name	SA-36
Allowable Stress, Ambient	114.5 MPa
Allowable Stress, Operating	114.5 MPa
Allowable Stress, Hydrotest	223.4 MPa
Material Density	0.00775 kg/cm³
P Number Thickness	30.988 mm
Yield Stress, Operating	222.8 MPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02600
Product Form	Plate; sheet
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No

Element From Node	10
Detail Type	Insulation
Detail ID	Fire proofing
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	50 mm
Density	0.0025 kg/cm³

Element From Node	10
Detail Type	Lining
Detail ID	Fire proofing
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	3000 mm
Thickness of Lining	50 mm
Density	0.0025 kg/cm³

Element From Node	10
Detail Type	Weight
Detail ID	TAILING LUG WT
Dist. from "FROM" Node / Offset dist	200 mm
Miscellaneous Weight	2451.5 N
Offset from Element Centerline	1745 mm

Element#	2/17
Element From Node	20
Element To Node	30
Element Type	Skirt Support



Description	Top skirt	
Distance "FROM" to "TO"	3000	mm
Skirt Outside Diameter	1860	mm
Diameter of Skirt at Base	2491	mm
Skirt Thickness	28	mm
Internal Corrosion Allowance	0.5	mm
Nominal Thickness	28	mm
External Corrosion Allowance	0.5	mm
Design Temperature Internal Pressure	125	°C
Design Temperature External Pressure	190	°C
Effective Diameter Multiplier	1.6	
Material Name	SA-516 70	
Allowable Stress, Ambient	137.9	MPa
Allowable Stress, Operating	137.9	MPa
Allowable Stress, Hydrotest	235.81	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	31.75	mm
Yield Stress, Operating	235.23	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.49	
Weld is pre-Heated	No	
Element From Node	20	
Detail Type	Platform	
Detail ID	SKRT PLT	
Dist. from "FROM" Node / Offset dist	500	mm
Platform Start Angle (degrees)	45.0	
Platform End Angle (degrees)	315.0	
Platform Wind Area	4320000	mm²
Platform Weight	21536	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm²
Platform Width	1500	mm
Platform Height	1200	mm
Platform Clearance or End Offset	0	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	30.0	
Ladder Start Elevation	-3500	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0.3998	N/mm
Platform Length (top head platform)	0	mm
Element From Node	20	
Detail Type	Insulation	
Detail ID	Fire proofing'	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	50	mm
Density	0.0025	kg/cm³
Element From Node	20	
Detail Type	Lining	
Detail ID	Fire proofing	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	50	mm
Density	0.0025	kg/cm³
Element From Node	20	
Detail Type	Weight	
Detail ID	SKIRT ACCESSORI	
Dist. from "FROM" Node / Offset dist	1500	mm
Miscellaneous Weight	3922.5	N
Offset from Element Centerline	0	mm



Element#	3/17
Element From Node	30
Element To Node	40
Element Type	Elliptical
Description	Bottom head
Distance "FROM" to "TO"	100 mm
Inside Diameter	1800 mm
Element Thickness	27.5 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	32 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.1575 MPa
Design Temperature Internal Pressure	125 °C
Design External Pressure	0.04903 MPa
Design Temperature External Pressure	190 °C
Effective Diameter Multiplier	1.6
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	30
Detail Type	Platform
Detail ID	BTM PTL
Dist. from "FROM" Node / Offset dist	0 mm
Platform Start Angle (degrees)	360.0
Platform End Angle (degrees)	0.0
Platform Wind Area	2160000 mm ²
Platform Weight	799.59 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	1500 mm
Platform Height	1200 mm
Platform Clearance or End Offset	85 mm
Platform Force Coefficient	1.2
Ladder Layout Angle	30.0
Ladder Start Elevation	-2000 mm
Ladder End Elevation	0 mm
Unit Weight of Ladder	0.3998 N/mm
Platform Length (top head platform)	0 mm
Element From Node	30
Detail Type	Liquid
Detail ID	Liquid 30
Dist. from "FROM" Node / Offset dist	-450 mm
Height/Length of Liquid	550 mm
Liquid Density	0.0005194 kg/cm ³
Element From Node	30
Detail Type	Insulation
Detail ID	Ins: 160
Dist. from "FROM" Node / Offset dist	-450 mm
Height/Length of Insulation	550 mm
Thickness of Insulation	60 mm
Density	0.00016 kg/cm ³
Element From Node	30
Detail Type	Nozzle
Detail ID	B2 10"
Dist. from "FROM" Node / Offset dist	400 mm
Nozzle Diameter	10 in.
Nozzle Schedule	80
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	1337.9 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B +IT



Element From Node	30
Detail Type	Nozzle
Detail ID	B1 4 "ASSUME #30
Dist. from "FROM" Node / Offset dist	200 mm
Nozzle Diameter	4 in.
Nozzle Schedule	120
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	286.84 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B +IT

Element From Node	30
Detail Type	Weight
Detail ID	VORTEX BREAKER
Dist. from "FROM" Node / Offset dist	-300 mm
Miscellaneous Weight	490.3 N
Offset from Element Centerline	500 mm

Element From Node	30
Detail Type	Weight
Detail ID	B2 OP
Dist. from "FROM" Node / Offset dist	0 mm
Miscellaneous Weight	8070.3 N
Offset from Element Centerline	2165 mm

Element#	4/17
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell 1
Distance "FROM" to "TO"	3000 mm
Inside Diameter	1800 mm
Element Thickness	30 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	30 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.1575 MPa
Design Temperature Internal Pressure	125 °C
Design External Pressure	0.04903 MPa
Design Temperature External Pressure	190 °C
Effective Diameter Multiplier	1.3
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	3000 mm
Liquid Density	0.0005194 kg/cm³

Element From Node	40
Detail Type	Insulation
Detail ID	Ins: 160
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	60 mm
Density	0.00016 kg/cm³

Element From Node	40
Detail Type	Nozzle
Detail ID	Noz L1B #300
Dist. from "FROM" Node / Offset dist	250 mm



Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 180.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 119.89 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID Noz L2B #300
 Dist. from "FROM" Node / Offset dist 250 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID Noz L3B #300
 Dist. from "FROM" Node / Offset dist 250 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 270.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID Noz L4B #300
 Dist. from "FROM" Node / Offset dist 250 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 119.89 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID Noz U2 #300
 Dist. from "FROM" Node / Offset dist 50 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 180.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 119.89 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID Noz U1 #300
 Dist. from "FROM" Node / Offset dist 50 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 0.0
 Blind Flange (Y/N) N



Weight of Nozzle (Used if > 0) 119.89 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID Noz MW5 24" #300
 Dist. from "FROM" Node / Offset dist 1500 mm
 Nozzle Diameter 24 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 180.0
 Blind Flange (Y/N) Y
 Weight of Nozzle (Used if > 0) 8442.4 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-516 70

Element From Node 40
 Detail Type Weight
 Detail ID MANWAY DAVIT MW
 Dist. from "FROM" Node / Offset dist 1500 mm
 Miscellaneous Weight 1470.9 N
 Offset from Element Centerline 1350 mm

Element From Node 40
 Detail Type Weight
 Detail ID BAFFELE& HOOD B
 Dist. from "FROM" Node / Offset dist 2000 mm
 Miscellaneous Weight 27114 N
 Offset from Element Centerline 0 mm

Element From Node 40
 Detail Type Weight
 Detail ID CLEAT WT-1
 Dist. from "FROM" Node / Offset dist -0.0002896 mm
 Miscellaneous Weight 16749 N
 Offset from Element Centerline 0 mm

Element# 5/17
 Element From Node 50
 Element To Node 60
 Element Type Cylinder
 Description Shell 2
 Distance "FROM" to "TO" 3000 mm
 Inside Diameter 1800 mm
 Element Thickness 30 mm
 Internal Corrosion Allowance 3 mm
 Nominal Thickness 30 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 2.1575 MPa
 Design Temperature Internal Pressure 125 °C
 Design External Pressure 0.04903 MPa
 Design Temperature External Pressure 190 °C
 Effective Diameter Multiplier 1.3
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 50
 Detail Type Platform
 Detail ID PTL 1
 Dist. from "FROM" Node / Offset dist 1000 mm
 Platform Start Angle (degrees) 180.0
 Platform End Angle (degrees) 0.0
 Platform Wind Area 4320000 mm²
 Platform Weight 15432 N
 Platform Railing Weight 0 N/mm



Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1200	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	30.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.3998	N/mm
Platform Length (top head platform)	0	mm

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1200	mm
Liquid Density	0.0005194	kg/cm ³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 160	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	60	mm
Density	0.00016	kg/cm ³

Element From Node	50	
Detail Type	Nozzle	
Detail ID	Noz L1A #300	
Dist. from "FROM" Node / Offset dist	1250	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	119.89	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B +IT	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	Noz L2A #300	
Dist. from "FROM" Node / Offset dist	1250	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	119.89	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B +IT	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	Noz L3A #300	
Dist. from "FROM" Node / Offset dist	1250	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	119.89	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B +IT	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	Noz L4A #300	
Dist. from "FROM" Node / Offset dist	1250	mm
Nozzle Diameter	2	in.



Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	119.89 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B +IT

Element#	6/17
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	Shell 3
Distance "FROM" to "TO"	3000 mm
Inside Diameter	1800 mm
Element Thickness	30 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	30 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.1575 MPa
Design Temperature Internal Pressure	125 °C
Design External Pressure	0.04903 MPa
Design Temperature External Pressure	190 °C
Effective Diameter Multiplier	1.3
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	60
Detail Type	Platform
Detail ID	PLAT2
Dist. from "FROM" Node / Offset dist	2000 mm
Platform Start Angle (degrees)	150.0
Platform End Angle (degrees)	330.0
Platform Wind Area	4320000 mm ²
Platform Weight	15432 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	1500 mm
Platform Height	1200 mm
Platform Clearance or End Offset	85 mm
Platform Force Coefficient	1.2
Ladder Layout Angle	30.0
Ladder Start Elevation	4000 mm
Ladder End Elevation	8000 mm
Unit Weight of Ladder	0.3998 N/mm
Platform Length (top head platform)	0 mm

Element From Node	60
Detail Type	Insulation
Detail ID	Ins: 160
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	60 mm
Density	0.00016 kg/cm ³

Element From Node	60
Detail Type	Nozzle
Detail ID	MW4 #300
Dist. from "FROM" Node / Offset dist	1500 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	8445.8 N
Grade of Attached Flange	GR 1.1



Nozzle Matl	SA-516 70
Element From Node	60
Detail Type	Nozzle
Detail ID	P1 #300
Dist. from "FROM" Node / Offset dist	1750 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	113.87 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B +IT
Element From Node	60
Detail Type	Nozzle
Detail ID	P2 #300
Dist. from "FROM" Node / Offset dist	1550 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	113.87 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B +IT
Element From Node	60
Detail Type	Nozzle
Detail ID	F3 #300
Dist. from "FROM" Node / Offset dist	900 mm
Nozzle Diameter	10 in.
Nozzle Schedule	80
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	1048.6 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B +IT
Element From Node	60
Detail Type	Tray
Detail ID	Tray 45
Dist. from "FROM" Node / Offset dist	1100 mm
Tray Spacing	600 mm
Number of Trays	1
Tray Weight per Unit Area	0.015 kg/cm ²
Height of Liquid on Tray	0 mm
Density of Liquid on Tray	0 kg/cm ³
Support Ring and Bolting Bar Weight	1050 N
Element From Node	60
Detail Type	Tray
Detail ID	Tray 44
Dist. from "FROM" Node / Offset dist	1700 mm
Tray Spacing	600 mm
Number of Trays	1
Tray Weight per Unit Area	0.015 kg/cm ²
Height of Liquid on Tray	0 mm
Density of Liquid on Tray	0 kg/cm ³
Support Ring and Bolting Bar Weight	1050 N
Element From Node	60
Detail Type	Tray
Detail ID	Tray 43
Dist. from "FROM" Node / Offset dist	2300 mm
Tray Spacing	600 mm
Number of Trays	1
Tray Weight per Unit Area	0.015 kg/cm ²



Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	1050	N

Element From Node	60	
Detail Type	Tray	
Detail ID	Tray 42	
Dist. from "FROM" Node / Offset dist	2900	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	1050	N

Element From Node	60	
Detail Type	Weight	
Detail ID	MANWAY DAVIT MW	
Dist. from "FROM" Node / Offset dist	1500	mm
Miscellaneous Weight	1470.9	N
Offset from Element Centerline	1200	mm

Element From Node	60	
Detail Type	Weight	
Detail ID	F3 PIPE WT	
Dist. from "FROM" Node / Offset dist	900	mm
Miscellaneous Weight	17788	N
Offset from Element Centerline	2198	mm

Element#	7/17	
Element From Node	70	
Element To Node	80	
Element Type	Cylinder	
Description	Shell 4	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	1800	mm
Element Thickness	30	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	30	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.1575	MPa
Design Temperature Internal Pressure	125	°C
Design External Pressure	0.04903	MPa
Design Temperature External Pressure	190	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	70	
Detail Type	Platform	
Detail ID	PLAT 4	
Dist. from "FROM" Node / Offset dist	2900	mm
Platform Start Angle (degrees)	180.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	4320000	mm ²
Platform Weight	15432	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1200	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	30.0	
Ladder Start Elevation	8000	mm
Ladder End Elevation	12000	mm
Unit Weight of Ladder	0.3998	N/mm



Platform Length (top head platform)	0	mm
Element From Node	70	
Detail Type	Insulation	
Detail ID	Ins: 160	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	60	mm
Density	0.00016	kg/cm ³
Element From Node	70	
Detail Type	Nozzle	
Detail ID	T3 #300	
Dist. from "FROM" Node / Offset dist	2660	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	120.43	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B +IT	
Element From Node	70	
Detail Type	Tray	
Detail ID	Tray 41	
Dist. from "FROM" Node / Offset dist	500	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	1050	N
Element From Node	70	
Detail Type	Tray	
Detail ID	Tray 40	
Dist. from "FROM" Node / Offset dist	1100	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	1050	N
Element From Node	70	
Detail Type	Tray	
Detail ID	Tray 39	
Dist. from "FROM" Node / Offset dist	1700	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	1050	N
Element From Node	70	
Detail Type	Tray	
Detail ID	Tray 38	
Dist. from "FROM" Node / Offset dist	2300	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	1050	N
Element From Node	70	
Detail Type	Tray	
Detail ID	Tray 37	



Dist. from "FROM" Node / Offset dist	2900	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	1050	N

Element#	8/17
Element From Node	80
Element To Node	90
Element Type	Cylinder
Description	Shell 5
Distance "FROM" to "TO"	3000 mm
Inside Diameter	1800 mm
Element Thickness	30 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	30 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.1575 MPa
Design Temperature Internal Pressure	125 °C
Design External Pressure	0.04903 MPa
Design Temperature External Pressure	190 °C
Effective Diameter Multiplier	1.3
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	80
Detail Type	Insulation
Detail ID	Ins: 160
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	60 mm
Density	0.00016 kg/cm ³

Element From Node	80
Detail Type	Nozzle
Detail ID	MW3 24" #300
Dist. from "FROM" Node / Offset dist	1800 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	270.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	8445.8 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70

Element From Node	80
Detail Type	Tray
Detail ID	Tray 36
Dist. from "FROM" Node / Offset dist	500 mm
Tray Spacing	600 mm
Number of Trays	1
Tray Weight per Unit Area	0.015 kg/cm ²
Height of Liquid on Tray	0 mm
Density of Liquid on Tray	0 kg/cm ³
Support Ring and Bolting Bar Weight	1050 N

Element From Node	80
Detail Type	Tray
Detail ID	Tray 35
Dist. from "FROM" Node / Offset dist	1100 mm
Tray Spacing	600 mm
Number of Trays	1
Tray Weight per Unit Area	0.015 kg/cm ²



Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	1050	N
Element From Node	80	
Detail Type	Tray	
Detail ID	Tray 34	
Dist. from "FROM" Node / Offset dist	1700	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	1050	N
Element From Node	80	
Detail Type	Tray	
Detail ID	Tray 33	
Dist. from "FROM" Node / Offset dist	2300	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	1050	N
Element From Node	80	
Detail Type	Tray	
Detail ID	Tray 32	
Dist. from "FROM" Node / Offset dist	2900	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	1050	N
Element From Node	80	
Detail Type	Weight	
Detail ID	MANWAY DAVIT MW	
Dist. from "FROM" Node / Offset dist	1800	mm
Miscellaneous Weight	1470.9	N
Offset from Element Centerline	1350	mm

Element#	9/17	
Element From Node	90	
Element To Node	100	
Element Type	Cylinder	
Description	Shell 6	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	1800	mm
Element Thickness	30	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	30	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.1575	MPa
Design Temperature Internal Pressure	125	°C
Design External Pressure	0.04903	MPa
Design Temperature External Pressure	190	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	90
Detail Type	Platform
Detail ID	PLAT:5



Dist. from "FROM" Node / Offset dist	900	mm
Platform Start Angle (degrees)	270.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	432000	mm ²
Platform Weight	15432	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1200	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	30.0	
Ladder Start Elevation	12000	mm
Ladder End Elevation	16000	mm
Unit Weight of Ladder	0.3998	N/mm
Platform Length (top head platform)	0	mm
Element From Node	90	
Detail Type	Insulation	
Detail ID	Ins: 160	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	60	mm
Density	0.00016	kg/cm ³
Element From Node	90	
Detail Type	Tray	
Detail ID	Tray 31	
Dist. from "FROM" Node / Offset dist	500	mm
Tray Spacing	1000	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	1050	N
Element From Node	90	
Detail Type	Tray	
Detail ID	Tray 30	
Dist. from "FROM" Node / Offset dist	1500	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	1050	N
Element From Node	90	
Detail Type	Tray	
Detail ID	Tray 29	
Dist. from "FROM" Node / Offset dist	2100	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	1050	N
Element From Node	90	
Detail Type	Tray	
Detail ID	Tray 28	
Dist. from "FROM" Node / Offset dist	2700	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	1050	N



Element#	10/17	
Element From Node	100	
Element To Node	110	
Element Type	Cylinder	
Description	Shell 7	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	1800	mm
Element Thickness	28	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	28	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.1575	MPa
Design Temperature Internal Pressure	125	°C
Design External Pressure	0.04903	MPa
Design Temperature External Pressure	190	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	100	
Detail Type	Platform	
Detail ID	PLAT 6	
Dist. from "FROM" Node / Offset dist	1900	mm
Platform Start Angle (degrees)	180.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	4320000	mm ²
Platform Weight	15440	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1200	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	30.0	
Ladder Start Elevation	16000	mm
Ladder End Elevation	20000	mm
Unit Weight of Ladder	0.3998	N/mm
Platform Length (top head platform)	0	mm
Element From Node	100	
Detail Type	Insulation	
Detail ID	Ins: 160	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	60	mm
Density	0.00016	kg/cm ³
Element From Node	100	
Detail Type	Tray	
Detail ID	Tray 27	
Dist. from "FROM" Node / Offset dist	300	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	1050	N
Element From Node	100	
Detail Type	Tray	
Detail ID	Tray 26	
Dist. from "FROM" Node / Offset dist	900	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³



Support Ring and Bolting Bar Weight	1050	N
Element From Node	100	
Detail Type	Tray	
Detail ID	Tray 25	
Dist. from "FROM" Node / Offset dist	1500	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	1050	N

Element From Node	100	
Detail Type	Tray	
Detail ID	Tray 24	
Dist. from "FROM" Node / Offset dist	2100	mm
Tray Spacing	1000	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	1050	N

Element#	11/17	
Element From Node	110	
Element To Node	120	
Element Type	Cylinder	
Description	Shell 8	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	1800	mm
Element Thickness	28	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	28	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.1575	MPa
Design Temperature Internal Pressure	125	°C
Design External Pressure	0.04903	MPa
Design Temperature External Pressure	190	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	110	
Detail Type	Platform	
Detail ID	PLAT 6	
Dist. from "FROM" Node / Offset dist	2900	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	4320000	mm ²
Platform Weight	15432	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1200	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	30.0	
Ladder Start Elevation	20000	mm
Ladder End Elevation	24000	mm
Unit Weight of Ladder	0.3998	N/mm
Platform Length (top head platform)	0	mm

Element From Node	110	
Detail Type	Insulation	
Detail ID	Ins: 160	



Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	60	mm
Density	0.00016	kg/cm ³

Element From Node	110	
Detail Type	Nozzle	
Detail ID	T2 #300	
Dist. from "FROM" Node / Offset dist	2650	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	118.27	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B +IT	

Element From Node	110	
Detail Type	Tray	
Detail ID	Tray 23	
Dist. from "FROM" Node / Offset dist	1100	mm
Tray Spacing	1000	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	1050	N

Element From Node	110	
Detail Type	Tray	
Detail ID	Tray 22	
Dist. from "FROM" Node / Offset dist	1700	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	810	N

Element From Node	110	
Detail Type	Tray	
Detail ID	Tray 21	
Dist. from "FROM" Node / Offset dist	2300	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	810	N

Element From Node	110	
Detail Type	Tray	
Detail ID	Tray 20	
Dist. from "FROM" Node / Offset dist	2900	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	810	N

Element From Node	110	
Detail Type	Weight	
Detail ID	CLEAT LOAD 2	
Dist. from "FROM" Node / Offset dist	2000	mm
Miscellaneous Weight	16749	N
Offset from Element Centerline	0	mm



Element#	12/17
Element From Node	120
Element To Node	130
Element Type	Cylinder
Description	Shell 9
Distance "FROM" to "TO"	3000 mm
Inside Diameter	1800 mm
Element Thickness	28 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	28 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.1575 MPa
Design Temperature Internal Pressure	125 °C
Design External Pressure	0.04903 MPa
Design Temperature External Pressure	190 °C
Effective Diameter Multiplier	1.3
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	120
Detail Type	Insulation
Detail ID	Ins: 160
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	60 mm
Density	0.00016 kg/cm ³
Element From Node	120
Detail Type	Nozzle
Detail ID	F1 4" #300
Dist. from "FROM" Node / Offset dist	900 mm
Nozzle Diameter	4 in.
Nozzle Schedule	160
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	259.62 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B +IT
Element From Node	120
Detail Type	Tray
Detail ID	Tray 19
Dist. from "FROM" Node / Offset dist	500 mm
Tray Spacing	600 mm
Number of Trays	1
Tray Weight per Unit Area	0.015 kg/cm ²
Height of Liquid on Tray	0 mm
Density of Liquid on Tray	0 kg/cm ³
Support Ring and Bolting Bar Weight	810 N
Element From Node	120
Detail Type	Tray
Detail ID	Tray 18
Dist. from "FROM" Node / Offset dist	1100 mm
Tray Spacing	600 mm
Number of Trays	1
Tray Weight per Unit Area	0.015 kg/cm ²
Height of Liquid on Tray	0 mm
Density of Liquid on Tray	0 kg/cm ³
Support Ring and Bolting Bar Weight	810 N
Element From Node	120
Detail Type	Tray
Detail ID	Tray 17
Dist. from "FROM" Node / Offset dist	1700 mm
Tray Spacing	600 mm



Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	810	N

Element From Node	120	
Detail Type	Tray	
Detail ID	Tray 16	
Dist. from "FROM" Node / Offset dist	2300	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	810	N

Element From Node	120	
Detail Type	Tray	
Detail ID	Tray 15	
Dist. from "FROM" Node / Offset dist	2900	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	810	N

Element From Node	120	
Detail Type	Weight	
Detail ID	MANWAY DAVIT MW	
Dist. from "FROM" Node / Offset dist	1000	mm
Miscellaneous Weight	1470.9	N
Offset from Element Centerline	2000	mm

Element From Node	120	
Detail Type	Weight	
Detail ID	F1 NOZZLE	
Dist. from "FROM" Node / Offset dist	900	mm
Miscellaneous Weight	607.97	N
Offset from Element Centerline	0	mm

Element#	13/17	
Element From Node	130	
Element To Node	140	
Element Type	Cylinder	
Description	Shell 10	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	1800	mm
Element Thickness	28	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	28	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.1575	MPa
Design Temperature Internal Pressure	125	°C
Design External Pressure	0.04903	MPa
Design Temperature External Pressure	190	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	130	
Detail Type	Platform	
Detail ID	PLAT: 6	
Dist. from "FROM" Node / Offset dist	900	mm
Platform Start Angle (degrees)	0.0	



Platform End Angle (degrees)	180.0	
Platform Wind Area	4320000	mm ²
Platform Weight	15424	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1200	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	30.0	
Ladder Start Elevation	24000	mm
Ladder End Elevation	28000	mm
Unit Weight of Ladder	0.3998	N/mm
Platform Length (top head platform)	0	mm
Element From Node	130	
Detail Type	Insulation	
Detail ID	Ins: 160	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	60	mm
Density	0.00016	kg/cm ³
Element From Node	130	
Detail Type	Tray	
Detail ID	Tray 14	
Dist. from "FROM" Node / Offset dist	500	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	810	N
Element From Node	130	
Detail Type	Tray	
Detail ID	Tray 13	
Dist. from "FROM" Node / Offset dist	1100	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	810	N
Element From Node	130	
Detail Type	Tray	
Detail ID	Tray 12	
Dist. from "FROM" Node / Offset dist	1700	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	810	N
Element From Node	130	
Detail Type	Tray	
Detail ID	Tray 11	
Dist. from "FROM" Node / Offset dist	2300	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	810	N
Element From Node	130	
Detail Type	Tray	
Detail ID	Tray 10	



Dist. from "FROM" Node / Offset dist	2900	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	810	N

Element#	14/17
Element From Node	140
Element To Node	150
Element Type	Cylinder
Description	Shell 11
Distance "FROM" to "TO"	3000 mm
Inside Diameter	1800 mm
Element Thickness	25 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	25 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.1575 MPa
Design Temperature Internal Pressure	125 °C
Design External Pressure	0.04903 MPa
Design Temperature External Pressure	190 °C
Effective Diameter Multiplier	1.3
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	140
Detail Type	Platform
Detail ID	PLAT:7
Dist. from "FROM" Node / Offset dist	1900 mm
Platform Start Angle (degrees)	30.0
Platform End Angle (degrees)	210.0
Platform Wind Area	432000 mm ²
Platform Weight	15432 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	1500 mm
Platform Height	1200 mm
Platform Clearance or End Offset	85 mm
Platform Force Coefficient	1.2
Ladder Layout Angle	30.0
Ladder Start Elevation	28000 mm
Ladder End Elevation	32000 mm
Unit Weight of Ladder	0.3998 N/mm
Platform Length (top head platform)	0 mm

Element From Node	140
Detail Type	Insulation
Detail ID	Ins: 160
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	60 mm
Density	0.00016 kg/cm ³

Element From Node	140
Detail Type	Tray
Detail ID	Tray 9
Dist. from "FROM" Node / Offset dist	500 mm
Tray Spacing	600 mm
Number of Trays	1
Tray Weight per Unit Area	0.015 kg/cm ²
Height of Liquid on Tray	0 mm
Density of Liquid on Tray	0 kg/cm ³
Support Ring and Bolting Bar Weight	810 N



Element From Node 140
 Detail Type Tray
 Detail ID Tray 8
 Dist. from "FROM" Node / Offset dist 1100 mm
 Tray Spacing 600 mm
 Number of Trays 1
 Tray Weight per Unit Area 0.015 kg/cm²
 Height of Liquid on Tray 0 mm
 Density of Liquid on Tray 0 kg/cm³
 Support Ring and Bolting Bar Weight 810 N

Element From Node 140
 Detail Type Tray
 Detail ID Tray 7
 Dist. from "FROM" Node / Offset dist 2100 mm
 Tray Spacing 1000 mm
 Number of Trays 1
 Tray Weight per Unit Area 0.015 kg/cm²
 Height of Liquid on Tray 0 mm
 Density of Liquid on Tray 0 kg/cm³
 Support Ring and Bolting Bar Weight 810 N

Element From Node 140
 Detail Type Tray
 Detail ID Tray 6
 Dist. from "FROM" Node / Offset dist 2700 mm
 Tray Spacing 600 mm
 Number of Trays 1
 Tray Weight per Unit Area 0.015 kg/cm²
 Height of Liquid on Tray 0 mm
 Density of Liquid on Tray 0 kg/cm³
 Support Ring and Bolting Bar Weight 810 N

Element From Node 140
 Detail Type Weight
 Detail ID CLEAT WT -3
 Dist. from "FROM" Node / Offset dist 0.002326 mm
 Miscellaneous Weight 16749 N
 Offset from Element Centerline 0 mm

Element# 15/17
 Element From Node 150
 Element To Node 160
 Element Type Cylinder
 Description Shell 12
 Distance "FROM" to "TO" 3000 mm
 Inside Diameter 1800 mm
 Element Thickness 25 mm
 Internal Corrosion Allowance 3 mm
 Nominal Thickness 25 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 2.1575 MPa
 Design Temperature Internal Pressure 125 °C
 Design External Pressure 0.04903 MPa
 Design Temperature External Pressure 190 °C
 Effective Diameter Multiplier 1.3
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 150
 Detail Type Platform
 Detail ID PLAT:8
 Dist. from "FROM" Node / Offset dist 2900 mm
 Platform Start Angle (degrees) 30.0
 Platform End Angle (degrees) 280.0
 Platform Wind Area 4320000 mm²



Platform Weight	20812	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1200	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	30.0	
Ladder Start Elevation	32000	mm
Ladder End Elevation	36000	mm
Unit Weight of Ladder	0.3998	N/mm
Platform Length (top head platform)	0	mm

Element From Node	150	
Detail Type	Insulation	
Detail ID	Ins: 160	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	60	mm
Density	0.00016	kg/cm ³

Element From Node	150	
Detail Type	Nozzle	
Detail ID	MW2 24" #300	
Dist. from "FROM" Node / Offset dist	1800	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	8291.9	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	

Element From Node	150	
Detail Type	Tray	
Detail ID	Tray 5	
Dist. from "FROM" Node / Offset dist	300	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	810	N

Element From Node	150	
Detail Type	Tray	
Detail ID	Tray 4	
Dist. from "FROM" Node / Offset dist	900	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	810	N

Element From Node	150	
Detail Type	Tray	
Detail ID	Tray 3	
Dist. from "FROM" Node / Offset dist	1500	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	810	N

Element From Node	150	
Detail Type	Tray	
Detail ID	Tray 2	



Dist. from "FROM" Node / Offset dist	2100	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	810	N

Element From Node	150	
Detail Type	Tray	
Detail ID	TRAY 1	
Dist. from "FROM" Node / Offset dist	2700	mm
Tray Spacing	600	mm
Number of Trays	1	
Tray Weight per Unit Area	0.015	kg/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg/cm ³
Support Ring and Bolting Bar Weight	810	N

Element#	16/17	
Element From Node	160	
Element To Node	170	
Element Type	Cylinder	
Description	Shell 13	
Distance "FROM" to "TO"	1800	mm
Inside Diameter	1800	mm
Element Thickness	22	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	22	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.1575	MPa
Design Temperature Internal Pressure	125	°C
Design External Pressure	0.04903	MPa
Design Temperature External Pressure	190	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	160	
Detail Type	Insulation	
Detail ID	Ins: 160	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1800	mm
Thickness of Insulation	60	mm
Density	0.00016	kg/cm ³

Element From Node	160	
Detail Type	Nozzle	
Detail ID	T1 #300	
Dist. from "FROM" Node / Offset dist	260	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	117.18	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B +IT	

Element From Node	160	
Detail Type	Nozzle	
Detail ID	MW1 24" #300	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	300	



Layout Angle 180.0
 Blind Flange (Y/N) Y
 Weight of Nozzle (Used if > 0) 7968 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B +IT

Element From Node 160
 Detail Type Nozzle
 Detail ID F2 6" #300
 Dist. from "FROM" Node / Offset dist 1600 mm
 Nozzle Diameter 6 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 395.82 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B +IT

Element From Node 160
 Detail Type Weight
 Detail ID F2 OP
 Dist. from "FROM" Node / Offset dist 1600 mm
 Miscellaneous Weight 12032 N
 Offset from Element Centerline 2114 mm

Element From Node 160
 Detail Type Weight
 Detail ID MANWAY DAVIT MW
 Dist. from "FROM" Node / Offset dist 1000 mm
 Miscellaneous Weight 1470.9 N
 Offset from Element Centerline 1400 mm

Element From Node 160
 Detail Type Weight
 Detail ID LIFITNG TRUNION
 Dist. from "FROM" Node / Offset dist 800 mm
 Miscellaneous Weight 14709 N
 Offset from Element Centerline 0 mm

Element From Node 160
 Detail Type Weight
 Detail ID F2 FEED PIPE
 Dist. from "FROM" Node / Offset dist 1600 mm
 Miscellaneous Weight 1068.9 N
 Offset from Element Centerline 0 mm

Element# 17/17
 Element From Node 170
 Element To Node 180
 Element Type Elliptical
 Description Top head
 Distance "FROM" to "TO" 100 mm
 Inside Diameter 1800 mm
 Element Thickness 19 mm
 Internal Corrosion Allowance 3 mm
 Nominal Thickness 22 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 2.1575 MPa
 Design Temperature Internal Pressure 125 °C
 Design External Pressure 0.04903 MPa
 Design Temperature External Pressure 190 °C
 Effective Diameter Multiplier 1.3
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Elliptical Head Factor 2.0
 Weld is pre-Heated No



Element From Node	170
Detail Type	Platform
Detail ID	TOP PTL
Dist. from "FROM" Node / Offset dist	450 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	0.0
Platform Wind Area	11289600 mm ²
Platform Weight	103457 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	7840 mm
Platform Height	1200 mm
Platform Clearance or End Offset	3920 mm
Platform Force Coefficient	1.2
Ladder Layout Angle	135.0
Ladder Start Elevation	-2350 mm
Ladder End Elevation	0 mm
Unit Weight of Ladder	0.3998 N/mm
Platform Length (top head platform)	7840 mm

Element From Node	170
Detail Type	Insulation
Detail ID	Ins: 160
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	550 mm
Thickness of Insulation	60 mm
Density	0.00016 kg/cm ³

Element From Node	170
Detail Type	Nozzle
Detail ID	T 8" #300
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	8 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	718.4 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B +IT

Element From Node	170
Detail Type	Nozzle
Detail ID	V1 4" #300
Dist. from "FROM" Node / Offset dist	500 mm
Nozzle Diameter	4 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	263.43 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B +IT

Element From Node	170
Detail Type	Nozzle
Detail ID	NV2 2" #300
Dist. from "FROM" Node / Offset dist	100 mm
Nozzle Diameter	2 in.
Nozzle Schedule	80
Nozzle Class	300
Layout Angle	270.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	100.3 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B +IT

Element From Node	170
Detail Type	Weight



Detail ID	T OP	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	13052	N
Offset from Element Centerline	2140	mm
Element From Node	170	
Detail Type	Weight	
Detail ID	PIPE DAVIT	
Dist. from "FROM" Node / Offset dist	200	mm
Miscellaneous Weight	5491.4	N
Offset from Element Centerline	1200	mm
Element From Node	170	
Detail Type	Weight	
Detail ID	PSV 1- 6"8"	
Dist. from "FROM" Node / Offset dist	450	mm
Miscellaneous Weight	24515	N
Offset from Element Centerline	2750	mm
Element From Node	170	
Detail Type	Weight	
Detail ID	PSV 2- 6"8"	
Dist. from "FROM" Node / Offset dist	450	mm
Miscellaneous Weight	24515	N
Offset from Element Centerline	2750	mm
Element From Node	170	
Detail Type	For./Mom.	
Detail ID	OHP	
Dist. from "FROM" Node / Offset dist	450	mm
Force in X Direction	11000	N
Force in Y Direction	-16500	N
Force in Z Direction	11000	N
Moment about X Axis	8072060	N-mm
Moment about Y Axis	0	N-mm
Moment about Z Axis	5707810	N-mm
Force/Moment Combination Method	SRSS	



XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Bottom skirt		...	3000	...	3000
Top skirt		...	6000	...	3000
Bottom head		...	6100	...	100
Shell 1		...	9100	...	3000
Shell 2		...	12100	...	3000
Shell 3		...	15100	...	3000
Shell 4		...	18100	...	3000
Shell 5		...	21100	...	3000
Shell 6		...	24100	...	3000
Shell 7		...	27100	...	3000
Shell 8		...	30100	...	3000
Shell 9		...	33100	...	3000
Shell 10		...	36100	...	3000
Shell 11		...	39100	...	3000
Shell 12		...	42100	...	3000
Shell 13		...	43900	...	1800
Top head		...	44000	...	100

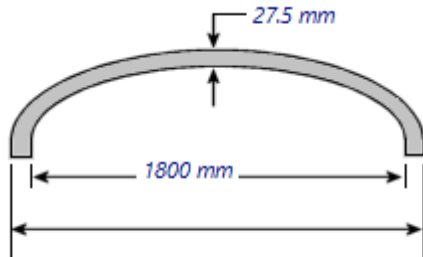
The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Bottom head

Elliptical Head Design Information:					
Design Pressure	P	2.182 MPa	Int. Design Temperature		125 °C
Material		SA-516 70	Ext. Design Temperature		190 °C
Length		100.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.9 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	137.9 MPa
Inside Diameter	Di	1800.000 mm	Actual Stress		80.3 MPa
Spec. Min. Thk t _e , or s _a		27.5000 mm	Spec. Nominal Thk.		32.000 mm
Surface Area		4351895.5 mm ²	SG of Contents		0.520
Element Volume		1018.1 ltr	Weight of Contents		5190.4 N
Empty Weight		11169.0 N	Operating Weight		16359.4 N
			Comp. All. Ope. Stress		124.2 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 22.500 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 11.250 mm

Required Thickness due to Internal Pressure [t_r]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (2.182 \cdot 1806.0 \cdot 0.996) / (2 \cdot 137.9 \cdot 1.0 - 0.2 \cdot 2.182)$$

$$= 14.2457 + 3.0000 = 17.2457 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.024 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 137.9 \cdot 1.0 \cdot 24.5) / (0.996 \cdot 1806.0 + 0.2 \cdot 24.5)$$

$$= 4 - 0 = 4 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 137.9 \cdot 1.0 \cdot 27.5) / (1.0 \cdot 1800.0 + 0.2 \cdot 27.5)$$

$$= 4 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [S_{act}]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (2.182 \cdot (0.996 \cdot 1806.0 + 0.2 \cdot 24.5)) / (2 \cdot 1.0 \cdot 24.5)$$

$$= 80.274 \text{ MPa}$$



Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (2.182 \cdot 903.0) / (137.9 \cdot 1.0 - 0.6 \cdot 2.182) + 3.0$
 $= 17.423 \text{ mm}$

Straight Flange Maximum Allowable Working Pressure:
 Less Operating Hydrostatic Head Pressure of 0.022 MPa
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (137.9 \cdot 1.0 \cdot 29.0) / (903.0 + 0.6 \cdot 29.0)$
 $= 4 - 0 = 4 \text{ MPa}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (1806.0 / (2 \cdot 453.0))^2) / 6$
 $= 0.995592$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 7.453 %
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 27.5$, $t_r = 16.158$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.66$, Temp. Reduction = 19 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -32 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 32.0$, $t_r = 16.377$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.565$, Temp. Reduction = 25 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	2.474 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-25 °C
Governing Thk.	27.500 mm	Basic MDMT	-32 °C
Stress/Thk. Ratio	0.660	Temperature Difference	16 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 22.500 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 11.250 mm

EXTERNAL PRESSURE RESULTS for: Bottom head

Elliptical Head From 30 to 40 Ext. Chart: CS-2 at 190 °C

Elastic Modulus from Chart: CS-2 at 190 °C: 194854 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
24.500	1855.00	75.71	0.0018344	92.89

$MAEP = B / (K_0 \cdot Do / t) = 92.8856 / (0.9 \cdot 75.7143) = 1.3631 \text{ MPa}$

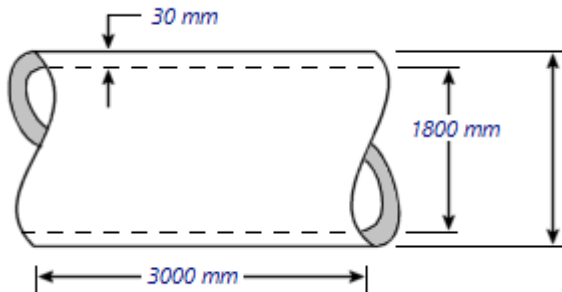
Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
3.350	1855.00	553.72	0.0002508	24.44

$$MAEP = B/(K0*Do/t) = 24.4376/(0.9 * 553.7189) = 0.049 \text{ MPa}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell 1

Cylindrical Shell Design Information:					
Design Pressure	P	2.179 MPa	Int. Design Temperature		125 °C
Material		SA-516 70	Ext. Design Temperature		190 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.9 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	137.9 MPa
Inside Diameter	Di	1800.000 mm	Actual Stress		74.2 MPa
Spec. Min. Thk t,e, or sa		30.0000 mm	Spec. Nominal Thk.		30.000 mm
Surface Area		17530088.0 mm²	SG of Contents		0.520
Element Volume		7635.4 ltr	Weight of Contents		38927.6 N
Empty Weight		39324.7 N	Operating Weight		78252.3 N
			Comp. All. Ope. Stress		126.0 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.179 \cdot 903.0) / (137.9 \cdot 1.0 - 0.6 \cdot 2.179)$$

$$= 14.4043 + 3.0000 = 17.4043 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.021 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 27.0) / (903.0 + 0.6 \cdot 27.0)$$

$$= 4 - 0 = 4 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 30.0) / (900.0 + 0.6 \cdot 30.0)$$

$$= 5 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.179 \cdot (903.0 + 0.6 \cdot 27.0)) / (1.0 \cdot 27.0)$$

$$= 74.179 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 1.639 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 30.0, tr = 16.358, c = 3.0 mm, E* = 1.0



Thickness Ratio = $tr * E / (tg - c) = 0.606$, Temp. Reduction = 22 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 4 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -18 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2.471 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-25 °C
Governing Thk.	30.000 mm	Basic MDMT	4 °C
Stress/Thk. Ratio	0.606	Temperature Difference	22 °C
UCS-66(c) red.	No	Computed Min. Temp.	-18 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell 1

Cylindrical Shell From 40 to 50 Ext. Chart: CS-2 at 190 °C

Elastic Modulus from Chart: CS-2 at 190 °C: 194854 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
27.000	1860.00	38300.00	68.89	20.5914	0.0002415	23.53

$$MAEP = (4*B)/(3*(Do/t)) = (4*23.5303)/(3*68.8889) = 0.4554 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.911	1860.00	38300.00	144.06	20.5914	0.0000544	5.30

$$MAEP = (4*B)/(3*(Do/t)) = (4*5.2979)/(3*144.0594) = 0.049 \text{ MPa}$$

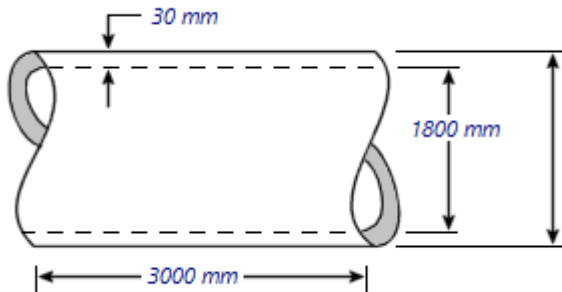
Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
27.000	1860.00	1073953.62	68.89	50.0000	0.0002392	23.31

$$MAEP = (4*B)/(3*(Do/t)) = (4*23.309)/(3*68.8889) = 0.4511 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell 2

Cylindrical Shell Design Information:					
Design Pressure	P	2.164 MPa	Int. Design Temperature		125 °C
Material		SA-516 70	Ext. Design Temperature		190 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.9 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	137.9 MPa
Inside Diameter	Di	1800.000 mm	Actual Stress		73.7 MPa
Spec. Min. Thk t _e , or s _a		30.0000 mm	Spec. Nominal Thk.		30.000 mm
Surface Area		17530088.0 mm ²	SG of Contents		0.520
Element Volume		7635.4 ltr	Weight of Contents		15571.1 N
Empty Weight		39324.7 N	Operating Weight		54895.7 N
			Comp. All. Ope. Stress		126.0 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.164 \cdot 903.0) / (137.9 \cdot 1.0 - 0.6 \cdot 2.164)$$

$$= 14.3022 + 3.0000 = 17.3022 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.006 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 27.0) / (903.0 + 0.6 \cdot 27.0)$$

$$= 4 - 0.01 = 4 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 30.0) / (900.0 + 0.6 \cdot 30.0)$$

$$= 5 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.164 \cdot (903.0 + 0.6 \cdot 27.0)) / (1.0 \cdot 27.0)$$

$$= 73.658 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.639 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 30.0, tr = 16.256, c = 3.0 mm, E* = 1.0



Thickness Ratio = $tr * E / (tg - c) = 0.602$, Temp. Reduction = 22 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 4 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -18 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2.456 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-25 °C
Governing Thk.	30.000 mm	Basic MDMT	4 °C
Stress/Thk. Ratio	0.602	Temperature Difference	22 °C
UCS-66(c) red.	No	Computed Min. Temp.	-18 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell 2

Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 190 °C

Elastic Modulus from Chart: CS-2 at 190 °C: 194854 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
27.000	1860.00	38300.00	68.89	20.5914	0.0002415	23.53

$$MAEP = (4*B)/(3*(Do/t)) = (4*23.5303)/(3*68.8889) = 0.4554 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.911	1860.00	38300.00	144.06	20.5914	0.0000544	5.30

$$MAEP = (4*B)/(3*(Do/t)) = (4*5.2979)/(3*144.0594) = 0.049 \text{ MPa}$$

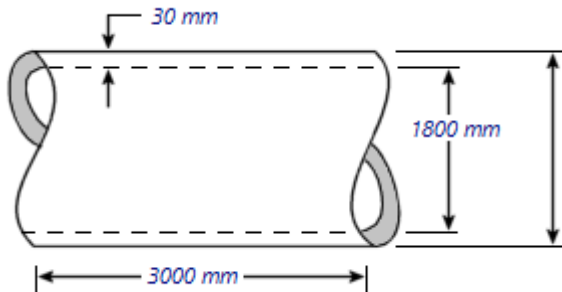
Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
27.000	1860.00	1073953.62	68.89	50.0000	0.0002392	23.31

$$MAEP = (4*B)/(3*(Do/t)) = (4*23.309)/(3*68.8889) = 0.4511 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell 3

Cylindrical Shell Design Information:					
Design Pressure	P	2.157 MPa	Int. Design Temperature		125 °C
Material		SA-516 70	Ext. Design Temperature		190 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.9 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	137.9 MPa
Inside Diameter	Di	1800.000 mm	Actual Stress		73.4 MPa
Spec. Min. Thk t _e , or s _a		30.0000 mm	Spec. Nominal Thk.		30.000 mm
Surface Area		17530088.0 mm ²	SG of Contents		0.000
Element Volume		7635.4 ltr	Weight of Contents		0.0 N
Empty Weight		39324.7 N	Operating Weight		39324.7 N
			Comp. All. Ope. Stress		126.0 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.157 \cdot 903.0) / (137.9 \cdot 1.0 - 0.6 \cdot 2.157)$$

$$= 14.2614 + 3.0000 = 17.2614 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 27.0) / (903.0 + 0.6 \cdot 27.0)$$

$$= 4 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 30.0) / (900.0 + 0.6 \cdot 30.0)$$

$$= 5 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.157 \cdot (903.0 + 0.6 \cdot 27.0)) / (1.0 \cdot 27.0)$$

$$= 73.450 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.639 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 30.0, t_r = 16.215, c = 3.0 mm, E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.601, Temp. Reduction = 22 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 4 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -18 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2.450 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-25 °C
Governing Thk.	30.000 mm	Basic MDMT	4 °C
Stress/Thk. Ratio	0.601	Temperature Difference	22 °C
UCS-66(c) red.	No	Computed Min. Temp.	-18 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell 3

Cylindrical Shell From 60 to 70 Ext. Chart: CS-2 at 190 °C

Elastic Modulus from Chart: CS-2 at 190 °C: 194854 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
27.000	1860.00	38300.00	68.89	20.5914	0.0002415	23.53

$$MAEP = (4*B)/(3*(Do/t)) = (4*23.5303)/(3*68.8889) = 0.4554 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.911	1860.00	38300.00	144.06	20.5914	0.0000544	5.30

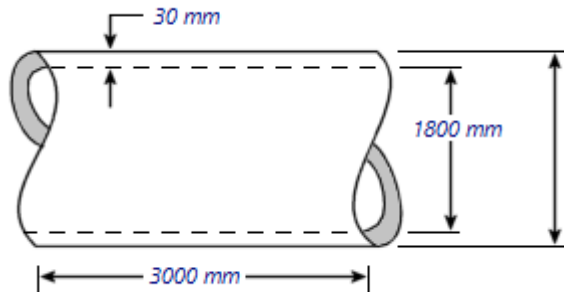
$$MAEP = (4*B)/(3*(Do/t)) = (4*5.2979)/(3*144.0594) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
27.000	1860.00	1073953.62	68.89	50.0000	0.0002392	23.31

$$MAEP = (4*B)/(3*(Do/t)) = (4*23.309)/(3*68.8889) = 0.4511 \text{ MPa}$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell 4

Cylindrical Shell Design Information:					
Design Pressure	P	2.157 MPa	Int. Design Temperature		125 °C
Material		SA-516 70	Ext. Design Temperature		190 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.9 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	137.9 MPa
Inside Diameter	Di	1800.000 mm	Actual Stress		73.4 MPa
Spec. Min. Thk t _e , or s _a		30.0000 mm	Spec. Nominal Thk.		30.000 mm
Surface Area		17530088.0 mm²	SG of Contents		0.000
Element Volume		7635.4 ltr	Weight of Contents		0.0 N
Empty Weight		39324.7 N	Operating Weight		39324.7 N
			Comp. All. Ope. Stress		126.0 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.157 \cdot 903.0) / (137.9 \cdot 1.0 - 0.6 \cdot 2.157)$$

$$= 14.2614 + 3.0000 = 17.2614 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 27.0) / (903.0 + 0.6 \cdot 27.0)$$

$$= 4 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 30.0) / (900.0 + 0.6 \cdot 30.0)$$

$$= 5 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.157 \cdot (903.0 + 0.6 \cdot 27.0)) / (1.0 \cdot 27.0)$$

$$= 73.450 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.639 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 30.0, t_r = 16.215, c = 3.0 mm, E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.601, Temp. Reduction = 22 °C



Min Metal Temp. w/o impact per UCS-66, Curve B 4 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -18 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2.450 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-25 °C
Governing Thk.	30.000 mm	Basic MDMT	4 °C
Stress/Thk. Ratio	0.601	Temperature Difference	22 °C
UCS-66(c) red.	No	Computed Min. Temp.	-18 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell 4

Cylindrical Shell From 70 to 80 Ext. Chart: CS-2 at 190 °C

Elastic Modulus from Chart: CS-2 at 190 °C: 194854 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
27.000	1860.00	38300.00	68.89	20.5914	0.0002415	23.53

$$MAEP = (4*B)/(3*(Do/t)) = (4*23.5303)/(3*68.8889) = 0.4554 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.911	1860.00	38300.00	144.06	20.5914	0.0000544	5.30

$$MAEP = (4*B)/(3*(Do/t)) = (4*5.2979)/(3*144.0594) = 0.049 \text{ MPa}$$

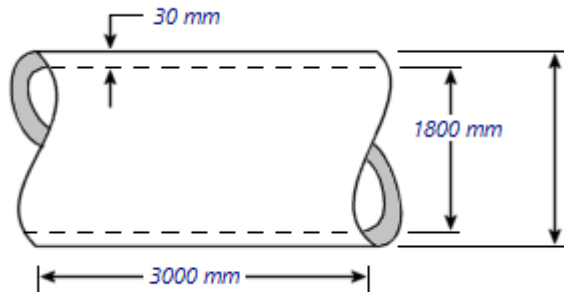
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
27.000	1860.00	1073953.62	68.89	50.0000	0.0002392	23.31

$$MAEP = (4*B)/(3*(Do/t)) = (4*23.309)/(3*68.8889) = 0.4511 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell 5

Cylindrical Shell Design Information:					
Design Pressure	P	2.157 MPa	Int. Design Temperature		125 °C
Material		SA-516 70	Ext. Design Temperature		190 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.9 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	137.9 MPa
Inside Diameter	Di	1800.000 mm	Actual Stress		73.4 MPa
Spec. Min. Thk t _e , or s _a		30.0000 mm	Spec. Nominal Thk.		30.000 mm
Surface Area		17530088.0 mm ²	SG of Contents		0.000
Element Volume		7635.4 ltr	Weight of Contents		0.0 N
Empty Weight		39324.7 N	Operating Weight		39324.7 N
			Comp. All. Ope. Stress		126.0 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.157 \cdot 903.0) / (137.9 \cdot 1.0 - 0.6 \cdot 2.157)$$

$$= 14.2614 + 3.0000 = 17.2614 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 27.0) / (903.0 + 0.6 \cdot 27.0)$$

$$= 4 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 30.0) / (900.0 + 0.6 \cdot 30.0)$$

$$= 5 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.157 \cdot (903.0 + 0.6 \cdot 27.0)) / (1.0 \cdot 27.0)$$

$$= 73.450 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.639 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 30.0, t_r = 16.215, c = 3.0 mm, E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.601, Temp. Reduction = 22 °C



Min Metal Temp. w/o impact per UCS-66, Curve B 4 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -18 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2.450 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-25 °C
Governing Thk.	30.000 mm	Basic MDMT	4 °C
Stress/Thk. Ratio	0.601	Temperature Difference	22 °C
UCS-66(c) red.	No	Computed Min. Temp.	-18 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell 5

Cylindrical Shell From 80 to 90 Ext. Chart: CS-2 at 190 °C

Elastic Modulus from Chart: CS-2 at 190 °C: 194854 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
27.000	1860.00	38300.00	68.89	20.5914	0.0002415	23.53

$$MAEP = (4*B)/(3*(Do/t)) = (4*23.5303)/(3*68.8889) = 0.4554 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.911	1860.00	38300.00	144.06	20.5914	0.0000544	5.30

$$MAEP = (4*B)/(3*(Do/t)) = (4*5.2979)/(3*144.0594) = 0.049 \text{ MPa}$$

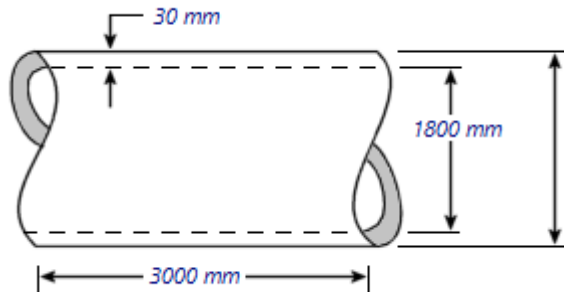
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
27.000	1860.00	1073953.62	68.89	50.0000	0.0002392	23.31

$$MAEP = (4*B)/(3*(Do/t)) = (4*23.309)/(3*68.8889) = 0.4511 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell 6

Cylindrical Shell Design Information:					
Design Pressure	P	2.157 MPa	Int. Design Temperature		125 °C
Material		SA-516 70	Ext. Design Temperature		190 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.9 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	137.9 MPa
Inside Diameter	Di	1800.000 mm	Actual Stress		73.4 MPa
Spec. Min. Thk t _e , or s _a		30.0000 mm	Spec. Nominal Thk.		30.000 mm
Surface Area		17530088.0 mm ²	SG of Contents		0.000
Element Volume		7635.4 ltr	Weight of Contents		0.0 N
Empty Weight		39324.7 N	Operating Weight		39324.7 N
			Comp. All. Ope. Stress		126.0 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.157 \cdot 903.0) / (137.9 \cdot 1.0 - 0.6 \cdot 2.157)$$

$$= 14.2614 + 3.0000 = 17.2614 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 27.0) / (903.0 + 0.6 \cdot 27.0)$$

$$= 4 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 30.0) / (900.0 + 0.6 \cdot 30.0)$$

$$= 5 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.157 \cdot (903.0 + 0.6 \cdot 27.0)) / (1.0 \cdot 27.0)$$

$$= 73.450 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.639 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 30.0, t_r = 16.215, c = 3.0 mm, E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.601, Temp. Reduction = 22 °C



Min Metal Temp. w/o impact per UCS-66, Curve B 4 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -18 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2.450 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-25 °C
Governing Thk.	30.000 mm	Basic MDMT	4 °C
Stress/Thk. Ratio	0.601	Temperature Difference	22 °C
UCS-66(c) red.	No	Computed Min. Temp.	-18 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell 6

Cylindrical Shell From 90 to 100 Ext. Chart: CS-2 at 190 °C

Elastic Modulus from Chart: CS-2 at 190 °C: 194854 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
27.000	1860.00	38300.00	68.89	20.5914	0.0002415	23.53

$$MAEP = (4*B)/(3*(Do/t)) = (4*23.5303)/(3*68.8889) = 0.4554 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.911	1860.00	38300.00	144.06	20.5914	0.0000544	5.30

$$MAEP = (4*B)/(3*(Do/t)) = (4*5.2979)/(3*144.0594) = 0.049 \text{ MPa}$$

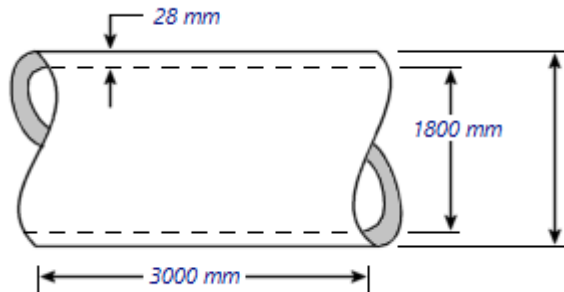
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
27.000	1860.00	1073953.62	68.89	50.0000	0.0002392	23.31

$$MAEP = (4*B)/(3*(Do/t)) = (4*23.309)/(3*68.8889) = 0.4511 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell 7

Cylindrical Shell Design Information:					
Design Pressure	P	2.157 MPa	Int. Design Temperature		125 °C
Material		SA-516 70	Ext. Design Temperature		190 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.9 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	137.9 MPa
Inside Diameter	Di	1800.000 mm	Actual Stress		79.2 MPa
Spec. Min. Thk t _e , or s _a		28.0000 mm	Spec. Nominal Thk.		28.000 mm
Surface Area		17492388.0 mm ²	SG of Contents		0.000
Element Volume		7635.4 ltr	Weight of Contents		0.0 N
Empty Weight		36663.0 N	Operating Weight		36663.0 N
			Comp. All. Ope. Stress		124.6 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.157 \cdot 903.0) / (137.9 \cdot 1.0 - 0.6 \cdot 2.157)$$

$$= 14.2614 + 3.0000 = 17.2614 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 25.0) / (903.0 + 0.6 \cdot 25.0)$$

$$= 4 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 28.0) / (900.0 + 0.6 \cdot 28.0)$$

$$= 4 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.157 \cdot (903.0 + 0.6 \cdot 25.0)) / (1.0 \cdot 25.0)$$

$$= 79.222 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.532 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 28.0, t_r = 16.215, c = 3.0 mm, E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.649, Temp. Reduction = 20 °C



Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -17 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2.450 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-25 °C
Governing Thk.	28.000 mm	Basic MDMT	2 °C
Stress/Thk. Ratio	0.649	Temperature Difference	20 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell 7

Cylindrical Shell From 100 to 110 Ext. Chart: CS-2 at 190 °C

Elastic Modulus from Chart: CS-2 at 190 °C: 194854 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	1856.00	38300.00	74.24	20.6358	0.0002073	20.20

$$MAEP = (4*B)/(3*(Do/t)) = (4*20.1954)/(3*74.24) = 0.3627 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.884	1856.00	38300.00	144.06	20.6358	0.0000544	5.30

$$MAEP = (4*B)/(3*(Do/t)) = (4*5.2978)/(3*144.0571) = 0.049 \text{ MPa}$$

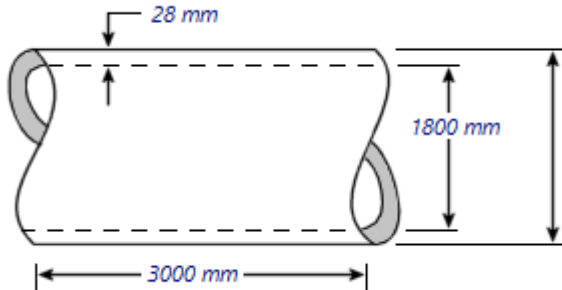
Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	1856.00	2063775.75	74.24	50.0000	0.0002052	19.99

$$MAEP = (4*B)/(3*(Do/t)) = (4*19.9899)/(3*74.24) = 0.359 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell 8

Cylindrical Shell Design Information:					
Design Pressure	P	2.157 MPa	Int. Design Temperature		125 °C
Material		SA-516 70	Ext. Design Temperature		190 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.9 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	137.9 MPa
Inside Diameter	Di	1800.000 mm	Actual Stress		79.2 MPa
Spec. Min. Thk t _e , or s _a		28.0000 mm	Spec. Nominal Thk.		28.000 mm
Surface Area		17492388.0 mm ²	SG of Contents		0.000
Element Volume		7635.4 ltr	Weight of Contents		0.0 N
Empty Weight		36663.0 N	Operating Weight		36663.0 N
			Comp. All. Ope. Stress		124.6 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.157 \cdot 903.0) / (137.9 \cdot 1.0 - 0.6 \cdot 2.157)$$

$$= 14.2614 + 3.0000 = 17.2614 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 25.0) / (903.0 + 0.6 \cdot 25.0)$$

$$= 4 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 28.0) / (900.0 + 0.6 \cdot 28.0)$$

$$= 4 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.157 \cdot (903.0 + 0.6 \cdot 25.0)) / (1.0 \cdot 25.0)$$

$$= 79.222 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.532 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 28.0, t_r = 16.215, c = 3.0 mm, E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.649, Temp. Reduction = 20 °C



Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -17 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2.450 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-25 °C
Governing Thk.	28.000 mm	Basic MDMT	2 °C
Stress/Thk. Ratio	0.649	Temperature Difference	20 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell 8

Cylindrical Shell From 110 to 120 Ext. Chart: CS-2 at 190 °C

Elastic Modulus from Chart: CS-2 at 190 °C: 194854 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	1856.00	38300.00	74.24	20.6358	0.0002073	20.20

$$MAEP = (4*B)/(3*(Do/t)) = (4*20.1954)/(3*74.24) = 0.3627 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.884	1856.00	38300.00	144.06	20.6358	0.0000544	5.30

$$MAEP = (4*B)/(3*(Do/t)) = (4*5.2978)/(3*144.0571) = 0.049 \text{ MPa}$$

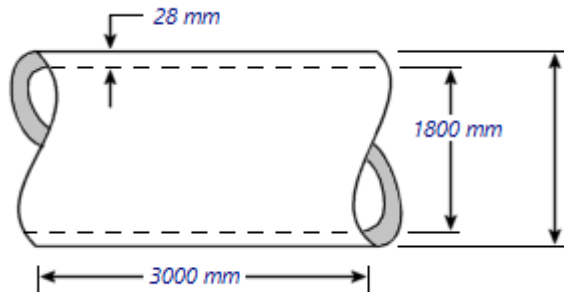
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	1856.00	2063775.75	74.24	50.0000	0.0002052	19.99

$$MAEP = (4*B)/(3*(Do/t)) = (4*19.9899)/(3*74.24) = 0.359 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell 9

Cylindrical Shell Design Information:					
Design Pressure	P	2.157 MPa	Int. Design Temperature		125 °C
Material		SA-516 70	Ext. Design Temperature		190 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.9 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	137.9 MPa
Inside Diameter	Di	1800.000 mm	Actual Stress		79.2 MPa
Spec. Min. Thk t _e , or s _a		28.0000 mm	Spec. Nominal Thk.		28.000 mm
Surface Area		17492388.0 mm ²	SG of Contents		0.000
Element Volume		7635.4 ltr	Weight of Contents		0.0 N
Empty Weight		36663.0 N	Operating Weight		36663.0 N
			Comp. All. Ope. Stress		124.6 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.157 \cdot 903.0) / (137.9 \cdot 1.0 - 0.6 \cdot 2.157)$$

$$= 14.2614 + 3.0000 = 17.2614 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 25.0) / (903.0 + 0.6 \cdot 25.0)$$

$$= 4 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 28.0) / (900.0 + 0.6 \cdot 28.0)$$

$$= 4 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.157 \cdot (903.0 + 0.6 \cdot 25.0)) / (1.0 \cdot 25.0)$$

$$= 79.222 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.532 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 28.0, t_r = 16.215, c = 3.0 mm, E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.649, Temp. Reduction = 20 °C



Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -17 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2.450 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-25 °C
Governing Thk.	28.000 mm	Basic MDMT	2 °C
Stress/Thk. Ratio	0.649	Temperature Difference	20 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell 9

Cylindrical Shell From 120 to 130 Ext. Chart: CS-2 at 190 °C

Elastic Modulus from Chart: CS-2 at 190 °C: 194854 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	1856.00	38300.00	74.24	20.6358	0.0002073	20.20

$$MAEP = (4*B)/(3*(Do/t)) = (4*20.1954)/(3*74.24) = 0.3627 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.884	1856.00	38300.00	144.06	20.6358	0.0000544	5.30

$$MAEP = (4*B)/(3*(Do/t)) = (4*5.2978)/(3*144.0571) = 0.049 \text{ MPa}$$

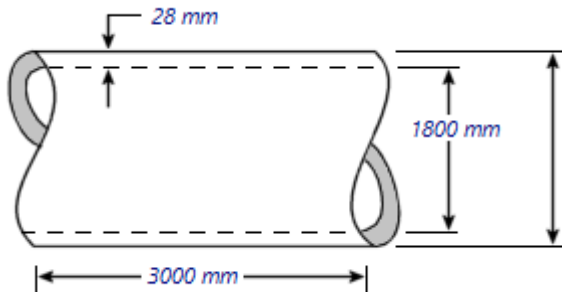
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	1856.00	2063775.75	74.24	50.0000	0.0002052	19.99

$$MAEP = (4*B)/(3*(Do/t)) = (4*19.9899)/(3*74.24) = 0.359 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell 10

Cylindrical Shell Design Information:					
Design Pressure	P	2.157 MPa	Int. Design Temperature		125 °C
Material		SA-516 70	Ext. Design Temperature		190 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.9 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	137.9 MPa
Inside Diameter	Di	1800.000 mm	Actual Stress		79.2 MPa
Spec. Min. Thk t _e , or s _a		28.0000 mm	Spec. Nominal Thk.		28.000 mm
Surface Area		17492388.0 mm²	SG of Contents		0.000
Element Volume		7635.4 ltr	Weight of Contents		0.0 N
Empty Weight		36663.0 N	Operating Weight		36663.0 N
			Comp. All. Ope. Stress		124.6 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.157 \cdot 903.0) / (137.9 \cdot 1.0 - 0.6 \cdot 2.157)$$

$$= 14.2614 + 3.0000 = 17.2614 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 25.0) / (903.0 + 0.6 \cdot 25.0)$$

$$= 4 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 28.0) / (900.0 + 0.6 \cdot 28.0)$$

$$= 4 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.157 \cdot (903.0 + 0.6 \cdot 25.0)) / (1.0 \cdot 25.0)$$

$$= 79.222 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.532 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 28.0, t_r = 16.215, c = 3.0 mm, E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.649, Temp. Reduction = 20 °C



Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -17 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2.450 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-25 °C
Governing Thk.	28.000 mm	Basic MDMT	2 °C
Stress/Thk. Ratio	0.649	Temperature Difference	20 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell 10

Cylindrical Shell From 130 to 140 Ext. Chart: CS-2 at 190 °C

Elastic Modulus from Chart: CS-2 at 190 °C: 194854 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	1856.00	38300.00	74.24	20.6358	0.0002073	20.20

$$MAEP = (4*B)/(3*(Do/t)) = (4*20.1954)/(3*74.24) = 0.3627 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.884	1856.00	38300.00	144.06	20.6358	0.0000544	5.30

$$MAEP = (4*B)/(3*(Do/t)) = (4*5.2978)/(3*144.0571) = 0.049 \text{ MPa}$$

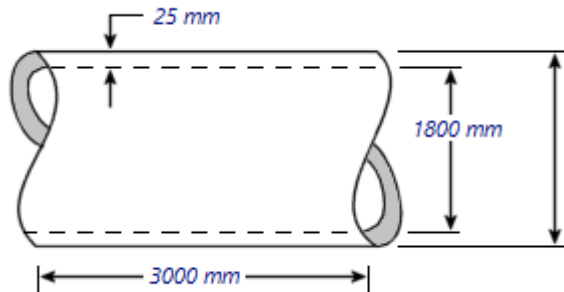
Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	1856.00	2063775.75	74.24	50.0000	0.0002052	19.99

$$MAEP = (4*B)/(3*(Do/t)) = (4*19.9899)/(3*74.24) = 0.359 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell 11

Cylindrical Shell Design Information:					
Design Pressure	P	2.157 MPa	Int. Design Temperature		125 °C
Material		SA-516 70	Ext. Design Temperature		190 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.9 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	137.9 MPa
Inside Diameter	Di	1800.000 mm	Actual Stress		89.8 MPa
Spec. Min. Thk t _e , or s _a		25.0000 mm	Spec. Nominal Thk.		25.000 mm
Surface Area		17435840.0 mm ²	SG of Contents		0.000
Element Volume		7635.4 ltr	Weight of Contents		0.0 N
Empty Weight		32681.0 N	Operating Weight		32681.0 N
			Comp. All. Ope. Stress		122.2 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.157 \cdot 903.0) / (137.9 \cdot 1.0 - 0.6 \cdot 2.157)$$

$$= 14.2614 + 3.0000 = 17.2614 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 22.0) / (903.0 + 0.6 \cdot 22.0)$$

$$= 3 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 25.0) / (900.0 + 0.6 \cdot 25.0)$$

$$= 4 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.157 \cdot (903.0 + 0.6 \cdot 22.0)) / (1.0 \cdot 22.0)$$

$$= 89.848 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.370 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 25.0, t_r = 16.215, c = 3.0 mm, E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.737, Temp. Reduction = 15 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -1 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -16 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2.450 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-25 °C
Governing Thk.	25.000 mm	Basic MDMT	-1 °C
Stress/Thk. Ratio	0.737	Temperature Difference	15 °C
UCS-66(c) red.	No	Computed Min. Temp.	-16 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell 11

Cylindrical Shell From 140 to 150 Ext. Chart: CS-2 at 190 °C

Elastic Modulus from Chart: CS-2 at 190 °C: 194854 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
22.000	1850.00	38300.00	84.09	20.7027	0.0001593	15.52

$$MAEP = (4*B)/(3*(Do/t)) = (4*15.517)/(3*84.0909) = 0.246 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.842	1850.00	38300.00	144.05	20.7027	0.0000544	5.30

$$MAEP = (4*B)/(3*(Do/t)) = (4*5.2976)/(3*144.0536) = 0.049 \text{ MPa}$$

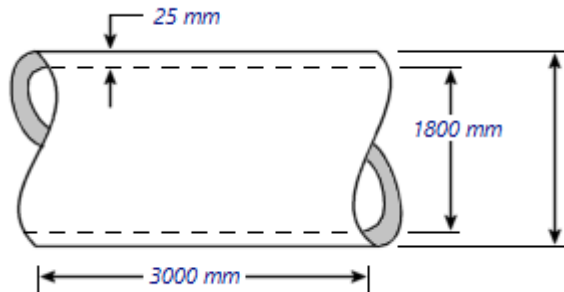
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
22.000	1850.00	948136.06	84.09	50.0000	0.0001575	15.34

$$MAEP = (4*B)/(3*(Do/t)) = (4*15.3431)/(3*84.0909) = 0.2433 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell 12

Cylindrical Shell Design Information:					
Design Pressure	P	2.157 MPa	Int. Design Temperature		125 °C
Material		SA-516 70	Ext. Design Temperature		190 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.9 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	137.9 MPa
Inside Diameter	Di	1800.000 mm	Actual Stress		89.8 MPa
Spec. Min. Thk t _e , or s _a		25.0000 mm	Spec. Nominal Thk.		25.000 mm
Surface Area		17435840.0 mm ²	SG of Contents		0.000
Element Volume		7635.4 ltr	Weight of Contents		0.0 N
Empty Weight		32681.0 N	Operating Weight		32681.0 N
			Comp. All. Ope. Stress		122.2 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.157 \cdot 903.0) / (137.9 \cdot 1.0 - 0.6 \cdot 2.157)$$

$$= 14.2614 + 3.0000 = 17.2614 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 22.0) / (903.0 + 0.6 \cdot 22.0)$$

$$= 3 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 25.0) / (900.0 + 0.6 \cdot 25.0)$$

$$= 4 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.157 \cdot (903.0 + 0.6 \cdot 22.0)) / (1.0 \cdot 22.0)$$

$$= 89.848 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.370 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 25.0, t_r = 16.215, c = 3.0 mm, E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.737, Temp. Reduction = 15 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -1 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -16 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2.450 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-25 °C
Governing Thk.	25.000 mm	Basic MDMT	-1 °C
Stress/Thk. Ratio	0.737	Temperature Difference	15 °C
UCS-66(c) red.	No	Computed Min. Temp.	-16 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell 12

Cylindrical Shell From 150 to 160 Ext. Chart: CS-2 at 190 °C

Elastic Modulus from Chart: CS-2 at 190 °C: 194854 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
22.000	1850.00	38300.00	84.09	20.7027	0.0001593	15.52

$$MAEP = (4*B)/(3*(Do/t)) = (4*15.517)/(3*84.0909) = 0.246 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.842	1850.00	38300.00	144.05	20.7027	0.0000544	5.30

$$MAEP = (4*B)/(3*(Do/t)) = (4*5.2976)/(3*144.0536) = 0.049 \text{ MPa}$$

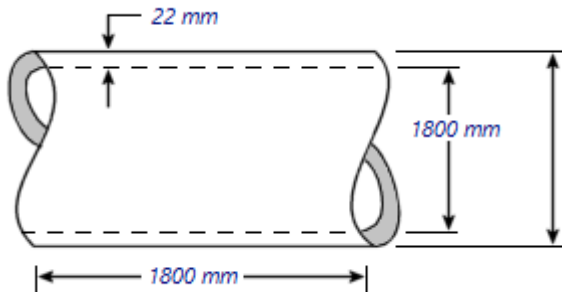
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
22.000	1850.00	948136.06	84.09	50.0000	0.0001575	15.34

$$MAEP = (4*B)/(3*(Do/t)) = (4*15.3431)/(3*84.0909) = 0.2433 \text{ MPa}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell 13

Cylindrical Shell Design Information:					
Design Pressure	P	2.157 MPa	Int. Design Temperature		125 °C
Material		SA-516 70	Ext. Design Temperature		190 °C
Length		1800.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.9 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	137.9 MPa
Inside Diameter	Di	1800.000 mm	Actual Stress		103.8 MPa
Spec. Min. Thk t _e or s _a		22.0000 mm	Spec. Nominal Thk.		22.000 mm
Surface Area		10427574.0 mm ²	SG of Contents		0.000
Element Volume		4581.3 ltr	Weight of Contents		0.0 N
Empty Weight		17227.2 N	Operating Weight		17227.2 N
			Comp. All. Ope. Stress		119.2 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.157 \cdot 903.0) / (137.9 \cdot 1.0 - 0.6 \cdot 2.157)$$

$$= 14.2614 + 3.0000 = 17.2614 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 19.0) / (903.0 + 0.6 \cdot 19.0)$$

$$= 3 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 22.0) / (900.0 + 0.6 \cdot 22.0)$$

$$= 3 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.157 \cdot (903.0 + 0.6 \cdot 19.0)) / (1.0 \cdot 19.0)$$

$$= 103.831 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.207 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 22.0, t_r = 16.215, c = 3.0 mm, E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.853, Temp. Reduction = 8 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -13 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2.450 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-25 °C
Governing Thk.	22.000 mm	Basic MDMT	-5 °C
Stress/Thk. Ratio	0.853	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-13 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell 13

Cylindrical Shell From 160 to 170 Ext. Chart: CS-2 at 190 °C

Elastic Modulus from Chart: CS-2 at 190 °C: 194854 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
19.000	1844.00	38300.00	97.05	20.7701	0.0001193	11.62

$$MAEP = (4*B)/(3*(Do/t)) = (4*11.6244)/(3*97.0526) = 0.1597 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.801	1844.00	38300.00	144.05	20.7701	0.0000544	5.30

$$MAEP = (4*B)/(3*(Do/t)) = (4*5.2976)/(3*144.049) = 0.049 \text{ MPa}$$

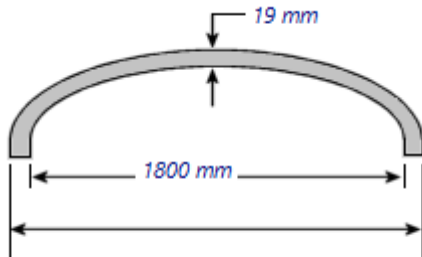
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
19.000	1844.00	1286165.38	97.05	50.0000	0.0001180	11.49

$$MAEP = (4*B)/(3*(Do/t)) = (4*11.4931)/(3*97.0526) = 0.1579 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Top head

Elliptical Head Design Information:					
Design Pressure	P	2.157 MPa	Int. Design Temperature		125 °C
Material		SA-516 70	Ext. Design Temperature		190 °C
Length		100.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.9 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	137.9 MPa
Inside Diameter	Di	1800.000 mm	Actual Stress		121.4 MPa
Spec. Min. Thk t _e , or s _a		19.0000 mm	Spec. Nominal Thk.		22.000 mm
Surface Area		4265224.0 mm ²	SG of Contents		0.000
Element Volume		1018.1 ltr	Weight of Contents		0.0 N
Empty Weight		7600.3 N	Operating Weight		7600.3 N
			Comp. All. Ope. Stress		115.5 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 22.500 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 11.250 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (2.157 \cdot 1806.0 \cdot 0.996) / (2 \cdot 137.9 \cdot 1.0 - 0.2 \cdot 2.157)$$

$$= 14.0873 + 3.0000 = 17.0873 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 137.9 \cdot 1.0 \cdot 16.0) / (0.996 \cdot 1806.0 + 0.2 \cdot 16.0)$$

$$= 2 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 137.9 \cdot 1.0 \cdot 19.0) / (1.0 \cdot 1800.0 + 0.2 \cdot 19.0)$$

$$= 3 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (2.157 \cdot (0.996 \cdot 1806.0 + 0.2 \cdot 16.0)) / (2 \cdot 1.0 \cdot 16.0)$$

$$= 121.441 \text{ MPa}$$

Straight Flange Required Thickness:



$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (2.157 \cdot 903.0) / (137.9 \cdot 1.0 - 0.6 \cdot 2.157) + 3.0$$

$$= 17.261 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.9 \cdot 1.0 \cdot 19.0) / (903.0 + 0.6 \cdot 19.0)$$

$$= 3 \text{ MPa}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (1806.0 / (2 \cdot 453.0))^2) / 6$$

$$= 0.995592$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.205 %
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 19.0, tr = 16.0, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 1.0, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -41 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 22.0, tr = 16.215, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.853, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -38 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -46 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	2.450 MPa
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-25 °C
Governing Thk.	19.000 mm	Basic MDMT	-41 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-41 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 22.500 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 11.250 mm

EXTERNAL PRESSURE RESULTS for: Top head

Elliptical Head From 170 to 180 Ext. Chart: CS-2 at 190 °C

Elastic Modulus from Chart: CS-2 at 190 °C: 194854 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
16.000	1838.00	114.88	0.0012090	83.75

$$MAEP = B / (K_0 \cdot Do / t) = 83.7547 / (0.9 \cdot 114.875) = 0.8101 \text{ MPa}$$



Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
3.319	1838.00	553.73	0.0002508	24.44

$$MAEP = B/(K0*Do/t) = 24.4373/(0.9 * 553.7277) = 0.049 \text{ MPa}$$

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	3.185 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2.805 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	3.771 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	2.695 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	3.085 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3.085 MPa
App 27-4	= M.A.W.P.	2.450 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 2.45 • 1.0
 = 3.185 MPa

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	131.2	235.8	0.556	3.57
Shell 1	121.2	235.8	0.514	3.56
Shell 2	120.2	235.8	0.510	3.53
Shell 3	119.2	235.8	0.506	3.50
Shell 4	118.2	235.8	0.501	3.47
Shell 5	117.2	235.8	0.497	3.44
Shell 6	116.2	235.8	0.493	3.41
Shell 7	124.3	235.8	0.527	3.38
Shell 8	123.2	235.8	0.522	3.35
Shell 9	122.1	235.8	0.518	3.33
Shell 10	121.0	235.8	0.513	3.30
Shell 11	136.0	235.8	0.577	3.27
Shell 12	134.8	235.8	0.572	3.24
Shell 13	154.4	235.8	0.655	3.21
Top head	179.6	235.8	0.762	3.19

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
B2 10"	Nozzle	118.00	118.00	1.000
B2 10"	Pad	138.00	138.00	1.000
B1 4 "ASSUME #30	Nozzle	118.00	118.00	1.000
B1 4 "ASSUME #30	Pad	138.00	138.00	1.000
Noz L1B #300	Nozzle	118.00	118.00	1.000
Noz L2B #300	Nozzle	118.00	118.00	1.000
Noz L3B #300	Nozzle	118.00	118.00	1.000
Noz L4B #300	Nozzle	118.00	118.00	1.000
Noz U2 #300	Nozzle	118.00	118.00	1.000
Noz U1 #300	Nozzle	118.00	118.00	1.000
Noz MW5 24" #300	Nozzle	137.90	137.90	1.000
Noz MW5 24" #300	Pad	137.90	137.90	1.000
Noz L1A #300	Nozzle	118.00	118.00	1.000
Noz L2A #300	Nozzle	118.00	118.00	1.000
Noz L3A #300	Nozzle	118.00	118.00	1.000
Noz L4A #300	Nozzle	118.00	118.00	1.000
MW4 #300	Nozzle	137.90	137.90	1.000
MW4 #300	Pad	138.00	138.00	1.000
P1 #300	Nozzle	118.00	118.00	1.000
P2 #300	Nozzle	118.00	118.00	1.000
F3 #300	Nozzle	118.00	118.00	1.000
F3 #300	Pad	138.00	138.00	1.000
T3 #300	Nozzle	118.00	118.00	1.000



MW3 24" #300	Nozzle	137.90	137.90	1.000
MW3 24" #300	Pad	138.00	138.00	1.000
T2 #300	Nozzle	118.00	118.00	1.000
F1 4" #300	Nozzle	118.00	118.00	1.000
F1 4" #300	Pad	138.00	138.00	1.000
MW2 24" #300	Nozzle	137.90	137.90	1.000
MW2 24" #300	Pad	138.00	138.00	1.000
T1 #300	Nozzle	118.00	118.00	1.000
MW1 24" #300	Nozzle	117.90	117.90	1.000
MW1 24" #300	Pad	138.00	138.00	1.000
F2 6" #300	Nozzle	118.00	118.00	1.000
F2 6" #300	Pad	138.00	138.00	1.000
T 8" #300	Nozzle	118.00	118.00	1.000
T 8" #300	Pad	138.00	138.00	1.000
V1 4" #300	Nozzle	118.00	118.00	1.000
V1 4" #300	Pad	138.00	138.00	1.000
NV2 2" #300	Nozzle	118.00	118.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom head	137.90	137.90	1.000
Shell 1	137.90	137.90	1.000
Shell 2	137.90	137.90	1.000
Shell 3	137.90	137.90	1.000
Shell 4	137.90	137.90	1.000
Shell 5	137.90	137.90	1.000
Shell 6	137.90	137.90	1.000
Shell 7	137.90	137.90	1.000
Shell 8	137.90	137.90	1.000
Shell 9	137.90	137.90	1.000
Shell 10	137.90	137.90	1.000
Shell 11	137.90	137.90	1.000
Shell 12	137.90	137.90	1.000
Shell 13	137.90	137.90	1.000
Top head	137.90	137.90	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
B2 10"	46.30	217.00	0.213
B1 4 "ASSUME #30	23.66	217.00	0.109
Noz L1B #300	9.46	217.00	0.044
Noz L2B #300	9.46	217.00	0.044
Noz L3B #300	9.46	217.00	0.044
Noz L4B #300	9.46	217.00	0.044
Noz U2 #300	9.46	217.00	0.044
Noz U1 #300	9.46	217.00	0.044
Noz MW5 24" #300	118.67	235.80	0.503
Noz L1A #300	9.35	217.00	0.043
Noz L2A #300	9.35	217.00	0.043
Noz L3A #300	9.35	217.00	0.043
Noz L4A #300	9.35	217.00	0.043
MW4 #300	116.71	235.81	0.495
P1 #300	9.26	217.00	0.043
P2 #300	9.27	217.00	0.043
F3 #300	45.35	217.00	0.209
T3 #300	9.16	217.00	0.042
MW3 24" #300	114.64	235.80	0.486
T2 #300	8.85	217.00	0.041
F1 4" #300	16.75	217.00	0.077
MW2 24" #300	107.75	235.81	0.457
T1 #300	8.52	217.00	0.039
MW1 24" #300	73.70	217.19	0.339



F2 6" #300	33.47	217.00	0.154
T 8" #300	34.74	217.00	0.160
V1 4" #300	16.73	217.00	0.077
NV2 2" #300	36.65	217.00	0.169

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	3.185 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2.805 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	3.771 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	2.695 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	3.085 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3.085 MPa
App 27-4	= M.A.W.P.	2.450 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 2.45 • 1.0
 = 3.185 MPa

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	131.2	235.8	0.556	3.57
Shell 1	121.2	235.8	0.514	3.56
Shell 2	120.2	235.8	0.510	3.53
Shell 3	119.2	235.8	0.506	3.50
Shell 4	118.2	235.8	0.501	3.47
Shell 5	117.2	235.8	0.497	3.44
Shell 6	116.2	235.8	0.493	3.41
Shell 7	124.3	235.8	0.527	3.38
Shell 8	123.2	235.8	0.522	3.35
Shell 9	122.1	235.8	0.518	3.33
Shell 10	121.0	235.8	0.513	3.30
Shell 11	136.0	235.8	0.577	3.27
Shell 12	134.8	235.8	0.572	3.24
Shell 13	154.4	235.8	0.655	3.21
Top head	179.6	235.8	0.762	3.19

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
B2 10"	Nozzle	118.00	118.00	1.000
B2 10"	Pad	138.00	138.00	1.000
B1 4 "ASSUME #30	Nozzle	118.00	118.00	1.000
B1 4 "ASSUME #30	Pad	138.00	138.00	1.000
Noz L1B #300	Nozzle	118.00	118.00	1.000
Noz L2B #300	Nozzle	118.00	118.00	1.000
Noz L3B #300	Nozzle	118.00	118.00	1.000
Noz L4B #300	Nozzle	118.00	118.00	1.000
Noz U2 #300	Nozzle	118.00	118.00	1.000
Noz U1 #300	Nozzle	118.00	118.00	1.000
Noz MW5 24" #300	Nozzle	137.90	137.90	1.000
Noz MW5 24" #300	Pad	137.90	137.90	1.000
Noz L1A #300	Nozzle	118.00	118.00	1.000
Noz L2A #300	Nozzle	118.00	118.00	1.000
Noz L3A #300	Nozzle	118.00	118.00	1.000
Noz L4A #300	Nozzle	118.00	118.00	1.000
MW4 #300	Nozzle	137.90	137.90	1.000
MW4 #300	Pad	138.00	138.00	1.000
P1 #300	Nozzle	118.00	118.00	1.000
P2 #300	Nozzle	118.00	118.00	1.000
F3 #300	Nozzle	118.00	118.00	1.000
F3 #300	Pad	138.00	138.00	1.000
T3 #300	Nozzle	118.00	118.00	1.000



MW3 24" #300	Nozzle	137.90	137.90	1.000
MW3 24" #300	Pad	138.00	138.00	1.000
T2 #300	Nozzle	118.00	118.00	1.000
F1 4" #300	Nozzle	118.00	118.00	1.000
F1 4" #300	Pad	138.00	138.00	1.000
MW2 24" #300	Nozzle	137.90	137.90	1.000
MW2 24" #300	Pad	138.00	138.00	1.000
T1 #300	Nozzle	118.00	118.00	1.000
MW1 24" #300	Nozzle	117.90	117.90	1.000
MW1 24" #300	Pad	138.00	138.00	1.000
F2 6" #300	Nozzle	118.00	118.00	1.000
F2 6" #300	Pad	138.00	138.00	1.000
T 8" #300	Nozzle	118.00	118.00	1.000
T 8" #300	Pad	138.00	138.00	1.000
V1 4" #300	Nozzle	118.00	118.00	1.000
V1 4" #300	Pad	138.00	138.00	1.000
NV2 2" #300	Nozzle	118.00	118.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom head	137.90	137.90	1.000
Shell 1	137.90	137.90	1.000
Shell 2	137.90	137.90	1.000
Shell 3	137.90	137.90	1.000
Shell 4	137.90	137.90	1.000
Shell 5	137.90	137.90	1.000
Shell 6	137.90	137.90	1.000
Shell 7	137.90	137.90	1.000
Shell 8	137.90	137.90	1.000
Shell 9	137.90	137.90	1.000
Shell 10	137.90	137.90	1.000
Shell 11	137.90	137.90	1.000
Shell 12	137.90	137.90	1.000
Shell 13	137.90	137.90	1.000
Top head	137.90	137.90	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
B2 10"	46.30	217.00	0.213
B1 4 "ASSUME #30	23.66	217.00	0.109
Noz L1B #300	9.46	217.00	0.044
Noz L2B #300	9.46	217.00	0.044
Noz L3B #300	9.46	217.00	0.044
Noz L4B #300	9.46	217.00	0.044
Noz U2 #300	9.46	217.00	0.044
Noz U1 #300	9.46	217.00	0.044
Noz MW5 24" #300	118.67	235.80	0.503
Noz L1A #300	9.35	217.00	0.043
Noz L2A #300	9.35	217.00	0.043
Noz L3A #300	9.35	217.00	0.043
Noz L4A #300	9.35	217.00	0.043
MW4 #300	116.71	235.81	0.495
P1 #300	9.26	217.00	0.043
P2 #300	9.27	217.00	0.043
F3 #300	45.35	217.00	0.209
T3 #300	9.16	217.00	0.042
MW3 24" #300	114.64	235.80	0.486
T2 #300	8.85	217.00	0.041
F1 4" #300	16.75	217.00	0.077
MW2 24" #300	107.75	235.81	0.457
T1 #300	8.52	217.00	0.039
MW1 24" #300	73.70	217.19	0.339



F2 6" #300	33.47	217.00	0.154
T 8" #300	34.74	217.00	0.160
V1 4" #300	16.73	217.00	0.077
NV2 2" #300	36.65	217.00	0.169

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	6000	...	...	No Calc	No Calc
20	30	6000	...	...	No Calc	No Calc
30	40	No Calc	1855	24.5	0.0018344	92.8856
40	50	38300	1860	27	0.00024152	23.5303
50	60	38300	1860	27	0.00024152	23.5303
60	70	38300	1860	27	0.00024152	23.5303
70	80	38300	1860	27	0.00024152	23.5303
80	90	38300	1860	27	0.00024152	23.5303
90	100	38300	1860	27	0.00024152	23.5303
100	110	38300	1856	25	0.00020729	20.1954
110	120	38300	1856	25	0.00020729	20.1954
120	130	38300	1856	25	0.00020729	20.1954
130	140	38300	1856	25	0.00020729	20.1954
140	150	38300	1850	22	0.00015927	15.517
150	160	38300	1850	22	0.00015927	15.517
160	170	38300	1844	19	0.00011931	11.6244
170	180	No Calc	1838	16	0.001209	83.7547

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	27.5	6.35007	0.049033	1.3631
40	50	30	15.9113	0.049033	0.45542
50	60	30	15.9113	0.049033	0.45542
60	70	30	15.9113	0.049033	0.45542
70	80	30	15.9113	0.049033	0.45542
80	90	30	15.9113	0.049033	0.45542
90	100	30	15.9113	0.049033	0.45542
100	110	28	15.8838	0.049033	0.36271
110	120	28	15.8838	0.049033	0.36271
120	130	28	15.8838	0.049033	0.36271
130	140	28	15.8838	0.049033	0.36271
140	150	25	15.8424	0.049033	0.24603
150	160	25	15.8424	0.049033	0.24603
160	170	22	15.8012	0.049033	0.1597
170	180	19	6.31932	0.049033	0.8101

Minimum

0

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	6000	No Calc	No Calc	No Calc
20	30	6000	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	38300	1073953	No Calc	No Calc
50	60	38300	1073953	No Calc	No Calc
60	70	38300	1073953	No Calc	No Calc
70	80	38300	1073953	No Calc	No Calc
80	90	38300	1073953	No Calc	No Calc



90	100	38300	1073953	No Calc	No Calc
100	110	38300	2063775	No Calc	No Calc
110	120	38300	2063775	No Calc	No Calc
120	130	38300	2063775	No Calc	No Calc
130	140	38300	2063775	No Calc	No Calc
140	150	38300	948136	No Calc	No Calc
150	160	38300	948136	No Calc	No Calc
160	170	38300	1286165	No Calc	No Calc
170	180	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	5965.38	...	5912.68	...	417.577
20	30	4416.21	...	4375.36	...	309.135
30	40	1139	1018.06	1032.22	1027.42	79.73
40	50	4010.27	7635.44	3615.17	7686.43	280.719
50	60	4010.27	7635.44	3615.17	7686.43	280.719
60	70	4010.27	7635.44	3615.17	7686.43	280.719
70	80	4010.27	7635.44	3615.17	7686.43	280.719
80	90	4010.27	7635.44	3615.17	7686.43	280.719
90	100	4010.27	7635.44	3615.17	7686.43	280.719
100	110	3738.83	7635.44	3343.73	7686.43	261.718
110	120	3738.83	7635.44	3343.73	7686.43	261.718
120	130	3738.83	7635.44	3343.73	7686.43	261.718
130	140	3738.83	7635.44	3343.73	7686.43	261.718
140	150	3332.76	7635.44	2937.66	7686.43	233.293
150	160	3332.76	7635.44	2937.66	7686.43	233.293
160	170	1756.8	4581.26	1519.74	4611.86	122.976
170	180	775.07	1018.06	669.379	1027.42	54.2549
Total		59734	98242.63	54450	98903.80	4181

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Insl	2996.79	...	1500	...	Fire proofing
10	Lini	2831.66	...	1500	...	Fire proofing
10	Wght	250	-1745	200	...	TAILING LUG WT
20	Plat	2349.92	-529.792	500	...	SKRT PLT
20	Insl	2252.44	...	1500	...	Fire proofing'
20	Lini	2068.44	...	1500	...	Fire proofing
20	Wght	400.007	...	1500	...	SKIRT ACCESSORI
30	Plat	87.2492	...	...	...	BTM PTL
30	Liqd	529.304	...	-225	...	Liquid 30
30	Insl	53.821	...	-175	...	Ins: 160
30	Nozl	145.983	-400	-614.369	...	B2 10"
30	Nozl	31.2993	200	-650.004	...	B1 4 "ASSUME #30
30	Wght	50	-500	-300	...	VORTEX BREAKER
30	Wght	823	2165	...	...	B2 OP
40	Liqd	3969.78	...	1500	...	Liquid 40
40	Insl	173.921	...	1500	...	Ins: 160
40	Nozl	13.0822	-1025.01	250	...	Noz L1B #300
40	Nozl	12.8627	...	250	-1025.01	Noz L2B #300
40	Nozl	12.8627	...	250	1025.01	Noz L3B #300
40	Nozl	13.0822	1025.01	250	...	Noz L4B #300
40	Nozl	13.0822	-1025.01	50	...	Noz U2 #300
40	Nozl	13.0822	1025.01	50	...	Noz U1 #300
40	Nozl	921.206	-1050	1500	...	Noz MW5 24" #300
40	Wght	150	...	1500	1350	MANWAY DAVIT MW
40	Wght	2765	...	2000	...	BAFFELE& HOOD B
40	Wght	1708	...	-0.00028956	...	CLEAT WT-1
50	Plat	1683.92	...	1000	1173.33	PTL 1
50	Liqd	1587.91	...	600	...	Liquid 50
50	Insl	173.921	...	1500	...	Ins: 160
50	Nozl	13.0822	-1025.01	1250	...	Noz L1A #300
50	Nozl	13.0822	...	1250	-1025.01	Noz L2A #300
50	Nozl	13.0822	...	1250	1025.01	Noz L3A #300
50	Nozl	13.0822	1025.01	1250	...	Noz L4A #300
60	Plat	1683.92	-586.667	2000	1016.14	PLAT2
60	Insl	173.921	...	1500	...	Ins: 160
60	Nozl	921.58	...	1500	-1050	MW4 #300
60	Nozl	12.4246	-1013.89	1750	...	P1 #300



60	Nozl	12.4246	1013.89	1550	...	P2 #300
60	Nozl	114.424	-1025.01	900	...	F3 #300
60	Tray	523.206	...	1100	...	Tray 45
60	Tray	523.206	...	1700	...	Tray 44
60	Tray	523.206	...	2300	...	Tray 43
60	Tray	523.206	...	2900	...	Tray 42
60	Wght	150	...	1500	-1200	MANWAY DAVIT MW
60	Wght	1814	-2198	900	...	F3 PIPE WT
70	Plat	1683.92	...	2900	1173.33	PLAT 4
70	Insl	173.921	...	1500	...	Ins: 160
70	Nozl	13.1414	-1025.01	2660	...	T3 #300
70	Tray	523.206	...	500	...	Tray 41
70	Tray	523.206	...	1100	...	Tray 40
70	Tray	523.206	...	1700	...	Tray 39
70	Tray	523.206	...	2300	...	Tray 38
70	Tray	523.206	...	2900	...	Tray 37
80	Insl	173.921	...	1500	...	Ins: 160
80	Nozl	921.58	...	1800	1050	MW3 24" #300
80	Tray	523.206	...	500	...	Tray 36
80	Tray	523.206	...	1100	...	Tray 35
80	Tray	523.206	...	1700	...	Tray 34
80	Tray	523.206	...	2300	...	Tray 33
80	Tray	523.206	...	2900	...	Tray 32
80	Wght	150	...	1800	1350	MANWAY DAVIT MW
90	Plat	1683.92	1173.33	900	...	PLAT:5
90	Insl	173.921	...	1500	...	Ins: 160
90	Tray	523.206	...	500	...	Tray 31
90	Tray	523.206	...	1500	...	Tray 30
90	Tray	523.206	...	2100	...	Tray 29
90	Tray	523.206	...	2700	...	Tray 28
100	Plat	1684.78	...	1900	1173.33	PLAT 6
100	Insl	173.558	...	1500	...	Ins: 160
100	Tray	523.206	...	300	...	Tray 27
100	Tray	523.206	...	900	...	Tray 26
100	Tray	523.206	...	1500	...	Tray 25
100	Tray	523.206	...	2100	...	Tray 24
110	Plat	1683.92	...	2900	-1173.33	PLAT 6
110	Insl	173.558	...	1500	...	Ins: 160
110	Nozl	12.9047	-1025.01	2650	...	T2 #300
110	Tray	523.206	...	1100	...	Tray 23
110	Tray	497.018	...	1700	...	Tray 22
110	Tray	497.018	...	2300	...	Tray 21
110	Tray	497.018	...	2900	...	Tray 20
110	Wght	1708	...	2000	...	CLEAT LOAD 2
120	Insl	173.558	...	1500	...	Ins: 160
120	Nozl	28.3291	-1025.01	900	...	F1 4" #300
120	Tray	497.018	...	499.997	...	Tray 19
120	Tray	497.018	...	1100	...	Tray 18
120	Tray	497.018	...	1700	...	Tray 17
120	Tray	497.018	...	2300	...	Tray 16
120	Tray	497.018	...	2900	...	Tray 15
120	Wght	150	-2000	999.997	...	MANWAY DAVIT MW
120	Wght	62	...	899.999	...	F1 NOZZLE
130	Plat	1683.06	...	900	-1173.33	PLAT: 6
130	Insl	173.558	...	1500	...	Ins: 160
130	Tray	497.018	...	500	...	Tray 14
130	Tray	497.018	...	1100	...	Tray 13
130	Tray	497.018	...	1700	...	Tray 12
130	Tray	497.018	...	2300	...	Tray 11
130	Tray	497.018	...	2900	...	Tray 10
140	Plat	1683.92	-586.667	1900	-1016.14	PLAT:7
140	Insl	173.015	...	1500	...	Ins: 160
140	Tray	497.018	...	500.002	...	Tray 9
140	Tray	497.018	...	1100	...	Tray 8
140	Tray	497.018	...	2100	...	Tray 7
140	Tray	497.018	...	2700	...	Tray 6
140	Wght	1708	...	0.0023256	...	CLEAT WT -3
150	Plat	2270.91	-627.183	2900	-292.46	PLAT:8
150	Insl	173.015	...	1500	...	Ins: 160
150	Nozl	904.79	...	1800	-1050	MW2 24" #300



150	Tray	497.018	...	300.003	...
150	Tray	497.018	...	900.003	...
150	Tray	497.018	...	1500	...
150	Tray	497.018	...	2100	...
150	Tray	497.018	...	2700	...
160	Insl	103.483	...	900	...
160	Nozl	12.7864	-1025.01	260	...
160	Nozl	869.439	-1050	1000	...
160	Nozl	43.1902	975.006	1600	...
160	Wght	1227	2114	1600	...
160	Wght	150	-1400	999.999	...
160	Wght	1500	...	800	...
160	Wght	109	...	1600	...
170	Plat	11288.9	...	450.003	...
170	Insl	52.7838	...	275	...
170	Nozl	78.3893	...	690.5	...
170	Nozl	28.7448	-500	589.672	...
170	Nozl	10.9443	...	662.72	100
170	Wght	1331	2140	...	...
170	Wght	560	-1200	200	...
170	Wght	2500	-2750	450	...
170	Wght	2500	2750	450	...
170	Forc	...	...	450	...
160	Trun	168.412	...	1350	...

Tray 5
 Tray 4
 Tray 3
 Tray 2
 TRAY 1
 Ins: 160
 T1 #300
 MW1 24" #300
 F2 6" #300
 F2 OP
 MANWAY DAVIT MW
 LIFITNG TRUNION
 F2 FEED PIPE
 TOP PTL
 Ins: 160
 T 8" #300
 V1 4" #300
 NV2 2" #300
 T OP
 PIPE DAVIT
 PSV 1- 6"8"
 PSV 2- 6"8"
 OHP
 Trunnions

Note: The individual tray liquid weights are listed below, but these weights are included in the tray weights above.

From	Tliq	Weight	X Offset	Y Offset
60	Tliq	0.00	0.000	1100.000
60	Tliq	0.00	0.000	1700.000
60	Tliq	0.00	0.000	2300.000
60	Tliq	0.00	0.000	2900.001
70	Tliq	0.00	0.000	500.000
70	Tliq	0.00	0.000	1100.000
70	Tliq	0.00	0.000	1700.000
70	Tliq	0.00	0.000	2300.000
70	Tliq	0.00	0.000	2900.000
80	Tliq	0.00	0.000	500.000
80	Tliq	0.00	0.000	1099.999
80	Tliq	0.00	0.000	1700.000
80	Tliq	0.00	0.000	2299.999
80	Tliq	0.00	0.000	2900.000
90	Tliq	0.00	0.000	500.000
90	Tliq	0.00	0.000	1500.000
90	Tliq	0.00	0.000	2100.000
90	Tliq	0.00	0.000	2700.000
100	Tliq	0.00	0.000	300.000
100	Tliq	0.00	0.000	900.000
100	Tliq	0.00	0.000	1500.000
100	Tliq	0.00	0.000	2100.001
110	Tliq	0.00	0.000	1100.000
110	Tliq	0.00	0.000	1700.001
110	Tliq	0.00	0.000	2300.001
110	Tliq	0.00	0.000	2900.000
120	Tliq	0.00	0.000	499.997
120	Tliq	0.00	0.000	1099.998
120	Tliq	0.00	0.000	1699.999
120	Tliq	0.00	0.000	2299.997
120	Tliq	0.00	0.000	2899.997
130	Tliq	0.00	0.000	500.000
130	Tliq	0.00	0.000	1100.000
130	Tliq	0.00	0.000	1699.999
130	Tliq	0.00	0.000	2299.999
130	Tliq	0.00	0.000	2900.000
140	Tliq	0.00	0.000	500.002
140	Tliq	0.00	0.000	1100.000
140	Tliq	0.00	0.000	2100.002



140 Tliq	0.00	0.000	2700.000
150 Tliq	0.00	0.000	300.003
150 Tliq	0.00	0.000	900.003
150 Tliq	0.00	0.000	1500.004
150 Tliq	0.00	0.000	2100.002
150 Tliq	0.00	0.000	2700.002

Total Weight of Each Detail Type:

Platforms	29468.3
Liquid	6087.0
Insulation	7543.1
Lining	4900.1
Nozzles	5214.0
Trays	18388.5
Tray Bolting Weight	4579.6
Weights	21765.0
Trunnions	168.4

Sum of the Detail Weights 98114.0 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	63916.4	63916.4	63916.4	63916.4	63916.4	63916.4
Nozzles	5214.0	5214.0	5214.0	5214.0	5214.0	5214.0
Tray Bar	4579.6	4579.6	4579.6	4579.6	4579.6	4579.6
Wld Weights	11170.0	11170.0	11170.0	11170.0	11170.0	11170.0
Trunnions	168.4	168.4	168.4	168.4	168.4	168.4
Platforms	...	...	29468.3	29468.3	29468.3	29468.3
Insulation	...	...	7543.1	7543.1	7543.1	7543.1
Lining	...	...	...	...	4900.1	4900.1
Trays	...	...	18388.5	18388.5	18388.5	18388.5
Empty Weights	...	...	...	5000.0	5000.0	...
Test Weights	...	5000.0	...	...	...	...
Ope Weights	...	...	...	...	...	10595.0
Ope. Liquid	...	...	...	...	...	6087.0
Test Liquid	...	98182.7	...	...	...	...
Totals	85048.4	188231.0	140448.3	145448.3	150348.4	162030.4

Field Installation Options:

* Lining installed after lifting.

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	85048.4 kgm
Shop Test Wt.	- Fabricated weight + water (full)	188231.0 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	140448.3 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	145448.3 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	150348.4 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	162030.4 kgm
Field Test Wt.	- Empty weight + test medium (full)	243537.6 kgm

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	60427.2 kgm
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	97093.6 kgm
Mass of the Upper 1/3 of the Vertical Vessel (empty)	60035.0 kgm

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:



From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	23475052	23098060
20	30	17527176	16999390
30	40	4351895	4077599
40	50	17530088	16964600
50	60	17530088	16964600
60	70	17530088	16964600
70	80	17530088	16964600
80	90	17530088	16964600
90	100	17530088	16964600
100	110	17492388	16964600
110	120	17492388	16964600
120	130	17492388	16964600
130	140	17492388	16964600
140	150	17435840	16964600
150	160	17435840	16964600
160	170	10427574	10178760
170	180	4265224	4077599

Total 270068672.0 262006624.0 mm²
 Total 270.1 262.0 m²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kgm	Total. Ele. Oper. Wgt. kgm	Total. Ele. Hydro. Wgt. kgm	Total Dtl. Offset Mom. N-mm	Oper. Wgt. No Liquid kgm
10	20	9629.75	12461.4	12405	4279601	12461.4
20	30	9327.7	11796.1	11352.4	12213122	11796.1
30	40	1587.08	2939.39	2499.62	16722682	2410.08
40	50	10087.2	14057	17346.1	9488851	10087.2
50	60	6201.16	7789.07	13460.1	...	6201.16
60	70	9452.5	11266.5	16711.5	49955868	11266.5
70	80	8778	8778	16037	132141	8778
80	90	8152.52	8152.52	15411.5	...	8152.52
90	100	8241.65	8241.65	15500.6	19382526	8241.65
100	110	7951.7	7951.7	15210.7	...	7951.7
110	120	9593.19	9593.19	16852.2	129761	9593.19
120	130	6899.52	6899.52	14158.5	3227848	6899.52
130	140	8342.25	8342.25	15601.2	...	8342.25
140	150	9119.05	9119.05	16378	9691263	9119.05
150	160	9399.86	9399.86	16658.8	13972142	9399.86
160	170	4836.09	6063.09	9191.47	14714683	6063.09
170	180	17849.1	19180.1	18762.8	21208846	19180.1

Empty Support Force + the Sum of the Y forces 1442765.50 N
 Operating Support Force + the Sum of the Y forces 1605369.38 N
 Hydro Support Force + the Sum of the Y forces 2404629.00 N

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kgm	Cumulative Oper. Wgt. kgm	Cumulative Hydro. Wgt. kgm
10	20	155943	162030	243538
20	30	143482	149569	231133
30	40	131686	137773	219780
40	50	129276	134833	217281
50	60	119189	120777	199934
60	70	112987	112987	186474
70	80	101721	101721	169763
80	90	92942.9	92942.9	153726
90	100	84790.4	84790.4	138314



100	110	76548.8	76548.8	122814
110	120	68597.1	68597.1	107603
120	130	59003.9	59003.9	90750.9
130	140	52104.4	52104.4	76592.4
140	150	43762.1	43762.1	60991.1
150	160	34643	34643	44613.1
160	170	25243.2	25243.2	27954.3
170	180	19180.1	19180.1	18762.8

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. N-mm	Cumulative Oper. Mom. N-mm	Cumulative Hydro. Mom. N-mm
10	20	175119328	175119328	175119328
20	30	170839712	170839712	170839712
30	40	158626592	158626592	158626592
40	50	141903920	141903920	141903920
50	60	132415072	132415072	132415072
60	70	132415072	132415072	132415072
70	80	82459216	82459216	82459216
80	90	82327072	82327072	82327072
90	100	82327072	82327072	82327072
100	110	62944544	62944544	62944544
110	120	62944544	62944544	62944544
120	130	62814784	62814784	62814784
130	140	59586936	59586936	59586936
140	150	59586936	59586936	59586936
150	160	49895672	49895672	49895672
160	170	35923528	35923528	35923528
170	180	21208846	21208846	21208846



Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
B2 10"	4.59	5.10	125	300	GR 1.1	2.307	4.579	4.590	4.579
B1 4 "ASSUME #	4.59	5.10	125	300	GR 1.1	4.321	2.564	4.590	2.564
Noz L1B #300	4.59	5.10	125	300	GR 1.1	...	...	...	...
Noz L2B #300	4.59	5.10	125	300	GR 1.1	...	...	...	...
Noz L3B #300	4.59	5.10	125	300	GR 1.1	...	...	...	...
Noz L4B #300	4.59	5.10	125	300	GR 1.1	...	...	...	...
Noz U2 #300	4.59	5.10	125	300	GR 1.1	...	...	...	...
Noz U1 #300	4.59	5.10	125	300	GR 1.1	...	...	...	...
Noz MW5 24" #3	4.59	5.10	125	300	GR 1.1	...	...	...	...
Noz L1A #300	4.59	5.10	125	300	GR 1.1	...	...	...	...
Noz L2A #300	4.59	5.10	125	300	GR 1.1	...	...	...	...
Noz L3A #300	4.59	5.10	125	300	GR 1.1	...	...	...	...
Noz L4A #300	4.59	5.10	125	300	GR 1.1	...	...	...	...
MW4 #300	4.59	5.10	125	300	GR 1.1	...	...	...	...
P1 #300	4.59	5.10	125	300	GR 1.1	...	...	...	...
P2 #300	4.59	5.10	125	300	GR 1.1	...	...	...	...
F3 #300	4.59	5.10	125	300	GR 1.1	2.316	4.570	4.590	4.570
T3 #300	4.59	5.10	125	300	GR 1.1	...	...	...	...
MW3 24" #300	4.59	5.10	125	300	GR 1.1	...	...	...	...
T2 #300	4.59	5.10	125	300	GR 1.1	...	...	...	...
F1 4" #300	4.59	5.10	125	300	GR 1.1	3.930	2.956	4.590	2.956
MW2 24" #300	4.59	5.10	125	300	GR 1.1	...	...	...	...
T1 #300	4.59	5.10	125	300	GR 1.1	...	...	...	...
MW1 24" #300	4.59	5.10	125	300	GR 1.1	...	...	...	...
F2 6" #300	4.59	5.10	125	300	GR 1.1	3.078	3.808	4.590	3.808
T 8" #300	4.59	5.10	125	300	GR 1.1	2.665	4.220	4.590	4.220
V1 4" #300	4.59	5.10	125	300	GR 1.1	...	...	...	...
NV2 2" #300	4.59	5.10	125	300	GR 1.1	...	...	...	...
Min Rating	2.564	5.100, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
B2 10"	302.387	0.50	11465998	14040	2.450	4.579	Pass
B1 4 "ASSUME #30	138.176	0.50	1979899	7500	2.450	2.564	Pass
F3 #300	302.387	0.50	11513660	14040	2.450	4.570	Pass
F1 4" #300	138.176	0.50	1842189	5616	2.450	2.956	Pass
F2 6" #300	196.088	0.50	4144917	8424	2.450	3.808	Pass
T 8" #300	248.539	0.50	7338235	11232	2.450	4.220	Pass

Evaluating Flange on Nozzle [B2 10"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(11465998) + 4 \cdot \text{abs}(14040) \cdot 302.387 \leq \pi \cdot 302.387^3 \cdot ((4.590 - 2.450) + .500 \cdot 4.590)$$

$$= 0.20044E+09 \leq 0.38542E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle B2 10" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.590, 0.500 \cdot 4.590 + 4.590 - 200444864.0 / (\pi \cdot 302.387^3))$$



= min(4.590, 4.579)
 = 4.579 MPa

Evaluating Flange on Nozzle [B1 4 "ASSUME #30]:

= $16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$
 = $16(1979899) + 4 \cdot \text{abs}(7500)138.176 \leq \pi \cdot 138.176^3 [(4.590 - 2.450) + .500 \cdot 4.590]$
 = $0.35825E+08 \leq 0.36774E+08$ Check Passes

Maximum Pressure for Flange on Nozzle B1 4 "ASSUME #30 [Pmax]:

= min(PR, FM • PR + PR - (16•ME + 4•FE • G) / (π • G³))
 = min(4.590, 0.500 • 4.590 + 4.590 - 35825308.0/(π • 138.176³))
 = min(4.590, 2.564)
 = 2.564 MPa

Evaluating Flange on Nozzle [F3 #300]:

= $16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$
 = $16(11513660) + 4 \cdot \text{abs}(14040)302.387 \leq \pi \cdot 302.387^3 [(4.590 - 2.450) + .500 \cdot 4.590]$
 = $0.20121E+09 \leq 0.38542E+09$ Check Passes

Maximum Pressure for Flange on Nozzle F3 #300 [Pmax]:

= min(PR, FM • PR + PR - (16•ME + 4•FE • G) / (π • G³))
 = min(4.590, 0.500 • 4.590 + 4.590 - 201207456.0/(π • 302.387³))
 = min(4.590, 4.570)
 = 4.570 MPa

Evaluating Flange on Nozzle [F1 4" #300]:

= $16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$
 = $16(1842189) + 4 \cdot \text{abs}(5616)138.176 \leq \pi \cdot 138.176^3 [(4.590 - 2.450) + .500 \cdot 4.590]$
 = $0.32580E+08 \leq 0.36774E+08$ Check Passes

Maximum Pressure for Flange on Nozzle F1 4" #300 [Pmax]:

= min(PR, FM • PR + PR - (16•ME + 4•FE • G) / (π • G³))
 = min(4.590, 0.500 • 4.590 + 4.590 - 32580248.0/(π • 138.176³))
 = min(4.590, 2.956)
 = 2.956 MPa

Evaluating Flange on Nozzle [F2 6" #300]:

= $16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$
 = $16(4144917) + 4 \cdot \text{abs}(8424)196.088 \leq \pi \cdot 196.088^3 [(4.590 - 2.450) + .500 \cdot 4.590]$
 = $0.72929E+08 \leq 0.10510E+09$ Check Passes

Maximum Pressure for Flange on Nozzle F2 6" #300 [Pmax]:

= min(PR, FM • PR + PR - (16•ME + 4•FE • G) / (π • G³))
 = min(4.590, 0.500 • 4.590 + 4.590 - 72928704.0/(π • 196.088³))
 = min(4.590, 3.808)
 = 3.808 MPa

Evaluating Flange on Nozzle [T 8" #300]:

= $16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$
 = $16(7338235) + 4 \cdot \text{abs}(11232)248.539 \leq \pi \cdot 248.539^3 [(4.590 - 2.450) + .500 \cdot 4.590]$
 = $0.12858E+09 \leq 0.21400E+09$ Check Passes

Maximum Pressure for Flange on Nozzle T 8" #300 [Pmax]:

= min(PR, FM • PR + PR - (16•ME + 4•FE • G) / (π • G³))
 = min(4.590, 0.500 • 4.590 + 4.590 - 128582552.0/(π • 248.539³))
 = min(4.590, 4.220)
 = 4.220 MPa

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 0.53116 Hz.

The Natural Frequency for the Vessel (Ope...) is 0.53109 Hz.

The Natural Frequency for the Vessel (Filled) is 0.44476 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	...	...
140	150	...	...
150	160	...	...
160	170	...	...
170	180	15556.3	16889412

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	...	...
140	150	...	...
150	160	...	...
160	170	...	...
170	180	...	...

User Input Forces and Moments:

From Node	Distance From	- - - - Fx	Forces - - - - Fy	- - - - Fz	- - - - Mx	Moments - - - - My	- - - - Mz
170	450.00	11000.	-16500.	11000.	0.807E+07	0.571E+07	

Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	4	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class B
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.0
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Use Gust Response Factor (Dynamic Analysis)		No

Calculated Values:

Checking the requirement for dynamic effect of wind (9.1):

Vessel Operating Natural Frequency: 0.531 Hz

Note: Vessel Natural frequency is < 1 Hz.

Note: Vessel height to diameter ratio (21.615) is > 5.

Note:

As per Section 9.1, Dynamic effects of wind should be considered. The Cross-Wind dynamic effects can be considered using the Vortex Shedding which can be invoked from Design Constraints tab. While the Along-Wind effects can be considered using Gust Response Factor method, which can be invoked from the Wind Load tab.

Design Wind Speed [Vz]:

$$= \text{Basic Wind Speed} \cdot k_1 \cdot k_{2,1} \cdot k_3 \cdot k_4 \quad (\text{for the 1st element})$$

$$= 50.0 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 1.0$$

$$= 49.997 \text{ meters/s}$$

Design Wind Pressure per 4.5.2 [pd]:

$$= \max(0.7 \cdot p_z, K_d \cdot K_a \cdot K_c \cdot 0.6 \cdot V_z^2)$$

$$= \max(0.7 \cdot 1499.814, 1.0 \cdot 0.8 \cdot 1.0 \cdot 0.6 \cdot 50.0^2)$$

$$= 1199.9 \text{ N/m}^2$$

Element Wind Load (operating condition):

$$= \text{Wind Area} \cdot C_f \cdot p_d$$

$$= 11.3 \cdot 0.7 \cdot 1199.851$$

$$= 9518.2 \text{ N}$$

From	Height mm	k1	k2	k3	Vz m/sec	Cf
10	1743.05	1.0000	1.0000	1.0000	50.00	0.7000
20	4725.56	1.0000	1.0000	1.0000	50.00	0.7000
30	6350.00	1.0000	1.0000	1.0000	50.00	0.7000
40	7900.00	1.0000	1.0000	1.0000	50.00	0.7000
50	10900.00	1.0000	1.0090	1.0000	50.45	0.7000
60	13900.00	1.0000	1.0390	1.0000	51.95	0.7000
70	16900.00	1.0000	1.0576	1.0000	52.88	0.7000
80	19900.00	1.0000	1.0696	1.0000	53.48	0.7000
90	22900.00	1.0000	1.0845	1.0000	54.22	0.7000
100	25900.00	1.0000	1.0995	1.0000	54.97	0.7000
110	28900.00	1.0000	1.1145	1.0000	55.72	0.7000
120	31900.00	1.0000	1.1248	1.0000	56.23	0.7000
130	34900.00	1.0000	1.1322	1.0000	56.61	0.7000

140	37900.00	1.0000	1.1398	1.0000	56.98	0.7000
150	40900.00	1.0000	1.1472	1.0000	57.36	0.7000
160	43300.00	1.0000	1.1532	1.0000	57.66	0.7000
170	44471.26	1.0000	1.1562	1.0000	57.81	0.7000



Note: Vortex shedding loads have been applied at the analyst's request, even though the vessel was determined to be stable and not susceptible to vortex shedding.

Vortex Shedding Loads, per NBC 1990 Supplement, pages 149-150

Compute the critical wind speed using the weighted average diameter of the top one-third of the vessel [Vh(ope)]
 $= F_n(ope) \cdot D_{avg} / 0.2 \cdot 0.6818$
 $= 0.5311 \cdot 6.4697 / 0.2 \cdot 0.6818$
 $= 11.714 \text{ mile/hr ; } 18.851 \text{ km/hr}$

Compute the aspect ratio (γ) for the entire structure [γ]
 $= (\text{Total Height})^2 / \text{Sum of the Wind Area}$
 $= 44000.0^2 / 91259424 = 21.2143$

Determine the Velocity Pressure Qh(ope) (English Units only)
 $= 0.5 \cdot 0.0668 \cdot \rho \cdot V_h(ope)^2$
 $= 0.5 \cdot 0.0668 \cdot 0.0749 \cdot 11.714^2 = 0.344 \text{ psf ; } 0.0164 \text{ kPa}$

Compute the equivalent force per unit area which will act on the top one-third of the vessel [F/area]:
 $= (q_h(ope) \cdot C_1) / ((\gamma \cdot (\beta(ope) - (C_2 \cdot \rho \cdot D_{avg}^2 / M_3))))^{1/2} \text{ [English Units]}$
 $= (0.344 \cdot 6.0) / ((21.21 \cdot (0.01 - (1.2 \cdot 0.0749 \cdot 6.4697^2 / 2768.5))))^{1/2}$
 $= 4.814417 \text{ psf or } 0.230509 \text{ kPa}$

Platform Load Calculations

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N	Cf
SKRT PLT	4320000.00	3800.00	0.00	0.00	1.20
BTM PTL	2160000.00	6300.00	0.00	0.00	1.20
PTL 1	4320000.00	10400.00	0.00	0.00	1.20
PLAT2	4320000.00	14400.00	0.00	0.00	1.20
PLAT 4	4320000.00	18300.00	0.00	0.00	1.20
PLAT:5	4320000.00	22300.00	0.00	0.00	1.20
PLAT 6	4320000.00	26300.00	0.10	443.92	1.20
PLAT 6	4320000.00	30300.00	0.16	701.45	1.20
PLAT: 6	4320000.00	34300.00	0.23	995.78	1.20
PLAT:7	4320000.00	38300.00	0.23	995.78	1.20
PLAT:8	4320000.00	42300.00	0.23	995.78	1.20
TOP PTL	11289600.00	44650.01	0.23	2602.30	1.20

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
TAILING LUG W	0.00	500.00	0.00	0.00
SKIRT ACCESSO	0.00	4800.00	0.00	0.00
VORTEX BREAK	0.00	6000.00	0.00	0.00
B2 OP	0.00	6300.00	0.00	0.00
MANWAY DAVIT	0.00	7900.00	0.00	0.00
BAFFLE& HOOD	0.00	8400.00	0.00	0.00
CLEAT WT-1	0.00	6400.00	0.00	0.00
MANWAY DAVIT	0.00	13900.00	0.00	0.00
F3 PIPE WT	0.00	13300.00	0.00	0.00
MANWAY DAVIT	0.00	20200.00	0.00	0.00
CLEAT LOAD 2	0.00	29400.00	0.12	0.00
MANWAY DAVIT	0.00	31400.00	0.21	0.00



F1 NOZZLE	0.00	31300.00	0.20	0.00
CLEAT WT -3	0.00	36400.00	0.23	0.00
F2 OP	0.00	44000.00	0.23	0.00
MANWAY DAVIT	0.00	43400.00	0.23	0.00
LIFITNG TRUNI	0.00	43200.00	0.23	0.00
F2 FEED PIPE	0.00	44000.00	0.23	0.00
T OP	0.00	44200.00	0.23	0.00
PIPE DAVIT	0.00	44400.00	0.23	0.00
PSV 1- 6"8"	0.00	44650.00	0.23	0.00
PSV 2- 6"8"	0.00	44650.00	0.23	0.00

The following table contains results for Vortex Shedding Case (Ope).

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	1743.05	3777.51	11332542	...	...
20	30	4725.56	3640.31	10920918	...	...
30	40	6350	3160	316000	...	...
40	50	7900	2574	7721999	...	...
50	60	10900	2574	7721999	...	...
60	70	13900	2574	7721999	...	...
70	80	16900	2574	7721999	...	...
80	90	19900	2574	7721999	...	...
90	100	22900	2574	7721999	...	...
100	110	25900	2568.8	7706399	...	450.776
110	120	28900	2568.8	7706399	0.10276	1289.77
120	130	31900	2568.8	7706399	0.23051	1243.45
130	140	34900	2568.8	7706399	0.23051	2222.11
140	150	37900	2561	7682999	0.23051	2245.96
150	160	40900	2561	7682999	0.23051	2269.54
160	170	43300	2553.2	4595760	0.23051	741.539
170	180	44471.3	2545.4	1312091	0.23051	2824.47

End of Vortex Shedding Calculations

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, mm
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, mm
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(104)$
- L - Total height of structure mm
- Lc - Total length of conical section(s) of vessel, mm
- tb - Uncorroded plate thickness at bottom of vessel, mm
- V30 - Design Wind Speed provided by user, km/hr
- Vc - Critical wind velocity, km/hr
- Vw - Maximum wind speed at top of structure, km/hr
- W - Total corroded weight of structure, N
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. N
- Z - Maximum amplitude of vibration at top of vessel, mm
- Dl - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, <= 0.000003 (High); 0.000003 < 0.000004 (Probable)
- P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

- :: Total Cone Length / Total Length < 0.5
- :: 0.0/44000.0 = 0.0



:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[6.76/144.36^2] \cdot 10^4 = 3.243$

Compute the vibration possibility. If $V_p > 0.000004$ no chance. $[V_p]$:
 = $W / (L \cdot Dr^2)$
 = $1533424 / (44000.0 \cdot 1933.682^2)$
 = $0.93205E-05$

Since V_p is > 0.000004 no further vibration analysis is required !

Platform Load Calculations

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N	Cf
SKRT PLT	4320000.00	3800.00	1.20	5185.22	1.20
BTM PTL	2160000.00	6300.00	1.20	2592.61	1.20
PTL 1	4320000.00	10400.00	1.22	5263.35	1.20
PLAT2	4320000.00	14400.00	1.30	5631.26	1.20
PLAT 4	4320000.00	18300.00	1.36	5861.53	1.20
PLAT:5	4320000.00	22300.00	1.40	6065.27	1.20
PLAT 6	4320000.00	26300.00	1.46	6291.38	1.20
PLAT 6	4320000.00	30300.00	1.50	6496.16	1.20
PLAT: 6	4320000.00	34300.00	1.53	6629.85	1.20
PLAT:7	4320000.00	38300.00	1.56	6747.62	1.20
PLAT:8	4320000.00	42300.00	1.59	6866.45	1.20
TOP PTL	11289600.00	44650.01	1.61	18127.90	1.20

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
TAILING LUG W	0.00	500.00	1.20	0.00
SKIRT ACCESSO	0.00	4800.00	1.20	0.00
VORTEX BREAK	0.00	6000.00	1.20	0.00
B2 OP	0.00	6300.00	1.20	0.00
MANWAY DAVIT	0.00	7900.00	1.20	0.00
BAFFLE& HOOD	0.00	8400.00	1.20	0.00
CLEAT WT-1	0.00	6400.00	1.20	0.00
MANWAY DAVIT	0.00	13900.00	1.30	0.00
F3 PIPE WT	0.00	13300.00	1.28	0.00
MANWAY DAVIT	0.00	20200.00	1.38	0.00
CLEAT LOAD 2	0.00	29400.00	1.50	0.00
MANWAY DAVIT	0.00	31400.00	1.51	0.00
F1 NOZZLE	0.00	31300.00	1.51	0.00
CLEAT WT -3	0.00	36400.00	1.55	0.00
F2 OP	0.00	44000.00	1.60	0.00
MANWAY DAVIT	0.00	43400.00	1.60	0.00
LIFITNG TRUNI	0.00	43200.00	1.60	0.00
F2 FEED PIPE	0.00	44000.00	1.60	0.00
T OP	0.00	44200.00	1.60	0.00
PIPE DAVIT	0.00	44400.00	1.60	0.00
PSV 1- 6"8"	0.00	44650.00	1.61	0.00
PSV 2- 6"8"	0.00	44650.00	1.61	0.00

The Natural Frequency for the Vessel (Ope...) is 0.53109 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	1743.05	3777.51	11332542	1.20031	10470
20	30	4725.56	3640.31	10920918	1.20031	14676.3
30	40	6350	3160	316000	1.20031	3121.37
40	50	7900	2574	7721999	1.20031	7134.24



50	60	10900	2574	7721999	1.22201	12787.3
60	70	13900	2574	7721999	1.29576	14118.8
70	80	16900	2574	7721999	1.34257	14961.6
80	90	19900	2574	7721999	1.37321	8161.88
90	100	22900	2574	7721999	1.41173	14887.9
100	110	25900	2568.8	7706399	1.45106	15634.6
110	120	28900	2568.8	7706399	1.49092	16335.5
120	130	31900	2568.8	7706399	1.51847	9007.02
130	140	34900	2568.8	7706399	1.53879	16295
140	150	37900	2561	7682999	1.55924	16721.5
150	160	40900	2561	7682999	1.57983	17154.2
160	170	43300	2553.2	4595760	1.5964	5647.05
170	180	44471.3	2545.4	1312091	1.60451	21641.3

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Input Values:

Seismic Design Code	Spectrum
Range Type	Period
Ordinate Type	Acceleration
Include Missing Mass Components	Yes
Shock Scale Factor X Direction	1.0000
Shock Scale Factor Y Direction	0.6700
Combination Method	Group
Spectrum Name	User Defined

User Defined Spectrum Data Points:

Acceleration	Period
0.1600	0.0010
0.3920	0.0500
0.5600	0.1000
0.5600	0.1300
0.5600	0.2000
0.5600	0.2700
0.5600	0.3000
0.5600	0.4000
0.5600	0.5000
0.5600	0.5500
0.5130	0.6000
0.4600	0.6700
0.3850	0.8000
0.3080	1.0000
0.2570	1.2000
0.2200	1.4000
0.1930	1.6000
0.1710	1.8000
0.1540	2.0000
0.1280	2.4000

Computed Natural Frequencies (OPE): 15

Mode	Freq (Hz)	Freq (Rad/Sec)	Period (Sec)
1	0.5326	3.3462	1.8777
2	3.4374	21.5976	0.2909
3	9.2188	57.9237	0.1085
4	16.0472	100.8276	0.0623
5	17.1612	107.8272	0.0583
6	26.6659	167.5470	0.0375
7	38.4300	241.4628	0.0260
8	45.8545	288.1124	0.0218
9	51.5403	323.8372	0.0194
10	64.4494	404.9477	0.0155
11	75.8905	476.8339	0.0132
12	77.7912	488.7763	0.0129
13	90.4156	568.0982	0.0111
14	99.7214	626.5682	0.0100
15	106.1398	666.8963	0.0094

Mass Participation Factors: 15

Mode	X	Y
1	127.78421140	0.00000000
2	6.10829871	-0.00000000
3	-0.58582671	-0.00000001
4	0.00000002	-0.29067117
5	-0.09650286	-0.00000001
6	-0.02536846	-0.00000017
7	0.00716257	-0.00000001
8	0.00000012	0.00844629
9	-0.00263628	0.00000006



10	-0.00128582	0.00000000
11	-0.00000001	0.00160226
12	0.00065537	0.00000001
13	0.00051199	0.00000011
14	0.00040829	0.00000000
15	0.00000000	0.00050579

Computed EigenVectors, Mass Normalized: 15

Node	Dx	Dy	Rz
Mode # 1 Freq (HZ) = 0.5326			
10	0.000000	0.000000	-0.000000
20	0.000235	0.000000	-0.000004
30	0.000980	0.000000	-0.000009
40	0.001016	0.000000	-0.000009
50	0.002541	0.000000	-0.000016
60	0.004902	0.000000	-0.000023
70	0.008007	0.000000	-0.000029
80	0.011766	0.000000	-0.000034
90	0.016089	0.000000	-0.000039
100	0.020890	0.000000	-0.000042
110	0.026103	0.000000	-0.000046
120	0.031657	0.000000	-0.000048
130	0.037473	0.000000	-0.000050
140	0.043476	0.000000	-0.000051
150	0.049610	0.000000	-0.000052
160	0.055818	0.000000	-0.000053
170	0.059558	0.000000	-0.000053
180	0.059765	0.000000	-0.000053
Mode # 2 Freq (HZ) = 3.4374			
10	0.000000	-0.000000	-0.000000
20	0.002069	-0.000000	-0.000025
30	0.007250	-0.000000	-0.000053
40	0.007480	-0.000000	-0.000054
50	0.016092	-0.000000	-0.000080
60	0.026695	-0.000000	-0.000089
70	0.037329	-0.000000	-0.000082
80	0.046285	-0.000000	-0.000062
90	0.052225	-0.000000	-0.000033
100	0.054195	-0.000000	0.000003
110	0.051477	-0.000000	0.000044
120	0.043771	-0.000000	0.000084
130	0.031454	-0.000000	0.000120
140	0.015289	-0.000000	0.000148
150	-0.003917	-0.000000	0.000170
160	-0.025025	-0.000000	0.000181
170	-0.038089	-0.000000	0.000183
180	-0.038812	-0.000000	0.000183
Mode # 3 Freq (HZ) = 9.2188			
10	-0.000000	0.000000	0.000000
20	-0.006925	0.000000	0.000069
30	-0.020755	0.000000	0.000119
40	-0.021312	0.000000	0.000121
50	-0.038209	0.000000	0.000120
60	-0.050623	0.000000	0.000061
70	-0.053144	0.000000	-0.000029
80	-0.043774	-0.000000	-0.000120
90	-0.024383	-0.000000	-0.000184
100	0.000541	-0.000000	-0.000206
110	0.025098	-0.000000	-0.000176
120	0.042858	-0.000000	-0.000100
130	0.049166	-0.000000	0.000003
140	0.042134	-0.000000	0.000110
150	0.021857	-0.000000	0.000209
160	-0.008075	-0.000000	0.000264
170	-0.028202	-0.000000	0.000274
180	-0.029310	-0.000000	0.000274



Mode # 4	Freq (HZ) = 16.0472		
10	0.000000	-0.000000	-0.000000
20	0.000000	-0.004192	-0.000000
30	0.000000	-0.008009	0.000000
40	0.000000	-0.008175	0.000000
50	-0.000000	-0.012601	0.000000
60	-0.000000	-0.016891	-0.000000
70	0.000000	-0.021020	-0.000000
80	0.000000	-0.024939	0.000000
90	-0.000000	-0.028647	0.000000
100	-0.000000	-0.032121	-0.000000
110	0.000000	-0.035597	-0.000000
120	0.000000	-0.038735	0.000000
130	-0.000000	-0.041529	0.000000
140	-0.000000	-0.043982	-0.000000
150	0.000000	-0.046302	-0.000000
160	0.000000	-0.048096	0.000000
170	-0.000000	-0.049027	0.000000
180	-0.000000	-0.049053	0.000000

Mode # 5	Freq (HZ) = 17.1612		
10	-0.000000	-0.000000	0.000000
20	-0.014084	-0.000000	0.000112
30	-0.035504	-0.000000	0.000141
40	-0.036226	-0.000000	0.000138
50	-0.049521	-0.000000	0.000022
60	-0.042118	-0.000000	-0.000139
70	-0.015652	-0.000000	-0.000247
80	0.018238	-0.000000	-0.000244
90	0.043957	-0.000000	-0.000130
100	0.049879	-0.000000	0.000043
110	0.032534	-0.000000	0.000207
120	-0.000712	-0.000000	0.000271
130	-0.033681	-0.000000	0.000203
140	-0.050427	0.000000	0.000033
150	-0.040178	0.000000	-0.000188
160	-0.004707	0.000000	-0.000332
170	0.022852	0.000000	-0.000362
180	0.024354	0.000000	-0.000362

Mode # 6	Freq (HZ) = 26.6659		
10	-0.000000	-0.000000	0.000000
20	-0.022513	-0.000000	0.000136
30	-0.046490	-0.000000	0.000089
40	-0.047017	-0.000000	0.000079
50	-0.040678	-0.000000	-0.000178
60	-0.002710	-0.000000	-0.000313
70	0.037393	-0.000000	-0.000206
80	0.048691	-0.000000	0.000054
90	0.023960	-0.000000	0.000270
100	-0.018584	-0.000000	0.000292
110	-0.048172	-0.000000	0.000091
120	-0.041201	-0.000000	-0.000179
130	-0.003441	-0.000000	-0.000311
140	0.036515	0.000000	-0.000207
150	0.046435	0.000000	0.000092
160	0.013968	0.000000	0.000339
170	-0.017576	0.000000	0.000394
180	-0.019277	0.000000	0.000394

Mode # 7	Freq (HZ) = 38.4300		
10	0.000000	-0.000000	-0.000000
20	0.027929	-0.000000	-0.000116
30	0.044027	-0.000000	0.000036
40	0.043898	-0.000000	0.000053
50	0.008578	-0.000000	0.000323
60	-0.039286	-0.000000	0.000202
70	-0.043440	-0.000000	-0.000165
80	0.002767	-0.000000	-0.000358
90	0.047866	-0.000000	-0.000162



100	0.044081	-0.000000	0.000206
110	-0.006581	-0.000000	0.000370
120	-0.050351	-0.000000	0.000131
130	-0.039974	0.000000	-0.000244
140	0.012050	0.000000	-0.000352
150	0.049240	0.000000	-0.000041
160	0.022979	0.000000	0.000344
170	-0.014810	0.000000	0.000442
180	-0.016822	0.000000	0.000443

Mode # 8 Freq (HZ) = 45.8545

10	0.000000	0.000001	-0.000000
20	0.000002	0.013614	-0.000000
30	0.000001	0.025098	0.000000
40	0.000001	0.025546	0.000000
50	-0.000002	0.035976	-0.000000
60	0.000003	0.043204	0.000000
70	-0.000002	0.047079	0.000000
80	-0.000001	0.047110	-0.000000
90	0.000002	0.043893	-0.000000
100	0.000000	0.037745	0.000000
110	-0.000001	0.028406	-0.000000
120	0.000000	0.016872	-0.000000
130	0.000000	0.004113	0.000000
140	-0.000000	-0.008923	0.000000
150	-0.000000	-0.022980	-0.000000
160	0.000001	-0.034903	0.000000
170	-0.000000	-0.041318	0.000000
180	-0.000000	-0.041501	0.000000

Mode # 9 Freq (HZ) = 51.5403

10	-0.000000	0.000000	0.000000
20	-0.033000	0.000000	0.000080
30	-0.035742	0.000001	-0.000164
40	-0.034813	0.000001	-0.000183
50	0.024476	0.000001	-0.000310
60	0.047897	0.000001	0.000100
70	-0.001622	0.000000	0.000372
80	-0.049688	0.000000	0.000095
90	-0.026596	-0.000000	-0.000323
100	0.034644	-0.000000	-0.000287
110	0.046363	-0.000001	0.000183
120	-0.013150	-0.000001	0.000387
130	-0.054849	-0.000001	0.000023
140	-0.019233	-0.000000	-0.000375
150	0.045974	-0.000000	-0.000192
160	0.032020	0.000000	0.000330
170	-0.013088	0.000000	0.000490
180	-0.015463	0.000000	0.000490

Mode #10 Freq (HZ) = 64.4494

10	-0.000000	0.000000	0.000000
20	-0.038703	0.000000	0.000039
30	-0.024271	0.000000	-0.000260
40	-0.022618	0.000000	-0.000277
50	0.046584	0.000000	-0.000162
60	0.022520	0.000000	0.000338
70	-0.045032	0.000000	0.000167
80	-0.022765	0.000000	-0.000328
90	0.043267	0.000000	-0.000208
100	0.033149	-0.000000	0.000290
110	-0.037206	-0.000000	0.000254
120	-0.035306	-0.000000	-0.000265
130	0.034011	-0.000000	-0.000291
140	0.042900	-0.000000	0.000227
150	-0.033725	-0.000000	0.000295
160	-0.037716	0.000000	-0.000274
170	0.011129	0.000000	-0.000490
180	0.013677	0.000000	-0.000491



Mode #11	Freq (HZ) = 75.8905		
10	-0.000000	0.000001	0.000000
20	-0.000000	0.023016	0.000000
30	-0.000000	0.039371	-0.000000
40	-0.000000	0.039851	0.000000
50	-0.000001	0.045339	-0.000000
60	0.000002	0.039777	0.000000
70	-0.000001	0.025758	0.000000
80	0.000002	0.005979	0.000000
90	-0.000004	-0.014930	-0.000000
100	0.000002	-0.033108	-0.000000
110	0.000001	-0.046294	0.000000
120	-0.000001	-0.049684	-0.000000
130	0.000001	-0.043189	0.000000
140	-0.000003	-0.028754	-0.000000
150	0.000004	-0.005430	-0.000000
160	-0.000002	0.019274	0.000000
170	-0.000000	0.033618	-0.000000
180	0.000000	0.034030	-0.000000

Mode #12	Freq (HZ) = 77.7912		
10	0.000000	0.000000	-0.000000
20	0.041074	0.000000	0.000010
30	0.007084	0.000000	0.000283
40	0.005142	0.000000	0.000288
50	-0.045720	0.000000	-0.000054
60	0.019648	0.000000	-0.000301
70	0.038853	0.000000	0.000211
80	-0.038334	0.000000	0.000256
90	-0.032260	0.000000	-0.000303
100	0.044756	0.000000	-0.000215
110	0.022712	0.000000	0.000356
120	-0.053658	0.000000	0.000110
130	-0.007675	0.000000	-0.000394
140	0.055201	-0.000000	-0.000002
150	-0.017594	-0.000000	0.000356
160	-0.045078	-0.000000	-0.000215
170	0.010169	-0.000000	-0.000509
180	0.013033	-0.000000	-0.000510

Mode #13	Freq (HZ) = 90.4156		
10	0.000001	0.000000	-0.000001
20	0.057540	0.000001	0.000073
30	-0.014438	0.000002	0.000322
40	-0.016742	0.000002	0.000307
50	-0.028797	0.000002	-0.000232
60	0.050412	0.000003	-0.000066
70	-0.009246	0.000003	0.000329
80	-0.042773	0.000003	-0.000163
90	0.036572	0.000003	-0.000233
100	0.022749	0.000002	0.000297
110	-0.050444	0.000002	0.000016
120	0.018140	0.000001	-0.000318
130	0.039788	0.000000	0.000207
140	-0.040391	-0.000000	0.000183
150	-0.003033	-0.000001	-0.000297
160	0.045395	-0.000002	0.000124
170	-0.008266	-0.000003	0.000455
180	-0.011037	-0.000003	0.000455

Mode #14	Freq (HZ) = 99.7214		
10	0.000001	0.000000	-0.000001
20	0.067096	0.000000	0.000129
30	-0.037755	0.000000	0.000258
40	-0.039523	0.000000	0.000214
50	0.018970	0.000000	-0.000201
60	0.022334	0.000000	0.000208
70	-0.041504	0.000000	-0.000028
80	0.030049	0.000000	-0.000198
90	0.016333	0.000000	0.000262



100	-0.044699	0.000000	-0.000004
110	0.023654	-0.000000	-0.000216
120	0.017909	-0.000000	0.000233
130	-0.043242	-0.000000	-0.000003
140	0.018607	0.000000	-0.000233
150	0.019684	0.000000	0.000231
160	-0.049222	-0.000000	-0.000067
170	0.007789	-0.000000	-0.000452
180	0.010731	-0.000000	-0.000453

Mode #15 Freq (HZ) = 106.1398

10	0.000000	0.000002	-0.000000
20	0.000000	0.030272	0.000000
30	-0.000000	0.045726	0.000000
40	-0.000000	0.045863	0.000000
50	-0.000000	0.032588	-0.000000
60	0.000000	0.003776	-0.000000
70	-0.000000	-0.026604	0.000000
80	-0.000000	-0.045345	-0.000000
90	0.000000	-0.047333	-0.000000
100	0.000000	-0.032387	0.000000
110	-0.000000	-0.003855	0.000000
120	0.000000	0.026272	-0.000000
130	0.000000	0.046177	0.000000
140	-0.000000	0.049477	0.000000
150	-0.000000	0.030031	-0.000000
160	0.000000	-0.004352	0.000000
170	-0.000000	-0.027022	0.000000
180	-0.000000	-0.027677	0.000000

Mass Percentages:

Included(X)	Included(Y)	Added(X)	Added(Y)
96.54	94.49	3.46	5.51

NOTE: In the following reports, "Max Contrib" displays the contribution of the Mode/Load combination having the maximum impact on the total (and names that Mode/Load combination).

Restraint Loads:

Node	Fx(N)	Fy(N)	Mz(N-mm)
10	237482.7	348097.5	5553454514.7
Max	166880.4	346181.5	5175372460.3
Contrib	2 (X)	4 (Y)	1 (X)

Element Forces and Moments:

Node	Fx(N)	Fy(N)	Mz(N-mm)
10	237234.3	347935.2	5553454515.0
Max	166880.4	346181.3	5175372460.6
Contrib	2 (X)	4 (Y)	1 (X)
20	237234.3	347935.2	4965482763.3
Max	166880.4	346181.3	4745150344.7
Contrib	2 (X)	4 (Y)	1 (X)
20	233692.9	344077.7	4965482763.3
Max	165204.4	342659.9	4745150344.7
Contrib	2 (X)	4 (Y)	1 (X)
30	233692.9	344077.7	4415463624.8
Max	165204.4	342659.9	4315215432.0
Contrib	2 (X)	4 (Y)	1 (X)
30	227261.8	339336.6	4415463624.8
Max	161339.1	338231.2	4315215432.0
Contrib	2 (X)	4 (Y)	1 (X)
40	227261.8	339336.6	4397981977.2
Max	161339.1	338231.2	4300910507.2



Contrib	2 (X)	4 (Y)	1 (X)
40	220472.7	333828.6	4397981977.2
Max	156739.5	333017.4	4300910507.2
Contrib	2 (X)	4 (Y)	1 (X)
50	220472.7	333828.6	3899776105.2
Max	156739.5	333017.4	3872704420.9
Contrib	2 (X)	4 (Y)	1 (X)
50	205810.4	323107.4	3899776105.2
Max	144019.6	322686.9	3872704420.9
Contrib	2 (X)	4 (Y)	1 (X)
60	205810.4	323107.4	3454234516.5
Max	144019.6	322686.9	3447525066.6
Contrib	2 (X)	4 (Y)	1 (X)
60	189277.3	310804.4	3454234516.5
Max	139971.8	310608.5	3447525066.6
Contrib	1 (X)	4 (Y)	1 (X)
70	189277.3	310804.4	3059638580.3
Max	139971.8	310608.5	3027439310.0
Contrib	1 (X)	4 (Y)	1 (X)
70	170262.3	294927.9	3059638580.3
Max	137055.7	294797.6	3027439310.0
Contrib	1 (X)	4 (Y)	1 (X)
80	170262.3	294927.9	2706831836.4
Max	137055.7	294797.6	2616105652.1
Contrib	1 (X)	4 (Y)	1 (X)
80	154661.4	279182.3	2706831836.4
Max	133436.3	278953.1	2616105652.1
Contrib	1 (X)	4 (Y)	1 (X)
90	154661.4	279182.3	2384921429.8
Max	133436.3	278953.1	2215634497.1
Contrib	1 (X)	4 (Y)	1 (X)
90	140963.5	261791.9	2384921429.8
Max	128643.8	261329.4	2215634497.1
Contrib	1 (X)	4 (Y)	1 (X)
100	140963.5	261791.9	2081589490.7
Max	128643.8	261329.4	1829546557.0
Contrib	1 (X)	4 (Y)	1 (X)
100	129873.0	242624.1	2081589490.7
Max	122497.4	241810.5	1829546557.0
Contrib	1 (X)	4 (Y)	1 (X)
110	129873.0	242624.1	1783863918.7
Max	122497.4	241810.5	1461905247.5
Contrib	1 (X)	4 (Y)	1 (X)
110	121943.8	219653.9	1783863918.7
Max	114176.3	218374.5	1461905247.5
Contrib	1 (X)	4 (Y)	1 (X)
120	121943.8	219653.9	1476182499.3
Max	114176.3	218374.5	1119237384.3
Contrib	1 (X)	4 (Y)	1 (X)
120	118694.5	196141.2	1476182499.3
Max	104689.8	194401.5	1119237384.3
Contrib	1 (X)	4 (Y)	1 (X)
130	118694.5	196141.2	1152890998.5
Max	104689.8	194401.5	805040566.8
Contrib	1 (X)	4 (Y)	1 (X)
130	117596.4	172763.0	1152890998.5
Max	94312.3	170648.8	805040566.8
Contrib	1 (X)	4 (Y)	1 (X)
140	117596.4	172763.0	817244656.0
Max	94312.3	170648.8	576249167.0



Contrib	1 (X)	4 (Y)	2 (X)
140	115212.0	144240.4	817244656.0
Max	80519.0	141830.1	576249167.0
Contrib	1 (X)	4 (Y)	2 (X)
150	115212.0	144240.4	476962696.2
Max	80519.0	141830.1	341107149.6
Contrib	1 (X)	4 (Y)	2 (X)
150	106078.0	112055.7	476962696.2
Max	75724.6	109653.6	341107149.6
Contrib	2 (X)	4 (Y)	2 (X)
160	106078.0	112055.7	159060701.5
Max	75724.6	109653.6	113841364.1
Contrib	2 (X)	4 (Y)	2 (X)
160	86240.3	83739.3	159060701.5
Max	61723.2	81746.0	113841364.1
Contrib	2 (X)	4 (Y)	2 (X)
170	86240.3	83739.3	3765551.8
Max	61723.2	81746.0	2694583.6
Contrib	2 (X)	4 (Y)	2 (X)
170	37640.2	36171.9	3765551.8
Max	26934.9	35305.3	2694583.6
Contrib	2 (X)	4 (Y)	2 (X)
180	37640.2	36171.9	0.0
Max	26934.9	35305.3	0.0
Contrib	2 (X)	4 (Y)	

Displacements:

Node	Dx(mm)	Dy(mm)	Rz(Deg)
10	0.0000	0.0000	0.0000
Max	0.0000	0.0000	0.0000
Contrib	2 (X)	4 (Y)	1 (X)
20	0.8357	0.0311	0.0278
Max	0.7637	0.0310	0.0262
Contrib	1 (X)	4 (Y)	1 (X)
30	3.3896	0.0594	0.0660
Max	3.1813	0.0591	0.0632
Contrib	1 (X)	4 (Y)	1 (X)
40	3.5101	0.0606	0.0682
Max	3.2961	0.0604	0.0653
Contrib	1 (X)	4 (Y)	1 (X)
50	8.6359	0.0934	0.1233
Max	8.2466	0.0930	0.1200
Contrib	1 (X)	4 (Y)	1 (X)
60	16.4581	0.1251	0.1719
Max	15.9103	0.1247	0.1690
Contrib	1 (X)	4 (Y)	1 (X)
70	26.6384	0.1555	0.2143
Max	25.9891	0.1552	0.2123
Contrib	1 (X)	4 (Y)	1 (X)
80	38.8635	0.1844	0.2511
Max	38.1886	0.1841	0.2501
Contrib	1 (X)	4 (Y)	1 (X)
90	52.8467	0.2117	0.2827
Max	52.2205	0.2115	0.2824
Contrib	1 (X)	4 (Y)	1 (X)
100	68.3234	0.2373	0.3096



Max	67.8039	0.2372	0.3095
Contrib	1 (X)	4 (Y)	1 (X)
110	85.0998	0.2629	0.3338
Max	84.7233	0.2628	0.3333
Contrib	1 (X)	4 (Y)	1 (X)
120	102.9772	0.2860	0.3533
Max	102.7511	0.2860	0.3521
Contrib	1 (X)	4 (Y)	1 (X)
130	121.7271	0.3066	0.3684
Max	121.6270	0.3066	0.3660
Contrib	1 (X)	4 (Y)	1 (X)
140	141.1337	0.3247	0.3792
Max	141.1123	0.3247	0.3756
Contrib	1 (X)	4 (Y)	1 (X)
150	161.0230	0.3419	0.3869
Max	161.0215	0.3419	0.3823
Contrib	1 (X)	4 (Y)	1 (X)
160	181.2122	0.3552	0.3906
Max	181.1706	0.3551	0.3853
Contrib	1 (X)	4 (Y)	1 (X)
170	193.3984	0.3621	0.3912
Max	193.3076	0.3620	0.3858
Contrib	1 (X)	4 (Y)	1 (X)
180	194.0756	0.3623	0.3912
Max	193.9816	0.3622	0.3858
Contrib	1 (X)	4 (Y)	1 (X)



Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support mm	Cumulative Shr Wind Cas N	Cumulative Shr Eqk Cas N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1500	15556.3	237234	699949696	...
20	30	4500	15556.3	233693	653261888	...
30	40	6050	15556.3	227262	606574080	...
40	50	7600	15556.3	220473	605017856	...
50	60	10600	15556.3	205810	558330048	...
60	70	13600	15556.3	189277	511642368	...
70	80	16600	15556.3	170262	464954656	...
80	90	19600	15556.3	154661	418267008	...
90	100	22600	15556.3	140964	371579424	...
100	110	25600	15556.3	129873	324892000	...
110	120	28600	15556.3	121944	278204704	...
120	130	31600	15556.3	118695	231517776	...
130	140	34600	15556.3	117596	184831312	...
140	150	37600	15556.3	115212	138145888	...
150	160	40600	15556.3	106078	91463072	...
160	170	43000	15556.3	86240.3	44791056	...
170	180	43950	15556.3	37640.2	16830272	...

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case + Vortex Shedding.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1500	232043	237234	5.967E+09	5.553E+09
20	30	4500	221573	233693	5.287E+09	4.965E+09
30	40	6050	206897	227262	4.644E+09	4.415E+09
40	50	7600	203775	220473	4.623E+09	4.398E+09
50	60	10600	196641	205810	4.022E+09	3.900E+09
60	70	13600	183854	189277	3.451E+09	3.454E+09
70	80	16600	169735	170262	2.921E+09	3.060E+09
80	90	19600	154773	154661	2.434E+09	2.707E+09
90	100	22600	146612	140964	1981411072	2.385E+09
100	110	25600	131724	129873	1563738624	2081589504
110	120	28600	115638	121944	1192545024	1783863936
120	130	31600	98013.1	118695	871937728	1476182528
130	140	34600	87762.6	117596	593161152	1152891008
140	150	37600	69245.5	115212	357553408	817244608
150	160	40600	50278	106078	178195328	476962720
160	170	43000	30854.3	86240.3	56447524	159060688
170	180	43950	24465.7	37640.2	6639216	3765551

The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1500	218755	237234	5.469E+09	5.553E+09
20	30	4500	208285	233693	4.828E+09	4.965E+09
30	40	6050	193609	227262	4.225E+09	4.415E+09
40	50	7600	190488	220473	4.206E+09	4.398E+09
50	60	10600	183354	205810	3.645E+09	3.900E+09
60	70	13600	170566	189277	3.114E+09	3.454E+09
70	80	16600	156447	170262	2.623E+09	3.060E+09
80	90	19600	141486	154661	2.176E+09	2.707E+09
90	100	22600	133324	140964	1763848320	2.385E+09
100	110	25600	118436	129873	1386055040	2081589504
110	120	28600	102802	121944	1054063872	1783863936
120	130	31600	86466	118695	770047232	1476182528
130	140	34600	77459	117596	524059968	1152891008
140	150	37600	61164	115212	316041088	817244608
150	160	40600	44442.5	106078	157567056	476962720
160	170	43000	27288.3	86240.3	49927292	159060688
170	180	43950	21641.3	37640.2	5872747	3765551

Note:
 The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	15556.3	0.023304	0.09217	0.60748E-04
20	30	15556.3	0.21694	0.40739	0.00014832
30	40	15556.3	0.41487	0.42248	0.0001534
40	50	15556.3	0.70355	1.08392	0.00028578
50	60	15556.3	1.55958	2.12656	0.00040754
60	70	15556.3	2.78087	3.51853	0.00051867
70	80	15556.3	4.33555	5.22795	0.00061917
80	90	15556.3	6.19175	7.22295	0.00070905
90	100	15556.3	8.31758	9.47164	0.00078831
100	110	15556.3	10.6834	11.951	0.00086267
110	120	15556.3	13.27	14.6362	0.00092552
120	130	15556.3	16.0451	17.4926	0.00097686
130	140	15556.3	18.9743	20.4858	0.0010167
140	150	15556.3	22.0246	23.5876	0.001049
150	160	15556.3	25.17	26.7667	0.0010682
160	170	15556.3	27.7299	28.6958	0.0010743
170	180	15556.3	28.7495	28.8032	0.0010743

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)
 Allowable deflection : 220.000 Actual deflection : 28.803 mm

The following table is for Wind Deflection; Operating Case + Vortex Shedding.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	232043	0.20218	0.79296	0.00051819
20	30	221573	1.84718	3.43214	0.001226
30	40	206897	3.49393	3.55673	0.0012659
40	50	203775	5.85156	8.90314	0.002275
50	60	196641	12.6593	17.0694	0.0031469
60	70	183854	22.084	27.656	0.0038902
70	80	169735	33.7397	40.2914	0.0045145
80	90	154773	47.2696	54.634	0.0050296
90	100	146612	62.3456	70.3669	0.0054427
100	110	131724	78.6731	87.24	0.0057904
110	120	115638	96.0335	105.022	0.0060505
120	130	98013.1	114.176	123.468	0.0062353
130	140	87762.6	132.873	142.368	0.0063544
140	150	69245.5	151.934	161.558	0.0064304
150	160	50278	171.22	180.907	0.0064629
160	170	30854.3	186.725	192.547	0.0064691
170	180	24465.7	192.87	193.194	0.0064691

Allowable deflection at the Tower Top (Emp)(6.000"/100ft. Criteria)
 Allowable deflection : 220.000 Actual deflection : 193.194 mm

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
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10	20	218755	0.1852	0.72597	0.00047414
20	30	208285	1.68995	3.1378	0.0011194
30	40	193609	3.19422	3.25156	0.0011557
40	50	190488	5.3452	8.12606	0.0020721
50	60	183354	11.5455	15.5562	0.0028606
60	70	170566	20.1125	25.1705	0.0035298
70	80	156447	30.6881	36.6253	0.0040894
80	90	141486	42.9437	49.6068	0.004549
90	100	133324	56.579	63.8262	0.0049159
100	110	118436	71.3261	79.0566	0.0052237
110	120	102802	86.9873	95.0902	0.0054534
120	130	86466	103.339	111.711	0.0056166
130	140	77459	120.182	128.732	0.0057219
140	150	61164	137.346	146.01	0.0057891
150	160	44442.5	154.708	163.428	0.0058179
160	170	27288.3	168.666	173.906	0.0058233
170	180	21641.3	174.198	174.489	0.0058233

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	36.0146	225.091
20	30	34.7065	216.916
30	40	30.1273	188.296
40	50	24.5404	153.377
50	60	24.5404	153.377
60	70	24.5404	153.377
70	80	24.5404	153.377
80	90	24.5404	153.377
90	100	24.5404	153.377
100	110	24.4908	153.068
110	120	24.4908	153.068
120	130	24.4908	153.068
130	140	24.4908	153.068
140	150	24.4165	152.603
150	160	24.4165	152.603
160	170	24.3421	152.138
170	180	24.2677	151.673

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 220.000 Actual deflection : 174.489 mm

Total Deflection in the Operating Condition + Applied Forces:
 Allowable deflection : 220.000 Actual deflection : 203.292 mm



Longitudinal Stress Constants:

From	To	Metal Area New mm ²	Metal Area Corroded mm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	156104	148320	95656824	90923304
20	30	162026	156256	73095784	70529512
30	40	157884	140892	71080192	63635384
40	50	172474	155481	77654824	70230032
50	60	172474	155481	77654824	70230032
60	70	172474	155481	77654824	70230032
70	80	172474	155481	77654824	70230032
80	90	172474	155481	77654824	70230032
90	100	172474	155481	77654824	70230032
100	110	160799	143807	72393728	64952936
110	120	160799	143807	72393728	64952936
120	130	160799	143807	72393728	64952936
130	140	160799	143807	72393728	64952936
140	150	143335	126342	64525020	57060092
150	160	143335	126342	64525020	57060092
160	170	125928	108935	56683988	49194768
170	180	108576	91583.7	48870072	41356404



Longitudinal Allowable Stresses:

From	To	Tensile MPa	Hydrotest Tensile MPa	Compressive MPa	Hydrotest Compressive MPa
10	20	96.18	187.656	-112.412	-123.217
20	30	81.0852	138.656	-125.971	-138.083
30	40	165.48	282.96	-124.206	-136.257
40	50	165.48	282.971	-125.958	-138.07
50	60	165.48	282.971	-125.958	-138.07
60	70	165.48	282.971	-125.958	-138.07
70	80	165.48	282.971	-125.958	-138.07
80	90	165.48	282.971	-125.958	-138.07
90	100	165.48	282.971	-125.958	-138.07
100	110	165.48	282.971	-124.577	-136.645
110	120	165.48	282.971	-124.577	-136.645
120	130	165.48	282.971	-124.577	-136.645
130	140	165.48	282.971	-124.577	-136.645
140	150	165.48	282.971	-122.163	-134.094
150	160	165.48	282.971	-122.163	-134.094
160	170	165.48	282.971	-119.209	-130.886
170	180	165.48	282.96	-115.494	-126.738

Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure MPa	Longitudinal Stress External Pressure MPa	Longitudinal Stress Hydrotest Pressure MPa
10	20	...	...	...
20	30	...	...	...
30	40	39.3274	-0.94055	58.0549
40	50	35.646	-0.8569	52.6204
50	60	35.646	-0.8569	52.6204
60	70	35.646	-0.8569	52.6204
70	80	35.646	-0.8569	52.6204
80	90	35.646	-0.8569	52.6204
90	100	35.646	-0.8569	52.6204
100	110	38.5322	-0.92248	56.881
110	120	38.5322	-0.92248	56.881
120	130	38.5322	-0.92248	56.881
130	140	38.5322	-0.92248	56.881
140	150	43.8455	-1.04322	64.7244
150	160	43.8455	-1.04322	64.7244
160	170	50.8366	-1.20208	75.0446
170	180	60.4493	-1.42054	89.2349

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty MPa	Wght. Str. Operating MPa	Wght. Str. Hydrotest MPa	Wght. Str. Emp. Mom. MPa	Wght. Str. Opr. Mom. MPa
10	20	-9.48692	-9.88939	-15.2823	1.87834	1.87834
20	30	-8.26479	-8.64682	-13.7937	2.24836	2.24836
30	40	-9.16605	-9.16605	-9.16605	2.49194	2.49194
40	50	-8.15397	-8.15397	-8.15397	2.01991	2.01991
50	60	-7.51773	-7.51773	-7.51773	1.88484	1.88484
60	70	-7.1266	-7.1266	-7.1266	1.88484	1.88484
70	80	-6.41597	-6.41597	-6.41597	1.17375	1.17375
80	90	-5.8623	-5.8623	-5.8623	1.17187	1.17187
90	100	-5.34809	-5.34809	-5.34809	1.17187	1.17187
100	110	-5.22021	-5.22021	-5.22021	0.96877	0.96877
110	120	-4.67794	-4.67794	-4.67794	0.96877	0.96877
120	130	-4.02374	-4.02374	-4.02374	0.96677	0.96677
130	140	-3.55323	-3.55323	-3.55323	0.91709	0.91709
140	150	-3.39686	-3.39686	-3.39686	1.04395	1.04395
150	160	-2.68903	-2.68903	-2.68903	0.87416	0.87416
160	170	-2.27251	-2.27251	-2.27251	0.73	0.73
170	180	-2.05382	-2.05382	-2.05382	0.51267	0.51267

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. MPa	Bend. Str. Oper. Wind MPa	Bend. Str. Oper. Equ. MPa	Bend. Str. Hyd. Wind MPa	Bend. Str. Hyd. Equ. MPa
10	20	1.87834	53.0872	54.5943	17.5188	...
20	30	2.24836	59.8895	62.5844	19.7635	...
30	40	2.49194	66.3778	69.3647	21.9047	...
40	50	2.01991	59.8714	62.6025	19.7576	...
50	60	1.88484	51.886	55.5108	17.1224	...



60	70	1.88484	44.3262	49.1688	14.6277	...
70	80	1.17375	37.3412	43.552	12.3226	...
80	90	1.17187	30.9772	38.53	10.2225	...
90	100	1.17187	25.1073	33.9478	8.2854	...
100	110	0.96877	21.3325	32.0374	7.03974	...
110	120	0.96877	16.2229	27.4552	5.35356	...
120	130	0.96677	11.8517	22.7197	3.91105	...
130	140	0.91709	8.06572	17.7439	2.66169	...
140	150	1.04395	5.53697	14.3179	1.8272	...
150	160	0.87416	2.76054	8.35628	0.91098	...
160	170	0.73	1.01457	3.23225	0.33481	...
170	180	0.51267	0.14196	0.091022	0.046846	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case MPa	Vortex Shedding Empty Case MPa	Vortex Shedding Test Case MPa	Earthquake Empty Case MPa
10	20	58.1247	...	...	53.225
20	30	65.8182	...	...	61.2525
30	40	72.9488	...	...	67.8885
40	50	65.8065	...	...	61.2759
50	60	57.2535	...	...	54.4225
60	70	49.1261	...	...	48.1676
70	80	41.5733	...	...	42.5466
80	90	34.6418	...	...	37.4923
90	100	28.2041	...	...	32.8658
100	110	24.0672	...	...	30.8216
110	120	18.3543	...	...	26.2156
120	130	13.4198	...	...	21.5111
130	140	9.12924	...	...	16.6401
140	150	6.26425	...	...	13.2772
150	160	3.12194	...	...	7.65615
160	170	1.14706	...	...	2.94369
170	180	0.16049	...	...	0.084421

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind MPa	Longitudinal Stress Y Forces Seismic MPa
10	20	...	-2.34604
20	30	...	-2.2022
30	40	...	-2.4087
40	50	...	-2.14726
50	60	...	-2.07829
60	70	...	-1.99916
70	80	...	-1.89704
80	90	...	-1.79576
90	100	...	-1.6839
100	110	...	-1.6873
110	120	...	-1.52755
120	130	...	-1.36404
130	140	...	-1.20146
140	150	...	-1.14176
150	160	...	-0.887
160	170	...	-0.76877
170	180	...	-0.39499

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded MPa	Earthquake For/Mom Corroded MPa	Wind For/Mom No Corrosion MPa	Earthquake For/Mom No Corrosion MPa
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10	20	7.18246	...	6.82704	...
20	30	8.59753	...	8.29569	...
30	40	9.52897	...	8.53092	...
40	50	8.61204	...	7.78862	...
50	60	7.94747	...	7.18759	...
60	70	7.2829	...	6.58656	...
70	80	6.61833	...	5.98554	...
80	90	5.95376	...	5.38451	...
90	100	5.2892	...	4.78348	...
100	110	5.00036	...	4.48641	...
110	120	4.2818	...	3.84171	...
120	130	3.56325	...	3.19701	...
130	140	2.84471	...	2.55232	...
140	150	2.42028	...	2.14028	...
150	160	1.60241	...	1.41703	...
160	170	0.91019	...	0.78994	...
170	180	0.40683	...	0.34428	...

Bending Stresses due to Blast Load:

From	To	Bending Stress MPa
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...
80	90	...
90	100	...
100	110	...
110	120	...
120	130	...
130	140	...
140	150	...
150	160	...
160	170	...
170	180	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	52.66	96.18	-71.63	112.41	0.5475	0.6373
20	62.47	81.09	-79.00	125.97	0.7704	0.6271
30	69.23	165.48	-87.56	124.21	0.4184	0.7050
40	62.35	165.48	-78.66	125.96	0.3768	0.6245
50	54.20	165.48	-69.24	125.96	0.3275	0.5497
60	46.37	165.48	-60.62	125.96	0.2802	0.4813
70	38.72	165.48	-51.55	125.96	0.2340	0.4093
80	32.24	165.48	-43.97	125.96	0.1948	0.3490
90	26.22	165.48	-36.92	125.96	0.1584	0.2931
100	22.08	165.48	-32.52	124.58	0.1334	0.2611
110	16.80	165.48	-26.15	124.58	0.1015	0.2099
120	12.36	165.48	-20.41	124.58	0.0747	0.1638
130	8.27	165.48	-15.38	124.58	0.0500	0.1235
140	5.60	165.48	-12.40	122.16	0.0339	0.1015
150	2.55	165.48	-7.93	122.16	0.0154	0.0649
160	0.38	165.48	-4.93	119.21	0.0023	0.0413
170		165.48	-3.12	115.49		0.0270

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	43.27	96.18	-66.94	112.41	0.4499	0.5955
20	53.03	81.09	-73.97	125.97	0.6541	0.5872



30	58.81	165.48	-81.96	124.21	0.3554	0.6598
40	52.99	165.48	-73.60	125.96	0.3202	0.5843
50	46.71	165.48	-65.90	125.96	0.2823	0.5232
60	40.93	165.48	-59.18	125.96	0.2473	0.4698
70	35.41	165.48	-52.03	125.96	0.2140	0.4131
80	31.01	165.48	-46.32	125.96	0.1874	0.3678
90	27.01	165.48	-41.07	125.96	0.1632	0.3261
100	24.88	165.48	-38.70	124.58	0.1504	0.3106
110	20.98	165.48	-33.39	124.58	0.1268	0.2680
120	17.09	165.48	-27.87	124.58	0.1033	0.2237
130	12.80	165.48	-22.31	124.58	0.0774	0.1791
140	9.78	165.48	-18.86	122.16	0.0591	0.1544
150	4.95	165.48	-12.11	122.16	0.0299	0.0991
160	0.63	165.48	-6.71	119.21	0.0038	0.0563
170		165.48	-3.05	115.49		0.0264

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	52.26	96.18	-72.04	112.41	0.5433	0.6408
20	62.09	81.09	-79.38	125.97	0.7657	0.6302
30	69.23	165.48	-87.56	124.21	0.4184	0.7050
40	62.35	165.48	-78.66	125.96	0.3768	0.6245
50	54.20	165.48	-69.24	125.96	0.3275	0.5497
60	46.37	165.48	-60.62	125.96	0.2802	0.4813
70	38.72	165.48	-51.55	125.96	0.2340	0.4093
80	32.24	165.48	-43.97	125.96	0.1948	0.3490
90	26.22	165.48	-36.92	125.96	0.1584	0.2931
100	22.08	165.48	-32.52	124.58	0.1334	0.2611
110	16.80	165.48	-26.15	124.58	0.1015	0.2099
120	12.36	165.48	-20.41	124.58	0.0747	0.1638
130	8.27	165.48	-15.38	124.58	0.0500	0.1235
140	5.60	165.48	-12.40	122.16	0.0339	0.1015
150	2.55	165.48	-7.93	122.16	0.0154	0.0649
160	0.38	165.48	-4.93	119.21	0.0023	0.0413
170		165.48	-3.12	115.49		0.0270

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	44.24	96.18	-68.71	112.41	0.4599	0.6112
20	53.98	81.09	-75.68	125.97	0.6658	0.6008
30	60.28	165.48	-83.43	124.21	0.3643	0.6717
40	54.32	165.48	-74.92	125.96	0.3283	0.5948
50	47.80	165.48	-66.99	125.96	0.2889	0.5319
60	41.93	165.48	-60.18	125.96	0.2534	0.4778
70	36.41	165.48	-53.04	125.96	0.2200	0.4211
80	32.04	165.48	-47.36	125.96	0.1936	0.3760
90	28.09	165.48	-42.15	125.96	0.1697	0.3346
100	26.10	165.48	-39.91	124.58	0.1577	0.3204
110	22.22	165.48	-34.63	124.58	0.1343	0.2780
120	18.30	165.48	-29.07	124.58	0.1106	0.2334
130	13.91	165.48	-23.42	124.58	0.0840	0.1880
140	10.82	165.48	-19.90	122.16	0.0654	0.1629
150	5.65	165.48	-12.81	122.16	0.0342	0.1048
160	0.92	165.48	-7.00	119.21	0.0056	0.0587
170		165.48	-3.05	115.49		0.0264

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	4.11	187.66	-34.68	123.22	0.0219	0.2815
20	8.22	138.66	-35.81	138.08	0.0593	0.2593
30	15.23	282.96	-33.56	136.26	0.0538	0.2463
40	13.62	282.97	-29.93	138.07	0.0481	0.2168
50	11.49	282.97	-26.52	138.07	0.0406	0.1921
60	9.39	282.97	-23.64	138.07	0.0332	0.1712



70	7.08	282.97	-19.91	138.07	0.0250	0.1442
80	5.53	282.97	-17.26	138.07	0.0195	0.1250
90	4.11	282.97	-14.81	138.07	0.0145	0.1072
100	2.79	282.97	-13.23	136.64	0.0099	0.0968
110	1.64	282.97	-11.00	136.64	0.0058	0.0805
120	0.85	282.97	-8.90	136.64	0.0030	0.0651
130	0.03	282.97	-7.13	136.64	0.0001	0.0522
140		282.97	-6.27	134.09		0.0467
150		282.97	-4.47	134.09		0.0334
160		282.97	-3.34	130.89		0.0255
170		282.96	-2.61	126.74		0.0206

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		187.66	-17.16	123.22		0.1393
20		138.66	-16.04	138.08		0.1162
30		282.96	-11.66	136.26		0.0856
40		282.97	-10.17	138.07		0.0737
50		282.97	-9.40	138.07		0.0681
60		282.97	-9.01	138.07		0.0653
70		282.97	-7.59	138.07		0.0550
80		282.97	-7.03	138.07		0.0509
90		282.97	-6.52	138.07		0.0472
100		282.97	-6.19	136.64		0.0453
110		282.97	-5.65	136.64		0.0413
120		282.97	-4.99	136.64		0.0365
130		282.97	-4.47	136.64		0.0327
140		282.97	-4.44	134.09		0.0331
150		282.97	-3.56	134.09		0.0266
160		282.97	-3.00	130.89		0.0229
170		282.96	-2.57	126.74		0.0203

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	52.26	96.18	-72.04	123.22	0.5433	0.5846
20	62.09	81.09	-79.38	138.08	0.7657	0.5749
30	108.56	165.48	-48.24	136.26	0.6560	0.3540
40	98.00	165.48	-43.01	138.07	0.5922	0.3115
50	89.85	165.48	-33.59	138.07	0.5429	0.2433
60	82.01	165.48	-24.97	138.07	0.4956	0.1809
70	74.36	165.48	-15.90	138.07	0.4494	0.1152
80	67.89	165.48	-8.32	138.07	0.4102	0.0603
90	61.87	165.48	-1.27	138.07	0.3739	0.0092
100	60.61	165.48		136.64	0.3663	
110	55.33	165.48		136.64	0.3343	
120	50.89	165.48		136.64	0.3075	
130	46.81	165.48		136.64	0.2829	
140	49.45	165.48		134.09	0.2988	
150	46.39	165.48		134.09	0.2804	
160	51.22	165.48		130.89	0.3095	
170	59.46	165.48		126.74	0.3593	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	44.24	96.18	-68.71	123.22	0.4599	0.5576
20	53.98	81.09	-75.68	138.08	0.6658	0.5481
30	99.61	165.48	-44.10	136.26	0.6019	0.3237
40	89.97	165.48	-39.28	138.07	0.5437	0.2845
50	83.45	165.48	-31.35	138.07	0.5043	0.2270
60	77.57	165.48	-24.53	138.07	0.4688	0.1777
70	72.06	165.48	-17.39	138.07	0.4355	0.1260
80	67.69	165.48	-11.71	138.07	0.4091	0.0848
90	63.73	165.48	-6.51	138.07	0.3851	0.0471
100	64.63	165.48	-1.38	136.64	0.3906	0.0101



110	60.75	165.48	136.64	0.3671
120	56.83	165.48	136.64	0.3434
130	52.44	165.48	136.64	0.3169
140	54.67	165.48	134.09	0.3304
150	49.50	165.48	134.09	0.2991
160	51.76	165.48	130.89	0.3128
170	58.60	165.48	126.74	0.3541

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	52.26	96.18	-72.04	112.41	0.5433	0.6408
20	62.09	81.09	-79.38	125.97	0.7657	0.6302
30	68.29	165.48	-88.51	124.21	0.4127	0.7126
40	61.49	165.48	-79.51	125.96	0.3716	0.6313
50	53.34	165.48	-70.09	125.96	0.3224	0.5565
60	45.51	165.48	-61.48	125.96	0.2750	0.4881
70	37.86	165.48	-52.41	125.96	0.2288	0.4161
80	31.38	165.48	-44.82	125.96	0.1897	0.3558
90	25.36	165.48	-37.77	125.96	0.1533	0.2999
100	21.16	165.48	-33.44	124.58	0.1279	0.2685
110	15.87	165.48	-27.07	124.58	0.0959	0.2173
120	11.44	165.48	-21.33	124.58	0.0691	0.1712
130	7.35	165.48	-16.30	124.58	0.0444	0.1309
140	4.56	165.48	-13.44	122.16	0.0276	0.1100
150	1.50	165.48	-8.97	122.16	0.0091	0.0734
160		165.48	-6.13	119.21		0.0514
170		165.48	-4.54	115.49		0.0393

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	44.24	96.18	-68.71	112.41	0.4599	0.6112
20	53.98	81.09	-75.68	125.97	0.6658	0.6008
30	59.34	165.48	-84.37	124.21	0.3586	0.6793
40	53.46	165.48	-75.78	125.96	0.3231	0.6016
50	46.94	165.48	-67.85	125.96	0.2837	0.5387
60	41.07	165.48	-61.04	125.96	0.2482	0.4846
70	35.56	165.48	-53.90	125.96	0.2149	0.4279
80	31.19	165.48	-48.22	125.96	0.1885	0.3828
90	27.23	165.48	-43.01	125.96	0.1646	0.3415
100	25.18	165.48	-40.84	124.58	0.1521	0.3278
110	21.30	165.48	-35.55	124.58	0.1287	0.2854
120	17.38	165.48	-30.00	124.58	0.1050	0.2408
130	12.98	165.48	-24.34	124.58	0.0785	0.1954
140	9.78	165.48	-20.94	122.16	0.0591	0.1714
150	4.61	165.48	-13.85	122.16	0.0279	0.1134
160		165.48	-8.21	119.21		0.0688
170		165.48	-4.47	115.49		0.0387

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	4.11	187.66	-34.68	123.22	0.0219	0.2815
20	8.22	138.66	-35.81	138.08	0.0593	0.2593
30	73.29	282.96		136.26	0.2590	
40	66.24	282.97		138.07	0.2341	
50	64.11	282.97		138.07	0.2266	
60	62.01	282.97		138.07	0.2191	
70	59.70	282.97		138.07	0.2110	
80	58.15	282.97		138.07	0.2055	
90	56.73	282.97		138.07	0.2005	
100	59.67	282.97		136.64	0.2109	
110	58.53	282.97		136.64	0.2068	
120	57.74	282.97		136.64	0.2040	
130	56.91	282.97		136.64	0.2011	
140	64.20	282.97		134.09	0.2269	



150	63.82	282.97	134.09	0.2255
160	73.84	282.97	130.89	0.2609
170	87.74	282.96	126.74	0.3101

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		187.66	-17.16	123.22		0.1393
20		138.66	-16.04	138.08		0.1162
30	51.38	282.96		136.26	0.1816	
40	46.49	282.97		138.07	0.1643	
50	46.99	282.97		138.07	0.1661	
60	47.38	282.97		138.07	0.1674	
70	47.38	282.97		138.07	0.1674	
80	47.93	282.97		138.07	0.1694	
90	48.44	282.97		138.07	0.1712	
100	52.63	282.97		136.64	0.1860	
110	53.17	282.97		136.64	0.1879	
120	53.82	282.97		136.64	0.1902	
130	54.24	282.97		136.64	0.1917	
140	62.37	282.97		134.09	0.2204	
150	62.91	282.97		134.09	0.2223	
160	73.50	282.97		130.89	0.2598	
170	87.69	282.96		126.74	0.3099	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		96.18	-11.37	123.22		0.0922
20		81.09	-10.51	138.08		0.0761
30	32.65	165.48		136.26	0.1973	
40	29.51	165.48		138.07	0.1783	
50	30.01	165.48		138.07	0.1814	
60	30.40	165.48		138.07	0.1837	
70	30.40	165.48		138.07	0.1837	
80	30.96	165.48		138.07	0.1871	
90	31.47	165.48		138.07	0.1902	
100	34.28	165.48		136.64	0.2072	
110	34.82	165.48		136.64	0.2104	
120	35.48	165.48		136.64	0.2144	
130	35.90	165.48		136.64	0.2169	
140	41.49	165.48		134.09	0.2507	
150	42.03	165.48		134.09	0.2540	
160	49.29	165.48		130.89	0.2979	
170	58.91	165.48		126.74	0.3560	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		96.18	-9.40	123.22		0.0763
20		81.09	-8.34	138.08		0.0604
30	31.15	165.48		136.26	0.1882	
40	28.30	165.48		138.07	0.1710	
50	28.87	165.48		138.07	0.1745	
60	29.22	165.48		138.07	0.1766	
70	29.86	165.48		138.07	0.1805	
80	30.36	165.48		138.07	0.1835	
90	30.82	165.48		138.07	0.1863	
100	33.86	165.48		136.64	0.2046	
110	34.35	165.48		136.64	0.2076	
120	34.93	165.48		136.64	0.2111	
130	35.35	165.48		136.64	0.2136	
140	40.85	165.48		134.09	0.2469	
150	41.48	165.48		134.09	0.2506	
160	48.87	165.48		130.89	0.2953	
170	58.72	165.48		126.74	0.3548	



Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	50.11	96.18	-69.89	123.22	0.5210	0.5672
20	59.42	81.09	-76.71	138.08	0.7328	0.5556
30	105.60	165.48	-45.28	136.26	0.6382	0.3323
40	95.32	165.48	-40.33	138.07	0.5760	0.2921
50	87.27	165.48	-31.01	138.07	0.5274	0.2246
60	79.53	165.48	-22.49	138.07	0.4806	0.1629
70	71.98	165.48	-13.52	138.07	0.4350	0.0979
80	65.60	165.48	-6.03	138.07	0.3964	0.0437
90	59.67	165.48		138.07	0.3606	
100	58.35	165.48		136.64	0.3526	
110	53.18	165.48		136.64	0.3214	
120	48.90	165.48		136.64	0.2955	
130	45.03	165.48		136.64	0.2721	
140	47.76	165.48		134.09	0.2886	
150	45.15	165.48		134.09	0.2729	
160	50.44	165.48		130.89	0.3048	
170	59.07	165.48		126.74	0.3570	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		96.18	-11.37	123.22		0.0922
20		81.09	-10.51	138.08		0.0761
30	32.65	165.48		136.26	0.1973	
40	29.51	165.48		138.07	0.1783	
50	30.01	165.48		138.07	0.1814	
60	30.40	165.48		138.07	0.1837	
70	30.40	165.48		138.07	0.1837	
80	30.96	165.48		138.07	0.1871	
90	31.47	165.48		138.07	0.1902	
100	34.28	165.48		136.64	0.2072	
110	34.82	165.48		136.64	0.2104	
120	35.48	165.48		136.64	0.2144	
130	35.90	165.48		136.64	0.2169	
140	41.49	165.48		134.09	0.2507	
150	42.03	165.48		134.09	0.2540	
160	49.29	165.48		130.89	0.2979	
170	58.91	165.48		126.74	0.3560	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	50.11	96.18	-69.89	112.41	0.5210	0.6218
20	59.42	81.09	-76.71	125.97	0.7328	0.6090
30	66.27	165.48	-84.61	124.21	0.4005	0.6812
40	59.67	165.48	-75.98	125.96	0.3606	0.6032
50	51.62	165.48	-66.66	125.96	0.3119	0.5292
60	43.88	165.48	-58.14	125.96	0.2652	0.4616
70	36.33	165.48	-49.16	125.96	0.2195	0.3903
80	29.95	165.48	-41.68	125.96	0.1810	0.3309
90	24.03	165.48	-34.72	125.96	0.1452	0.2757
100	19.82	165.48	-30.26	124.58	0.1197	0.2429
110	14.65	165.48	-24.00	124.58	0.0885	0.1927
120	10.36	165.48	-18.41	124.58	0.0626	0.1478
130	6.49	165.48	-13.60	124.58	0.0392	0.1092
140	3.91	165.48	-10.71	122.16	0.0236	0.0876
150	1.31	165.48	-6.69	122.16	0.0079	0.0547
160		165.48	-4.15	119.21		0.0348
170		165.48	-2.73	115.49		0.0236

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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10		80.15	-14.11	102.68	0.1375
20		67.57	-13.10	115.07	0.1138
30	30.24	137.90		113.55	0.2193
40	27.36	137.90		115.06	0.1984
50	27.93	137.90		115.06	0.2026
60	28.41	137.90		115.06	0.2060
70	28.51	137.90		115.06	0.2067
80	29.16	137.90		115.06	0.2115
90	29.79	137.90		115.06	0.2160
100	32.59	137.90		113.87	0.2364
110	33.30	137.90		113.87	0.2414
120	34.11	137.90		113.87	0.2474
130	34.69	137.90		113.87	0.2516
140	40.35	137.90		111.75	0.2926
150	41.14	137.90		111.75	0.2984
160	48.53	137.90		109.07	0.3519
170	58.51	137.90		105.62	0.4243

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		80.15	-14.11	93.68		0.1507
20		67.57	-13.10	104.98		0.1248
30		137.90	-15.01	103.50		0.1450
40		137.90	-13.18	104.97		0.1255
50		137.90	-12.34	104.97		0.1175
60		137.90	-11.87	104.97		0.1131
70		137.90	-10.34	104.97		0.0985
80		137.90	-9.69	104.97		0.0923
90		137.90	-9.06	104.97		0.0863
100		137.90	-8.80	103.81		0.0848
110		137.90	-8.10	103.81		0.0780
120		137.90	-7.28	103.81		0.0701
130		137.90	-6.59	103.81		0.0635
140		137.90	-6.63	101.80		0.0651
150		137.90	-5.49	101.80		0.0540
160		137.90	-4.97	99.34		0.0501
170		137.90	-4.38	96.25		0.0455

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		96.18	-11.77	123.22		0.0955
20		81.09	-10.90	138.08		0.0789
30	32.65	165.48		136.26	0.1973	
40	29.51	165.48		138.07	0.1783	
50	30.01	165.48		138.07	0.1814	
60	30.40	165.48		138.07	0.1837	
70	30.40	165.48		138.07	0.1837	
80	30.96	165.48		138.07	0.1871	
90	31.47	165.48		138.07	0.1902	
100	34.28	165.48		136.64	0.2072	
110	34.82	165.48		136.64	0.2104	
120	35.48	165.48		136.64	0.2144	
130	35.90	165.48		136.64	0.2169	
140	41.49	165.48		134.09	0.2507	
150	42.03	165.48		134.09	0.2540	
160	49.29	165.48		130.89	0.2979	
170	58.91	165.48		126.74	0.3560	

Absolute Maximum of the all of the Stress Ratio's 0.7704

Governing Element: Top skirt

Governing Load Case 1 : NP+EW+WI+FW+BW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
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Bottom skirt	...	-9.5	1.9	60.3	52.7	+ Tens
Bottom skirt	...	-9.5	-1.9	-60.3	-71.6	- Comp
Top skirt	...	-8.3	2.2	68.5	62.5	+ Tens
Top skirt	...	-8.3	-2.2	-68.5	-79.0	- Comp
Bottom head	...	-9.2	2.5	75.9	69.2	+ Tens
Bottom head	...	-9.2	-2.5	-75.9	-87.6	- Comp
Shell 1	...	-8.2	2.0	68.5	62.3	+ Tens
Shell 1	...	-8.2	-2.0	-68.5	-78.7	- Comp
Shell 2	...	-7.5	1.9	59.8	54.2	+ Tens
Shell 2	...	-7.5	-1.9	-59.8	-69.2	- Comp
Shell 3	...	-7.1	1.9	51.6	46.4	+ Tens
Shell 3	...	-7.1	-1.9	-51.6	-60.6	- Comp
Shell 4	...	-6.4	1.2	44.0	38.7	+ Tens
Shell 4	...	-6.4	-1.2	-44.0	-51.5	- Comp
Shell 5	...	-5.9	1.2	36.9	32.2	+ Tens
Shell 5	...	-5.9	-1.2	-36.9	-44.0	- Comp
Shell 6	...	-5.3	1.2	30.4	26.2	+ Tens
Shell 6	...	-5.3	-1.2	-30.4	-36.9	- Comp
Shell 7	...	-5.2	1.0	26.3	22.1	+ Tens
Shell 7	...	-5.2	-1.0	-26.3	-32.5	- Comp
Shell 8	...	-4.7	1.0	20.5	16.8	+ Tens
Shell 8	...	-4.7	-1.0	-20.5	-26.2	- Comp
Shell 9	...	-4.0	1.0	15.4	12.4	+ Tens
Shell 9	...	-4.0	-1.0	-15.4	-20.4	- Comp
Shell 10	...	-3.6	0.9	10.9	8.3	+ Tens
Shell 10	...	-3.6	-0.9	-10.9	-15.4	- Comp
Shell 11	...	-3.4	1.0	8.0	5.6	+ Tens
Shell 11	...	-3.4	-1.0	-8.0	-12.4	- Comp
Shell 12	...	-2.7	0.9	4.4	2.5	+ Tens
Shell 12	...	-2.7	-0.9	-4.4	-7.9	- Comp
Shell 13	...	-2.3	0.7	1.9	0.4	+ Tens
Shell 13	...	-2.3	-0.7	-1.9	-4.9	- Comp
Top head	...	-2.1	0.5	0.5	...	+ Tens
Top head	...	-2.1	-0.5	-0.5	-3.1	- Comp

Fatigue Stress Calculations due to Wind Induced Vortex Shedding Loads

From	Elevation	Alternating Stress	Sif	Number of Cycles	Hours of Total Vibration
10	3000.00	122.30	2.0	154107	80.60
20	6000.00	207.74	3.0	21829	11.42
30	6100.00	153.50	2.0	58780	30.74
40	9100.00	138.47	2.0	97987	51.25
50	12100.00	120.47	2.0	162725	85.11
60	15100.00	103.37	2.0	309825	162.05
70	18100.00	87.48	2.0	1000000	523.03
80	21100.00	72.89	2.0	1000000	523.03
90	24100.00	59.35	2.0	1000000	523.03
100	27100.00	50.64	2.0	1000000	523.03
110	30100.00	38.62	2.0	1000000	523.03
120	33100.00	28.24	2.0	1000000	523.03



130	36100.00	19.21	2.0	1000000	523.03
140	39100.00	13.18	2.0	1000000	523.03
150	42100.00	6.57	2.0	1000000	523.03
160	43900.00	2.41	2.0	1000000	523.03
170	44000.00	0.34	2.0	1000000	523.03

Cycle Life Determination based on ASME Div. II, Figure 3-F.1

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Top skirt

Mean skirt diameter at the critical height is computed later.
 This is a Conical type Support Skirt , Wider at the Bottom

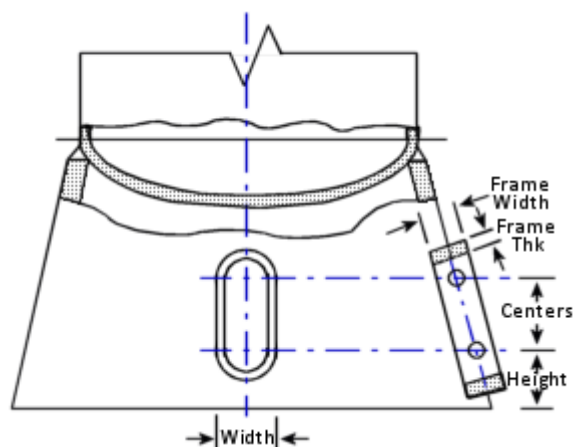
Half Apex Angle of the skirt	a	6.004 deg
Total vessel (empty) weight	Fg	1427378.625 N
Head Weight below section 2-2	DeltaFg	23633.268 N
Weight of vessel contents (liquid)	Ff	59689.043 N
Vessel Gross Weight	Fg + Ff	1487067.750 N
Skirt Mean Diameter Top End	DzTop	1831.846 mm
Skirt Mean Diameter Bottom End	DzBot	2462.846 mm
Skirt Analysis Thickness (Corroded)	ez	27.000 mm
Skirt Height	H	3000.000 mm
Additional Axial Force at section 1-1	F1	-16500.000 N
Moment at section 1-1	M1	5250188800.000 N-mm
Applied Moment - Bottom of Skirt	Mb	5939841536.000 N-mm
Weight at Bottom of Skirt	W	1466673.000 N

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical
Head Mean Diameter at Skirt Top	Db	1830.500 mm
Head Corroded Thickness	eb	24.500 mm

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 453.0 \text{ mm} \\
 K &= D_i / (2 * h_i) = 1806.0 / (2 * 453.0) = 1.9934 \\
 R &= D_i (0.44 * K + 0.02) = 1806.0 * (0.44 * 1.993 + 0.02) = 1620.1373 \text{ mm} \\
 r &= D_i ((0.5 / K) - 0.08) = 1806.0 * ((0.5 / 1.993) - 0.08) = 308.52 \text{ mm}
 \end{aligned}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	65535770624.000 mm ⁴
Moment of Inertia	Iyy =	65535770624.000 mm ⁴
Section Modulus	W =	71551696.000 mm ³
Section Metal Area	A =	156239.578 mm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:
 $= \sqrt{(\min(lxx, lyy) / \text{Area})}$
 $= \sqrt{(\min(0.65536E+11, 0.65536E+11) / 156239.578)}$
 $= 647.655 \text{ mm}$

Skirt Slenderness Ratio [rg/L]:
 $= rg / L$
 $= 647.655 / 3000.0$
 $= 0.216$

Correct the Moment [M4]:
 bottom of skirt is the critical end
 Controlling Moment [M4] = 0.59398E+10 N-mm

Axial Stress at Residual (Weakest) Surface Area [Sigma]:
 Downwind Side:
 $= -M / Z - W / A$
 $= -0.59398E+10 / 51734E+09 - 1466673 / 156239.57812$
 $= -20.866 \text{ MPa}$

Upwind Side:
 $= +(M) / Z - W / A$
 $= +.59398E+10 / 51734E+09 - 1466673 / 156239.57812$
 $= 2.09 \text{ MPa}$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:
 $= F1 - Fg - Ff + 4 \cdot M1 / Dz$
 $= -16500.0 - 1427379 - 59689.043 + 4 \cdot 0.52502E+10 / 1831.846$
 $= 9956046 \text{ N}$

Force at the joint, 16.12-2 [Fzq]:
 $= F1 - Fg - Ff - 4 \cdot M1 / Dz$
 $= -16500.0 - 1427379 - 59689.043 - 4 \cdot 0.52502E+10 / 1831.846$
 $= -12963181.0 \text{ N}$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:
 $= (Fzp + \Delta Fq) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (9956046 + 23633.268 + 59689.043) / (\pi \cdot 1830.5 \cdot 24.5) + 2.157 \cdot 1830.5 / (4 \cdot 24.5)$
 $= 111.56 \text{ MPa}$

Stress [σ_{1qm}]:
 $= (Fzq + \Delta Fq + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (-0.1E+08 + 23633.268 + 59689.043) / (\pi \cdot 1830.5 \cdot 24.5) + 2.157 \cdot 1830.5 / (4 \cdot 24.5)$
 $= -51.13 \text{ MPa}$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:
 $= (Ff + \Delta Fq) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (59689.043 + 23633.268) / (\pi \cdot 1830.5 \cdot 24.5) + 2.157 \cdot 1830.5 / (4 \cdot 24.5)$
 $= 40.89 \text{ MPa}$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:
 $= Fzp / (\pi \cdot Dz \cdot ez)$
 $= 9956046 / (\pi \cdot 1831.846 \cdot 27.0)$
 $= 64.08 \text{ MPa}$

Stress [σ_{3qm}]:
 $= Fzq / (\pi \cdot Dz \cdot ez)$
 $= -0.1E+08 / (\pi \cdot 1831.846 \cdot 27.0)$
 $= -83.43 \text{ MPa}$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:
 $= 1 - 0.5 (Db + eb - Dz + ez) / (r + eb)$
 $= 1 - 0.5 (1830.5 + 24.5 - 2462.85 + 27.0) / (308.52 + 24.5)$
 $= 0.9247$

$\gamma = \arccos(\cos(\gamma)) = \arccos(0.925) = 22.377 \text{ deg}$

Dimension [a]:
 $= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$
 $= 0.5 \cdot \sqrt{(24.5^2 + 27.0^2 + 2 \cdot 24.5 \cdot 27.0 \cdot 0.925)}$
 $= 25.262 \text{ mm}$

Bending moment at verification points p and q [Mp & Mq]:
 $M_p = a \cdot F_{zp} = 25.26 \cdot 9956046.0 = 251.6092 \times 10^6 \text{ N-mm}$
 $M_q = a \cdot F_{zq} = 25.26 \cdot -12963181.0 = -327.4711 \times 10^6 \text{ N-mm}$

Correction factor [C]:
 $= 0.63 - 0.057 (eb/ez)^2$
 $= 0.63 - 0.057 (24.5/27.0)^2$
 $= 0.583$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:
 $= C \cdot 6 \cdot M_p / (\pi \cdot Db \cdot eb^2)$
 $= 0.583 \cdot 6 \cdot 251.6092 \times 10^6 / (\pi \cdot 1830.5 \cdot 24.5^2)$
 $= 254.92 \text{ MPa}$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:
 $= C \cdot 6 \cdot M_q / (\pi \cdot Db \cdot eb^2)$
 $= 0.583 \cdot 6 \cdot -327.4711 \times 10^6 / (\pi \cdot 1830.5 \cdot 24.5^2)$
 $= -331.92 \text{ MPa}$

Stress [$\sigma_{3pb(o)}$]:
 $= C \cdot 6 \cdot M_p / (\pi \cdot Dz \cdot ez^2)$
 $= 0.583 \cdot 6 \cdot 251.6092 \times 10^6 / (\pi \cdot 1831.846 \cdot 27.0^2)$
 $= 209.74 \text{ MPa}$

Stress [$\sigma_{3qb(o)}$]:
 $= C \cdot 6 \cdot M_q / (\pi \cdot Dz \cdot ez^2)$
 $= 0.583 \cdot 6 \cdot -327.4711 \times 10^6 / (\pi \cdot 1831.846 \cdot 27.0^2)$
 $= -273.10 \text{ MPa}$

Geometry variables, 16.12-32 to 36:
 $eb/Db = 0.965/72.067 = 0.0134$
 $y = 125 \cdot eb/Db = 125 \cdot 24.5/1830.5 = 1.673$

As this an elliptical head, it will be treated as a Korbbogen type head

Geometric value [α]:
 $= 4.2 - 0.2 \cdot y = 4.2 - 0.2 \cdot 1.673 = 3.8654$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:
 $= (P + Ph) \cdot Db \cdot ((y / y_A) \cdot \alpha - 1) / (4 \cdot eb)$
 $= (2.157 + 0.0) \cdot 1830.5 \cdot ((22.38/40.0) \cdot 3.865 - 1) / (4 \cdot 24.5)$
 $= 46.84 \text{ MPa}$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:
 $= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$
 $= 111.56 - 254.921 + 46.843$
 $= -96.52 \text{ MPa}$

Stress [σ_{1potot}]:
 $= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$



$$= 111.56 + 254.921 - 46.843$$

$$= 319.64 \text{ MPa}$$

Stress [σ_{1qitot}]:

$$= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$$

$$= -51.126 - -331.917 + 46.843$$

$$= 327.63 \text{ MPa}$$

Stress [σ_{1qotot}]:

$$= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$$

$$= -51.126 + -331.917 - 46.843$$

$$= -429.89 \text{ MPa}$$

Stress [σ_{2pitot}]:

$$= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$$

$$= 40.89 + 254.921 + 46.843$$

$$= 342.65 \text{ MPa}$$

Stress [σ_{2potot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 40.89 - 254.921 - 46.843$$

$$= -260.87 \text{ MPa}$$

Stress [σ_{2qitot}]:

$$= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$$

$$= 40.89 + -331.917 + 46.843$$

$$= -244.18 \text{ MPa}$$

Stress [σ_{2qotot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 40.89 - -331.917 - 46.843$$

$$= 325.96 \text{ MPa}$$

Stress [σ_{3pitot}]:

$$= \sigma_{3pm} - \sigma_{3pb}$$

$$= 64.08 - 209.745$$

$$= -145.66 \text{ MPa}$$

Stress [σ_{3potot}]:

$$= \sigma_{3pm} + \sigma_{3pb}$$

$$= 64.08 + 209.745$$

$$= 273.82 \text{ MPa}$$

Stress [σ_{3qitot}]:

$$= \sigma_{3qm} - \sigma_{3qb}$$

$$= -83.435 - -273.096$$

$$= 189.66 \text{ MPa}$$

Stress [σ_{3qotot}]:

$$= \sigma_{3qm} + \sigma_{3qb}$$

$$= -83.435 + -273.096$$

$$= -356.53 \text{ MPa}$$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2$$

$$= 137.9(3 - (1/1.5)) \cdot (111.56/137.9)^2$$

$$= 353.53 \text{ MPa}$$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2$$

$$= 137.9(3 - (1/1.5)) \cdot (-51.126/137.9)^2$$

$$= 401.06 \text{ MPa}$$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2$$

$$= 137.9(3 - (1/1.5)) \cdot (40.89/137.9)^2$$

$$= 405.62 \text{ MPa}$$



For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{2qm} / f)^2)$$

$$= 137.9(3 - (1/1.5) \cdot (40.89/137.9)^2)$$

$$= 405.62 \text{ MPa}$$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2)$$

$$= 137.9(3 - (1/1.5) \cdot (64.08/137.9)^2)$$

$$= 393.85 \text{ MPa}$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2)$$

$$= 137.9(3 - (1/1.5) \cdot (-83.435/137.9)^2)$$

$$= 380.05 \text{ MPa}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (MPa)

Stress	Actual	Allowed	Result	Equation
Stot1pi	-96.52	353.53	Pass	46
Stot1po	319.64	353.53	Pass	47
Stot1qi	327.63	401.06	Pass	48
Stot1qo	-429.89	401.06	Fail	49
Stot2pi	342.65	405.62	Pass	50
Stot2po	-260.87	405.62	Pass	51
Stot2qi	-244.18	405.62	Pass	52
Stot2qo	325.96	405.62	Pass	53
Stot3pi	-145.66	393.85	Pass	54
Stot3po	273.82	393.85	Pass	55
Stot3qi	189.66	380.05	Pass	56
Stot3qo	-356.53	380.05	Pass	57

WARNING - One of the Stresses is over the limit in an area without openings!

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Shop/Field Installation Options :

Platform(s) installed in the Shop.
 Trays are installed in the Shop.
 Insulation is installed in the Shop.
 Lining is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	31506.088 mm
Center of Gravity of the Liquid	7989.130 mm
Center of Gravity of the Insulation	9533.155 mm
Center of Gravity of the Lining	2766.364 mm
Center of Gravity of the Nozzles	24042.020 mm
Center of Gravity of the Trays	27249.213 mm
Center of Gravity of the Added Weights (Operating)	28562.318 mm
Center of Gravity of the Added Weights (Empty)	29640.580 mm
Center of Gravity of the Trunnions	43450.000 mm
Center of Gravity of Bare Shell New and Cold	20401.404 mm
Center of Gravity of Bare Shell Corroded	19993.740 mm
Vessel CG in the Operating Condition	23070.000 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	24353.316 mm
Vessel CG in the Test Condition	24613.938 mm

Rigging Analysis Results:

• Lifting weight based on the user selected calculated empty weight

Total Effective Length of Vessel for this analysis	44000.00 mm
Total vessel weight (No Liquid)	Twt 1474316.00 N
Impact weight multiplication factor	Imp 1.50
Design lifting weight, DWT = Imp * Twt	2211474.00 N
Elevation of the Tailing Lug (bottom)	250.00 mm
Elevation of the Lifting Lug (top)	43450.00 mm
Design Reaction force at the tailing lug	977588.44 N
Design Reaction force at the Trunnions	1233885.50 N
CG distance during field lifting operation	24353.32 mm
CG distance from tailing lug	24103.32 mm
CG distance from the nearer lifting lug/trunnion	19096.69 mm
Compressive Allowable Increase Factor	1.00

Critical Values: (MPa, mm)

	Max Stress	Elevation	Allowables
Bending	135.12	24100.01	115.64 (UG-23)
Shear	8.51	43180.02	96.53 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm	Bending Moment N-mm	Bending Stress MPa	Shear Force N	Shear Stress MPa
0.00	0.0	0.0	256115.8	1.3
6700.00	5152327168.0	66.3	495890.4	2.9
16300.00	9315918848.0	119.9	81801.4	0.5
25900.01	9534374912.0	131.7	320085.0	2.0
35500.02	5876995584.0	81.2	720682.1	4.5
43900.02	68308728.0	1.4	36185.0	0.3

Unity Check (Actual Stress / Allowable Stress):



Maximum Unity Check is 1.1684 at elevation 24100.0078 mm - Must be ≤ 1

Note: The rigging analysis is performed using a uniformly distributed load.

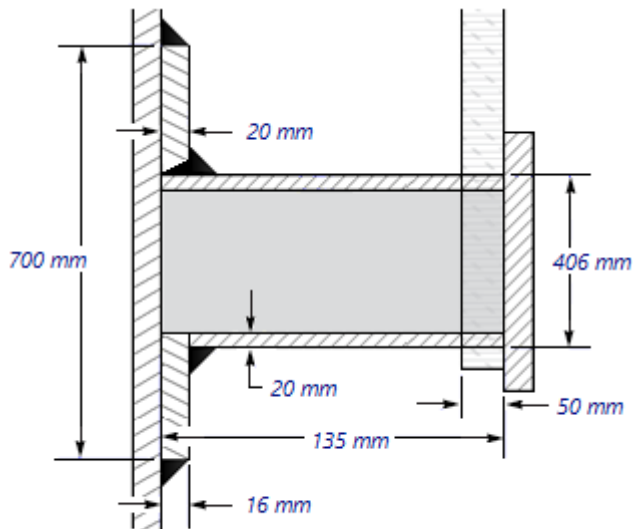
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Trunnion Analysis:

Input Data:

Lifting Lug Material	SA-516 70
Lifting Lug Yield Stress	Yield 262.01 MPa
Trunnion Outside Diameter	Tod 406.000 mm
Trunnion Thickness	Tthk 20.0000 mm
Trunnion Length	Tl 135.0000 mm
Reinforcing Pad Used	Yes
Pad Outside Diameter	Dp 700.000 mm
Pad Thickness	Thkp 20.0000 mm
Pad Edge Fillet Weld Thk	Tfp 16.0000 mm
Bail/Sling Width	Tt 50.0000 mm
Sling Angle from Horizontal	0.0 degrees
Occasional Load Factor	Occfac 1.0
Impact Factor	Impfac 1.50
Number of Lugs in Group	2



Trunnion may be inserted or abutting

Trunnion Force and Stress Results for different Lift Angles (N&MPa):

Bending Stress Allowable:	172.9 MPa
Shear Stress Allowable:	104.8 MPa
Bearing Stress Allowable:	196.5 MPa

Angle	Trunnion Lifting Load	Bending Stress	Shear Stress	Bearing Stress	Unity Check	Pass Fail
0	616942.8	30.4	25.4	30.4	0.419	Pass
5	618462.9	30.5	25.5	30.5	0.420	Pass
10	619996.7	30.6	25.6	30.5	0.421	Pass
15	621568.7	30.6	25.6	30.6	0.422	Pass
20	623205.1	30.7	25.7	30.7	0.423	Pass
25	624937.2	30.8	25.8	30.8	0.424	Pass
30	626802.4	30.9	25.8	30.9	0.425	Pass
35	628849.3	31.0	25.9	31.0	0.427	Pass
40	631142.6	31.1	26.0	31.1	0.428	Pass
45	633771.8	31.2	26.1	31.2	0.430	Pass
50	636867.2	31.4	26.3	31.4	0.432	Pass
55	640627.9	31.6	26.4	31.6	0.435	Pass
60	645374.8	31.8	26.6	31.8	0.438	Pass
65	651664.4	32.1	26.9	32.1	0.442	Pass
70	660556.0	32.6	27.2	32.5	0.448	Pass
75	674349.4	33.3	27.8	33.2	0.458	Pass
80	699161.2	34.5	28.8	34.4	0.474	Pass



85	758474.3	37.4	31.3	37.4	0.515	Pass
90	1105737.0	54.5	45.6	54.5	0.750	Pass

Trunnion Calculation Results:

Trunnion Lifting Load [Tlf]:
 = Resultant Load • Impact Factor
 = 737158 • 1.5
 = 1105737.0 N

Trunnion Section Modulus [Z]:
 = $\pi (T_{od}^4 - T_{id}^4) / (32 \cdot T_{od})$
 = $\pi (406.0^4 - 366.0^4) / (32 \cdot 406.0)$
 = 2231105.8 mm³

Trunnion Bending Moment [Tmom]:
 = $(T_l - 0.5 \cdot T_t) / 1.000 \cdot T_{lf}$
 = $(135.0 - 0.5 \cdot 50.0) / 1.000 \cdot 1105737$
 = 121680376.0 N-mm

Trunnion Calculated Bending Stress [Sbc]:
 = $T_{mom} \cdot 1.000 / Z$
 = $0.12168E+09 \cdot 1.000 / 2231106$
 = 54.52 MPa [Passed] $Sbc < \min(0.66 \cdot S_Y \cdot Occfac, 0.75 \cdot S_y) (172.93)$

Trunnion Cross Section Area [Txsa]:
 = $\pi/4 (T_{od}^2 - T_{id}^2)$
 = $\pi/4 (406.0^2 - 366.0^2)$
 = 24253.1 mm²

Trunnion Calculated Shear Stress [Ssc]:
 = T_{lf} / T_{xsa}
 = 1105737/24253.1
 = 45.60 MPa [Passed] $Ssc < 0.4 \cdot S_y \cdot Occfac (104.80)$

Trunnion Unity Check [Uchk]:
 = $Sbc/S_{ba} + Ssc/S_{sa}$
 = $54.52/172.93 + 45.6/104.8$
 = 0.750 [Passed]

Weld Section Modulus at Edge of Trunnion [Sw]:
 = $\pi (T_{od}/2)^2$
 = $\pi (406.0/2)^2$
 = 129461.8 mm²

Shear Stress in Weld due to Bending Moment [fs]:
 = T_{mom} / S_w
 = $0.12168E+09 / 129461.8$
 = 939.513 N/mm

Required Fillet Weld Leg Size at Edge of Trunnion [Wsr]:
 = $\max(10\text{mm}, fs / (0.4 \cdot S_y))$
 = $\max(10\text{mm}, 939.513 / (0.4 \cdot 262.0))$
 = 10.000 mm

Trunnion Bearing Contact Area [Tbca]:
 = $T_{od} \cdot T_t$
 = $406.0 \cdot 50.0$
 = 20300.0 mm²

Trunnion Bearing Stress [Sbec]:
 = T_{lf} / T_{bca}
 = $1105737 / 20300.0$
 = 54.47 MPa [Passed] $Sbec < \min(0.75 \cdot S_Y \cdot Occfac, 0.9 \cdot S_y) (196.51)$

Summary of Results:

Trunnion Unity Check Value: 0.7503 Must be less than or equal to 1



Comparison of Stress Values versus Allowables: (MPa)

Type of Stress	Actual	Allowable	Pass/Fail
Bending Stress	54.52	172.93	Pass
Shear Stress	45.60	104.80	Pass
Bearing Stress	54.47	196.51	Pass

Results of local shell stress analysis due to loads on the trunnion:

This following Stress Analysis assumes that the Trunnion passes through the pad and is welded to the shell via a groove weld. Additionally the re-pad is welded to the shell at the outside edge via a fillet weld.

In the local stress results below, the 'Nozzle' is the 'Trunnion'.

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).

Loads on shell and WRC 107 results at different Lift Angles (N & N-mm):

Angle	PT	PL	MC	ML	Rat1	Rat2	P/F
0	616942.8	0.0	67891216.0	0.0	0.300	0.437	Pass
2	617174.4	21552.2	67916712.0	2371704.0	0.300	0.437	Pass
5	616109.4	53902.6	67799520.0	5931689.5	0.300	0.437	Pass
10	610577.5	107661.3	67190760.0	11847543.0	0.297	0.433	Pass
15	600389.2	160873.8	66069592.0	17703294.0	0.293	0.427	Pass
20	585621.2	213148.7	64444452.0	23455860.0	0.288	0.417	Pass
25	566385.4	264109.8	62327652.0	29063860.0	0.280	0.405	Pass
30	542826.8	313401.2	59735152.0	34488108.0	0.271	0.390	Pass
35	515123.2	360693.2	56686524.0	39692332.0	0.260	0.373	Pass
40	483483.2	405690.7	53204720.0	44644064.0	0.247	0.353	Pass
45	448144.3	448144.3	49315860.0	49315860.0	0.233	0.331	Pass
50	409370.3	487868.6	45048992.0	53687296.0	0.218	0.306	Pass
55	367449.1	524771.7	40435784.0	57748284.0	0.202	0.281	Pass
60	322687.4	558910.9	35510004.0	61505124.0	0.202	0.253	Pass
65	275405.3	590608.6	30306860.0	64993280.0	0.213	0.234	Pass
70	225923.4	620719.6	24861652.0	68306832.0	0.224	0.243	Pass
75	174534.5	651371.6	19206576.0	71679920.0	0.235	0.252	Pass
80	121408.1	688539.4	13360304.0	75770040.0	0.249	0.265	Pass
85	66105.4	755588.1	7274542.0	83148384.0	0.273	0.289	Pass
90	0.0	1105737.0	0.0	*****	0.399	0.423	Pass

- PT - Transverse load per lug
- PL - Longitudinal load per lug
- MC - Circumferential moment in shell
- ML - Longitudinal moment in shell
- Rat1 - Computed WRC 107 stress ratio at trunnion edge
- Rat2 - Computed WRC 107 stress ratio at pad edge

Worst case combination loads were at: 2.0 degrees from horizontal.

Input Echo, WRC107/537 Item 1, Description: Trunnion

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	0.0000 mm
Vessel Diameter	Dv	1800.000 mm
Vessel Thickness	Tv	22.000 mm
Design Temperature	T1	37.8 C
Attachment Type	Type	Round
Diameter Basis for Nozzle	Nbasis	OD
Corrosion Allowance for Nozzle	Can	0.0000 mm
Nozzle Diameter	Dn	406.000 mm
Nozzle Thickness	Tn	20.000 mm

Nozzle Cold S.I. Allowable	SNmc	137.90	MPa
Nozzle Hot S.I. Allowable	SNmh	137.90	MPa
Thickness of Reinforcing Pad	Tpad	20.000	mm
Diameter of Reinforcing Pad	Dpad	732.000	mm
Design Internal Pressure	Dp	0.000	MPa
Include Pressure Thrust		No	

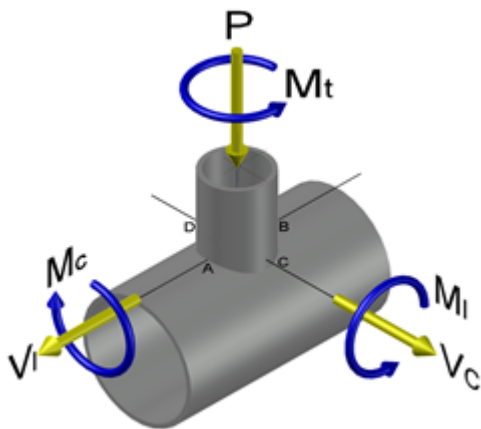
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	0.0	N
Longitudinal Shear (SUS)	VL	21552.2	N
Circumferential Shear (SUS)	Vc	617174.4	N
Circumferential Moment (SUS)	Mc	67916712.0	N-mm
Longitudinal Moment (SUS)	ML	2371703.8	N-mm
Torsional Moment (SUS)	Mt	0.0	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



Worst case combination loads were at: 2.0 degrees from horizontal.

WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 406.0 + 2 \cdot 1.65 \cdot \sqrt{911.0 (22.0 - 0)}$$

$$= 873.180 \text{ mm}$$

WRC 107 Stress Calculation for SUStained loads:

Radial Load	P	0.0	N
Circumferential Shear	VC	617174.4	N
Longitudinal Shear	VL	21552.2	N
Circumferential Moment	MC	67916712.0	N-mm
Longitudinal Moment	ML	2371703.8	N-mm
Torsional Moment	MT	0.0	N-mm

Dimensionless Parameters used : Gamma = 21.93

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
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N(PHI) / (P/Rm)	0.193	4C	3.415	(A,B)
N(PHI) / (P/Rm)	0.193	3C	2.566	(C,D)
M(PHI) / (P)	0.193	2C1	0.056	(A,B)
M(PHI) / (P)	0.193	1C	0.087	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.193	3A	0.843	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.193	1A	0.089	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.193	3B	2.456	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.193	1B	0.036	(A,B,C,D)
N(x) / (P/Rm)	0.193	3C	2.566	(A,B)
N(x) / (P/Rm)	0.193	4C	3.415	(C,D)
M(x) / (P)	0.193	1C1	0.091	(A,B)
M(x) / (P)	0.193	2C	0.056	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.193	4A	1.344	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.193	2A	0.046	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.193	4B	0.798	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.193	2B	0.058	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-8.3	-8.3	8.3	8.3
Circ. Bend. MC		0.0	0.0	0.0	0.0	-115.7	115.7	115.7	-115.7
Circ. Memb. ML		-0.8	-0.8	0.8	0.8	0.0	0.0	0.0	0.0
Circ. Bend. ML		-1.6	1.6	1.6	-1.6	0.0	0.0	0.0	0.0
Tot. Circ. Str.		-2.5	0.8	2.5	-0.8	-124.1	107.4	124.1	-107.4
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-13.3	-13.3	13.3	13.3
Long. Bend. MC		0.0	0.0	0.0	0.0	-59.7	59.7	59.7	-59.7
Long. Memb. ML		-0.3	-0.3	0.3	0.3	0.0	0.0	0.0	0.0
Long. Bend. ML		-2.6	2.6	2.6	-2.6	0.0	0.0	0.0	0.0
Tot. Long. Str.		-2.9	2.3	2.9	-2.3	-73.0	46.4	73.0	-46.4
Shear VC		23.0	23.0	-23.0	-23.0	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		23.0	23.0	-23.0	-23.0	-0.8	-0.8	0.8	0.8
Str. Int.		46.1	46.1	46.1	46.1	124.1	107.4	124.1	107.4

Dimensionless Parameters used : Gamma = 41.41

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.352	4C	3.974	(A,B)
N(PHI) / (P/Rm)	0.352	3C	1.745	(C,D)
M(PHI) / (P)	0.352	2C1	0.012	(A,B)
M(PHI) / (P)	0.352	1C	0.063	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.352	3A	1.284	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.352	1A	0.064	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.352	3B	2.457	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.352	1B	0.009	(A,B,C,D)
N(x) / (P/Rm)	0.352	3C	1.745	(A,B)



N(x)	/ (P/Rm)	0.352	4C	3.974	(C,D)
M(x)	/ (P)	0.352	1C1	0.028	(A,B)
M(x)	/ (P)	0.352	2C !	0.032	(C,D)
N(x)	/ (MC/(Rm**2 * Beta))	0.352	4A	3.994	(A,B,C,D)
M(x)	/ (MC/(Rm * Beta))	0.352	2A	0.025	(A,B,C,D)
N(x)	/ (ML/(Rm**2 * Beta))	0.352	4B	1.361	(A,B,C,D)
M(x)	/ (ML/(Rm * Beta))	0.352	2B	0.015	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-13.6	-13.6	13.6	13.6
Circ. Bend. MC		0.0	0.0	0.0	0.0	-167.3	167.3	167.3	-167.3
Circ. Memb. ML		-0.9	-0.9	0.9	0.9	0.0	0.0	0.0	0.0
Circ. Bend. ML		-0.9	0.9	0.9	-0.9	0.0	0.0	0.0	0.0
Tot. Circ. Str.		-1.8	-0.1	1.8	0.1	-180.9	153.7	180.9	-153.7
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-42.3	-42.3	42.3	42.3
Long. Bend. MC		0.0	0.0	0.0	0.0	-66.5	66.5	66.5	-66.5
Long. Memb. ML		-0.5	-0.5	0.5	0.5	0.0	0.0	0.0	0.0
Long. Bend. ML		-1.4	1.4	1.4	-1.4	0.0	0.0	0.0	0.0
Tot. Long. Str.		-1.9	0.9	1.9	-0.9	-108.7	24.2	108.7	-24.2
Shear VC		24.4	24.4	-24.4	-24.4	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.9	-0.9	0.9	0.9
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		24.4	24.4	-24.4	-24.4	-0.9	-0.9	0.9	0.9
Str. Int.		48.8	48.8	48.8	48.8	180.9	153.7	180.9	153.7

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		-0.8	-0.8	0.8	0.8	-8.3	-8.3	8.3	8.3
Circ. Q (SUS)		-1.6	1.6	1.6	-1.6	-115.7	115.7	115.7	-115.7
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		-0.3	-0.3	0.3	0.3	-13.3	-13.3	13.3	13.3
Long. Q (SUS)		-2.6	2.6	2.6	-2.6	-59.7	59.7	59.7	-59.7
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		23.0	23.0	-23.0	-23.0	-0.8	-0.8	0.8	0.8
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		46.1	46.1	46.1	46.1	13.4	13.4	13.4	13.4



Pm+Pl+Q (Total)	46.1	46.1	46.1	46.1	124.1	107.4	124.1	107.4
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Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	137.90	Passed
Pm+Pl (SUS)	46.09	206.85	Passed
Pm+Pl+Q (TOTAL)	124.08	413.70	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		-0.9	-0.9	0.9	0.9	-13.6	-13.6	13.6	13.6
Circ. Q (SUS)		-0.9	0.9	0.9	-0.9	-167.3	167.3	167.3	-167.3
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		-0.5	-0.5	0.5	0.5	-42.3	-42.3	42.3	42.3
Long. Q (SUS)		-1.4	1.4	1.4	-1.4	-66.5	66.5	66.5	-66.5
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		24.4	24.4	-24.4	-24.4	-0.9	-0.9	0.9	0.9
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		48.8	48.8	48.8	48.8	42.3	42.3	42.3	42.3
Pm+Pl+Q (Total)		48.8	48.8	48.8	48.8	180.9	153.7	180.9	153.7

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	137.90	Passed
Pm+Pl (SUS)	48.80	206.85	Passed
Pm+Pl+Q (TOTAL)	180.88	413.70	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	3120.7805	mm
Skirt Thickness	STHK	20.0000	mm
Skirt Internal Corrosion Allowance	SCA	0.5000	mm
Skirt External Corrosion Allowance		0.5000	mm
Skirt Material	SA-36		

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	55.0000	mm
Design Temperature of the Basing		40.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	137.90	MPa
Basing Yield Stress		260.74	MPa
Inside Diameter of Basing	DI, D4	3000.9998	mm
Outside Diameter of Basing	DOU, D5	3451.0002	mm
Nominal Diameter of Bolts	BND, d(b0)	64.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Root Area of a Single Bolt	Area	2316.3999	mm ²
Bolt Material		SA-36	
Bolt Operating Allowable Stress	SA, fB	114.50	MPa
Number of Bolts	RN, nb	20	
Diameter of Bolt Circle	DC, DBC	3299.0002	mm
Bolt Allowable Shear Stress		107.873	MPa
Ultimate Comp. Strength of Concrete	FPC	35.0	MPa
Allowable Comp. Strength of Concrete	FC	8.0	MPa
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	20.0000	mm
Width of Gussets at Top Plate	TWDT	165.0000	mm
Width of Gussets at Base Plate	BWDT	165.0000	mm
Gusset Plate Elastic Modulus	E	201631.0	MPa
Gusset Plate Yield Stress	SY	260.7	MPa
Height of Gussets	HG, h1	300.0000	mm
Distance between Gussets	RG, b6	110.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	55.0000	mm
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	45.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	165.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	62.0000	mm
External Corrosion Allowance	CA	3.0000	mm
Dead Weight of Vessel	DW	1490816.0	N
Operating Weight of Vessel	ROW	1605369.4	N
Test Weight of Vessel	TW	2388129.0	N
Earthquake Moment on Basing	EQMOM	5569087488.0	N-mm
Wind Moment on Basing	WIMOM	5484868096.0	N-mm
Vortex Shedding Moment on Basing	Vormom	5967279616.0	N-mm
User Moment at the Base [WindLoad Case]		699949696.0	N-mm (UWIMOM)
Test Moment on Basing	TM	2051464064.0	N-mm
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0.55691E+10/3299.0) - 1605369) / 20$$

$$= 257217.5 \text{ N}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0.61848E+10/3299.0) - 1490816) / 20$$

$$= 300258.3 \text{ N}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0.59673E+10/3299.0) - 1605369) / 20$$

$$= 281347.8 \text{ N}$$

Bolt Load, Test Moment Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0.20515E+10/3299.0) - 2388129) / 20$$

$$= 4912.0 \text{ N}$$

Governing Bolt Load Condition, Wind + Dead Weight Condition:

Area Available in 1 Bolt (corroded) Abss : 2316.3999 mm²
 Area Available in all the Bolts Abss * RN : 46327.9961 mm²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.181	0.272	1.157	2.702	0.463	0.774	1529449.2	3020265.2	49524.7
4	0.317	0.340	1.555	2.405	0.434	0.782	1569434.4	3060250.5	57107.3
6	0.329	0.323	1.585	2.380	0.431	0.782	1573282.8	3064098.8	57840.1
8	0.326	0.327	1.578	2.386	0.432	0.782	1572313.8	3063129.5	57655.2
10	0.328	0.327	1.583	2.382	0.432	0.782	1573040.4	3063856.2	57793.8
12	0.328	0.328	1.583	2.382	0.432	0.782	1572979.6	3063795.8	57782.2

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 1572980 / (4.47 \cdot 3299.0 \cdot 2.382)$$

$$= 89.57 \text{ MPa, Should be } \leq 114.5 \text{ MPa}$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot Bolt \text{ Area} / (\pi \cdot Dc)$$

$$= 20 \cdot 2316.4 / (\pi \cdot 3299.0)$$

$$= 4.47 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 4.438[(2 \cdot 0.3276 \cdot 3299.0 + 225.0) / (2 \cdot 0.3276 \cdot 3299.0)]$$

$$= 4.9 \text{ MPa, Should be } \leq 8.0 \text{ MPa}$$

Values for table 10.3, l = 165.11, b = 259.103, l/b = 0.637236

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(16814.322, 32487.531) = 32487.531 \text{ N}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 32487.531 / 172.1)} + 3.0$$

$$= 36.657 \text{ mm}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 165.0, b = 110.0, b/l = 0.666667

As b/l (0.667) is less than 1, inverting b/l = 1.5.

Moment Term, based on geometry [Mo]:

$$= P/(4 \cdot \pi) [1.3(\ln((2\sin(\pi \cdot a/l))/(\pi \cdot g))) + 1] - [(0.7 - g^2)P/(4\pi)]$$

$$= 265205.25/(4 \cdot \pi) [1.3(\ln((2 \cdot 165.0 \cdot \sin(\pi \cdot 89.11/165.0))/$$

$$(\pi \cdot 53.975))) + 1] - [(0.7 - 0.071) \cdot 265205.25/(4 \cdot \pi)]$$

$$= 25873.5742 \text{ N}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(\text{Mo}) / \text{Fb}) + \text{Ca}}$$

$$= \sqrt{(6 \cdot \text{Abs}(25873.57)/173.83) + 3.0}$$

$$= 32.8855 \text{ mm}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot l_{\text{avg}}) + \text{Ca}$$

$$= 265205.25/(124.11 \cdot 165.0) + 3.0$$

$$= 15.952 \text{ (not less than } 9.525 + 3.0 \text{) mm}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 3299.0/20 = 518.206 \text{ mm}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76(P \cdot a/(m \cdot (h + t_{ba})) \cdot 1.5 \cdot \text{Sktope})^{2/3} \cdot r^{1/3} + \text{Ca}$$

$$= 1.76(265205 \cdot 89.11/(518.21 \cdot 355.0 \cdot 172))^{2/3} \cdot 1560.391^{1/3} + \text{Ca}$$

$$= 16.822 + 1.0 = 17.822 \text{ mm}$$

Shear Stress in a Single Bolt [rb]:

$$= \text{Shear Force}/(\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 237234.3/(2316.4 \cdot 20)$$

$$= 5.1 \text{ MPa. Must be less than } 107.9 \text{ MPa.}$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	36.6568	mm
Actual Basing Thickness as entered by user	55.0000	mm
Required Thickness of Chair Cap	32.8855	mm
Actual Top Ring Thickness as entered by user	45.0000	mm
Required Gusset thickness, + CA	15.9517	mm
Actual Gusset Thickness as entered by user	20.0000	mm
Required Thickness of Skirt for Local Stress	17.8216	mm
Given Thickness of Skirt	20.0000	mm
Required Gusset Height to meet local stress	294.4280	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA})$$

$$= 72.037 \cdot (20.0 - 1.0)$$

$$= 1368.59 \text{ N/mm}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 1369/[(0.4 \cdot 223) \cdot 2 \cdot 0.707]$$

$$= 10.861 \text{ mm}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 265205.2/(210.0 + 2 \cdot (300.0 + 45.0))$$

$$= 294.672 \text{ N/mm}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 265205.2 \cdot 89.11/(210.0 \cdot (345.0) + 0.6667 \cdot (345.0)^2)$$

$$= 155.681 \text{ N/mm}$$



Resultant Weld Load [Wr]:

$$= \sqrt{W_v^2 + W_h^2}$$

$$= \sqrt{294.67^2 + 155.68^2}$$

$$= 333.269 \text{ N/mm}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 333.27 / [(0.4 \cdot 223) \cdot 2 \cdot 0.707]$$

$$= 2.645 \text{ mm}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$

$$= 265205.2 / (2 \cdot 165.0)$$

$$= 803.652 \text{ N/mm}$$

Weld Load [Wh]:

$$= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth})$$

$$= 265205.2 \cdot 89.11 / (2 \cdot 345.0 \cdot 165.0)$$

$$= 207.575 \text{ N/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{W_v^2 + W_h^2}$$

$$= \sqrt{803.65^2 + 207.57^2}$$

$$= 830.027 \text{ N/mm}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 830.03 / [(0.4 \cdot 223) \cdot 2 \cdot 0.707]$$

$$= 6.587 \text{ mm}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	10.8614	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	6.5873	mm
Required Gusset to Top Plate Double Fillet Weld Size	6.5873	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: B2 10"

From: 30

Pressure for Reinforcement Calculations	P	2.1814	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Elliptical Head	D	1800.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	27.5000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	400.0000	mm
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User Entered Minimum Design Metal Temperature		-25.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		10.0000	in.

Size and Thickness Basis		Minimum	
Nominal Thickness		80	

Flange Material		SA-105 +IT	
Flange Type		Weld Neck Flange	

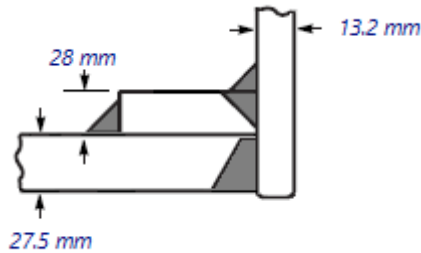
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	540.0000	mm
Thickness of Pad	te	28.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm
Reinforcing Pad Width		133.4750	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: B2 10"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	10.750 in.
Actual Thickness Used in Calculation	0.520 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 2.18 \cdot 1806.0 \cdot 0.8972 / (2 \cdot 137.9 \cdot 1.0 - 0.2 \cdot 2.18)$
 $= 12.8358 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2.18 \cdot 136.525 / (118 \cdot 1.0 + 0.4 \cdot 2.18)$
 $= 2.5054 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.6332 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	505.2934 mm
Parallel to Vessel Wall, opening length	d	252.6467 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		53.5041 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:

$= \min(1, Sn/Sv)$
 $= \min(1, 118.0/137.9)$
 $= 0.856$

Weld Strength Reduction Factor [fr2]:

$= \min(1, Sn/Sv)$
 $= \min(1, 118.0/137.9)$
 $= 0.856$

Weld Strength Reduction Factor [fr4]:

$= \min(1, Sp/Sv)$
 $= \min(1, 138.0/137.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.856, 1.0)$
 $= 0.856$

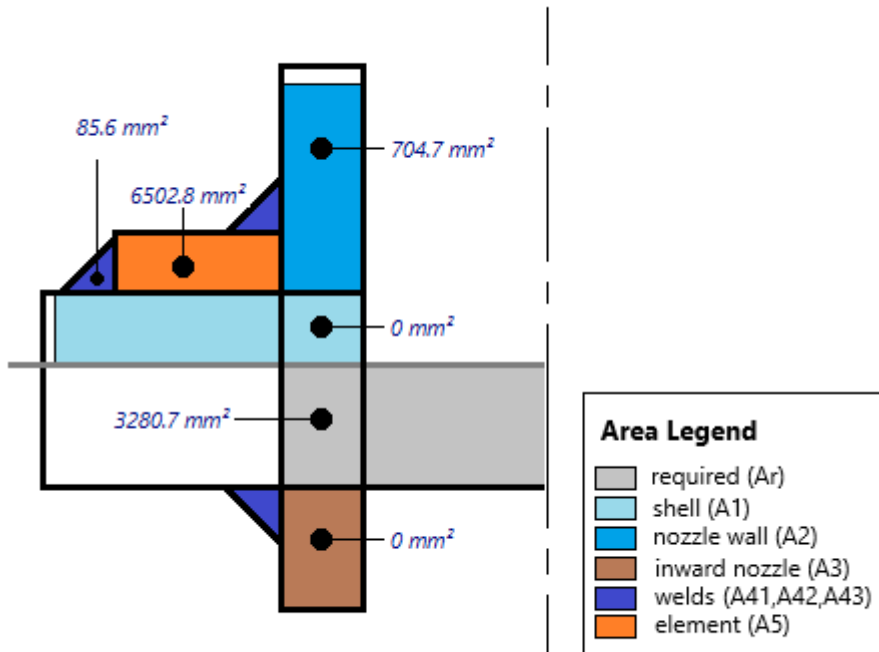


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	3280.709	428.125	NA
Area in Shell	A1	0.000	0.000	NA
Area in Nozzle Wall	A2	704.717	876.150	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	85.569	85.569	NA
Area in Element	A5	6502.815	6502.815	NA
TOTAL AREA AVAILABLE	Atot	7293.102	7464.535	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 361.9937 28.0000 mm
 Based on given Pad Diameter: 540.0000 10.7233 mm
 Based on Shell or Nozzle Thickness: 461.6947 13.2016 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (252.6467 \cdot 12.8358 \cdot 1.0 + 2 \cdot 10.2016 \cdot 12.8358 \cdot 1.0 \cdot (1 - 0.86))$
 $= 3280.709 \text{ mm}^2$

The area in the shell has not been calculated by user request.

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 53.5) \cdot (10.2 - 2.51) \cdot 0.8557$
 $= 704.717 \text{ mm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi - can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$
 $= 85.569 \text{ mm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 $= (505.2934 - 273.05) \cdot 28.0 \cdot 1.0$
 $= 6502.815 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 5.5054 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 17.2442 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 17.2442 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 3.3197 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 11.1026 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[11.103, \max(17.2442, 4.5)]$
 $= 11.1026 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(5.5054, 11.1026)$
 $= 11.1026 \text{ mm}$

Available Nozzle Neck Thickness = 13.2016 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	32.8,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	262.2 MPa	Passed
Occasional	:	13.0,	Allowable	:	156.9 MPa	Passed
Shear	:	14.6,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, $t_g = 27.5$, $t_r = 12.836$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.524$, Temp. Reduction = 30 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-32 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D

Govrn. thk, $t_g = 13.202$, $t_r = 2.505$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.246$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -18 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -62 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = 2.18/5.10 = 0.428

Weld Size Calculations, Description: B2 10"

Intermediate Calc. for nozzle/shell Welds Tmin 10.2016 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	9.8980 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (3280.7087 - 0.0 + 2 \cdot 10.2016 \cdot 0.8557 \cdot (1.0 \cdot 24.5 - 12.8358)) \cdot 138)$
 $= 480451.47 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (704.7173 + 6502.8149 + 85.5692 - 0.0 \cdot 0.86) \cdot 138$
 $= 1005633.06 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (704.7173 + 0.0 + 85.5692 + (427.7443)) \cdot 138$
 $= 167952.17 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (704.7173 + 0.0 + 85.5692 + 6502.8149 + (427.7443)) \cdot 138$
 $= 1064614.00 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 273.05 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 247972. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 540.0 \cdot 14.0 \cdot 0.49 \cdot 138$
 $= 802352. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 131.4242) \cdot (13.2016 - 3.0) \cdot 0.7 \cdot 118$
 $= 347887. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 273.05 \cdot 28.0 \cdot 0.74 \cdot 138$
 $= 1226292. \text{ N}$



Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 273.05 \cdot (24.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 1050346. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SPEW} + \text{SNW}) = (802352 + 347887) = 1150240 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (247972 + 1226292 + 1050346 + 0) = 2524610 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Spew} + \text{Tngw} + \text{Sinw})$
 $= (802352 + 1050346 + 0) = 1852698 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 1150239 N , must exceed W = 480451 N or W1 = 1005633 N
 Path 2-2 = 2524610 N , must exceed W = 480451 N or W2 = 167952 N
 Path 3-3 = 1852698 N , must exceed W = 480451 N or W3 = 1064614 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 5.6855 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 283.1856 mm

Input Echo, WRC107/537 Item 1, Description: B2 10" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	3240.000	mm
Vessel Thickness	Tv	27.500	mm
Design Temperature	T1	125.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	137.90	MPa
Vessel Hot S.I. Allowable	Smh	137.90	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	273.050	mm
Nozzle Thickness	Tn	13.202	mm
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa
Thickness of Reinforcing Pad	Tpad	28.000	mm
Diameter of Reinforcing Pad	Dpad	540.000	mm
Design Internal Pressure	Dp	2.181	MPa
Include Pressure Thrust		No	

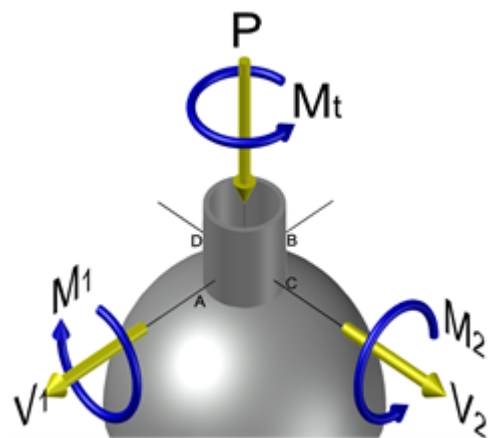
External Forces and Moments in WRC 107/537 Convention:

Radial Load	(SUS)	P	-14040.0	N
Longitudinal Shear	(SUS)	(Vl) V1	17550.0	N
Circumferential Shear	(SUS)	(Vc) V2	0.0	N
Circumferential Moment	(SUS)	(Mc) M1	11465999.0	N-mm
Longitudinal Moment	(SUS)	(Ml) M2	0.0	N-mm
Torsional Moment	(SUS)	Mt	10530002.0	N-mm

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 Yes
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{1635.25 (27.5 - 3.0)}$$

$$= 933.575 \text{ mm}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load		P	-14040.0	N
Circumferential Shear	(VC)	V2	0.0	N
Longitudinal Shear	(VL)	V1	17550.0	N
Circumferential Moment	(MC)	M1	11465999.0	N-mm
Longitudinal Moment	(ML)	M2	0.0	N-mm
Torsional Moment		MT	10530002.0	N-mm

Stress Indices for Nozzles in Spherical Shells and Portions of Formed Heads per Table 5-D.1

Stress	Inside Corner	Outside Corner
sn	2.0000	2.0000
st	-0.2000	2.0000

Dimensionless Param: U = 0.46 TAU = 12.88 RHO = 4.00 (5.15)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 4	0.04519	(A,B,C,D)
$M(x) / P$	SP 4	0.01943	(A,B,C,D)



$N(x) * T * \sqrt{R_m * T} / MC$ SM 4 0.11760 (A,B,C,D)
 $M(x) * \sqrt{R_m * T} / MC$ SM 4 0.05449 (A,B,C,D)
 $N(x) * T * \sqrt{R_m * T} / ML$ SM 4 0.11760 (A,B,C,D)
 $M(x) * \sqrt{R_m * T} / ML$ SM 4 0.05449 (A,B,C,D)

 $N(y) * T / P$ SP 4 0.30941 (A,B,C,D)
 $M(y) / P$ SP 4 0.08531 (A,B,C,D)
 $N(y) * T * \sqrt{R_m * T} / MC$ SM 4 0.35475 (A,B,C,D)
 $M(y) * \sqrt{R_m * T} / MC$ SM 4 0.27874 (A,B,C,D)
 $N(y) * T * \sqrt{R_m * T} / ML$ SM 4 0.35475 (A,B,C,D)
 $M(y) * \sqrt{R_m * T} / ML$ SM 4 0.27874 (A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Rad.	Bend. P	0.6	-0.6	0.6	-0.6	0.6	-0.6	0.6	-0.6
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-1.7	-1.7	1.7	1.7
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-4.6	4.6	4.6	-4.6
Rad.	Memb. ML	-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Rad. Str.		0.8	-0.4	0.8	-0.4	-5.5	2.6	7.1	-3.3
Tang.	Memb. P	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6
Tang.	Bend. P	2.6	-2.6	2.6	-2.6	2.6	-2.6	2.6	-2.6
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-5.0	-5.0	5.0	5.0
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-23.6	23.6	23.6	-23.6
Tang.	Memb. ML	-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.		4.2	-1.0	4.2	-1.0	-24.5	17.6	32.8	-19.7
Shear VC		0.0	0.0	-0.0	-0.0	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Shear MT		1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7
Tot. Shear		1.7	1.7	1.7	1.7	0.9	0.9	2.5	2.5
Str. Int.		4.9	3.5	4.9	3.5	24.5	17.6	33.1	20.0

Unitless Prm: $U = 1.35$ TAU = 0.00 (25.97) RHO = 0.00 (2.40)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SR 2	0.04408	(A,B,C,D)
$M(x) / P$	SR 2	0.02217	(A,B,C,D)
$N(x) * T * \sqrt{R_m * T} / MC$	SR 3	0.04689	(A,B,C,D)
$M(x) * \sqrt{R_m * T} / MC$	SR 3	0.03682	(A,B,C,D)
$N(x) * T * \sqrt{R_m * T} / ML$	SR 3	0.04689	(A,B,C,D)
$M(x) * \sqrt{R_m * T} / ML$	SR 3	0.03682	(A,B,C,D)
$N(y) * T / P$	SR 2	0.01325	(A,B,C,D)
$M(y) / P$	SR 2	0.00665	(A,B,C,D)
$N(y) * T * \sqrt{R_m * T} / MC$	SR 3	0.01411	(A,B,C,D)
$M(y) * \sqrt{R_m * T} / MC$	SR 3	0.01107	(A,B,C,D)
$N(y) * T * \sqrt{R_m * T} / ML$	SR 3	0.01411	(A,B,C,D)
$M(y) * \sqrt{R_m * T} / ML$	SR 3	0.01107	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)



Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Rad. Bend. P		3.1	-3.1	3.1	-3.1	3.1	-3.1	3.1	-3.1
Rad. Memb. MC		0.0	0.0	0.0	0.0	-4.5	-4.5	4.5	4.5
Rad. Bend. MC		0.0	0.0	0.0	0.0	-21.1	21.1	21.1	-21.1
Rad. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Rad. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Rad. Str.		4.1	-2.1	4.1	-2.1	-21.4	14.5	29.7	-18.7
Tang. Memb. P		0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
Tang. Bend. P		0.9	-0.9	0.9	-0.9	0.9	-0.9	0.9	-0.9
Tang. Memb. MC		0.0	0.0	0.0	0.0	-1.3	-1.3	1.3	1.3
Tang. Bend. MC		0.0	0.0	0.0	0.0	-6.3	6.3	6.3	-6.3
Tang. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tang. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.		1.2	-0.6	1.2	-0.6	-6.4	4.4	8.9	-5.6
Shear VC		0.0	0.0	-0.0	-0.0	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Shear MT		0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Tot. Shear		0.9	0.9	0.9	0.9	0.1	0.1	1.8	1.8
Str. Int.		4.4	2.5	4.4	2.5	21.4	14.5	29.8	18.9

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		67.5	67.5	67.5	67.5	67.5	-6.7	67.5	-6.7
Rad. Pl (SUS)		0.2	0.2	0.2	0.2	-1.4	-1.4	1.9	1.9
Rad. Q (SUS)		0.6	-0.6	0.6	-0.6	-4.0	4.0	5.2	-5.2
Long. Pm (SUS)		67.5	-6.7	67.5	-6.7	67.5	67.5	67.5	67.5
Long. Pl (SUS)		1.6	1.6	1.6	1.6	-3.4	-3.4	6.6	6.6
Long. Q (SUS)		2.6	-2.6	2.6	-2.6	-21.0	21.0	26.2	-26.2
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Shear Q (SUS)		1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7
Pm (SUS)		67.5	74.2	67.5	74.2	67.5	74.2	67.5	74.2
Pm+Pl (SUS)		69.1	72.9	69.1	72.9	66.3	72.2	74.2	78.9
Pm+Pl+Q (Total)		72.4	75.0	72.4	75.0	62.1	89.2	100.6	58.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	74.23	137.90	Passed
Pm+Pl (SUS)	78.95	206.85	Passed
Pm+Pl+Q (TOTAL)	100.56	413.70	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	72.3	72.3	72.3	72.3	72.3	72.3	72.3	72.3
Rad.	Pl (SUS)	1.0	1.0	1.0	1.0	-3.4	-3.4	5.5	5.5
Rad.	Q (SUS)	3.1	-3.1	3.1	-3.1	-18.0	18.0	24.2	-24.2
Long.	Pm (SUS)	72.3	72.3	72.3	72.3	72.3	72.3	72.3	72.3
Long.	Pl (SUS)	0.3	0.3	0.3	0.3	-1.0	-1.0	1.7	1.7
Long.	Q (SUS)	0.9	-0.9	0.9	-0.9	-5.4	5.4	7.3	-7.3
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Shear	Q (SUS)	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Pm (SUS)		72.3	72.3	72.3	72.3	72.3	72.3	72.3	72.3
Pm+Pl (SUS)		73.3	73.3	73.3	73.3	71.5	71.5	77.9	77.9
Pm+Pl+Q (Total)		76.7	72.1	76.7	72.1	65.8	86.8	102.1	66.9

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	72.27	137.90	Passed
Pm+Pl (SUS)	77.95	206.85	Passed
Pm+Pl+Q (TOTAL)	102.11	413.70	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: B1 4 "ASSUME #30

From: 30

Pressure for Reinforcement Calculations	P	2.1816	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Elliptical Head	D	1800.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	27.5000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	200.0000	mm
-------------------------------	----	----------	----

User Entered Minimum Design Metal Temperature		-25.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		4.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		120	

Flange Material		SA-105 +IT	
Flange Type		Weld Neck Flange	

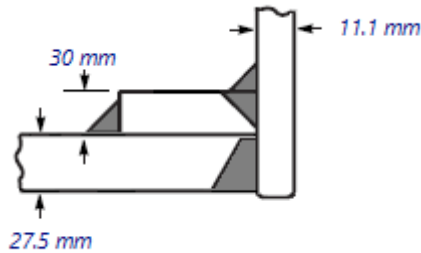
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	240.0000	mm
Thickness of Pad	te	30.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	30.0000	mm
Reinforcing Pad Width		62.8500	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: B1 4 "ASSUME #30

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500 in.
Actual Thickness Used in Calculation	0.438 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 2.18 \cdot 1806.0 \cdot 0.8972 / (2 \cdot 137.9 \cdot 1.0 - 0.2 \cdot 2.18)$
 $= 12.8368 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2.18 \cdot 57.15 / (118 \cdot 1.0 + 0.4 \cdot 2.18)$
 $= 1.0489 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3814 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	196.0992 mm
Parallel to Vessel Wall, opening length	d	98.0496 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		50.3130 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:

$= \min(1, Sn/Sv)$
 $= \min(1, 118.0/137.9)$
 $= 0.856$

Weld Strength Reduction Factor [fr2]:

$= \min(1, Sn/Sv)$
 $= \min(1, 118.0/137.9)$
 $= 0.856$

Weld Strength Reduction Factor [fr4]:

$= \min(1, Sp/Sv)$
 $= \min(1, 138.0/137.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.856, 1.0)$
 $= 0.856$

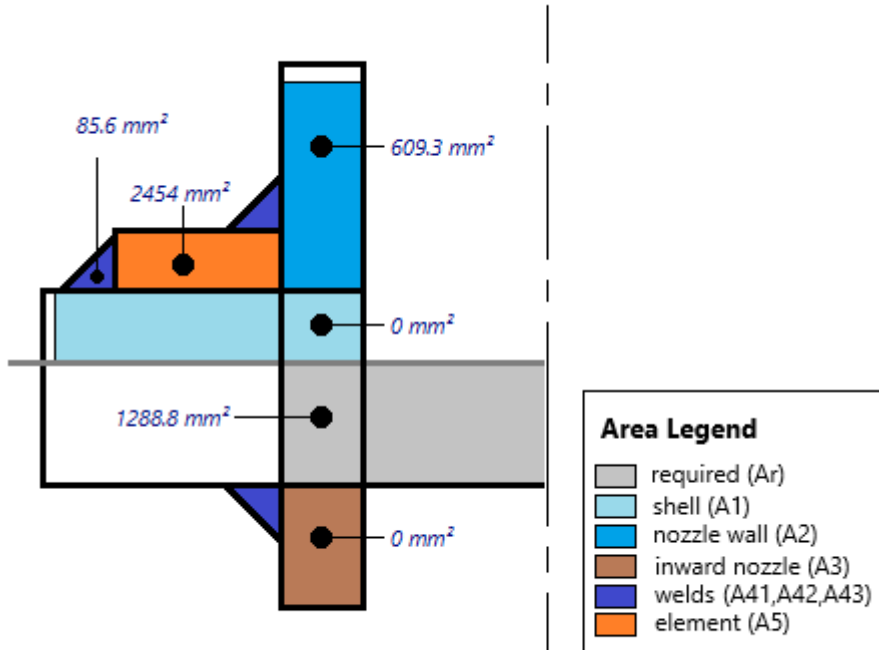


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1288.750	168.165	NA
Area in Shell	A1	0.000	0.000	NA
Area in Nozzle Wall	A2	609.308	666.782	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	85.569	85.569	NA
Area in Element	A5	2453.976	2453.976	NA
TOTAL AREA AVAILABLE	Atot	3148.853	3206.327	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 134.0958 30.0000 mm
 Based on given Pad Diameter: 240.0000 7.2601 mm
 Based on Shell or Nozzle Thickness: 167.6809 11.1252 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (98.0496 \cdot 12.8368 \cdot 1.0 + 2 \cdot 8.1252 \cdot 12.8368 \cdot 1.0 \cdot (1 - 0.86))$
 $= 1288.750 \text{ mm}^2$

The area in the shell has not been calculated by user request.

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 50.31) \cdot (8.13 - 1.05) \cdot 0.8557$
 $= 609.308 \text{ mm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi - can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$
 $= 85.569 \text{ mm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot f_{r4}$
 $= (196.0992 - 114.3) \cdot 30.0 \cdot 1.0$
 $= 2453.976 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 4.0489 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 17.2453 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr_{16b}) = 17.2453 \text{ mm}$
Wall Thickness, shell/head, external pressure	$trb2 = 3.3197 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 8.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.258, \max(17.2453, 4.5)]$
 $= 8.2578 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(4.0489, 8.2578)$
 $= 8.2578 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 11.125 = 9.735 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	32.8,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	262.2 MPa	Passed
Occasional	:	6.1,	Allowable	:	156.9 MPa	Passed
Shear	:	21.9,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, $t_g = 27.5$, $t_r = 12.837$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.524$, Temp. Reduction = $30 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve D	-32 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-48 $^\circ\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: D

Govrn. thk, $t_g = 9.735$, $t_r = 1.049$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.156$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-104 $^\circ\text{C}$



Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -18 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -62 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = 2.18/5.10 = 0.428

Weld Size Calculations, Description: B1 4 "ASSUME #3

Intermediate Calc. for nozzle/shell Welds Tmin 8.1252 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.6876 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5 * TminPad	11.3120 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (1288.7504 - 0.0 + 2 \cdot 8.1252 \cdot 0.8557 \cdot (1.0 \cdot 24.5 - 12.8368)) \cdot 138)$
 $= 200066.31 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (609.3083 + 2453.9758 + 85.5692 - 0.0 \cdot 0.86) \cdot 138$
 $= 434189.94 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (609.3083 + 0.0 + 85.5692 + (340.681)) \cdot 138$
 $= 142791.38 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (609.3083 + 0.0 + 85.5692 + 2453.9758 + (340.681)) \cdot 138$
 $= 481165.88 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 103802. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 240.0 \cdot 16.0 \cdot 0.49 \cdot 138$
 $= 407544. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 53.0874) \cdot (11.1252 - 3.0) \cdot 0.7 \cdot 118$
 $= 111923. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 114.3 \cdot 30.0 \cdot 0.74 \cdot 138$
 $= 549998. \text{ N}$



Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 114.3 \cdot (24.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 439680. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SPEW} + \text{SNW}) = (407544 + 111923) = 519467 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (103802 + 549998 + 439680 + 0) = 1093480 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Spew} + \text{Tngw} + \text{Sinw})$
 $= (407544 + 439680 + 0) = 847224 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 519466 N , must exceed W = 200066 N or W1 = 434189 N
 Path 2-2 = 1093480 N , must exceed W = 200066 N or W2 = 142791 N
 Path 3-3 = 847223 N , must exceed W = 200066 N or W3 = 481165 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 0.9949 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 278.4950 mm

Input Echo, WRC107/537 Item 1, Description: B1 4 "ASSUME #:

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	3240.000	mm
Vessel Thickness	Tv	27.500	mm
Design Temperature	T1	125.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	137.90	MPa
Vessel Hot S.I. Allowable	Smh	137.90	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	114.300	mm
Nozzle Thickness	Tn	11.125	mm
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa
Thickness of Reinforcing Pad	Tpad	30.000	mm
Diameter of Reinforcing Pad	Dpad	240.000	mm
Design Internal Pressure	Dp	2.182	MPa
Include Pressure Thrust		No	

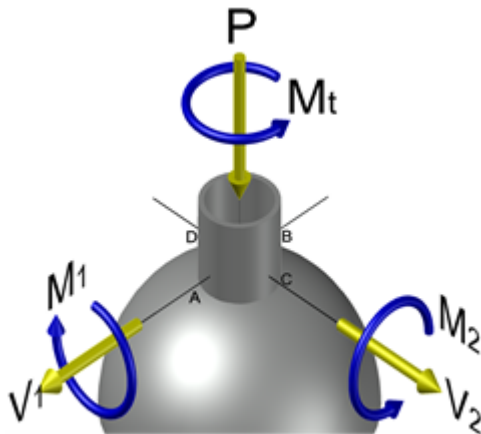
External Forces and Moments in WRC 107/537 Convention:

Radial Load	(SUS)	P	-7500.0	N
Longitudinal Shear	(SUS)	(Vl) V1	5000.0	N
Circumferential Shear	(SUS)	(Vc) V2	5000.0	N
Circumferential Moment	(SUS)	(Mc) M1	1400000.0	N-mm
Longitudinal Moment	(SUS)	(Ml) M2	1400000.0	N-mm
Torsional Moment	(SUS)	Mt	2000000.0	N-mm

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(1635.25 (27.5 - 3.0))}$$

$$= 774.825 \text{ mm}$$

WRC 107 Stress Calculation for SUStained loads:

Radial Load		P	-7500.0	N
Circumferential Shear	(VC)	V2	5000.0	N
Longitudinal Shear	(VL)	V1	5000.0	N
Circumferential Moment	(MC)	M1	1400000.0	N-mm
Longitudinal Moment	(ML)	M2	1400000.0	N-mm
Torsional Moment		MT	2000000.0	N-mm

Dimensionless Param: U = 0.19 TAU = 6.53 RHO = 4.00 (6.71)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 4	0.08776	(A,B,C,D)
$M(x) / P$	SP 4	0.03221	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.30775	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 4	0.12840	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.30775	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 4	0.12840	(A,B,C,D)
$N(y) * T / P$	SP 4	0.34539	(A,B,C,D)
$M(y) / P$	SP 4	0.20367	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.11063	(A,B,C,D)



M(y) * SQRT(Rm * T) / MC SM 4 0.92470 (A,B,C,D)
 N(y) * T * SQRT(Rm * T) / ML SM 4 0.11063 (A,B,C,D)
 M(y) * SQRT(Rm * T) / ML SM 4 0.92470 (A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Rad. Bend. P		0.5	-0.5	0.5	-0.5	0.5	-0.5	0.5	-0.5
Rad. Memb. MC		0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Rad. Bend. MC		0.0	0.0	0.0	0.0	-1.2	1.2	1.2	-1.2
Rad. Memb. ML		-0.5	-0.5	0.5	0.5	0.0	0.0	0.0	0.0
Rad. Bend. ML		-1.2	1.2	1.2	-1.2	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-1.0	0.5	2.4	-1.0	-1.0	0.5	2.4	-1.0
Tang. Memb. P		0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Tang. Bend. P		3.1	-3.1	3.1	-3.1	3.1	-3.1	3.1	-3.1
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Tang. Bend. MC		0.0	0.0	0.0	0.0	-8.7	8.7	8.7	-8.7
Tang. Memb. ML		-0.2	-0.2	0.2	0.2	0.0	0.0	0.0	0.0
Tang. Bend. ML		-8.7	8.7	8.7	-8.7	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-4.9	6.3	12.8	-10.8	-4.9	6.3	12.8	-10.8
Shear VC		0.5	0.5	-0.5	-0.5	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Shear MT		1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8
Tot. Shear		2.3	2.3	1.3	1.3	1.3	1.3	2.3	2.3
Str. Int.		6.1	7.5	13.0	10.9	5.3	6.6	13.3	11.3

Unitless Prm: U = 0.60 TAU = 0.00 (14.27) RHO = 0.00 (3.02)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.11507	(A,B,C,D)
M(x) / P	SR 2	0.06744	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.12850	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.18622	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.12850	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.18622	(A,B,C,D)
N(y) * T / P	SR 2	0.03455	(A,B,C,D)
M(y) / P	SR 2	0.02058	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.03886	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.05543	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.03886	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.05543	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		1.4	1.4	1.4	1.4	1.4	1.4	1.4	1.4
Rad. Bend. P		5.1	-5.1	5.1	-5.1	5.1	-5.1	5.1	-5.1
Rad. Memb. MC		0.0	0.0	0.0	0.0	-1.5	-1.5	1.5	1.5
Rad. Bend. MC		0.0	0.0	0.0	0.0	-13.0	13.0	13.0	-13.0



Rad. Memb. ML	-1.5	-1.5	1.5	1.5	0.0	0.0	0.0	0.0
Rad. Bend. ML	-13.0	13.0	13.0	-13.0	0.0	0.0	0.0	0.0
Tot. Rad. Str.	-8.0	7.9	21.0	-15.1	-8.0	7.9	21.0	-15.1
Tang. Memb. P	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
Tang. Bend. P	1.5	-1.5	1.5	-1.5	1.5	-1.5	1.5	-1.5
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Tang. Bend. MC	0.0	0.0	0.0	0.0	-3.9	3.9	3.9	-3.9
Tang. Memb. ML	-0.5	-0.5	0.5	0.5	0.0	0.0	0.0	0.0
Tang. Bend. ML	-3.9	3.9	3.9	-3.9	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-2.4	2.3	6.3	-4.5	-2.4	2.3	6.3	-4.5
Shear VC	0.5	0.5	-0.5	-0.5	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Shear MT	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Tot. Shear	1.4	1.4	0.4	0.4	0.4	0.4	1.4	1.4
Str. Int.	8.4	8.3	21.0	15.1	8.0	7.9	21.1	15.3

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		32.5	32.5	32.5	32.5	32.5	32.5	32.5	32.5
Rad. Pl (SUS)		-0.3	-0.3	0.7	0.7	-0.3	-0.3	0.7	0.7
Rad. Q (SUS)		-0.7	0.7	1.7	-1.7	-0.7	0.7	1.7	-1.7
Long. Pm (SUS)		32.5	32.5	32.5	32.5	32.5	32.5	32.5	32.5
Long. Pl (SUS)		0.7	0.7	1.0	1.0	0.7	0.7	1.0	1.0
Long. Q (SUS)		-5.6	5.6	11.8	-11.8	-5.6	5.6	11.8	-11.8
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.5	0.5	-0.5	-0.5	-0.5	-0.5	0.5	0.5
Shear Q (SUS)		1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8
Pm (SUS)		32.5	32.5	32.5	32.5	32.5	32.5	32.5	32.5
Pm+Pl (SUS)		33.4	33.4	33.9	33.9	33.4	33.4	33.9	33.9
Pm+Pl+Q (Total)		32.6	39.6	45.5	31.7	31.9	39.1	45.8	32.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	32.51	137.90	Passed
Pm+Pl (SUS)	33.92	206.85	Passed
Pm+Pl+Q (TOTAL)	45.84	413.70	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Rad. Pm (SUS)	72.3	72.3	72.3	72.3	72.3	72.3	72.3	72.3
Rad. Pl (SUS)	-0.1	-0.1	2.9	2.9	-0.1	-0.1	2.9	2.9
Rad. Q (SUS)	-8.0	8.0	18.1	-18.1	-8.0	8.0	18.1	-18.1
Long. Pm (SUS)	72.3	72.3	72.3	72.3	72.3	72.3	72.3	72.3
Long. Pl (SUS)	-0.0	-0.0	0.9	0.9	-0.0	-0.0	0.9	0.9
Long. Q (SUS)	-2.3	2.3	5.4	-5.4	-2.3	2.3	5.4	-5.4
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)	0.5	0.5	-0.5	-0.5	-0.5	-0.5	0.5	0.5
Shear Q (SUS)	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Pm (SUS)	72.3	72.3	72.3	72.3	72.3	72.3	72.3	72.3
Pm+Pl (SUS)	72.8	72.8	75.3	75.3	72.8	72.8	75.3	75.3
Pm+Pl+Q (Total)	70.3	80.5	93.3	67.8	69.9	80.2	93.4	67.9

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	72.27	137.90	Passed
Pm+Pl (SUS)	75.34	206.85	Passed
Pm+Pl+Q (TOTAL)	93.42	413.70	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: Noz L1B #300

From: 40

Pressure for Reinforcement Calculations	P	2.1776	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Cylindrical Shell	D	1800.00	mm
Design Length of Section	L	38300.0039	mm
Shell Finished (Minimum) Thickness	t	30.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		350.00	mm
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User Entered Minimum Design Metal Temperature		-25.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

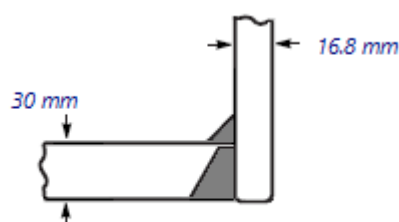
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: Noz L1B #300

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.18 \cdot 903.0 / (138 \cdot 1.0 - 0.6 \cdot 2.18)$
 $= 14.3958 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.6313 and the shell thk. is 30.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.18 \cdot 28.4 / (118 \cdot 1.0 - 0.6 \cdot 2.18)$
 $= 0.5300 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3189 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	138.4000	mm
Parallel to Vessel Wall	Rn+tn+t	69.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz L1B #300.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.5300	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 17.3958	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 17.3958	mm
Wall Thickness, shell/head, external pressure	trb2	= 3.3211	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(17.3958, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.53, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16.8, tr = 0.53, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.038$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

Weld Size Calculations, Description: Noz L1B #300

Intermediate Calc. for nozzle/shell Welds Tmin 13.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

[Skipping the nozzle attachment weld strength calculations. Per UW-15\(b\)\(2\) the nozzles exempted by UG-36\(c\)\(3\)\(a\) \(small nozzles\) do not require a weld strength check.](#)

The Drop for this Nozzle is : 0.9899 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 280.9899 mm

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Input, Nozzle Desc: Noz L2B #300

From: 40

Pressure for Reinforcement Calculations	P	2.1776	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Cylindrical Shell	D	1800.00	mm
Design Length of Section	L	38300.0039	mm
Shell Finished (Minimum) Thickness	t	30.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		350.00	mm
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User Entered Minimum Design Metal Temperature		-25.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

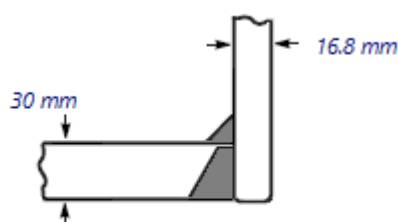
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: Noz L2B #300

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.18 \cdot 903.0 / (138 \cdot 1.0 - 0.6 \cdot 2.18)$
 $= 14.3958 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.6313 and the shell thk. is 30.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.18 \cdot 28.4 / (118 \cdot 1.0 - 0.6 \cdot 2.18)$
 $= 0.5300 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3189 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	138.4000	mm
Parallel to Vessel Wall	Rn+tn+t	69.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz L2B #300.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.5300	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 17.3958	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 17.3958	mm
Wall Thickness, shell/head, external pressure	trb2	= 3.3211	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(17.3958, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.53, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B



Govrn. thk, tg = 16.8, tr = 0.53, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E^* / (tg - c) = 0.038$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -13 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

Weld Size Calculations, Description: Noz L2B #300

Intermediate Calc. for nozzle/shell Welds Tmin 13.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.9899 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 280.9899 mm

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Input, Nozzle Desc: Noz L3B #300

From: 40

Pressure for Reinforcement Calculations	P	2.1776	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Cylindrical Shell	D	1800.00	mm
Design Length of Section	L	38300.0039	mm
Shell Finished (Minimum) Thickness	t	30.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		350.00	mm
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User Entered Minimum Design Metal Temperature		-25.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

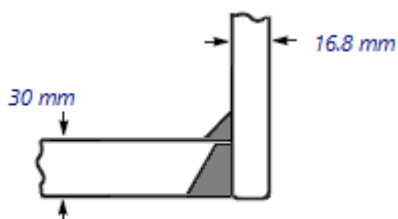
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: Noz L3B #300

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.18 \cdot 903.0 / (138 \cdot 1.0 - 0.6 \cdot 2.18)$
 $= 14.3958 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.6313 and the shell thk. is 30.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.18 \cdot 28.4 / (118 \cdot 1.0 - 0.6 \cdot 2.18)$
 $= 0.5300 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3189 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	138.4000	mm
Parallel to Vessel Wall	Rn+tn+t	69.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz L3B #300.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.5300	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 17.3958	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 17.3958	mm
Wall Thickness, shell/head, external pressure	trb2	= 3.3211	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(17.3958, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.53, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B



Govrn. thk, tg = 16.8, tr = 0.53, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E* / (tg - c) = 0.038$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -13 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

Weld Size Calculations, Description: Noz L3B #300

Intermediate Calc. for nozzle/shell Welds Tmin 13.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.9899 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 280.9899 mm

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Input, Nozzle Desc: Noz L4B #300

From: 40

Pressure for Reinforcement Calculations	P	2.1776	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Cylindrical Shell	D	1800.00	mm
Design Length of Section	L	38300.0039	mm
Shell Finished (Minimum) Thickness	t	30.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		350.00	mm
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User Entered Minimum Design Metal Temperature		-25.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

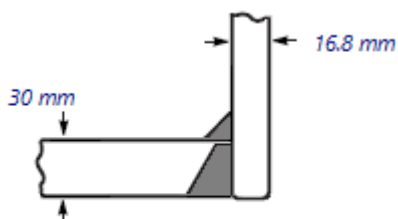
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: Noz L4B #300

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.18 \cdot 903.0 / (138 \cdot 1.0 - 0.6 \cdot 2.18)$
 $= 14.3958 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.6313 and the shell thk. is 30.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.18 \cdot 28.4 / (118 \cdot 1.0 - 0.6 \cdot 2.18)$
 $= 0.5300 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3189 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	138.4000	mm
Parallel to Vessel Wall	Rn+tn+t	69.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz L4B #300.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.5300	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 17.3958	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 17.3958	mm
Wall Thickness, shell/head, external pressure	trb2	= 3.3211	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(17.3958, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.53, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16.8, tr = 0.53, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.038$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

Weld Size Calculations, Description: Noz L4B #300

Intermediate Calc. for nozzle/shell Welds Tmin 13.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.9899 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 280.9899 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 69.20 mm

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Input, Nozzle Desc: Noz U2 #300

From: 40

Pressure for Reinforcement Calculations	P	2.1786	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Cylindrical Shell	D	1800.00	mm
Design Length of Section	L	38300.0039	mm
Shell Finished (Minimum) Thickness	t	30.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		150.00	mm
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User Entered Minimum Design Metal Temperature		-25.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

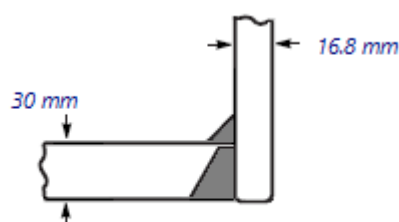
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: Noz U2 #300

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.18 \cdot 903.0 / (138 \cdot 1.0 - 0.6 \cdot 2.18)$
 $= 14.4026 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.6313 and the shell thk. is 30.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.18 \cdot 28.4 / (118 \cdot 1.0 - 0.6 \cdot 2.18)$
 $= 0.5302 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3189 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	138.4000	mm
Parallel to Vessel Wall	Rn+tn+t	69.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz U2 #300.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.5302	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 17.4026	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 17.4026	mm
Wall Thickness, shell/head, external pressure	trb2	= 3.3211	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(17.4026, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.5302, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16.8, tr = 0.53, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.038$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

Weld Size Calculations, Description: Noz U2 #300

Intermediate Calc. for nozzle/shell Welds Tmin 13.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.9899 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 280.9899 mm

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Input, Nozzle Desc: Noz U1 #300

From: 40

Pressure for Reinforcement Calculations	P	2.1786	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Cylindrical Shell	D	1800.00	mm
Design Length of Section	L	38300.0039	mm
Shell Finished (Minimum) Thickness	t	30.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		150.00	mm
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User Entered Minimum Design Metal Temperature		-25.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

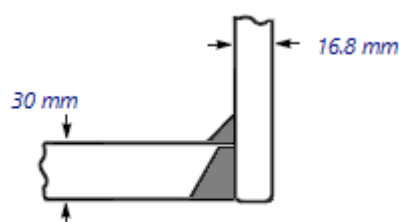
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: Noz U1 #300

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.18 \cdot 903.0 / (138 \cdot 1.0 - 0.6 \cdot 2.18)$
 $= 14.4026 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.6313 and the shell thk. is 30.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.18 \cdot 28.4 / (118 \cdot 1.0 - 0.6 \cdot 2.18)$
 $= 0.5302 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3189 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	138.4000	mm
Parallel to Vessel Wall	Rn+tn+t	69.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz U1 #300.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.5302	mm
Wall Thickness per UG16(b),	tr16b = 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1 = 17.4026	mm
Wall Thickness	tb1 = max(trb1, tr16b) = 17.4026	mm
Wall Thickness, shell/head, external pressure	trb2 = 3.3211	mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000	mm
Wall Thickness per table UG-45	tb3 = 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(17.4026, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.5302, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16.8, tr = 0.53, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.038$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

Weld Size Calculations, Description: Noz U1 #300

Intermediate Calc. for nozzle/shell Welds Tmin 13.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.9899 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 280.9899 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 69.20 mm

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Input, Nozzle Desc: Noz MW5 24" #300

From: 40

Pressure for Reinforcement Calculations	P	2.1712	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Cylindrical Shell	D	1800.00	mm
Design Length of Section	L	38300.0039	mm
Shell Finished (Minimum) Thickness	t	30.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1600.00	mm
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User Entered Minimum Design Metal Temperature		-25.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	137.90	MPa
Allowable Stress At Ambient	Sna	137.90	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105 +IT	
Flange Type		Weld Neck Flange	

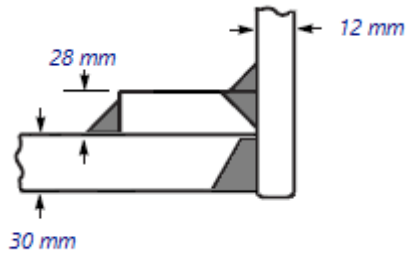
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	28.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	137.90	MPa
Pad Allowable Stress At Ambient	Spa	137.90	MPa
Diameter of Pad along vessel surface	Dp	1220.0000	mm
Thickness of Pad	te	28.0000	mm
Weld leg size between Pad and Shell	Wp	15.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm
Reinforcing Pad Width		305.2000	mm
This is a Manway, Coupling or Access Opening.			

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: Noz MW5 24" #300

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.472 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.17 \cdot 903.0 / (138 \cdot 1.0 - 0.6 \cdot 2.17)$
 $= 14.3533 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.6313 and the shell thk. is 30.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2.17 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 2.17)$
 $= 4.7690 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0955 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1183.2001 mm
Parallel to Vessel Wall, opening length	d	591.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		50.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 137.9 / 137.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 137.9 / 137.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 137.9 / 137.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.)$
 $= 1.000$

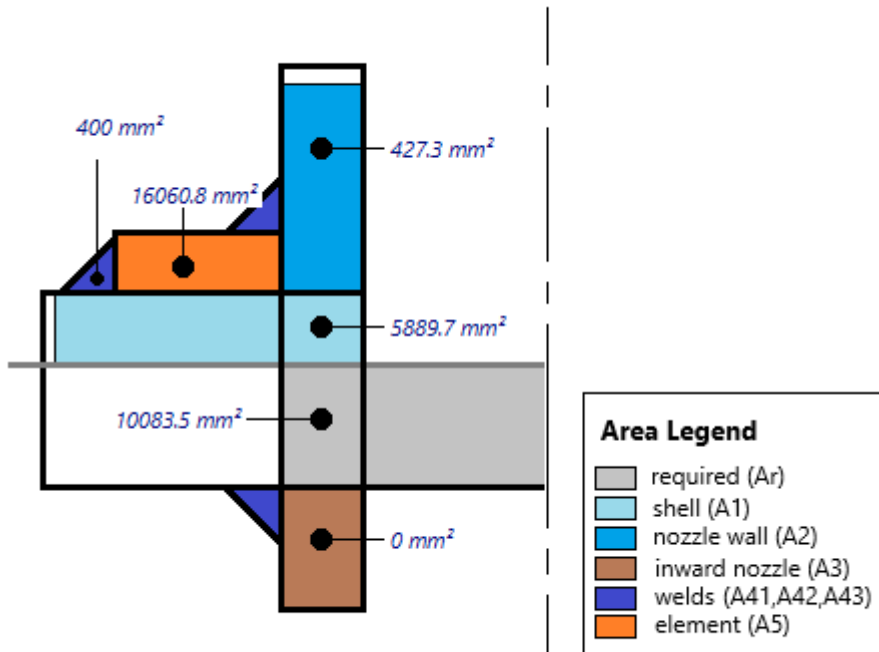


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	10083.454	3819.173	NA
Area in Shell	A1	5889.746	8334.853	NA
Area in Nozzle Wall	A2	427.328	798.357	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		400.000	400.000	NA
Area in Element	A5	16060.800	16060.800	NA
TOTAL AREA AVAILABLE	Atot	22777.875	25594.010	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 729.8278 28.0000 mm
 Based on given Pad Diameter: 1220.0000 5.8689 mm
 Based on Shell or Nozzle Thickness: 890.1316 12.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (591.6 \cdot 17.0444 \cdot 1.0 + 2 \cdot 9.0 \cdot 17.0444 \cdot 1.0 \cdot (1 - 1.))$
 $= 10083.454 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 591.6(1.0 \cdot 27.0 - 1.0 \cdot 17.044) - 2 \cdot 9.0$
 $(1.0 \cdot 27.0 - 1.0 \cdot 17.044) \cdot (1 - 1.)$
 $= 5889.746 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 50.5) \cdot (9.0 - 4.77) \cdot 1.$



= 427.328 mm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 = $20.0^2 \cdot 1. + (0.0)^2 \cdot 1. + 0.0^2 \cdot 1.$
 = 400.000 mm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 = $(1183.2001 - 609.6) \cdot 28.0 \cdot 1.$
 = 16060.800 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 7.7690$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = t_a
 = 7.7690 mm

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: D

Govrn. thk, $t_g = 12.0$, $t_r = 4.769$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.53$, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

Govrn. thk, $t_g = 12.0$, $t_r = 4.769$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.53$, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, $t_g = 12.0$, $t_r = 4.769$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.53$, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:D, Curve:B)

Govrn. thk, $t_g = 28.0$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.532$, Temp. Reduction = 29 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -27 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:D, Curve:B)

Govrn. thk, $t_g = 12.0$, $t_r = 4.769$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.53$, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -48 °C
 Gov. MDMT of the Reinforcement Pad : -27 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -27 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -18 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -48 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $2.17/5.10 = 0.426$

Weld Size Calculations, Description: Noz MW5 24" #30

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = $0.7 * W_o$ mm
Pad Weld	9.5000 = $0.5 * T_{minPad}$	10.6050 = $0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (10083.4541 - 5889.7456 + 2 \cdot 9.0 \cdot 1.0 \cdot (1.0 \cdot 27.0 - 17.0444)) \cdot 138)$
 $= 602972.94 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - C_{an} / 7.07)^2 \cdot f_{r2}) \cdot S_v$
 $= (427.3282 + 16060.7998 + 400.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 2328675.00 \text{ N}$

Weld Load [W2]:
 $= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (427.3282 + 0.0 + 400.0 + (486.0)) \cdot 138$
 $= 181092.55 \text{ N}$

Weld Load [W3]:
 $= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (427.3282 + 0.0 + 400.0 + 16060.7998 + (486.0)) \cdot 138$
 $= 2395688.50 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [S_{onw}]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 609.6 \cdot 20.0 \cdot 0.49 \cdot 138$
 $= 1293952. \text{ N}$

Shear, Pad Element Weld [S_{pew}]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 1220.0 \cdot 15.0 \cdot 0.49 \cdot 138$
 $= 1942202. \text{ N}$

Shear, Nozzle Wall [S_{nw}]:
 $= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - C_{an}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 300.3) \cdot (12.0 - 3.0) \cdot 0.7 \cdot 138$
 $= 819546. \text{ N}$

Tension, Pad Groove Weld [T_{pgw}]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 609.6 \cdot 28.0 \cdot 0.74 \cdot 138$
 $= 2735785. \text{ N}$

Tension, Shell Groove Weld [T_{ngw}]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 609.6 \cdot (30.0 - 3.0) \cdot 0.74 \cdot 138$
 $= 2638078. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:



Failure Path [Path 1-1]:
= (SPEW + SNW) = (1942202 + 819546) = 2761748 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (1293952 + 2735785 + 2638078 + 0) = 6667815 N

Failure Path [Path 3-3]:
= (Spew + Tngw + Sinw)
= (1942202 + 2638078 + 0) = 4580280 N

Summary of Failure Path Calculations:

Path 1-1 = 2761747 N	, must exceed W = 602972 N	or W1 = 2328675 N
Path 2-2 = 6667814 N	, must exceed W = 602972 N	or W2 = 181092 N
Path 3-3 = 4580280 N	, must exceed W = 602972 N	or W3 = 2395688 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 53.1843 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 383.1843 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 2.008 %

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 220.64 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 591.60 mm

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Input, Nozzle Desc: Noz L1A #300

From: 50

Pressure for Reinforcement Calculations	P	2.1575	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Cylindrical Shell	D	1800.00	mm
Design Length of Section	L	38300.0039	mm
Shell Finished (Minimum) Thickness	t	30.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		4350.00	mm
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User Entered Minimum Design Metal Temperature		-25.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

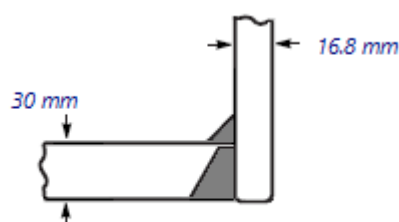
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: Noz L1A #300

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.16 \cdot 903.0 / (138 \cdot 1.0 - 0.6 \cdot 2.16)$
 $= 14.2614 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.5565 and the shell thk. is 30.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.16 \cdot 28.4 / (118 \cdot 1.0 - 0.6 \cdot 2.16)$
 $= 0.5250 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3189 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	138.4000	mm
Parallel to Vessel Wall	Rn+tn+t	69.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz L1A #300.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.5250	mm
Wall Thickness per UG16(b),	tr16b = 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1 = 17.2614	mm
Wall Thickness	tb1 = max(trb1, tr16b) = 17.2614	mm
Wall Thickness, shell/head, external pressure	trb2 = 3.3211	mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000	mm
Wall Thickness per table UG-45	tb3 = 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(17.2614, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.525, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16.8, tr = 0.525, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.038$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

Weld Size Calculations, Description: Noz L1A #300

Intermediate Calc. for nozzle/shell Welds Tmin 13.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.9899 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 280.9899 mm

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Input, Nozzle Desc: Noz L2A #300

From: 50

Pressure for Reinforcement Calculations	P	2.1575	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Cylindrical Shell	D	1800.00	mm
Design Length of Section	L	38300.0039	mm
Shell Finished (Minimum) Thickness	t	30.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		4350.00	mm
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User Entered Minimum Design Metal Temperature		-25.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

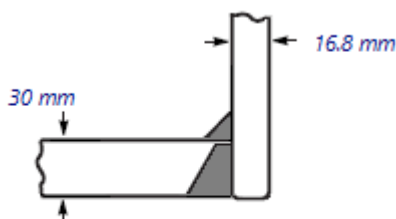
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: Noz L2A #300

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.16 \cdot 903.0 / (138 \cdot 1.0 - 0.6 \cdot 2.16)$
 $= 14.2614 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.5565 and the shell thk. is 30.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.16 \cdot 28.4 / (118 \cdot 1.0 - 0.6 \cdot 2.16)$
 $= 0.5250 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3189 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	138.4000	mm
Parallel to Vessel Wall	Rn+tn+t	69.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz L2A #300.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.5250	mm
Wall Thickness per UG16(b),	tr16b = 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1 = 17.2614	mm
Wall Thickness	tb1 = max(trb1, tr16b) = 17.2614	mm
Wall Thickness, shell/head, external pressure	trb2 = 3.3211	mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000	mm
Wall Thickness per table UG-45	tb3 = 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(17.2614, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.525, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16.8, tr = 0.525, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.038$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

Weld Size Calculations, Description: Noz L2A #300

Intermediate Calc. for nozzle/shell Welds Tmin 13.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.9899 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 280.9899 mm

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Input, Nozzle Desc: Noz L3A #300

From: 50

Pressure for Reinforcement Calculations	P	2.1575	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Cylindrical Shell	D	1800.00	mm
Design Length of Section	L	38300.0039	mm
Shell Finished (Minimum) Thickness	t	30.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		4350.00	mm
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User Entered Minimum Design Metal Temperature		-25.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

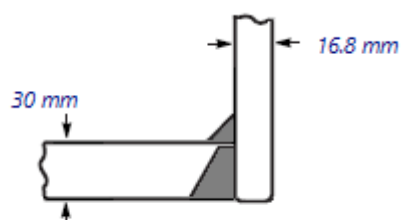
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

This is a Manway, Coupling or Access Opening.

Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.





ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.16 \cdot 903.0 / (138 \cdot 1.0 - 0.6 \cdot 2.16)$
 $= 14.2614 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.5565 and the shell thk. is 30.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.16 \cdot 28.4 / (118 \cdot 1.0 - 0.6 \cdot 2.16)$
 $= 0.5250 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3189 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	138.4000	mm
Parallel to Vessel Wall	Rn+tn+t	69.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz L3A #300.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 3.5250 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= t_a$
 $= 3.5250 \text{ mm}$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govern. thk, $t_g = 16.8$, $t_r = 0.525$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.038$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

Weld Size Calculations, Description: Noz L3A #300

Intermediate Calc. for nozzle/shell welds T_{min} 13.8000 mm

Results Per UW-16.1:



	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.9899 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 280.9899 mm

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Input, Nozzle Desc: Noz L4A #300

From: 50

Pressure for Reinforcement Calculations	P	2.1575	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Cylindrical Shell	D	1800.00	mm
Design Length of Section	L	38300.0039	mm
Shell Finished (Minimum) Thickness	t	30.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		4350.00	mm
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User Entered Minimum Design Metal Temperature		-25.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

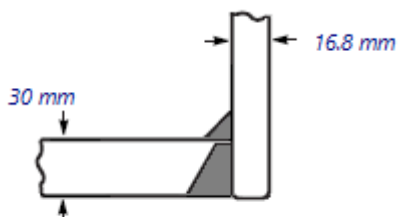
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

This is a Manway, Coupling or Access Opening.

Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.





ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.16 \cdot 903.0 / (138 \cdot 1.0 - 0.6 \cdot 2.16)$
 $= 14.2614 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.5565 and the shell thk. is 30.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.16 \cdot 28.4 / (118 \cdot 1.0 - 0.6 \cdot 2.16)$
 $= 0.5250 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3189 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	138.4000	mm
Parallel to Vessel Wall	Rn+tn+t	69.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz L4A #300.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 3.5250 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= t_a$
 $= 3.5250 \text{ mm}$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govern. thk, $t_g = 16.8$, $t_r = 0.525$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.038$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

Weld Size Calculations, Description: Noz L4A #300

Intermediate Calc. for nozzle/shell welds T_{min} 13.8000 mm

Results Per UW-16.1:



	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.9899 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 280.9899 mm

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Input, Nozzle Desc: MW4 #300
From: 60

Pressure for Reinforcement Calculations	P	2.1575	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Cylindrical Shell	D	1800.00	mm
Design Length of Section	L	38300.0039	mm
Shell Finished (Minimum) Thickness	t	30.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		7600.00	mm
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User Entered Minimum Design Metal Temperature		-25.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	137.90	MPa
Allowable Stress At Ambient	Sna	137.90	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105 +IT	
Flange Type		Weld Neck Flange	

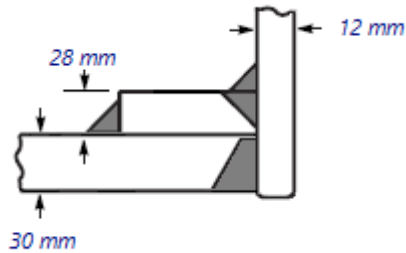
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1220.0000	mm
Thickness of Pad	te	28.0000	mm
Weld leg size between Pad and Shell	Wp	15.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm
Reinforcing Pad Width		305.2000	mm
This is a Manway, Coupling or Access Opening.			

Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: MW4 #300

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.472 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.16 \cdot 903.0 / (138 \cdot 1.0 - 0.6 \cdot 2.16)$
 $= 14.2614 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2.16 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 2.16)$
 $= 4.7390 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0955 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1183.2001 mm
Parallel to Vessel Wall, opening length	d	591.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		50.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 137.9 / 137.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 137.9 / 137.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 137.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

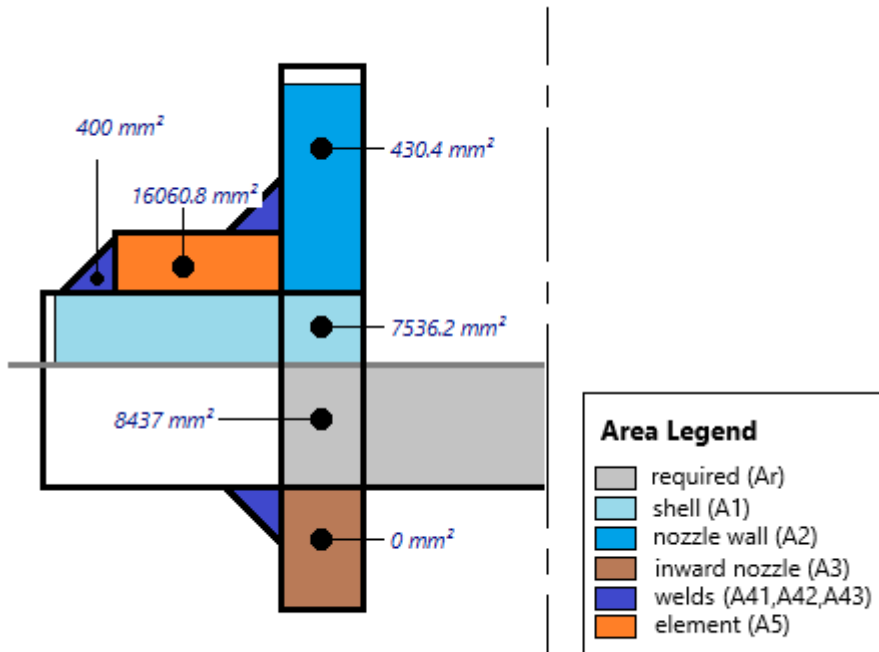


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	8437.046	3819.173	NA
Area in Shell	A1	7536.154	8334.853	NA
Area in Nozzle Wall	A2	430.364	798.357	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		400.000	400.000	NA
Area in Element	A5	16060.803	16060.803	NA
TOTAL AREA AVAILABLE	Atot	24427.320	25594.012	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 612.1189 28.0000 mm
 Based on given Pad Diameter: 1220.0000 0.1230 mm
 Based on Shell or Nozzle Thickness: 615.4774 12.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (591.6 \cdot 14.2614 \cdot 1.0 + 2 \cdot 9.0 \cdot 14.2614 \cdot 1.0 \cdot (1 - 1.0))$
 $= 8437.046 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 591.6(1.0 \cdot 27.0 - 1.0 \cdot 14.2614) - 2 \cdot 9.0$
 $(1.0 \cdot 27.0 - 1.0 \cdot 14.2614) \cdot (1 - 1.0)$
 $= 7536.154 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 50.5) \cdot (9.0 - 4.74) \cdot 1.0$

= 430.364 mm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $20.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 0.0^2 \cdot 1.0$
 = 400.000 mm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
 = $(1183.2001 - 609.6) \cdot 28.0 \cdot 1.0$
 = 16060.803 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 7.7390$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= t_a
 = 7.7390 mm

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 4.739$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.527$, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 4.739$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.527$, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 28.0$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.528$, Temp. Reduction = 29 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 4.739$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.527$, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -48 °C
Gov. MDMT of the Reinforcement Pad	: -27 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -27 °C

Flange MDMT including Temperature reduction per UCS-66.1:



MDMT of ASME B16.5/47 flange per Matl. Specification -18 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -48 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = 2.16/5.10 = 0.423

Weld Size Calculations, Description: MW4 #300

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	10.6050 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (8437.0459 - 7536.1543 + 2 \cdot 9.0 \cdot 1.0 \cdot (1.0 \cdot 27.0 - 14.2614)) \cdot 138)$
 $= 155839.48 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (430.3639 + 16060.8027 + 400.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 2329093.75 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (430.3639 + 0.0 + 400.0 + (486.0)) \cdot 138$
 $= 181511.16 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (430.3639 + 0.0 + 400.0 + 16060.8027 + (486.0)) \cdot 138$
 $= 2396107.50 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 609.6 \cdot 20.0 \cdot 0.49 \cdot 138$
 $= 1293952. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 1220.0 \cdot 15.0 \cdot 0.49 \cdot 138$
 $= 1942202. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 300.3) \cdot (12.0 - 3.0) \cdot 0.7 \cdot 138$
 $= 819546. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 609.6 \cdot 28.0 \cdot 0.74 \cdot 138$
 $= 2737769. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 609.6 \cdot (30.0 - 3.0) \cdot 0.74 \cdot 138$
 $= 2638078. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.



Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (1942202 + 819546) = 2761748 \text{ N}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (1293952 + 2737769 + 2638078 + 0) = 6669800 \text{ N}$

Failure Path [Path 3-3]:
 $= (Spew + Tngw + Sinw)$
 $= (1942202 + 2638078 + 0) = 4580280 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 2761747 N	, must exceed W = 155839 N	or W1 = 2329093 N
Path 2-2 = 6669799 N	, must exceed W = 155839 N	or W2 = 181511 N
Path 3-3 = 4580280 N	, must exceed W = 155839 N	or W3 = 2396107 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 53.1843 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 383.1843 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 2.008 %

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 591.60 mm

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Input, Nozzle Desc: P1 #300

From: 60

Pressure for Reinforcement Calculations	P	2.1575	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Cylindrical Shell	D	1800.00	mm
Design Length of Section	L	38300.0039	mm
Shell Finished (Minimum) Thickness	t	30.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		7850.00	mm
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User Entered Minimum Design Metal Temperature		-25.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

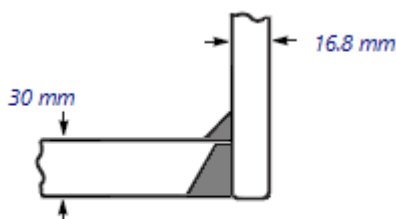
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	227.7750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		28.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: P1 #300

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.16 \cdot 903.0 / (138 \cdot 1.0 - 0.6 \cdot 2.16)$
 $= 14.2614 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.16 \cdot 28.4 / (118 \cdot 1.0 - 0.6 \cdot 2.16)$
 $= 0.5250 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3065 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	138.4000	mm
Parallel to Vessel Wall	Rn+tn+t	69.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: P1 #300.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.5250	mm
Wall Thickness per UG16(b),	tr16b = 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1 = 17.2614	mm
Wall Thickness	tb1 = max(trb1, tr16b) = 17.2614	mm
Wall Thickness, shell/head, external pressure	trb2 = 3.3211	mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000	mm
Wall Thickness per table UG-45	tb3 = 7.8000	mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(17.2614, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(3.525, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 16.8, tr = 0.525, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.038$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.



Weld Size Calculations, Description: P1 #300

Intermediate Calc. for nozzle/shell welds Tmin 13.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.9899 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 258.7649 mm

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Input, Nozzle Desc: P2 #300

From: 60

Pressure for Reinforcement Calculations	P	2.1575	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Cylindrical Shell	D	1800.00	mm
Design Length of Section	L	38300.0039	mm
Shell Finished (Minimum) Thickness	t	30.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		7650.00	mm
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User Entered Minimum Design Metal Temperature		-25.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

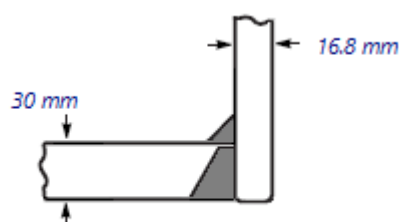
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	227.7750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		28.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: P2 #300

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.661 in.

Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.16 \cdot 903.0 / (138 \cdot 1.0 - 0.6 \cdot 2.16)$
 $= 14.2614 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.16 \cdot 28.4 / (118 \cdot 1.0 - 0.6 \cdot 2.16)$
 $= 0.5250 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3065 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	138.4000	mm
Parallel to Vessel Wall	Rn+tn+t	69.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: P2 #300.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.5250	mm
Wall Thickness per UG16(b),	tr16b = 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1 = 17.2614	mm
Wall Thickness	tb1 = max(trb1, tr16b) = 17.2614	mm
Wall Thickness, shell/head, external pressure	trb2 = 3.3211	mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000	mm
Wall Thickness per table UG-45	tb3 = 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(17.2614, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.525, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16.8, tr = 0.525, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.038$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.



Weld Size Calculations, Description: P2 #300

Intermediate Calc. for nozzle/shell welds Tmin 13.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.9899 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 258.7649 mm

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Input, Nozzle Desc: F3 #300

From: 60

Pressure for Reinforcement Calculations	P	2.1575	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Cylindrical Shell	D	1800.00	mm
Design Length of Section	L	38300.0039	mm
Shell Finished (Minimum) Thickness	t	30.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		7000.00	mm
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User Entered Minimum Design Metal Temperature		-25.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		10.0000	in.

Size and Thickness Basis		Minimum	
Nominal Thickness		80	

Flange Material		SA-105 +IT	
Flange Type		Weld Neck Flange	

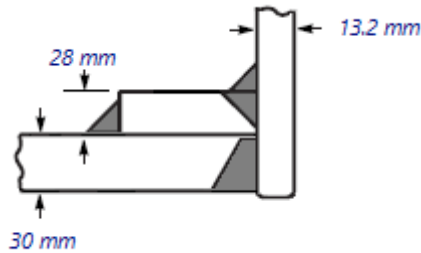
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	28.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	540.0000	mm
Thickness of Pad	te	28.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm
Reinforcing Pad Width		133.4750	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: F3 #300

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	10.750 in.
Actual Thickness Used in Calculation	0.520 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.16 \cdot 903.0 / (138 \cdot 1.0 - 0.6 \cdot 2.16)$
 $= 14.2614 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2.16 \cdot 136.525 / (118 \cdot 1.0 + 0.4 \cdot 2.16)$
 $= 2.4780 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.6332 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	505.2934 mm
Parallel to Vessel Wall, opening length	d	252.6467 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		53.5041 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 137.9)$
 $= 0.856$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 137.9)$
 $= 0.856$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 137.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.856, 1.0)$
 $= 0.856$

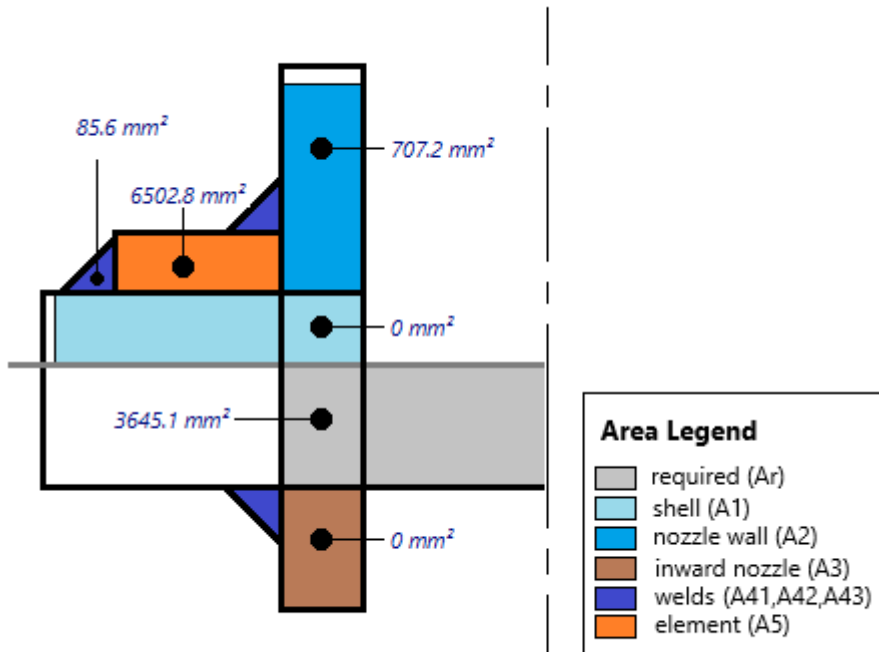


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	3645.086	1650.011	NA
Area in Shell	A1	0.000	0.000	NA
Area in Nozzle Wall	A2	707.221	876.150	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	85.569	85.569	NA
Area in Element	A5	6502.815	6502.815	NA
TOTAL AREA AVAILABLE	Atot	7295.605	7464.535	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 374.9177 28.0000 mm
 Based on given Pad Diameter: 540.0000 12.2815 mm
 Based on Shell or Nozzle Thickness: 489.1060 13.2016 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (252.6467 \cdot 14.2614 \cdot 1.0 + 2 \cdot 10.2016 \cdot 14.2614 \cdot 1.0 \cdot (1 - 0.86))$
 $= 3645.086 \text{ mm}^2$

The area in the shell has not been calculated by user request.

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 53.5) \cdot (10.2 - 2.48) \cdot 0.8557$
 $= 707.221 \text{ mm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi - can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$
 $= 85.569 \text{ mm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot f_{r4}$
 $= (505.2934 - 273.05) \cdot 28.0 \cdot 1.0$
 $= 6502.815 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 5.4780 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 17.2614 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr_{16b}) = 17.2614 \text{ mm}$
Wall Thickness, shell/head, external pressure	$trb2 = 3.3211 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 11.1026 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[11.103, \max(17.2614, 4.5)]$
 $= 11.1026 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(5.478, 11.1026)$
 $= 11.1026 \text{ mm}$

Available Nozzle Neck Thickness = 13.2016 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	32.7,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	262.3 MPa	Passed
Occasional	:	12.8,	Allowable	:	156.9 MPa	Passed
Shear	:	14.6,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 28.0$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.528$, Temp. Reduction = 29 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 13.202$, $tr = 2.478$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.243$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-20 °C
--	--------

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -27 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -27 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -18 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -63 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = 2.16/5.10 = 0.423

Weld Size Calculations, Description: F3 #300

Intermediate Calc. for nozzle/shell Welds Tmin 10.2016 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	9.8980 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (3645.0864 - 0.0 + 2 \cdot 10.2016 \cdot 0.8557 \cdot (1.0 \cdot 27.0 - 14.2614)) \cdot 138)$
 $= 533281.44 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (707.2212 + 6502.8149 + 85.5692 - 0.0 \cdot 0.86) \cdot 138$
 $= 1005978.44 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (707.2212 + 0.0 + 85.5692 + (471.3916)) \cdot 138$
 $= 174315.91 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (707.2212 + 0.0 + 85.5692 + 6502.8149 + (471.3916)) \cdot 138$
 $= 1070977.88 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 273.05 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 247972. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 540.0 \cdot 14.0 \cdot 0.49 \cdot 138$
 $= 802352. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 131.4242) \cdot (13.2016 - 3.0) \cdot 0.7 \cdot 118$
 $= 347887. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 273.05 \cdot 28.0 \cdot 0.74 \cdot 138$



= 1226292. N

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 273.05 \cdot (30.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 1181639. N$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (802352 + 347887) = 1150240 N$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (247972 + 1226292 + 1181639 + 0) = 2655904 N$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (802352 + 1181639 + 0) = 1983992 N$$

Summary of Failure Path Calculations:

Path 1-1 = 1150239 N	, must exceed W = 533281 N	or W1 = 1005978 N
Path 2-2 = 2655903 N	, must exceed W = 533281 N	or W2 = 174315 N
Path 3-3 = 1983991 N	, must exceed W = 533281 N	or W3 = 1070977 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 10.4153 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 290.4153 mm

Input Echo, WRC297 Item 1, Description: F3 #300 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1800.000	mm
Shell Thickness	30.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	137.90 MPa
Vessel Hot S.I. Allowable	Smh	137.90 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	273.050	mm
Nozzle Thickness	13.2016	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	540.000	mm
Thickness of Reinforcing Pad	28.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.



Design Internal Pressure	Dp	2.16	MPa
Radial Load	P	-14039.98	N
Circumferential Shear	Vc	10530.02	N
Longitudinal Shear	Vl	14039.98	N
Circumferential Moment	Mc	0.702E+07	N-mm
Longitudinal Moment	Ml	0.913E+07	N-mm
Torsional Moment	Mt	0.105E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(930.5(30.0 - 3.0))}$
 $= 796.113 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1861.000 mm
Nozzle Outside Diameter	(d) =	273.050 mm
Vessel Thickness used	(T) =	55.000 mm
Nozzle Thickness used	(t) =	10.202 mm
T / t	=	5.391
d / t	=	26.765

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.853$$

Nr/P	=	0.072
Mr/P	=	0.052
M0/P	=	0.138
N0/P	=	0.325
MrD/Mc	=	0.090
NrDL/Mc	=	0.103
M0d/Mc	=	0.285
N0DL/Mc	=	0.189
MrD/Ml	=	0.084
NrDL/Ml	=	0.097
M0D/Ml	=	0.239
N0DL/Ml	=	0.376

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-4	-4	4	4
Outplane Bending (Ml)	-15	15	15	-15
Normal Pressure Stress	34	36	34	36
Outplane Stress Summary	19	45	59	22

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (Ml)	-1	-1	1	1



Inplane Bending (ML)	-5	5	5	-5
Inplane Pressure Stress	17	17	17	17

Inplane Stress Summary	12	20	25	11
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	2	2	1	1
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	20	46	59	22
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-14	14	14	-14
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	17	17	17	17

Outplane Stress Summary	6	27	38	1
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-4	4	4	-4
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	34	36	34	36

Inplane Stress Summary	30	39	41	31
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	1	1	2	2
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	30	39	42	31
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	30	33	30	33
Hoop Stress Summary	28	30	36	39

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	12	-12	12	-12
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-17	-17	17	17
Axial Bending (ML)	-69	69	69	-69
Axial Pressure Stress	15	15	15	15
Axial Stress Summary	-56	56	116	-47

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	8	8	8	8
Shear Stress Summary	11	11	6	6

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	87	60	116	87

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	30	33	30	33
Hoop Stress Summary	30	33	33	36

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	12	-12	12	-12
Axial Membrane (Mc)	-13	-13	13	13



Axial Bending (Mc)	-56	56	56	-56
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	15	15	15	15

Axial Stress Summary	-39	47	99	-38
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	8	8	8	8

Shear Stress Summary	5	5	12	12
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Two * Max Shear Stress	71	49	101	79
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(916.5(30.0 - 3.0))}$
 $= 792.163 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	1833.000 mm
Nozzle Outside Diameter (d)	=	540.000 mm
Vessel Thickness used (T)	=	27.000 mm
Nozzle Thickness used (t)	=	143.677 mm
T / t	=	0.188
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.427$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.065
Mr/P	=	0.126
M0/P	=	0.037
N0/P	=	0.068
MrD/Mc	=	0.280
NrDL/Mc	=	0.093
M0d/Mc	=	0.085
N0DL/Mc	=	0.081
MrD/Ml	=	0.097
NrDL/Ml	=	0.081
M0D/Ml	=	0.029
N0DL/Ml	=	0.088

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-3	3	3	-3
Normal Pressure Stress	71	73	71	73



Outplane Stress Summary 70 72 82 68

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Stresses parallel to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	14	-14	14	-14
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-13	13	13	-13
Inplane Pressure Stress	35	35	35	35

Inplane Stress Summary 35 33 66 10

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Shear stress normal to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary 1 1 0 0

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress 70 72 82 68

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-9	9	9	-9
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	35	35	35	35

Outplane Stress Summary 30 40 51 24

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses parallel to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	14	-14	14	-14
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-29	29	29	-29
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	71	73	71	73

Inplane Stress Summary 55 88 118 31

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	1	1
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	55	88	118	31

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	34	36	34	36	34	36	34	36
Circ. Pl (SUS)	-3	-3	5	5	0	0	0	0
Circ. Q (SUS)	-12	12	18	-18	-3	3	5	-5
Long. Pm (SUS)	17	17	17	17	17	17	17	17
Long. Pl (SUS)	-1	-1	1	1	0	0	2	2
Long. Q (SUS)	-4	4	6	-6	-11	11	17	-17
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	34.0	36.0	34.0	36.0	34.0	36.0	34.0	36.0
Pm+Pl (SUS)	31.0	33.0	39.0	41.0	34.0	36.0	34.0	36.0
Pm+Pl+Q (Total)	19.1	45.0	57.0	23.1	31.0	39.1	39.3	31.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	36.00	137.90	Passed
Pm+Pl (SUS)	41.00	206.85	Passed
Pm+Pl+Q (TOTAL)	57.03	413.70	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	71	73	71	73	71	73	71	73
Circ. Pl (SUS)	-1	-1	3	3	0	0	2	2
Circ. Q (SUS)	1	-1	7	-7	-15	15	43	-43
Long. Pm (SUS)	35	35	35	35	35	35	35	35
Long. Pl (SUS)	0	0	2	2	0	0	2	2
Long. Q (SUS)	1	-1	27	-27	-5	5	13	-13
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0



Pm (SUS)	71.0	73.0	71.0	73.0	71.0	73.0	71.0	73.0
Pm+Pl (SUS)	70.0	72.0	74.0	76.0	71.0	73.0	73.0	75.0
Pm+Pl+Q (Total)	71.0	71.0	81.0	69.0	56.0	88.0	116.0	32.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	73.00	137.90	Passed
Pm+Pl (SUS)	76.00	206.85	Passed
Pm+Pl+Q (TOTAL)	116.00	413.70	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	30	33	30	33	30	33	30	33
Circ. Pl (SUS)	-3	-3	5	5	0	0	2	2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	15	15	15	15	15	15	15	15
Long. Pl (SUS)	-16	-16	18	18	-12	-12	14	14
Long. Q (SUS)	-57	57	81	-81	-44	44	68	-68
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-3	-3	3	3
Shear Q (SUS)	8	8	8	8	8	8	8	8
Pm (SUS)	30.0	33.0	30.0	33.0	30.0	33.0	30.0	33.0
Pm+Pl (SUS)	28.3	31.3	36.2	38.7	30.3	33.3	33.9	36.2
Pm+Pl+Q (Total)	87.3	59.4	114.5	86.8	71.7	48.6	98.8	77.2

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	33.00	118.00	Passed
Pm+Pl (SUS)	38.70	177.00	Passed
Pm+Pl+Q (TOTAL)	114.45	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: T3 #300

From: 70

Pressure for Reinforcement Calculations	P	2.1575	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Cylindrical Shell	D	1800.00	mm
Design Length of Section	L	38300.0039	mm
Shell Finished (Minimum) Thickness	t	30.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		11760.00	mm
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User Entered Minimum Design Metal Temperature		-25.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

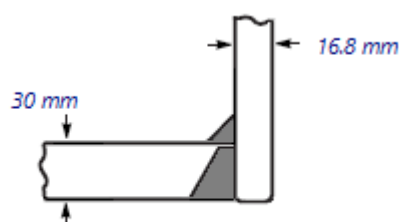
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: T3 #300

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.661 in.

Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.16 \cdot 903.0 / (138 \cdot 1.0 - 0.6 \cdot 2.16)$
 $= 14.2614 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.16 \cdot 28.4 / (118 \cdot 1.0 - 0.6 \cdot 2.16)$
 $= 0.5250 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3189 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	138.4000	mm
Parallel to Vessel Wall	Rn+tn+t	69.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: T3 #300.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.5250 mm
Wall Thickness per UG16(b),	tr16b	= 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 17.2614 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 17.2614 mm
Wall Thickness, shell/head, external pressure	trb2	= 3.3211 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000 mm
Wall Thickness per table UG-45	tb3	= 7.8000 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(17.2614, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.525, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 16.8, tr = 0.525, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.038$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.



Weld Size Calculations, Description: T3 #300

Intermediate Calc. for nozzle/shell Welds Tmin 13.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.9899 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 280.9899 mm

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Input, Nozzle Desc: MW3 24" #300

From: 80

Pressure for Reinforcement Calculations	P	2.1575	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Cylindrical Shell	D	1800.00	mm
Design Length of Section	L	38300.0039	mm
Shell Finished (Minimum) Thickness	t	30.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		13900.00	mm
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User Entered Minimum Design Metal Temperature		-25.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	137.90	MPa
Allowable Stress At Ambient	Sna	137.90	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105 +IT	
Flange Type		Weld Neck Flange	

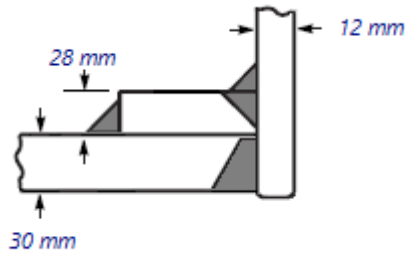
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1220.0000	mm
Thickness of Pad	te	28.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm
Reinforcing Pad Width		305.2000	mm
This is a Manway, Coupling or Access Opening.			

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: MW3 24" #300

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.472 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.16 \cdot 903.0 / (138 \cdot 1.0 - 0.6 \cdot 2.16)$
 $= 14.2614 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2.16 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 2.16)$
 $= 4.7390 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0955 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1183.2001 mm
Parallel to Vessel Wall, opening length	d	591.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		50.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 137.9 / 137.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 137.9 / 137.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 137.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1., 1.0)$
 $= 1.000$

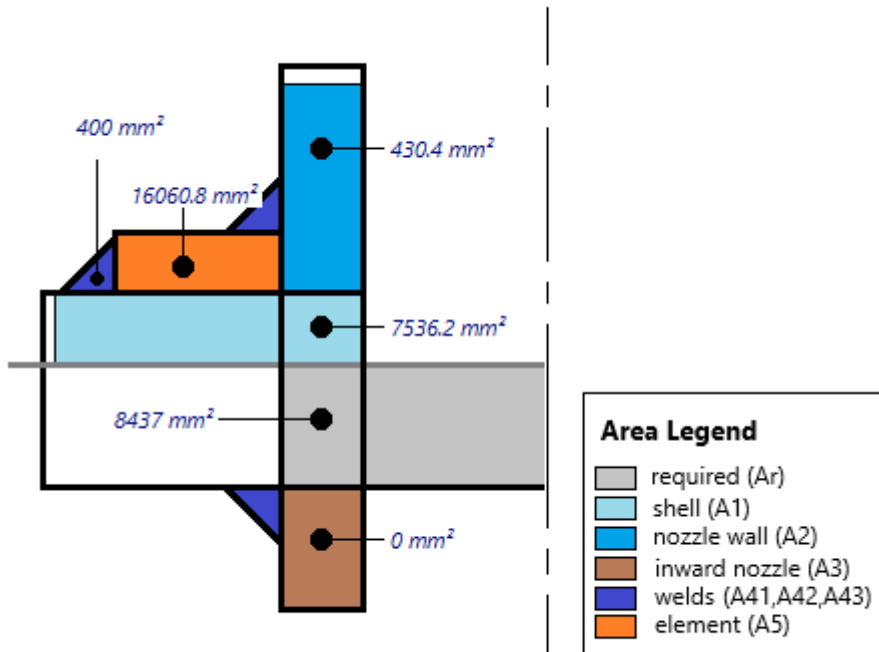


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	8437.046	3819.173	NA
Area in Shell	A1	7536.154	8334.853	NA
Area in Nozzle Wall	A2	430.364	798.357	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		400.000	400.000	NA
Area in Element	A5	16060.803	16060.803	NA
TOTAL AREA AVAILABLE	Atot	24427.320	25594.012	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 612.1189 28.0000 mm
 Based on given Pad Diameter: 1220.0000 0.1230 mm
 Based on Shell or Nozzle Thickness: 615.4774 12.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (591.6 \cdot 14.2614 \cdot 1.0 + 2 \cdot 9.0 \cdot 14.2614 \cdot 1.0 \cdot (1 - 1.))$
 $= 8437.046 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 591.6(1.0 \cdot 27.0 - 1.0 \cdot 14.261) - 2 \cdot 9.0$
 $(1.0 \cdot 27.0 - 1.0 \cdot 14.2614) \cdot (1 - 1.)$
 $= 7536.154 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 50.5) \cdot (9.0 - 4.74) \cdot 1.$

= 430.364 mm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - c_{an}/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $20.0^2 \cdot 1. + (0.0)^2 \cdot 1. + 0.0^2 \cdot 1.0$
 = 400.000 mm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lp}, t_e)) \cdot fr_4$
 = $(1183.2001 - 609.6) \cdot 28.0 \cdot 1.0$
 = 16060.803 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 7.7390$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= t_a
 = 7.7390 mm

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: D

Govrn. thk, $t_g = 12.0$, $t_r = 4.739$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.527$, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

Govrn. thk, $t_g = 12.0$, $t_r = 4.739$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.527$, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, $t_g = 12.0$, $t_r = 4.739$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.527$, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:D, Curve:B)

Govrn. thk, $t_g = 28.0$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.528$, Temp. Reduction = 29 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -27 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:D, Curve:B)

Govrn. thk, $t_g = 12.0$, $t_r = 4.739$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.527$, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -48 °C
 Gov. MDMT of the Reinforcement Pad : -27 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -27 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -18 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -48 °C



Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $2.16/5.10 = 0.423$

Weld Size Calculations, Description: MW3 24" #300

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = $0.7 * W_o$ mm
Pad Weld	9.5000 = $0.5 * T_{minPad}$	9.8980 = $0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_r1 \cdot (E_1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (8437.0459 - 7536.1543 + 2 \cdot 9.0 \cdot 1. \cdot (1.0 \cdot 27.0 - 14.2614)) \cdot 138)$
 $= 155839.47 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - Can / .707)^2 \cdot f_r2) \cdot S_v$
 $= (430.3638 + 16060.8027 + 400.0 - 0.0 \cdot 1. \cdot 138$
 $= 2329093.75 \text{ N}$

Weld Load [W2]:
 $= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (430.3638 + 0.0 + 400.0 + (486.0)) \cdot 138$
 $= 181511.11 \text{ N}$

Weld Load [W3]:
 $= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (430.3638 + 0.0 + 400.0 + 16060.8027 + (486.0)) \cdot 138$
 $= 2396107.50 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 609.6 \cdot 20.0 \cdot 0.49 \cdot 138$
 $= 1293952. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 1220.0 \cdot 14.0 \cdot 0.49 \cdot 138$
 $= 1812722. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 300.3) \cdot (12.0 - 3.0) \cdot 0.7 \cdot 138$
 $= 819546. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 609.6 \cdot 28.0 \cdot 0.74 \cdot 138$
 $= 2737769. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 609.6 \cdot (30.0 - 3.0) \cdot 0.74 \cdot 138$
 $= 2638078. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:



Failure Path [Path 1-1]:
 $= (\text{SPEW} + \text{SNW}) = (1812722 + 819546) = 2632268 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (1293952 + 2737769 + 2638078 + 0) = 6669799 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Spew} + \text{Tngw} + \text{Sinw})$
 $= (1812722 + 2638078 + 0) = 4450800 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 2632267 N	, must exceed W = 155839 N	or W1 = 2329093 N
Path 2-2 = 6669798 N	, must exceed W = 155839 N	or W2 = 181511 N
Path 3-3 = 4450800 N	, must exceed W = 155839 N	or W3 = 2396107 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 53.1843 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 383.1843 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 2.008 %

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Input, Nozzle Desc: T2 #300

From: 110

Pressure for Reinforcement Calculations	P	2.1575	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Cylindrical Shell	D	1800.00	mm
Design Length of Section	L	38300.0039	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		23750.00	mm
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User Entered Minimum Design Metal Temperature		-25.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

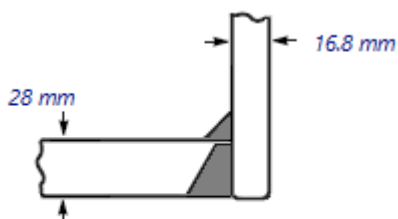
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		28.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: T2 #300

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 2.16 \cdot 903.0 / (138 \cdot 1.0 - 0.6 \cdot 2.16)$$

$$= 14.2614 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 2.16 \cdot 28.4 / (118 \cdot 1.0 - 0.6 \cdot 2.16)$$

$$= 0.5250 \text{ mm}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.3189 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	134.4000	mm
Parallel to Vessel Wall	Rn+tn+t	67.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: T2 #300.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.5250	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 17.2614	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 17.2614	mm
Wall Thickness, shell/head, external pressure	trb2	= 3.3211	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[7.8, \max(17.2614, 4.5)]$$

$$= 7.8000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.525, 7.8)$$

$$= 7.8000 \text{ mm}$$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 16.8, tr = 0.525, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.038, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.



Weld Size Calculations, Description: T2 #300

Intermediate Calc. for nozzle/shell welds Tmin 13.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.9899 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 278.9899 mm

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Input, Nozzle Desc: F1 4" #300

From: 120

Pressure for Reinforcement Calculations	P	2.1575	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Cylindrical Shell	D	1800.00	mm
Design Length of Section	L	38300.0039	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		25000.00	mm
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User Entered Minimum Design Metal Temperature		-25.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		4.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-105 +IT	
Flange Type		Weld Neck Flange	

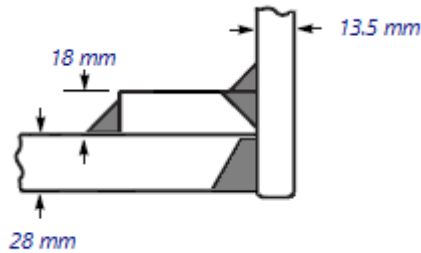
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	224.0000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		54.8500	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: F1 4" #300

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500 in.
Actual Thickness Used in Calculation	0.531 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, t_r [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.16 \cdot 903.0 / (138 \cdot 1.0 - 0.6 \cdot 2.16)$
 $= 14.2614 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, t_{rn} [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2.16 \cdot 57.15 / (118 \cdot 1.0 + 0.4 \cdot 2.16)$
 $= 1.0373 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3814 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	186.6504 mm
Parallel to Vessel Wall, opening length	d	93.3252 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		44.2185 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 137.9)$
 $= 0.856$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 137.9)$
 $= 0.856$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 137.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.856, 1.0)$
 $= 0.856$

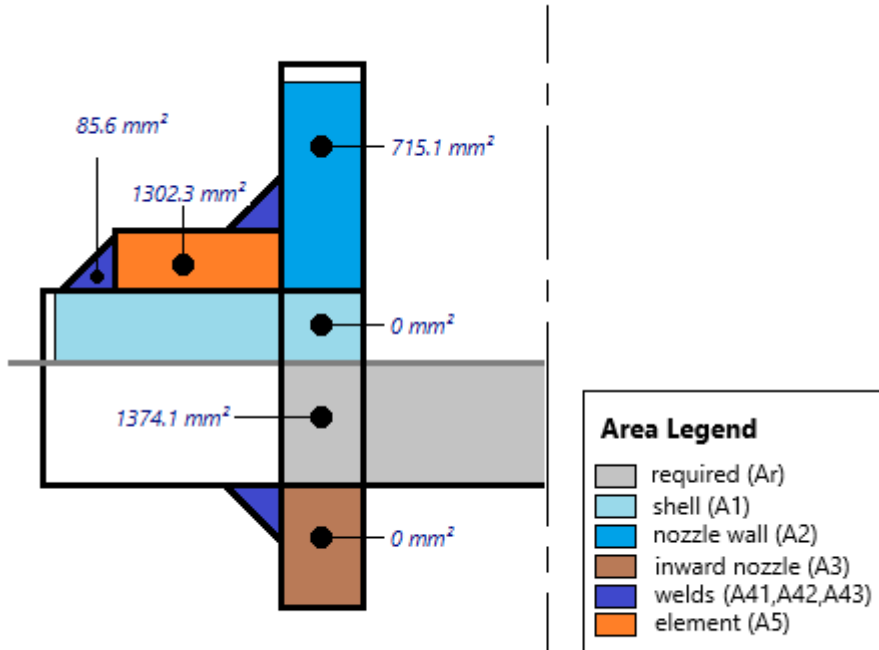


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1374.115	620.689	NA
Area in Shell	A1	0.000	0.000	NA
Area in Nozzle Wall	A2	715.134	764.773	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	85.569	85.569	NA
Area in Element	A5	1302.307	1302.307	NA
TOTAL AREA AVAILABLE	Atot	2103.010	2152.650	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 146.1562 18.0000 mm
 Based on given Pad Diameter: 224.0000 7.9255 mm
 Based on Shell or Nozzle Thickness: 156.8146 13.4874 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (93.3252 \cdot 14.2614 \cdot 1.0 + 2 \cdot 10.4874 \cdot 14.2614 \cdot 1.0 \cdot (1 - 0.86))$
 $= 1374.115 \text{ mm}^2$

The area in the shell has not been calculated by user request.

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 44.22) \cdot (10.49 - 1.04) \cdot 0.8557$
 $= 715.134 \text{ mm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi - can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$
 $= 85.569 \text{ mm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot f_r$
 $= (186.6504 - 114.3) \cdot 18.0 \cdot 1.0$
 $= 1302.307 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 4.0373 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 17.2614 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr_{16b}) = 17.2614 \text{ mm}$
Wall Thickness, shell/head, external pressure	$trb2 = 3.3211 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 8.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.258, \max(17.2614, 4.5)]$
 $= 8.2578 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(4.0373, 8.2578)$
 $= 8.2578 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 13.487 = 11.801 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	25.3,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	269.7 MPa	Passed
Occasional	:	4.3,	Allowable	:	156.9 MPa	Passed
Shear	:	16.3,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 18.0$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.57$, Temp. Reduction = 25°C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-35 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 11.801$, $tr = 1.037$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.118$, Temp. Reduction = 78°C



Min Metal Temp. w/o impact per UCS-66, Curve B -24 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -35 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -35 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -18 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -63 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = 2.16/5.10 = 0.423

Weld Size Calculations, Description: F1 4" #300

Intermediate Calc. for nozzle/shell Welds Tmin 10.4874 mm
 Intermediate Calc. for pad/shell Welds TminPad 18.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.0000 = 0.5*TminPad	9.8980 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (1374.1149 - 0.0 + 2 \cdot 10.4874 \cdot 0.8557 \cdot (1.0 \cdot 25.0 - 14.2614)) \cdot 138)$
 $= 216050.38 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (715.1337 + 1302.307 + 85.5692 - 0.0 \cdot 0.86) \cdot 138$
 $= 289980.41 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (715.1337 + 0.0 + 85.5692 + (448.6995)) \cdot 138$
 $= 172277.95 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (715.1337 + 0.0 + 85.5692 + 1302.307 + (448.6995)) \cdot 138$
 $= 351850.81 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 103802. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 224.0 \cdot 14.0 \cdot 0.49 \cdot 138$
 $= 332828. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 51.9063) \cdot (13.4874 - 3.0) \cdot 0.7 \cdot 118$
 $= 141248. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$



$$= (3.1416/2) \cdot 114.3 \cdot 18.0 \cdot 0.74 \cdot 138$$

$$= 329999. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 114.3 \cdot (18.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 329760. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (332828 + 141248) = 474075 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (103802 + 329999 + 329760 + 0) = 763561 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (332828 + 329760 + 0) = 662587 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 474075 N	, must exceed W = 216050 N	or W1 = 289980 N
Path 2-2 = 763561 N	, must exceed W = 216050 N	or W2 = 172277 N
Path 3-3 = 662587 N	, must exceed W = 216050 N	or W3 = 351850 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3 MPa

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 1.8164 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 279.8164 mm

Input Echo, WRC297 Item 1, Description: F1 4" #300 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1800.000	mm
Shell Thickness	28.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	137.90 MPa
Vessel Hot S.I. Allowable	Smh	137.90 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	114.300	mm
Nozzle Thickness	13.4874	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	224.000	mm
Thickness of Reinforcing Pad	18.0000	mm

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.



Design Internal Pressure	Dp	2.16	MPa
Radial Load	P	-5616.00	N
Circumferential Shear	Vc	4212.00	N
Longitudinal Shear	Vl	5616.00	N
Circumferential Moment	Mc	0.112E+07	N-mm
Longitudinal Moment	Ml	0.146E+07	N-mm
Torsional Moment	Mt	0.168E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (4.082) is < 5
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tn (8.475) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(924.5(28.0 - 3.0))}$
 $= 615.992 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1849.000 mm
Nozzle Outside Diameter	(d) =	114.300 mm
Vessel Thickness used	(T) =	43.000 mm
Nozzle Thickness used	(t) =	10.487 mm
T / t	=	4.100
d / t	=	10.899

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.405$$

Nr/P	=	0.191
Mr/P	=	0.142
M0/P	=	0.225
N0/P	=	0.240
MrD/Mc	=	0.147
NrDL/Mc	=	0.166
M0d/Mc	=	0.263
N0DL/Mc	=	-0.114
MrD/Ml	=	0.143
NrDL/Ml	=	0.167
M0D/Ml	=	0.254
N0DL/Ml	=	0.084

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	-10	10	10	-10
Normal Pressure Stress	44	46	44	46
Outplane Stress Summary	37	52	60	33

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2



Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-5	5	5	-5
Inplane Pressure Stress	22	22	22	22

Inplane Stress Summary	18	24	32	15
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	2	2	1	1
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	38	53	60	33
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-8	8	8	-8
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	22	22	22	22

Outplane Stress Summary	19	27	34	9
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-4	4	4	-4
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	44	46	44	46

Inplane Stress Summary	41	48	52	40
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	1	1	2	2
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left	Left	Right	Right



	Outside	Inside	Outside	Inside
Two * Max Shear Stress	41	48	53	40

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10	12	10	12
Hoop Stress Summary	10	13	11	14

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	14	-14	14	-14
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-17	-17	17	17
Axial Bending (ML)	-41	41	41	-41
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-38	15	80	-30

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	7	7	7	7
Shear Stress Summary	10	10	5	5

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	52	24	80	46

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10	12	10	12
Hoop Stress Summary	11	14	10	13

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	1	1	1	1
Axial Bending (P)	14	-14	14	-14
Axial Membrane (Mc)	-13	-13	13	13
Axial Bending (Mc)	-34	34	34	-34
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5

Axial Stress Summary	-26	12	68	-27
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	7	7	7	7
Shear Stress Summary	4	4	10	10

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	39	18	70	46

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(915.5(28.0 - 3.0))}$
 $= 613.544 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	1831.000 mm
Nozzle Outside Diameter (d)	=	224.000 mm
Vessel Thickness used (T)	=	25.000 mm
Nozzle Thickness used (t)	=	65.337 mm
T / t	=	0.383
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.047$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.103
Mr/P	=	0.219
M0/P	=	0.064
N0/P	=	0.125
MrD/Mc	=	0.308
NrDL/Mc	=	0.083
M0d/Mc	=	0.096
N0DL/Mc	=	0.044
MrD/Ml	=	0.189
NrDL/Ml	=	0.090
M0D/Ml	=	0.060
N0DL/Ml	=	0.117

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1	-1	1	1



Outplane Bending (ML)	-3	3	3	-3
Normal Pressure Stress	76	79	76	79

Outplane Stress Summary	76	79	86	74
-------------------------	----	----	----	----

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	11	-11	11	-11
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-11	11	11	-11
Inplane Pressure Stress	38	38	38	38
Inplane Stress Summary	38	38	63	16

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	1	1	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	76	79	86	74

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-4	4	4	-4
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	38	38	38	38
Outplane Stress Summary	38	40	47	31

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	11	-11	11	-11
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-14	14	14	-14
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	76	79	76	79
Inplane Stress Summary	74	82	105	53

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
-----------------------	----	----	----	----



(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	0	0	1	1

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	74	82	105	54

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	44	46	44	46	44	46	44	46
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	-6	6	14	-14	-2	2	6	-6
Long. Pm (SUS)	22	22	22	22	22	22	22	22
Long. Pl (SUS)	-1	-1	1	1	0	0	0	0
Long. Q (SUS)	-3	3	7	-7	-4	4	12	-12
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	44.0	46.0	44.0	46.0	44.0	46.0	44.0	46.0
Pm+Pl (SUS)	44.0	46.0	44.0	46.0	44.0	46.0	44.0	46.0
Pm+Pl+Q (Total)	38.0	52.0	58.0	32.1	42.0	48.0	50.1	40.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	46.00	137.90	Passed
Pm+Pl (SUS)	46.00	206.85	Passed
Pm+Pl+Q (TOTAL)	58.04	413.70	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	76	79	76	79	76	79	76	79
Circ. Pl (SUS)	0	0	2	2	0	0	0	0
Circ. Q (SUS)	0	0	6	-6	-3	3	25	-25
Long. Pm (SUS)	38	38	38	38	38	38	38	38
Long. Pl (SUS)	0	0	0	0	1	1	1	1
Long. Q (SUS)	0	0	22	-22	-1	1	7	-7



Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	76.0	79.0	76.0	79.0	76.0	79.0	76.0	79.0
Pm+Pl (SUS)	76.0	79.0	78.0	81.0	76.0	79.0	76.0	79.0
Pm+Pl+Q (Total)	76.0	79.0	84.0	75.0	73.0	82.0	101.0	54.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	79.00	137.90	Passed
Pm+Pl (SUS)	81.00	206.85	Passed
Pm+Pl+Q (TOTAL)	101.00	413.70	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	10	12	10	12	10	12	10	12
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-16	-16	18	18	-12	-12	14	14
Long. Q (SUS)	-27	27	55	-55	-20	20	48	-48
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	10.0	12.0	10.0	12.0	10.0	12.0	10.0	12.0
Pm+Pl (SUS)	21.4	23.3	23.3	23.4	17.5	19.4	19.4	19.5
Pm+Pl+Q (Total)	51.3	23.2	78.4	45.1	38.3	17.5	68.4	44.8

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	12.00	118.00	Passed
Pm+Pl (SUS)	23.35	177.00	Passed
Pm+Pl+Q (TOTAL)	78.37	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: MW2 24" #300

From: 150

Pressure for Reinforcement Calculations	P	2.1575	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Cylindrical Shell	D	1800.00	mm
Design Length of Section	L	38300.0039	mm
Shell Finished (Minimum) Thickness	t	25.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		34900.00	mm
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User Entered Minimum Design Metal Temperature		-25.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	137.90	MPa
Allowable Stress At Ambient	Sna	137.90	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105 +IT	
Flange Type		Weld Neck Flange	

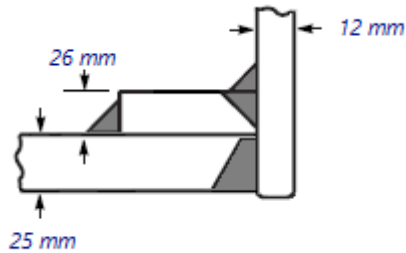
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1220.0000	mm
Thickness of Pad	te	26.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	26.0000	mm
Reinforcing Pad Width		305.2000	mm
This is a Manway, Coupling or Access Opening.			

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: MW2 24" #300

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.472 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.16 \cdot 903.0 / (138 \cdot 1.0 - 0.6 \cdot 2.16)$
 $= 14.2614 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2.16 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 2.16)$
 $= 4.7390 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0955 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1183.2001 mm
Parallel to Vessel Wall, opening length	d	591.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		48.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 137.9 / 137.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 137.9 / 137.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 137.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

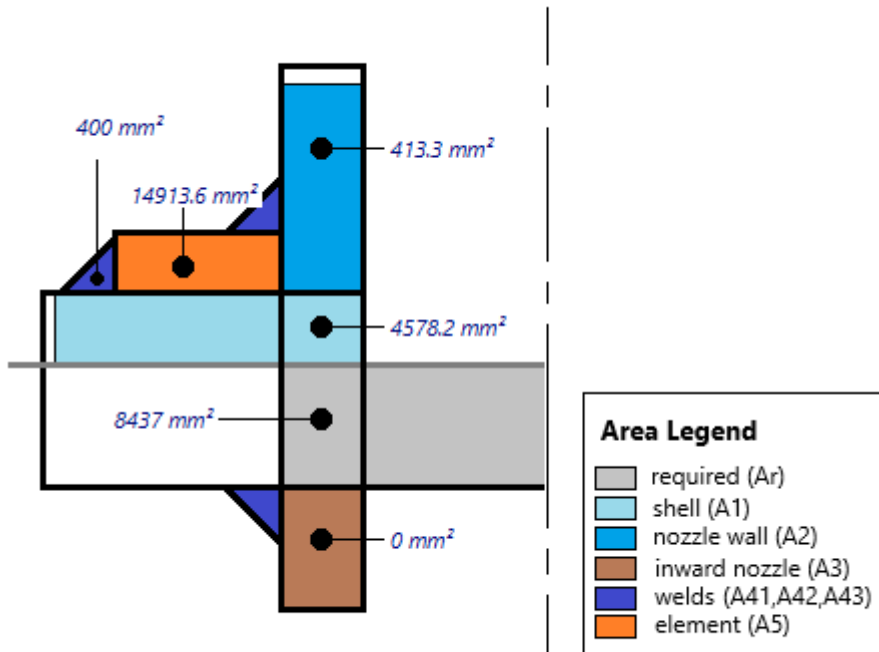


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	8437.046	3798.793	NA
Area in Shell	A1	4578.154	5417.614	NA
Area in Nozzle Wall	A2	413.320	766.739	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	400.000	400.000	NA
Area in Element	A5	14913.602	14913.602	NA
TOTAL AREA AVAILABLE	Atot	20305.076	21497.953	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 726.7374 26.0000 mm
 Based on given Pad Diameter: 1220.0000 5.3096 mm
 Based on Shell or Nozzle Thickness: 863.3976 12.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (591.6 \cdot 14.2614 \cdot 1.0 + 2 \cdot 9.0 \cdot 14.2614 \cdot 1.0 \cdot (1 - 1.0))$
 $= 8437.046 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 591.6(1.0 \cdot 22.0 - 1.0 \cdot 14.2614) - 2 \cdot 9.0$
 $(1.0 \cdot 22.0 - 1.0 \cdot 14.2614) \cdot (1 - 1.0)$
 $= 4578.154 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 48.5) \cdot (9.0 - 4.74) \cdot 1.0$



= 413.320 mm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $20.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 0.0^2 \cdot 1.0$
 = 400.000 mm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
 = $(1183.2001 - 609.6) \cdot 26.0 \cdot 1.0$
 = 14913.602 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 7.7390$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= t_a
 = 7.7390 mm

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 4.739$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.527$, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 4.739$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.527$, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 25.0$, $t_r = 14.261$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.648$, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-21 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 4.739$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.527$, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	:	-48 °C
Gov. MDMT of the Reinforcement Pad	:	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-29 °C

Flange MDMT including Temperature reduction per UCS-66.1:



MDMT of ASME B16.5/47 flange per Matl. Specification -18 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -48 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = 2.16/5.10 = 0.423

Weld Size Calculations, Description: MW2 24" #300

Intermediate Calc. for nozzle/shell Welds T_{min} 9.0000 mm
 Intermediate Calc. for pad/shell Welds T_{minPad} 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = 0.7 * W _o mm
Pad Weld	9.5000 = 0.5*T _{minPad}	9.8980 = 0.7 * W _p mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (8437.0459 - 4578.1543 + 2 \cdot 9.0 \cdot 1.0 \cdot (1.0 \cdot 22.0 - 14.2614)) \cdot 138)$
 $= 551303.00 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (413.3198 + 14913.6016 + 400.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 2168558.00 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (413.3198 + 0.0 + 400.0 + (396.0)) \cdot 138$
 $= 166751.02 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (413.3198 + 0.0 + 400.0 + 14913.6016 + (396.0)) \cdot 138$
 $= 2223161.75 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [S_{onw}]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 609.6 \cdot 20.0 \cdot 0.49 \cdot 138$
 $= 1293952. \text{ N}$

Shear, Pad Element Weld [S_{pew}]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 1220.0 \cdot 14.0 \cdot 0.49 \cdot 138$
 $= 1812722. \text{ N}$

Shear, Nozzle Wall [S_{nw}]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 300.3) \cdot (12.0 - 3.0) \cdot 0.7 \cdot 138$
 $= 819546. \text{ N}$

Tension, Pad Groove Weld [T_{pgw}]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 609.6 \cdot 26.0 \cdot 0.74 \cdot 138$
 $= 2542214. \text{ N}$

Tension, Shell Groove Weld [T_{ngw}]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 609.6 \cdot (18.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 1758719. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.



Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (1812722 + 819546) = 2632268 \text{ N}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (1293952 + 2542214 + 1758719 + 0) = 5594885 \text{ N}$

Failure Path [Path 3-3]:
 $= (Spew + Tngw + Sinw)$
 $= (1812722 + 1758719 + 0) = 3571441 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 2632267 N	, must exceed W = 551303 N	or W1 = 2168558 N
Path 2-2 = 5594884 N	, must exceed W = 551303 N	or W2 = 166751 N
Path 3-3 = 3571441 N	, must exceed W = 551303 N	or W3 = 2223161 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 53.1843 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 378.1843 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 2.008 %

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Input, Nozzle Desc: T1 #300

From: 160

Pressure for Reinforcement Calculations	P	2.1575	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Cylindrical Shell	D	1800.00	mm
Design Length of Section	L	38300.0039	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		36360.00	mm
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User Entered Minimum Design Metal Temperature		-25.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

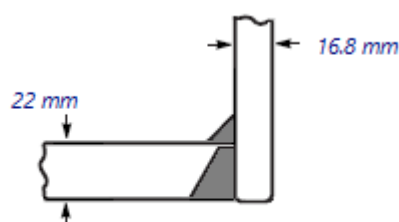
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: T1 #300

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.16 \cdot 903.0 / (138 \cdot 1.0 - 0.6 \cdot 2.16)$
 $= 14.2614 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.16 \cdot 28.4 / (118 \cdot 1.0 - 0.6 \cdot 2.16)$
 $= 0.5250 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3189 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	122.4000 mm
Parallel to Vessel Wall	Rn+tn+t	61.2000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: T1 #300.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.5250 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 17.2614 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 17.2614 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.3211 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(17.2614, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(3.525, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 16.8, tr = 0.525, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.038$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.



Weld Size Calculations, Description: T1 #300

Intermediate Calc. for nozzle/shell welds Tmin 13.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.9899 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 272.9899 mm

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Input, Nozzle Desc: MW1 24" #300

From: 160

Pressure for Reinforcement Calculations	P	2.1575	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Cylindrical Shell	D	1800.00	mm
Design Length of Section	L	38300.0039	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		37100.00	mm
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User Entered Minimum Design Metal Temperature		-25.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	117.90	MPa
Allowable Stress At Ambient	Sna	117.90	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm

Flange Material		SA-105 +IT	
Flange Type		Weld Neck Flange	

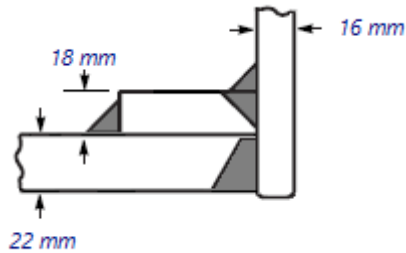
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1220.0000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		305.2000	mm
This is a Manway, Coupling or Access Opening.			

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: MW1 24" #300

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.630 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.16 \cdot 903.0 / (138 \cdot 1.0 - 0.6 \cdot 2.16)$
 $= 14.2614 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2.16 \cdot 304.8 / (118 \cdot 1.0 + 0.4 \cdot 2.16)$
 $= 5.5368 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0955 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1167.2001 mm
Parallel to Vessel Wall, opening length	d	583.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		47.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 117.9 / 137.9)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 117.9 / 137.9)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 137.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

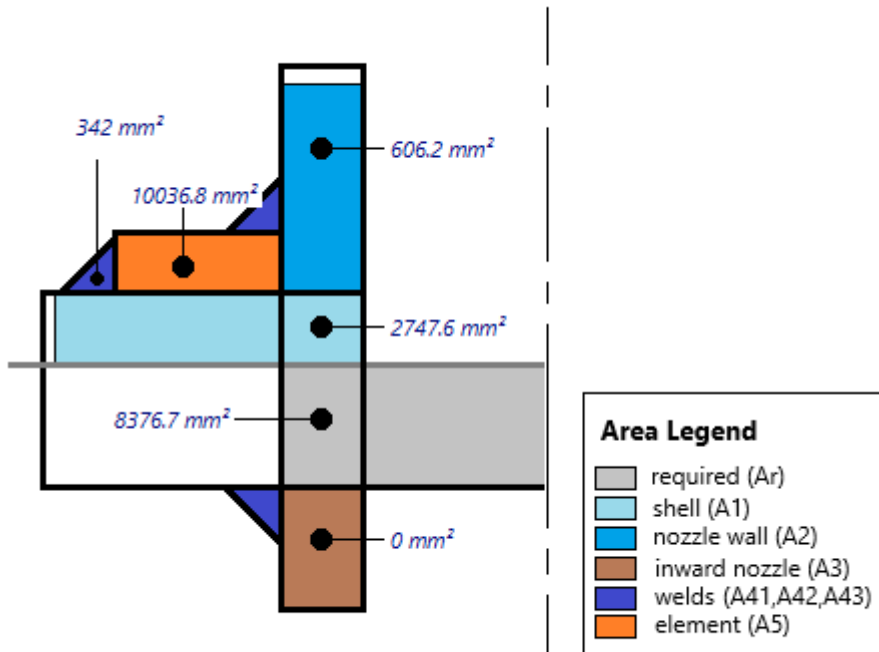


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	8376.719	3759.519	NA
Area in Shell	A1	2747.582	3594.252	NA
Area in Nozzle Wall	A2	606.201	966.948	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		342.001	342.001	NA
Area in Element	A5	10036.801	10036.801	NA
TOTAL AREA AVAILABLE	Atot	13732.586	14940.003	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 869.6519 18.0000 mm
 Based on given Pad Diameter: 1220.0000 8.3948 mm
 Based on Shell or Nozzle Thickness: 902.1584 16.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (583.6 \cdot 14.2614 \cdot 1.0 + 2 \cdot 13.0 \cdot 14.2614 \cdot 1.0 \cdot (1 - 0.86))$
 $= 8376.719 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 583.6(1.0 \cdot 19.0 - 1.0 \cdot 14.261) - 2 \cdot 13.0$
 $(1.0 \cdot 19.0 - 1.0 \cdot 14.2614) \cdot (1 - 0.855)$
 $= 2747.582 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 47.5) \cdot (13.0 - 5.54) \cdot 0.855$



= 606.201 mm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $20.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$
 = 342.001 mm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
 = $(1167.2001 - 609.6) \cdot 18.0 \cdot 1.0$
 = 10036.801 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 8.5368$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = t_a
 = 8.5368 mm

Available Nozzle Neck Thickness = 16.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 18.0$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.751$, Temp. Reduction = 14 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-24 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 16.0$, $tr = 5.537$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.426$, Temp. Reduction = 45 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	:	-48 °C
Gov. MDMT of the Reinforcement Pad	:	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-18 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-63 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $2.16/5.10 = 0.423$

Weld Size Calculations, Description: MW1 24" #300



Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 18.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.000 = 0.7 * Wo mm
Pad Weld	9.0000 = 0.5*TminPad	11.3120 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (8376.7188 - 2747.5815 + 2 \cdot 13.0 \cdot 0.855 \cdot (1.0 \cdot 19.0 - 14.2614)) \cdot 138)$
 $= 790717.25 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (606.2015 + 10036.8008 + 342.0014 - 0.0 \cdot 0.86) \cdot 138$
 $= 1514703.25 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (606.2015 + 0.0 + 342.0014 + (422.3718)) \cdot 138$
 $= 188986.19 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (606.2015 + 0.0 + 342.0014 + 10036.8008 + (422.3718)) \cdot 138$
 $= 1572943.50 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 609.6 \cdot 20.0 \cdot 0.49 \cdot 118$
 $= 1106334. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 1220.0 \cdot 16.0 \cdot 0.49 \cdot 138$
 $= 2071682. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 298.3) \cdot (16.0 - 3.0) \cdot 0.7 \cdot 118$
 $= 1005402. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 609.6 \cdot 18.0 \cdot 0.74 \cdot 138$
 $= 1759994. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 609.6 \cdot (22.0 - 3.0) \cdot 0.74 \cdot 138$
 $= 1856426. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (S_{PEW} + S_{NW}) = (2071682 + 1005402) = 3077085 \text{ N}$

Failure Path [Path 2-2]:
 $= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$



$$= (1106334 + 1759994 + 1856426 + 0) = 4722754 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (2071682 + 1856426 + 0) = 3928108 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 3077084 N , must exceed W = 790717 N or W1 = 1514703 N

Path 2-2 = 4722753 N , must exceed W = 790717 N or W2 = 188986 N

Path 3-3 = 3928107 N , must exceed W = 790717 N or W3 = 1572943 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3 MPa

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 53.1843 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 375.1843 mm

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Input, Nozzle Desc: F2 6" #300

From: 160

Pressure for Reinforcement Calculations	P	2.1575	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Cylindrical Shell	D	1800.00	mm
Design Length of Section	L	38300.0039	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		37700.00	mm
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User Entered Minimum Design Metal Temperature		-25.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		6.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

Flange Material		SA-105 +IT	
Flange Type		Weld Neck Flange	

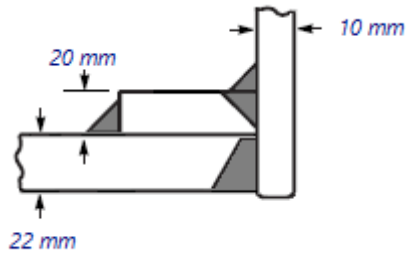
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	345.0000	mm
Thickness of Pad	te	20.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		96.3000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: F2 6" #300

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.000 in.
Actual Thickness Used in Calculation	0.394 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.16 \cdot 903.0 / (138 \cdot 1.0 - 0.6 \cdot 2.16)$
 $= 14.2614 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2.16 \cdot 76.2 / (118 \cdot 1.0 + 0.4 \cdot 2.16)$
 $= 1.3831 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3644 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	276.8000 mm
Parallel to Vessel Wall, opening length	d	138.4000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 137.9)$
 $= 0.856$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 137.9)$
 $= 0.856$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 137.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.856, 1.0)$
 $= 0.856$

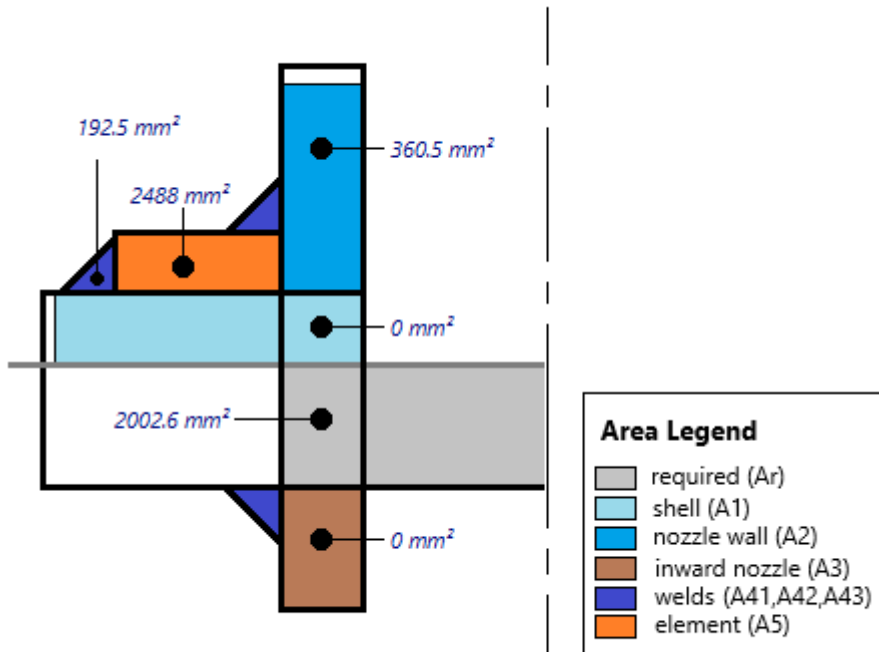


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	2002.590	898.774	NA
Area in Shell	A1	0.000	0.000	NA
Area in Nozzle Wall	A2	360.476	425.855	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	192.531	192.531	NA
Area in Element	A5	2488.000	2488.000	NA
TOTAL AREA AVAILABLE	Atot	3041.006	3106.385	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 224.8792 20.0000 mm
 Based on given Pad Diameter: 345.0000 11.6526 mm
 Based on the Estimated Diameter Limit: 275.2125 11.8032 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (138.4 \cdot 14.2614 \cdot 1.0 + 2 \cdot 7.0 \cdot 14.2614 \cdot 1.0 \cdot (1 - 0.86))$
 $= 2002.590 \text{ mm}^2$

The area in the shell has not been calculated by user request.

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 37.5) \cdot (7.0 - 1.38) \cdot 0.8557$
 $= 360.476 \text{ mm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi - can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 15.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$
 $= 192.531 \text{ mm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lwp}, t_e)) \cdot f_{r4}$
 $= (276.8 - 152.4) \cdot 20.0 \cdot 1.0$
 $= 2488.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 4.3831 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 17.2614 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr_{16b}) = 17.2614 \text{ mm}$
Wall Thickness, shell/head, external pressure	$trb2 = 3.3211 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 9.2200 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[9.22, \max(17.2614, 4.5)]$
 $= 9.2200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(4.3831, 9.22)$
 $= 9.2200 \text{ mm}$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	44.8,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	250.2 MPa	Passed
Occasional	:	10.2,	Allowable	:	156.9 MPa	Passed
Shear	:	24.9,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 20.0$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.751$, Temp. Reduction = 14 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-21 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 10.0$, $tr = 1.383$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.198$, Temp. Reduction = 78 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -29 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -18 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -63 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = 2.16/5.10 = 0.423

Weld Size Calculations, Description: F2 6" #300

Intermediate Calc. for nozzle/shell Welds Tmin 7.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.9000 = 0.7 * tmin.	10.6050 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	9.8980 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (2002.5903 - 0.0 + 2 \cdot 7.0 \cdot 0.8557 \cdot (1.0 \cdot 19.0 - 14.2614)) \cdot 138)$
 $= 283961.22 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (360.4759 + 2488.0 + 192.5308 - 0.0 \cdot 0.86) \cdot 138$
 $= 419319.16 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (360.4759 + 0.0 + 192.5308 + (227.6142)) \cdot 138$
 $= 107638.46 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (360.4759 + 0.0 + 192.5308 + 2488.0 + (227.6142)) \cdot 138$
 $= 450704.47 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 152.4 \cdot 15.0 \cdot 0.49 \cdot 118$
 $= 207605. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 345.0 \cdot 14.0 \cdot 0.49 \cdot 138$
 $= 512614. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 72.7) \cdot (10.0 - 3.0) \cdot 0.7 \cdot 118$
 $= 132046. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 152.4 \cdot 18.0 \cdot 0.74 \cdot 138$



= 439999. N

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 152.4 \cdot (18.0 - 0.0) \cdot 0.74 \cdot 138$
 = 439680. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (512614 + 132046) = 644660 \text{ N}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (207605 + 439999 + 439680 + 0) = 1087283 \text{ N}$

Failure Path [Path 3-3]:
 $= (Spew + Tngw + Sinw)$
 $= (512614 + 439680 + 0) = 952294 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 644660 N	, must exceed W = 283961 N	or W1 = 419319 N
Path 2-2 = 1087283 N	, must exceed W = 283961 N	or W2 = 107638 N
Path 3-3 = 952293 N	, must exceed W = 283961 N	or W3 = 450704 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 3.2316 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 175.2316 mm

Input Echo, WRC297 Item 1, Description: F2 6" #300 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1800.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 137.90	MPa
Vessel Hot S.I. Allowable	Smh 137.90	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	152.400	mm
Nozzle Thickness	10.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	MPa
Nozzle Hot S.I. Allowable	SNmh 118.00	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	345.000	mm
Thickness of Reinforcing Pad	20.0000	mm

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.



Design Internal Pressure	Dp	2.16	MPa
Radial Load	P	-8423.98	N
Circumferential Shear	Vc	6317.98	N
Longitudinal Shear	Vl	8423.98	N
Circumferential Moment	Mc	0.253E+07	N-mm
Longitudinal Moment	Ml	0.329E+07	N-mm
Torsional Moment	Mt	0.379E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 152.4 + 2 \cdot 1.65 \cdot \sqrt{(922.5(22.0 - 3.0))}$
 $= 589.292 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1845.000 mm
Nozzle Outside Diameter	(d) =	152.400 mm
Vessel Thickness used	(T) =	39.000 mm
Nozzle Thickness used	(t) =	7.000 mm
T / t	=	5.571
d / t	=	21.771

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.568$$

Nr/P	=	0.093
Mr/P	=	0.065
M0/P	=	0.188
N0/P	=	0.343
MrD/Mc	=	0.098
NrDL/Mc	=	0.113
M0d/Mc	=	0.303
N0DL/Mc	=	0.115
MrD/Ml	=	0.097
NrDL/Ml	=	0.110
M0D/Ml	=	0.283
N0DL/Ml	=	0.233

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-3	-3	3	3
Outplane Bending (Ml)	-24	24	24	-24
Normal Pressure Stress	48	51	48	51
Outplane Stress Summary	29	67	84	25

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (Ml)	-1	-1	1	1



Inplane Bending (ML)	-8	8	8	-8
Inplane Pressure Stress	24	24	24	24

Inplane Stress Summary	17	29	36	16
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	3	3	1	1
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	30	67	84	26
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-19	19	19	-19
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	24	24	24	24

Outplane Stress Summary	11	38	53	1
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-6	6	6	-6
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	48	51	48	51

Inplane Stress Summary	43	54	59	44
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	44	54	60	44
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-3	-3	3	3
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	24	26	24	26
Hoop Stress Summary	22	25	29	32

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	18	-18	18	-18
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-29	-29	29	29
Axial Bending (ML)	-111	111	111	-111
Axial Pressure Stress	12	12	12	12
Axial Stress Summary	-107	77	174	-85

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	14	14	14	14
Shear Stress Summary	18	18	11	11

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	135	83	175	119

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	24	26	24	26
Hoop Stress Summary	25	27	27	30

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	18	-18	18	-18
Axial Membrane (Mc)	-22	-22	22	22



Axial Bending (Mc)	-84	84	84	-84
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	12	12	12	12

Axial Stress Summary	-73	57	140	-65
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-5	-5	5	5
Shear due to Torsion	14	14	14	14

Shear Stress Summary	9	9	19	19
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Two * Max Shear Stress	100	60	143	103
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 152.4 + 2 \cdot 1.65 \cdot \sqrt{(912.5(22.0 - 3.0))}$
 $= 586.917 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	1825.000 mm
Nozzle Outside Diameter (d)	=	345.000 mm
Vessel Thickness used (T)	=	19.000 mm
Nozzle Thickness used (t)	=	103.300 mm
T / t	=	0.184
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.853$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.076
Mr/P	=	0.154
M0/P	=	0.045
N0/P	=	0.079
MrD/Mc	=	0.292
NrDL/Mc	=	0.091
M0d/Mc	=	0.090
N0DL/Mc	=	0.072
MrD/Ml	=	0.118
NrDL/Ml	=	0.094
M0D/Ml	=	0.037
N0DL/Ml	=	0.101

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-5	5	5	-5
Normal Pressure Stress	101	103	101	103



Outplane Stress Summary 101 102 118 96

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Stresses parallel to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	21	-21	21	-21
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-18	18	18	-18
Inplane Pressure Stress	50	50	50	50

Inplane Stress Summary 52 47 95 14

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Shear stress normal to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary 1 1 0 0

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress 101 102 118 96

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-10	10	10	-10
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	50	50	50	50

Outplane Stress Summary 46 55 71 36

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses parallel to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	21	-21	21	-21
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-35	35	35	-35
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	101	103	101	103

Inplane Stress Summary 87 117 162 50

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	1	1

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	87	117	162	50

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	48	51	48	51	48	51	48	51
Circ. Pl (SUS)	-2	-2	4	4	-1	-1	1	1
Circ. Q (SUS)	-18	18	30	-30	-4	4	8	-8
Long. Pm (SUS)	24	24	24	24	24	24	24	24
Long. Pl (SUS)	-1	-1	1	1	0	0	2	2
Long. Q (SUS)	-6	6	10	-10	-13	13	25	-25
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	48.0	51.0	48.0	51.0	48.0	51.0	48.0	51.0
Pm+Pl (SUS)	46.0	49.0	52.0	55.0	47.0	50.0	49.0	52.0
Pm+Pl+Q (Total)	28.4	67.1	82.1	25.4	43.1	54.2	57.6	44.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	51.00	137.90	Passed
Pm+Pl (SUS)	55.00	206.85	Passed
Pm+Pl+Q (TOTAL)	82.08	413.70	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	101	103	101	103	101	103	101	103
Circ. Pl (SUS)	-1	-1	3	3	0	0	2	2
Circ. Q (SUS)	1	-1	11	-11	-14	14	56	-56
Long. Pm (SUS)	50	50	50	50	50	50	50	50
Long. Pl (SUS)	-1	-1	3	3	0	0	2	2
Long. Q (SUS)	3	-3	39	-39	-4	4	16	-16
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1



Pm (SUS)	101.0	103.0	101.0	103.0	101.0	103.0	101.0	103.0
Pm+Pl (SUS)	100.0	102.0	104.0	106.0	101.0	103.0	103.0	105.0
Pm+Pl+Q (Total)	101.0	101.0	115.0	95.0	87.0	117.0	159.0	49.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	103.00	137.90	Passed
Pm+Pl (SUS)	106.00	206.85	Passed
Pm+Pl+Q (TOTAL)	159.01	413.70	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	24	26	24	26	24	26	24	26
Circ. Pl (SUS)	-2	-2	4	4	0	0	2	2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	12	12	12	12	12	12	12	12
Long. Pl (SUS)	-27	-27	31	31	-20	-20	24	24
Long. Q (SUS)	-93	93	129	-129	-66	66	102	-102
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-5	-5	5	5
Shear Q (SUS)	14	14	14	14	14	14	14	14
Pm (SUS)	24.0	26.0	24.0	26.0	24.0	26.0	24.0	26.0
Pm+Pl (SUS)	37.5	39.5	43.6	43.7	33.5	35.4	38.1	38.4
Pm+Pl+Q (Total)	134.4	82.9	172.8	118.1	99.6	60.4	141.1	101.4

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	26.00	118.00	Passed
Pm+Pl (SUS)	43.66	177.00	Passed
Pm+Pl+Q (TOTAL)	172.84	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: T 8" #300

From: 170

Pressure for Reinforcement Calculations	P	2.1575	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Elliptical Head	D	1800.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	19.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
-------------------------------	----	--------	----

User Entered Minimum Design Metal Temperature		-25.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		8.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105 +IT	
Flange Type		Weld Neck Flange	

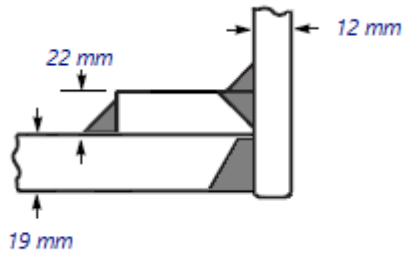
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	19.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	440.0000	mm
Thickness of Pad	te	22.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm
Reinforcing Pad Width		118.4000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: T 8" #300

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	8.000 in.
Actual Thickness Used in Calculation	0.472 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 2.16 \cdot 1806.0 \cdot 0.8972 / (2 \cdot 137.9 \cdot 1.0 - 0.2 \cdot 2.16)$
 $= 12.6944 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2.16 \cdot 101.6 / (118 \cdot 1.0 + 0.4 \cdot 2.16)$
 $= 1.8441 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5714 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	370.4000 mm
Parallel to Vessel Wall, opening length	d	185.2000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		40.0000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:

$= \min(1, Sn/Sv)$
 $= \min(1, 118.0/137.9)$
 $= 0.856$

Weld Strength Reduction Factor [fr2]:

$= \min(1, Sn/Sv)$
 $= \min(1, 118.0/137.9)$
 $= 0.856$

Weld Strength Reduction Factor [fr4]:

$= \min(1, Sp/Sv)$
 $= \min(1, 138.0/137.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.856, 1.0)$
 $= 0.856$

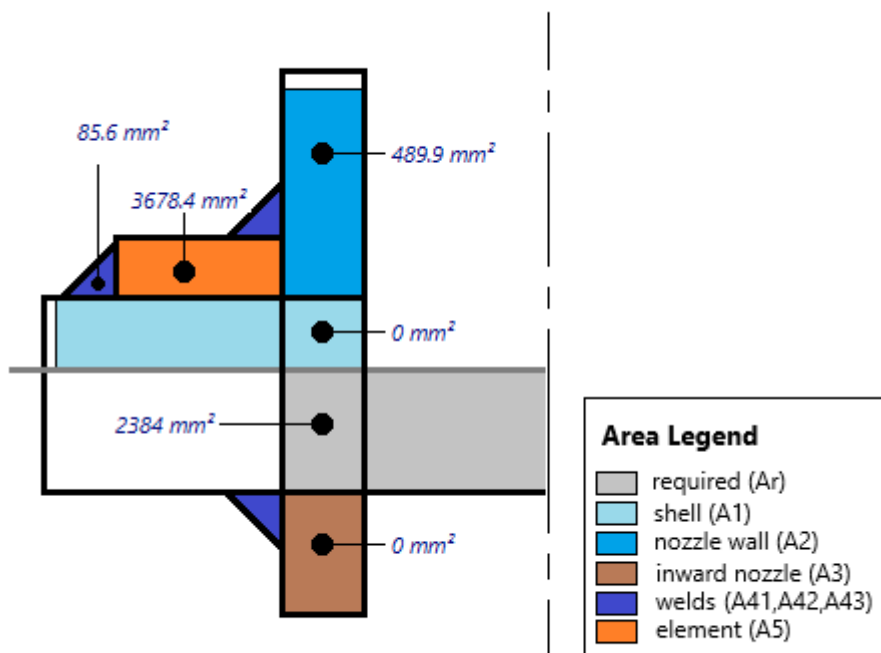


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	2383.982	311.680	NA
Area in Shell	A1	0.000	0.000	NA
Area in Nozzle Wall	A2	489.858	576.982	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	85.569	85.569	NA
Area in Element	A5	3678.400	3678.400	NA
TOTAL AREA AVAILABLE	Atot	4253.828	4340.951	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 285.4070 22.0000 mm
 Based on given Pad Diameter: 440.0000 10.8167 mm
 Based on Shell or Nozzle Thickness: 353.9128 12.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (185.2 \cdot 12.6944 \cdot 1.0 + 2 \cdot 9.0 \cdot 12.6944 \cdot 1.0 \cdot (1 - 0.86))$
 $= 2383.982 \text{ mm}^2$

The area in the shell has not been calculated by user request.

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 40.0) \cdot (9.0 - 1.84) \cdot 0.8557$
 $= 489.858 \text{ mm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi - can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$
 $= 85.569 \text{ mm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lwp}, t_e)) \cdot f_{r4}$
 $= (370.4 - 203.2) \cdot 22.0 \cdot 1.0$
 $= 3678.400 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 4.8441 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 17.0873 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr_{16b}) = 17.0873 \text{ mm}$
Wall Thickness, shell/head, external pressure	$trb2 = 3.3197 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 10.1600 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[10.16, \max(17.0873, 4.5)]$
 $= 10.1600 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(4.8441, 10.16)$
 $= 10.1600 \text{ mm}$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	37.3,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	257.7 MPa	Passed
Occasional	:	10.6,	Allowable	:	156.9 MPa	Passed
Shear	:	19.3,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, $t_g = 19.0$, $t_r = 12.694$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.793$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-41 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: D

Govrn. thk, $t_g = 12.0$, $t_r = 1.844$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.205$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -18 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -63 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = 2.16/5.10 = 0.423

Weld Size Calculations, Description: T 8" #300

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	9.8980 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (2383.9819 - 0.0 + 2 \cdot 9.0 \cdot 0.8557 \cdot (1.0 \cdot 16.0 - 12.6944)) \cdot 138)$
 $= 335743.59 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (489.8585 + 3678.3999 + 85.5692 - 0.0 \cdot 0.86) \cdot 138$
 $= 586552.88 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (489.8585 + 0.0 + 85.5692 + (246.4395)) \cdot 138$
 $= 113325.84 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (489.8585 + 0.0 + 85.5692 + 3678.3999 + (246.4395)) \cdot 138$
 $= 620534.00 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 203.2 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 184538. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 440.0 \cdot 14.0 \cdot 0.49 \cdot 138$
 $= 653769. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 97.1) \cdot (12.0 - 3.0) \cdot 0.7 \cdot 118$
 $= 226754. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 203.2 \cdot 22.0 \cdot 0.74 \cdot 138$
 $= 717035. \text{ N}$



Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 203.2 \cdot (19.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 521102. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (653769 + 226754) = 880522 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (184538 + 717035 + 521102 + 0) = 1422674 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (653769 + 521102 + 0) = 1174870 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 880522 N , must exceed W = 335743 N or W1 = 586552 N
 Path 2-2 = 1422674 N , must exceed W = 335743 N or W2 = 113325 N
 Path 3-3 = 1174870 N , must exceed W = 335743 N or W3 = 620534 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 3.1593 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 322.1593 mm

Input Echo, WRC107/537 Item 1, Description: T 8" #300 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	3240.000	mm
Vessel Thickness	Tv	19.000	mm
Design Temperature	T1	125.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	137.90	MPa
Vessel Hot S.I. Allowable	Smh	137.90	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	203.200	mm
Nozzle Thickness	Tn	12.000	mm
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa
Thickness of Reinforcing Pad	Tpad	22.000	mm
Diameter of Reinforcing Pad	Dpad	440.000	mm
Design Internal Pressure	Dp	2.157	MPa
Include Pressure Thrust		No	

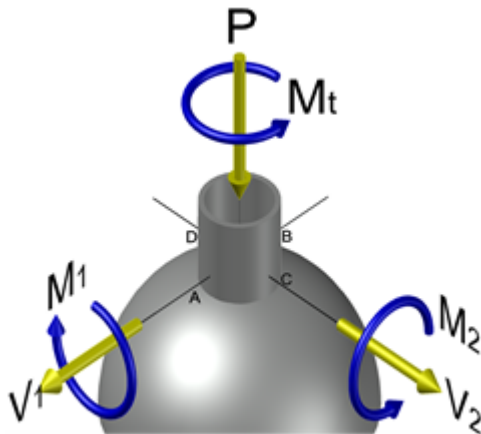
External Forces and Moments in WRC 107/537 Convention:

Radial Load	(SUS)	P	-11232.0	N
Longitudinal Shear	(SUS)	(Vl) V1	14040.0	N
Circumferential Shear	(SUS)	(Vc) V2	0.0	N
Circumferential Moment	(SUS)	(Mc) M1	7338234.0	N-mm
Longitudinal Moment	(SUS)	(Ml) M2	0.0	N-mm
Torsional Moment	(SUS)	Mt	6739197.5	N-mm

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 203.2 + 2 \cdot 1.65 \cdot \sqrt{(1631.0 (19.0 - 3.0))}$$

$$= 736.290 \text{ mm}$$

WRC 107 Stress Calculation for SUStained loads:

Radial Load		P	-11232.0	N
Circumferential Shear	(VC)	V2	0.0	N
Longitudinal Shear	(VL)	V1	14040.0	N
Circumferential Moment	(MC)	M1	7338234.0	N-mm
Longitudinal Moment	(ML)	M2	0.0	N-mm
Torsional Moment		MT	6739197.5	N-mm

Dimensionless Param: U = 0.41 TAU = 10.79 RHO = 4.00 (4.22)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 4	0.05518	(A,B,C,D)
$M(x) / P$	SP 4	0.02048	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.15392	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 4	0.06157	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.15392	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 4	0.06157	(A,B,C,D)
$N(y) * T / P$	SP 4	0.31795	(A,B,C,D)
$M(y) / P$	SP 4	0.09944	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.31988	(A,B,C,D)



M(y) * SQRT(Rm * T) / MC SM 4 0.34367 (A,B,C,D)
 N(y) * T * SQRT(Rm * T) / ML SM 4 0.31988 (A,B,C,D)
 M(y) * SQRT(Rm * T) / ML SM 4 0.34367 (A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
Rad. Bend. P		1.0	-1.0	1.0	-1.0	1.0	-1.0	1.0	-1.0
Rad. Memb. MC		0.0	0.0	0.0	0.0	-3.1	-3.1	3.1	3.1
Rad. Bend. MC		0.0	0.0	0.0	0.0	-7.5	7.5	7.5	-7.5
Rad. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Rad. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Rad. Str.		1.4	-0.5	1.4	-0.5	-9.3	3.9	12.0	-4.9
Tang. Memb. P		2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5
Tang. Bend. P		4.6	-4.6	4.6	-4.6	4.6	-4.6	4.6	-4.6
Tang. Memb. MC		0.0	0.0	0.0	0.0	-6.5	-6.5	6.5	6.5
Tang. Bend. MC		0.0	0.0	0.0	0.0	-41.9	41.9	41.9	-41.9
Tang. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tang. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.		7.1	-2.2	7.1	-2.2	-41.3	33.3	55.6	-37.6
Shear VC		0.0	0.0	-0.0	-0.0	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.2	-1.2	1.2	1.2
Shear MT		2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7
Tot. Shear		2.7	2.7	2.7	2.7	1.6	1.6	3.9	3.9
Str. Int.		8.2	5.7	8.2	5.7	41.4	33.3	55.9	38.1

Unitless Prm: U = 1.36 TAU = 0.00 (23.94) RHO = 0.00 (1.78)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.04353	(A,B,C,D)
M(x) / P	SR 2	0.02182	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.04610	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.03600	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.04610	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.03600	(A,B,C,D)
N(y) * T / P	SR 2	0.01309	(A,B,C,D)
M(y) / P	SR 2	0.00654	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.01386	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.01082	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.01386	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.01082	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		1.9	1.9	1.9	1.9	1.9	1.9	1.9	1.9
Rad. Bend. P		5.7	-5.7	5.7	-5.7	5.7	-5.7	5.7	-5.7
Rad. Memb. MC		0.0	0.0	0.0	0.0	-8.2	-8.2	8.2	8.2
Rad. Bend. MC		0.0	0.0	0.0	0.0	-38.3	38.3	38.3	-38.3



Rad. Memb. ML	-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Rad. Bend. ML	-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Rad. Str.	7.7	-3.8	7.7	-3.8	-38.8	26.3	54.2	-34.0
Tang. Memb. P	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Tang. Bend. P	1.7	-1.7	1.7	-1.7	1.7	-1.7	1.7	-1.7
Tang. Memb. MC	0.0	0.0	0.0	0.0	-2.5	-2.5	2.5	2.5
Tang. Bend. MC	0.0	0.0	0.0	0.0	-11.5	11.5	11.5	-11.5
Tang. Memb. ML	-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tang. Bend. ML	-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.	2.3	-1.1	2.3	-1.1	-11.7	7.9	16.3	-10.2
Shear VC	0.0	0.0	-0.0	-0.0	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-1.3	-1.3	1.3	1.3
Shear MT	1.4	1.4	1.4	1.4	1.4	1.4	1.4	1.4
Tot. Shear	1.4	1.4	1.4	1.4	0.1	0.1	2.7	2.7
Str. Int.	8.0	4.4	8.0	4.4	38.8	26.3	54.3	34.3

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		46.1	46.1	46.1	46.1	46.1	46.1	46.1	46.1
Rad. Pl (SUS)		0.4	0.4	0.4	0.4	-2.7	-2.7	3.6	3.6
Rad. Q (SUS)		1.0	-1.0	1.0	-1.0	-6.6	6.6	8.5	-8.5
Long. Pm (SUS)		46.1	46.1	46.1	46.1	46.1	46.1	46.1	46.1
Long. Pl (SUS)		2.5	2.5	2.5	2.5	-4.0	-4.0	9.0	9.0
Long. Q (SUS)		4.6	-4.6	4.6	-4.6	-37.3	37.3	46.6	-46.6
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.0	0.0	0.0	0.0	-1.2	-1.2	1.2	1.2
Shear Q (SUS)		2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7
Pm (SUS)		46.1	46.1	46.1	46.1	46.1	46.1	46.1	46.1
Pm+Pl (SUS)		48.6	48.6	48.6	48.6	44.1	44.1	55.3	55.3
Pm+Pl+Q (Total)		54.3	47.6	54.3	47.6	36.9	79.4	102.0	41.6

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	46.09	137.90	Passed
Pm+Pl (SUS)	55.31	206.85	Passed
Pm+Pl+Q (TOTAL)	101.99	413.70	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Rad. Pm (SUS)	109.4	109.4	109.4	109.4	109.4	109.4	109.4	109.4
Rad. Pl (SUS)	1.9	1.9	1.9	1.9	-6.3	-6.3	10.1	10.1
Rad. Q (SUS)	5.7	-5.7	5.7	-5.7	-32.6	32.6	44.1	-44.1
Long. Pm (SUS)	109.4	109.4	109.4	109.4	109.4	109.4	109.4	109.4
Long. Pl (SUS)	0.6	0.6	0.6	0.6	-1.9	-1.9	3.0	3.0
Long. Q (SUS)	1.7	-1.7	1.7	-1.7	-9.8	9.8	13.2	-13.2
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)	0.0	0.0	0.0	0.0	-1.3	-1.3	1.3	1.3
Shear Q (SUS)	1.4	1.4	1.4	1.4	1.4	1.4	1.4	1.4
Pm (SUS)	109.4	109.4	109.4	109.4	109.4	109.4	109.4	109.4
Pm+Pl (SUS)	111.3	111.3	111.3	111.3	107.9	107.9	119.7	119.7
Pm+Pl+Q (Total)	117.4	108.9	117.4	108.9	97.7	135.7	163.8	99.5

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	109.43	137.90	Passed
Pm+Pl (SUS)	119.74	206.85	Passed
Pm+Pl+Q (TOTAL)	163.77	413.70	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: V1 4" #300

From: 170

Pressure for Reinforcement Calculations	P	2.1575	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.90	MPa
Parent Allowable Stress At Ambient	Sva	137.90	MPa

Inside Diameter of Elliptical Head	D	1800.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	19.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	500.0000	mm
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User Entered Minimum Design Metal Temperature		-25.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		4.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105 +IT	
Flange Type		Weld Neck Flange	

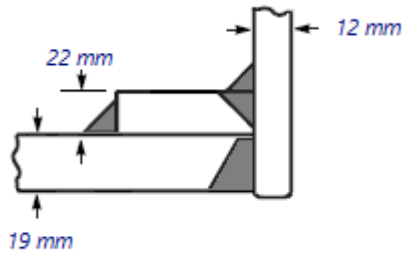
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	19.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	240.0000	mm
Thickness of Pad	te	22.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm
Reinforcing Pad Width		69.2000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: V1 4" #300

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.000 in.
Actual Thickness Used in Calculation	0.472 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 2.16 \cdot 1806.0 \cdot 0.8972 / (2 \cdot 137.9 \cdot 1.0 - 0.2 \cdot 2.16)$
 $= 12.6944 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2.16 \cdot 50.8 / (118 \cdot 1.0 + 0.4 \cdot 2.16)$
 $= 0.9221 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3552 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	176.2758 mm
Parallel to Vessel Wall, opening length	d	88.1379 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		40.0000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118.0/137.9)$
 $= 0.856$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118.0/137.9)$
 $= 0.856$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138.0/137.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.856, 1.0)$
 $= 0.856$

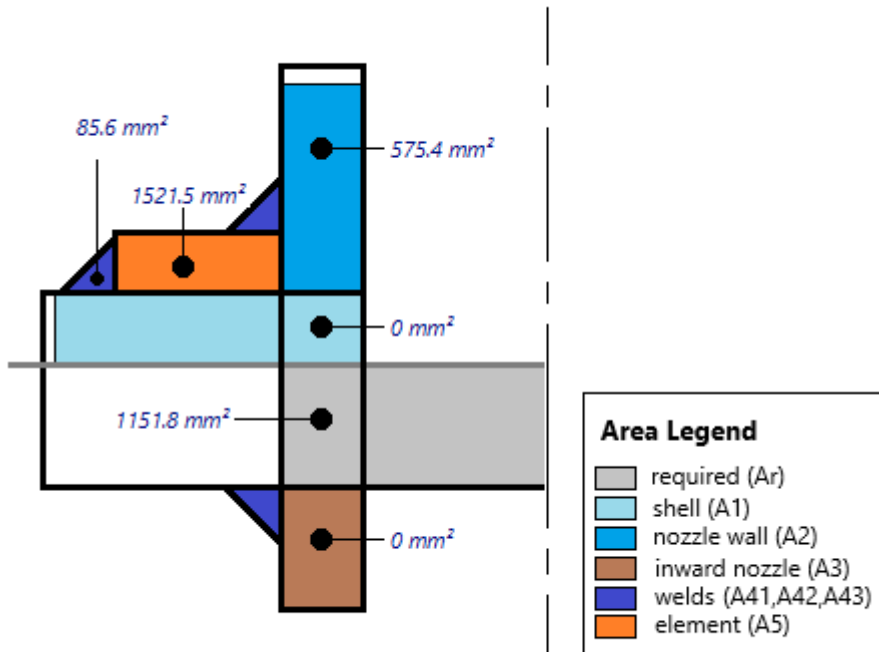


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1151.834	150.590	NA
Area in Shell	A1	0.000	0.000	NA
Area in Nozzle Wall	A2	575.392	615.768	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	85.569	85.569	NA
Area in Element	A5	1521.538	1521.538	NA
TOTAL AREA AVAILABLE	Atot	2182.499	2222.876	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 71.53 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 123.9124 22.0000 mm
 Based on given Pad Diameter: 240.0000 7.0976 mm
 Based on Shell or Nozzle Thickness: 148.0211 12.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (88.1379 \cdot 12.6944 \cdot 1.0 + 2 \cdot 9.0 \cdot 12.6944 \cdot 1.0 \cdot (1 - 0.86))$
 $= 1151.834 \text{ mm}^2$

The area in the shell has not been calculated by user request.

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 / \sin(\alpha3)$
 $= (2 \cdot 40.0) \cdot (9.0 - 0.92) \cdot 0.8557 / \sin(74.0)$
 $= 575.392 \text{ mm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi - can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$



$$= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$$

$$= 85.569 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{wp}, t_e)) \cdot f_{r4}$$

$$= (176.2758 - 107.1149) \cdot 22.0 \cdot 1.0$$

$$= 1521.538 \text{ mm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, $t_g = 19.0$, $t_r = 12.694$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.793$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-41 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D

Govrn. thk, $t_g = 12.0$, $t_r = 0.922$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.102$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -48 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-18 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-63 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $2.16/5.10 = 0.423$

Weld Size Calculations, Description: V1 4" #300

Intermediate Calc. for nozzle/shell Welds	T_{min}	9.0000 mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$9.8980 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - t_r)) S_v)$$

$$= \max(0, (1151.8342 - 0.0 + 2 \cdot 9.0 \cdot 0.8557 \cdot (1.0 \cdot 16.0 - 12.6944)) 138)$$

$$= 165844.86 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (575.3916 + 1521.5383 + 85.5692 - 0.0 \cdot 0.86) \cdot 138$$

$$= 300941.00 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (575.3916 + 0.0 + 85.5692 + (246.4395)) \cdot 138$$

$$= 125119.86 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (575.3916 + 0.0 + 85.5692 + 1521.5383 + (246.4395)) \cdot 138$$

$$= 334922.12 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 107.1149 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 97277. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 240.0 \cdot 14.0 \cdot 0.49 \cdot 138$$

$$= 356601. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 48.8132) \cdot (12.0 - 3.0) \cdot 0.7 \cdot 118$$

$$= 113992. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 107.1149 \cdot 22.0 \cdot 0.74 \cdot 138$$

$$= 377978. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 107.1149 \cdot (19.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 274694. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (356601 + 113992) = 470593 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (97277 + 377978 + 274694 + 0) = 749949 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (356601 + 274694 + 0) = 631295 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 470592 N	, must exceed W = 165844 N	or W1 = 300941 N
Path 2-2 = 749949 N	, must exceed W = 165844 N	or W2 = 125119 N
Path 3-3 = 631294 N	, must exceed W = 165844 N	or W3 = 334922 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: V1 4" #300

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 4.000 in.
 Actual Thickness Used in Calculation 0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 2.16 \cdot 1806.0 \cdot 0.8972 / (2 \cdot 137.9 \cdot 1.0 - 0.2 \cdot 2.16)$
 $= 12.6944 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2.16 \cdot 50.8 / (118 \cdot 1.0 + 0.4 \cdot 2.16)$
 $= 0.9221 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3552 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 167.2000 mm
 Parallel to Vessel Wall, opening length d 83.6000 mm
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 40.0000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

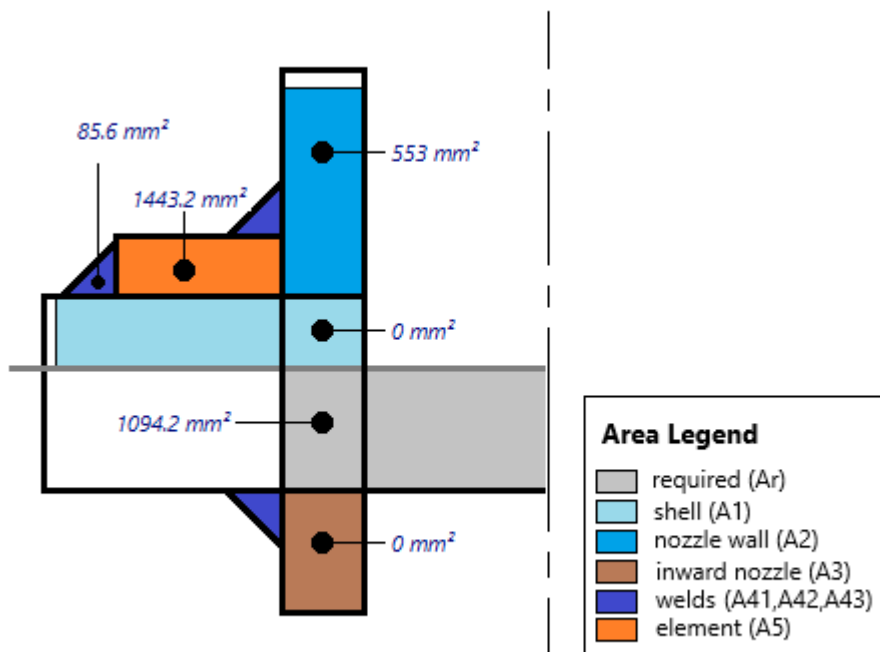


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1094.228	143.059	NA
Area in Shell	A1	0.000	0.000	NA
Area in Nozzle Wall	A2	552.979	591.783	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	85.569	85.569	NA



Area in Element	A5	1443.200	1443.200	NA
TOTAL AREA AVAILABLE	Atot	2081.748	2120.552	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 122.3127 22.0000 mm
 Based on given Pad Diameter: 240.0000 6.9464 mm
 Based on Shell or Nozzle Thickness: 139.5734 12.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \cdot UG-37(c)$
 $= (83.6 \cdot 12.6944 \cdot 1.0 + 2 \cdot 9.0 \cdot 12.6944 \cdot 1.0 \cdot (1 - 0.86))$
 $= 1094.228 \text{ mm}^2$

The area in the shell has not been calculated by user request.

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 40.0) \cdot (9.0 - 0.92) \cdot 0.8557$
 $= 552.979 \text{ mm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi - can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$
 $= 85.569 \text{ mm}^2$

Area Available in Element [A5]:
 $= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$
 $= (167.2 - 101.6) \cdot 22.0 \cdot 1.0$
 $= 1443.200 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 3.9221 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 17.0873 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 17.0873 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 3.3197 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 8.0200 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.02, \max(17.0873, 4.5)]$
 $= 8.0200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.9221, 8.02)$
 $= 8.0200 \text{ mm}$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 35.3,	Allowable	: 118.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 259.7 MPa	Passed
Occasional	: 4.5,	Allowable	: 156.9 MPa	Passed
Shear	: 23.0,	Allowable	: 82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.



Weld Size Calculations, Description: V1 4" #300

Intermediate Calc. for nozzle/shell Welds T_{min} 9.0000 mm
 Intermediate Calc. for pad/shell Welds T_{min}Pad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm
Pad Weld	9.5000 = 0.5*T _{min} Pad	9.8980 = 0.7 * W _p mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (1094.2283 - 0.0 + 2 \cdot 9.0 \cdot 0.8557 \cdot (1.0 \cdot 16.0 - 12.6944)) \cdot 138)$$

$$= 157901.69 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (552.9785 + 1443.1998 + 85.5692 - 0.0 \cdot 0.86) \cdot 138$$

$$= 287048.56 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (552.9785 + 0.0 + 85.5692 + (246.4395)) \cdot 138$$

$$= 122029.35 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (552.9785 + 0.0 + 85.5692 + 1443.1998 + (246.4395)) \cdot 138$$

$$= 321029.62 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 101.6 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 92269. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 240.0 \cdot 14.0 \cdot 0.49 \cdot 138$$

$$= 356601. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 46.3) \cdot (12.0 - 3.0) \cdot 0.7 \cdot 118$$

$$= 108123. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 101.6 \cdot 22.0 \cdot 0.74 \cdot 138$$

$$= 358517. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 101.6 \cdot (19.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 260551. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (356601 + 108123) = 464724 \text{ N}$$

Failure Path [Path 2-2]:



$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (92269 + 358517 + 260551 + 0) = 711337 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (356601 + 260551 + 0) = 617152 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 464723 N , must exceed W = 157901 N or W1 = 287048 N
 Path 2-2 = 711337 N , must exceed W = 157901 N or W2 = 122029 N
 Path 3-3 = 617151 N , must exceed W = 157901 N or W3 = 321029 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 17.2389 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 287.1829 mm

Warning:
The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: V1 4" #300 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	3240.000	mm
Vessel Thickness	Tv	19.000	mm
Design Temperature	T1	125.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	137.90	MPa
Vessel Hot S.I. Allowable	Smh	137.90	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

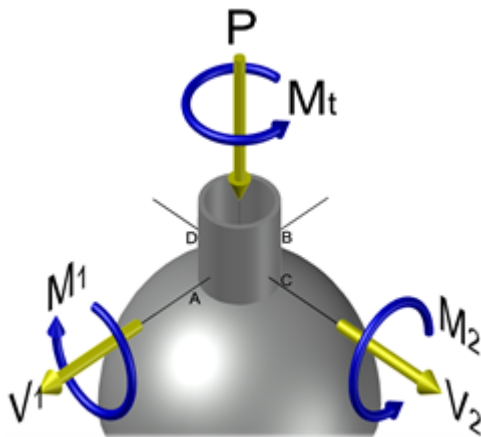
Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	101.600	mm
Nozzle Thickness	Tn	12.000	mm
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa
Thickness of Reinforcing Pad	Tpad	22.000	mm
Diameter of Reinforcing Pad	Dpad	240.000	mm
Design Internal Pressure	Dp	2.157	MPa
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-5616.0	N
Longitudinal Shear (SUS)	(V1) V1	7020.0	N
Circumferential Shear (SUS)	(Vc) V2	0.0	N
Circumferential Moment (SUS)	(Mc) M1	1834558.5	N-mm
Longitudinal Moment (SUS)	(Ml) M2	0.0	N-mm
Torsional Moment (SUS)	Mt	1684795.9	N-mm

Use Interactive Control No
 WRC107 Version Version March 1979
 Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(\text{t} - \text{ca}))}$$

$$= 101.6 + 2 \cdot 1.65 \cdot \sqrt{(1631.0 (19.0 - 3.0))}$$

$$= 634.690 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	-5616.0	N
Circumferential Shear	(VC) V2	0.0	N
Longitudinal Shear	(VL) V1	7020.0	N
Circumferential Moment	(MC) M1	1834558.5	N-mm
Longitudinal Moment	(ML) M2	0.0	N-mm
Torsional Moment	MT	1684795.9	N-mm

Dimensionless Param: U = 0.20 TAU = 5.14 RHO = 4.00 (4.22)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 4	0.08973	(A,B,C,D)
$M(x) / P$	SP 4	0.02924	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.30901	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 4	0.11493	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.30901	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 4	0.11493	(A,B,C,D)
$N(y) * T / P$	SP 4	0.34303	(A,B,C,D)
$M(y) / P$	SP 4	0.19278	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.10451	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 4	0.85378	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.10451	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 4	0.85378	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)



Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
Rad. Bend. P		0.7	-0.7	0.7	-0.7	0.7	-0.7	0.7	-0.7
Rad. Memb. MC		0.0	0.0	0.0	0.0	-1.6	-1.6	1.6	1.6
Rad. Bend. MC		0.0	0.0	0.0	0.0	-3.5	3.5	3.5	-3.5
Rad. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Rad. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Rad. Str.		1.0	-0.3	1.0	-0.3	-4.0	1.6	6.1	-2.3
Tang. Memb. P		1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3
Tang. Bend. P		4.5	-4.5	4.5	-4.5	4.5	-4.5	4.5	-4.5
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Tang. Bend. MC		0.0	0.0	0.0	0.0	-26.0	26.0	26.0	-26.0
Tang. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tang. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.		5.8	-3.2	5.8	-3.2	-20.7	22.4	32.4	-28.7
Shear VC		0.0	0.0	-0.0	-0.0	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.2	-1.2	1.2	1.2
Shear MT		2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7
Tot. Shear		2.7	2.7	2.7	2.7	1.6	1.6	3.9	3.9
Str. Int.		7.3	6.2	7.3	6.2	20.9	22.5	33.0	29.2

Unitless Prm: U = 0.74 TAU = 0.00 (12.83) RHO = 0.00 (1.78)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.09337	(A,B,C,D)
M(x) / P	SR 2	0.05143	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.10817	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.12946	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.10817	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.12946	(A,B,C,D)
N(y) * T / P	SR 2	0.02802	(A,B,C,D)
M(y) / P	SR 2	0.01591	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.03269	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.03871	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.03269	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.03871	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Rad. Bend. P		6.8	-6.8	6.8	-6.8	6.8	-6.8	6.8	-6.8
Rad. Memb. MC		0.0	0.0	0.0	0.0	-4.8	-4.8	4.8	4.8
Rad. Bend. MC		0.0	0.0	0.0	0.0	-34.4	34.4	34.4	-34.4
Rad. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Rad. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Rad. Str.		8.8	-4.7	8.8	-4.7	-30.4	24.9	48.1	-34.4
Tang. Memb. P		0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Tang. Bend. P		2.1	-2.1	2.1	-2.1	2.1	-2.1	2.1	-2.1
Tang. Memb. MC		0.0	0.0	0.0	0.0	-1.4	-1.4	1.4	1.4



Tang. Bend. MC	0.0	0.0	0.0	0.0	-10.3	10.3	10.3	-10.3
Tang. Memb. ML	-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tang. Bend. ML	-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.	2.7	-1.5	2.7	-1.5	-9.0	7.4	14.5	-10.3
Shear VC	0.0	0.0	-0.0	-0.0	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-1.2	-1.2	1.2	1.2
Shear MT	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
Tot. Shear	1.2	1.2	1.2	1.2	-0.0	-0.0	2.3	2.3
Str. Int.	9.0	5.1	9.0	5.1	30.4	24.9	48.2	34.6

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		46.1	46.1	46.1	46.1	46.1	46.1	46.1	46.1
Rad. Pl (SUS)		0.3	0.3	0.3	0.3	-1.2	-1.2	1.9	1.9
Rad. Q (SUS)		0.7	-0.7	0.7	-0.7	-2.8	2.8	4.2	-4.2
Long. Pm (SUS)		46.1	46.1	46.1	46.1	46.1	46.1	46.1	46.1
Long. Pl (SUS)		1.3	1.3	1.3	1.3	0.8	0.8	1.9	1.9
Long. Q (SUS)		4.5	-4.5	4.5	-4.5	-21.5	21.5	30.5	-30.5
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.0	0.0	0.0	0.0	-1.2	-1.2	1.2	1.2
Shear Q (SUS)		2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7
Pm (SUS)		46.1	46.1	46.1	46.1	46.1	46.1	46.1	46.1
Pm+Pl (SUS)		47.4	47.4	47.4	47.4	47.4	47.4	49.1	49.1
Pm+Pl+Q (Total)		53.2	47.4	53.2	47.4	42.2	68.6	79.1	44.4

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	46.09	137.90	Passed
Pm+Pl (SUS)	49.14	206.85	Passed
Pm+Pl+Q (TOTAL)	79.06	413.70	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		109.4	109.4	109.4	109.4	109.4	109.4	109.4	109.4
Rad. Pl (SUS)		2.0	2.0	2.0	2.0	-2.7	-2.7	6.8	6.8
Rad. Q (SUS)		6.8	-6.8	6.8	-6.8	-27.7	27.7	41.2	-41.2
Long. Pm (SUS)		109.4	109.4	109.4	109.4	109.4	109.4	109.4	109.4
Long. Pl (SUS)		0.6	0.6	0.6	0.6	-0.8	-0.8	2.1	2.1
Long. Q (SUS)		2.1	-2.1	2.1	-2.1	-8.2	8.2	12.4	-12.4



Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)	0.0	0.0	0.0	0.0	-1.2	-1.2	1.2	1.2
Shear Q (SUS)	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
Pm (SUS)	109.4	109.4	109.4	109.4	109.4	109.4	109.4	109.4
Pm+Pl (SUS)	111.5	111.5	111.5	111.5	109.1	109.1	116.5	116.5
Pm+Pl+Q (Total)	118.5	108.3	118.5	108.3	100.4	134.4	157.7	99.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	109.43	137.90	Passed
Pm+Pl (SUS)	116.54	206.85	Passed
Pm+Pl+Q (TOTAL)	157.65	413.70	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: NV2 2" #300

Parent: V1 4" #300

Pressure for Reinforcement Calculations	P	2.1575	MPa
Temperature for Internal Pressure	Temp	125	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	190	°C

Parent Material		SA-106 B	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	77.60	mm
Design Length of Section	L	250.0000	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

User Entered Minimum Design Metal Temperature		-25.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		80	

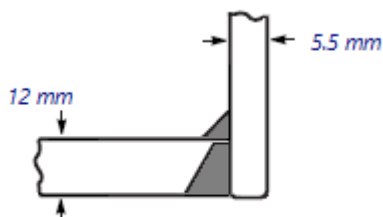
Flange Material		SA-105 +IT	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: NV2 2" #300

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.375	in.
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Actual Thickness Used in Calculation 0.218 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.16 \cdot 41.8 / (118 \cdot 1.0 - 0.6 \cdot 2.16)$
= 0.7727 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $2.16 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 2.16)$
= 0.5475 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2621 mm

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	110.5012	mm
Parallel to Vessel Wall, opening length	d	55.2506	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	6.3430	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: NV2 2" #300.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.5475	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 3.7727	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 4.5000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[6.42, max(4.5, 4.5)]
= 4.5000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.5475, 4.5)
= 4.5000 mm

Available Nozzle Neck Thickness = $0.875 \cdot 5.537 = 4.845$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-18 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-63 °C



Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = 2.16/5.10 = 0.423

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 82.8759 mm
 Parallel to Vessel Wall, opening length d 41.4380 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 6.3430 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: NV2 2" #300.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (mm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	28.464	28.464
Area Available in Shell	A1	0.000	0.000
Area Available in Nozzle Wall	A2	25.242	25.242
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	86.626	86.626
Area Available in Pad	A5	0.000	0.000
Area Available in Hub	A6	0.000	0.000
TOTAL AREA AVAILABLE	Atot	111.868	111.868

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements
 of App. 1-7(b). Therefore, the Membrane and Bending stress checks
 are not required.

Weld Size Calculations, Description: NV2 2" #300

Intermediate Calc. for nozzle/shell Welds Tmin 2.5372 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 1.7760 = 0.7 * tmin. 7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 14.3938 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 276.3938 mm

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: Unity Check for Rigging Exceeded 1.0!**

**** Error: Skirt Overstress Failure!**

Please review all reports carefully!

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PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: